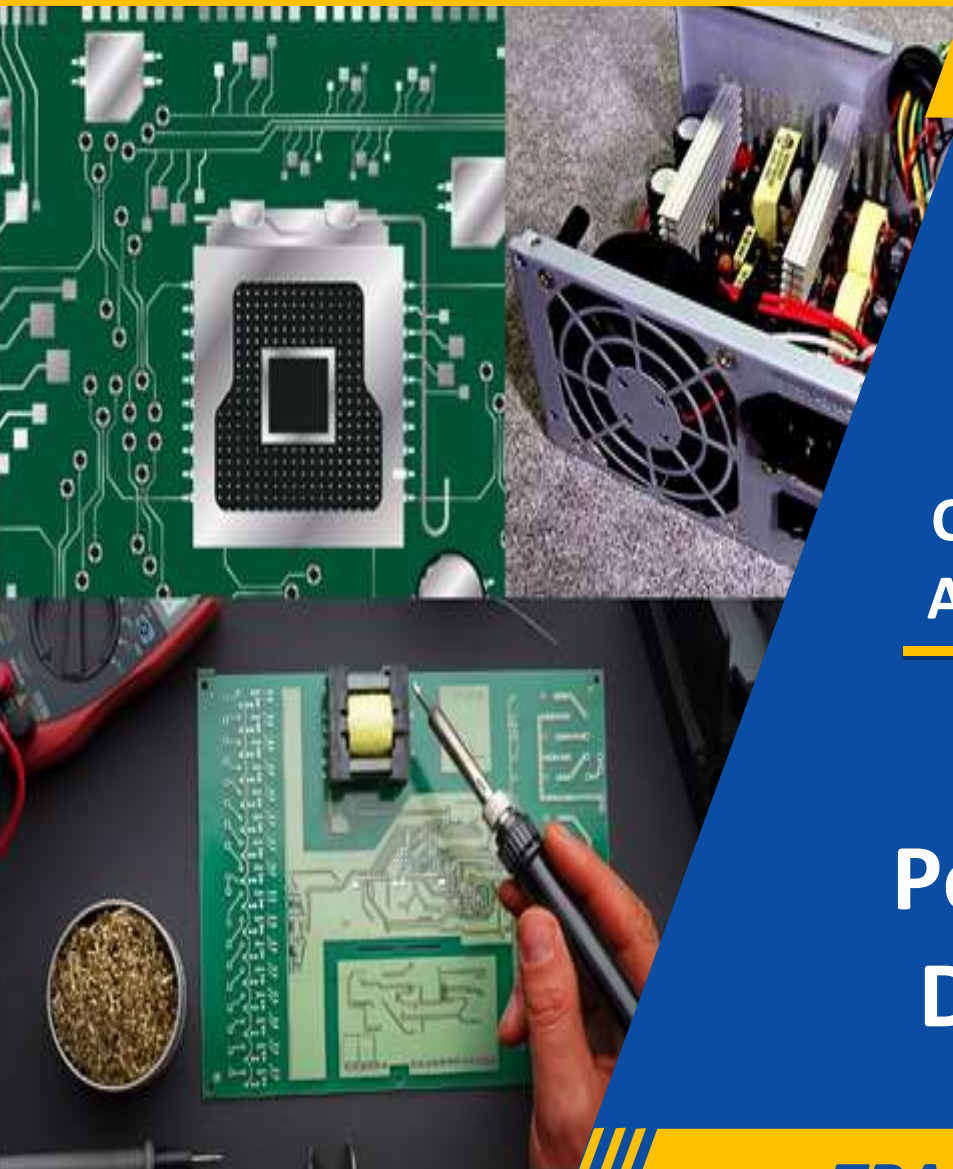




RQF LEVEL 5



CSAPS501

**COMPUTER SYSTEM
AND ARCHITECTURE**

**Computer
Power System
Development**

TRAINEE'S MANUAL

October, 2024



COMPUTER POWER SYSTEM DEVELOPMENT



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ACRONYMS

AC:	Alternating current
ACPI:	Advanced Configuration and Power Interface
ANSI:	American National Standards Institute
API:	Application Programming Interface
ASIC:	Application-Specific Integrated Circuit
ATS:	Automatic Transfer Switch
ATX:	Advanced Technology Extended
AVFS:	Adaptive Voltage and Frequency Scaling
AVR:	Automatic Voltage Regulator
BMS:	Battery Management System
CPU:	Central Processing Unit
DC:	Direct current
DVFS:	Dynamic Voltage and Frequency Scaling
ESD:	Electro-Static Discharge
GPU:	Graphical Processing Unit
IEEE:	Institute of Electrical and Electronics Engineers
IoT:	Internet of Things
KPI:	Key Performance Indicator
Li-ion:	Lithium-ion (batteries)
PC:	Personal Computer
PCB:	Printed Circuit Board
PFC:	Power Factor Correction
PSU:	Power Supply Unit
PWM:	Pulse Width Modulation
RAM:	Random Access Memory

RGB: Read Green Blue

RMS: Root Mean Square

ROM: Read Only Memory

RTB: Rwanda TVET Board

SFX: Small Form Factor

SMD: Surface Mount Device

SMPS: Switched Mode Power Supply

SNMP: Simple Network Management Protocol

SOC: State of Charge

SOF: State of Function

SOH: State of Health

TFX: Thin Form Factor Extended

TIA: Telecommunications Industry Association

TQUM Project: TVET Quality Management Project

UPS: Uninterruptible Power Supply

VCC: Voltage Common Collector (used in the context of power supply)

VRM: Voltage Regulator Module

INTRODUCTION

This trainee's manual includes all the knowledge and skills required in Computer System and Architecture specifically for the module of "**Computer Power System Development**". Trainees enrolled in this module will engage in practical activities designed to develop and enhance their competencies.

The development of this training manual followed the Competency-Based Training and Assessment (CBT/A) approach, offering ample practical opportunities that mirror real-life situations.

The trainee's manual is organized into Learning Outcomes, which is broken down into indicative content that includes both theoretical and practical activities. It provides detailed information on the key competencies required for each learning outcome, along with the objectives to be achieved.

As a trainee, you will start by addressing questions related to the activities, which are designed to foster critical thinking and guide you towards practical applications in the labor market. The manual also provides essential information, including learning hours, required materials, and key tasks to complete throughout the learning process.

All activities included in this training manual are designed to facilitate both individual and group work. After completing the activities, you will conduct a formative assessment, referred to as the end learning outcome assessment. Ensure that you thoroughly review the key readings and the 'Points to Remember' section.

MODULE CODE AND TITLE: CSAPS501 COMPUTER POWER SYSTEM DEVELOPMENT

Learning Outcome 1: Prepare the working environment.

Learning Outcome 2: Implement power electronic converters.

Learning Outcome 3: Implement a power supply unit (PSU) of a computer system.

Learning Outcome 4: Implement computer power management.

Learning Outcome 1: Prepare the Working Environment



Indicative contents

1.1 Introduction to computer power system

1.2 Analysing computer system power requirement

1.3 Calculation of system power Consumption

1.4 Selection of tools, materials and equipment

Key Competencies for Learning Outcome 1: Prepare the Working Environment

Knowledge	Skills	Attitudes
• Definition of key concepts • Description of computer power system • Description of computer system parts 'power consumption • Identification of tools, materials and equipment used in computer power system.	• Drawing a computer power system architecture. • Analysing power supply unit features • Calculating computer power system requirements. • Selecting tools, materials and equipment.	• Having attention to details. • Being rapid. • Having team work spirit.



Duration: 15 hrs

Learning outcome 1 objectives:



By the end of the learning outcome, the trainees will be able to:

1. Define correctly key concepts related to computer power system.
2. Describe clearly computer power system according to their types.
3. Draw correctly computer power system architecture according to standards.
4. Calculate accurately computer power consumption according to components power rating.
5. Select properly tools, materials and equipment according to work to be done.



Resources

Equipment	Tools	Materials
• PPEs • Oscilloscope • Multi-meter • Function generator • SMD rework station • ESD workstation	• Universal plier • Tweezers • Flashlight or headlamp • Cable testers • ESD Tools applicator • Wire cutter/striper Pliers • Soldering iron • Di-soldering pump • Glue gun	• Isopropyl alcohol and cotton swabs • Electrical tapes and zip ties • Spare screws • Labelling materials • Connectors • Foam cleaner • Battery/cells • Jumper wires • Breadboard • PCB • Power cables • Glue stick • Drawing materials (papers and pencils)



Indicative content 1.1: Introduction to Computer Power System



Duration: 3 hrs



Theoretical Activity 1.1.1: Description of computer power system concepts



Tasks:

1: Answer the following questions related to description of computer power system concepts.

- i. What do you understand by the following terms?
 - a) Uninterruptable Power Supply
 - b) Power management
 - c) Ripple voltage and Current
 - d) Harmonic Distortion
 - e) Battery Management System (BMS)
 - f) Power converter
- ii. What are the types of computer power supply?
- iii. Identify the features of computer power supply.
- iv. Give the advantages and disadvantages of computer power supply.
- v. Discuss on the working principle of computer power system

2: Write the findings/answers on papers or flipchart.

3: Present your findings to the class.

4: Ask questions for more clarifications.

5: Read the key reading 1.1.1 from your manual to get more clarifications.



Key readings 1.1.1.:

Description of computer power system

1. Definition of key concepts

- a) **A computer power system** is a broader term that encompasses all the components involved in delivering electrical power to a computer. This includes:
- **Power Supply:** The primary component that converts AC power from the wall outlet to DC power suitable for the computer's internal components.
 - **Cables:** Connect the power supply to the motherboard and other components.
 - **Circuitry:** The internal circuitry within the computer that distributes power to various components.
 - **Power Management:** The system's ability to regulate power consumption and efficiency
- b) **Uninterruptible Power Supply (UPS):** A UPS provides a continuous, uninterrupted power supply to critical loads during power outages or voltage fluctuations. It typically consists of a battery bank and an inverter.
- c) **Power Management:** This refers to the efficient and effective use of electrical power. It involves techniques like power saving modes, dynamic frequency scaling, and voltage scaling to reduce energy consumption.
- d) **Efficiency:** The efficiency of a power system is the ratio of output power to input power. A higher efficiency means less energy is wasted as heat.
- e) **Ripple Voltage and Current:** These are variations in the voltage or current levels of a power supply. Excessive ripple can cause instability and damage to sensitive components.
- f) **Harmonic Distortion:** This occurs when non-sinusoidal currents or voltages are introduced into a power system. Harmonics can cause overheating, reduced efficiency, and interference with other equipment.
- g) **Battery Management System (BMS):** A BMS monitors and controls the battery bank in a UPS or other battery-powered system. It ensures safe and efficient operation by balancing cells, preventing overcharging or over-discharging, and detecting faults.
- h) **Power Converter:** A power converter is a device that changes the voltage, current, or frequency of electrical power. They are essential components in power systems, allowing for compatibility between different devices and power sources.

2. Types of computer power supply

Computer power supplies can be classified based on several factors, including their form factor, efficiency rating, output wattage, and type of power regulation. Here's a breakdown of the different classifications:

a) **Form Factor**

- **ATX (Advanced Technology Extended):** The most common form factor for desktop computers. Standard size and mounting specifications.
- **SFX (Small Form Factor):** A smaller version of ATX, used in compact builds or mini-ATX systems.
- **TFX (Thin Form Factor):** Used in slim desktop cases, often in home theater PCs.
- **Flex ATX:** Used in ultra-compact systems where space is critical.
- **Server PSUs:** Custom power supplies designed for server cases, often much larger or built to fit specific server enclosures.

b) Efficiency Rating (80 PLUS Certification)

Power supplies are rated based on their efficiency under typical load conditions. The higher the certification, the less power is wasted as heat.

- **80 PLUS:** Standard efficiency certification.
- **80 PLUS Bronze:** 82% efficiency at 50% load.
- **80 PLUS Silver:** 85% efficiency at 50% load.
- **80 PLUS Gold:** 87% efficiency at 50% load.
- **80 PLUS Platinum:** 90% efficiency at 50% load.
- **80 PLUS Titanium:** 94% efficiency at 50% load.

c) Wattage Output

- Power supplies are classified based on their maximum wattage output, which determines how much power they can provide to the computer components.
 - **Low Wattage (300-500W):** Suitable for basic or office desktops with integrated graphics and low power requirements.
 - **Mid-Range Wattage (500-750W):** Ideal for standard desktop PCs with mid-range graphics cards and multiple peripherals.
 - **High Wattage (750-1200W):** Needed for gaming PCs, workstations, or systems with multiple GPUs.
 - **Extreme Wattage (1200W and above):** Required for extreme gaming setups, server racks, or heavily overclocked systems.

d) Power Regulation (Design)

- **Non-Modular:** All cables are permanently attached to the power supply unit.
- **Semi-Modular:** Some cables are permanently attached, while others can be connected as needed.
- **Fully Modular:** All cables are detachable, allowing for better cable management and airflow.

e) Cooling Method

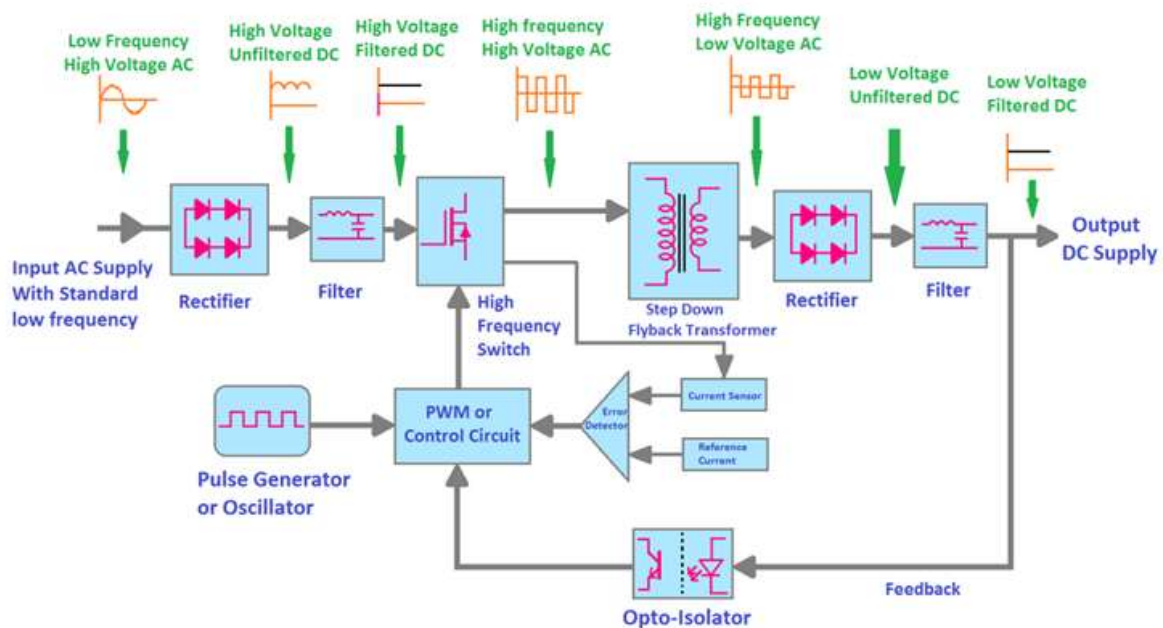
- **Fan-Cooled:** Uses a fan to dissipate heat, common in most power supplies.
- **Fanless (Passive Cooling):** Designed for silent operation, suitable for low-power systems that don't generate much heat.

f) Specialized Features

- **Redundant Power Supplies:** Often used in servers where continuous uptime is critical. These PSUs have a backup module that takes over if the main one fails.
- **Uninterruptible Power Supply (UPS):** A device that provides battery backup to protect against power outages.

3. Block diagram

A Switched-Mode Power Supply (SMPS) block diagram typically consists of several key components that work together to convert electrical power efficiently. The diagram starts with an AC input from the mains or DC input from a battery, followed by an EMI filter to reduce electromagnetic interference. The input is then passed through a rectifier and filter circuit, which converts the AC input to DC. This DC voltage is supplied to a high-frequency switching transistor or MOSFET, controlled by a PWM controller that regulates the switching frequency. The transistor chops the DC voltage into a high-frequency AC signal, which is passed through a transformer to step up or step down the voltage as required. The output from the transformer is rectified again using diodes and filtered with capacitors to produce a stable DC output. Additionally, a feedback circuit monitors the output voltage and sends signals back to the PWM controller to adjust the switching duty cycle, ensuring accurate voltage regulation. An isolator may also be included for safety and noise reduction, separating the input and output stages.



A Switched-Mode Power Supply (SMPS) block diagram typically consists of several key components that work together to convert electrical power efficiently. The diagram starts with an AC input from the mains or DC input from a battery, followed by an EMI filter to reduce electromagnetic interference. The input is then passed through a rectifier and filter circuit, which converts the AC input to DC. This DC voltage is supplied to a high-frequency switching transistor or MOSFET, controlled by a PWM controller that regulates the switching frequency. The transistor chops the DC voltage into a high-frequency AC

signal, which is passed through a transformer to step up or step down the voltage as required. The output from the transformer is rectified again using diodes and filtered with capacitors to produce a stable DC output. Additionally, a feedback circuit monitors the output voltage and sends signals back to the PWM controller to adjust the switching duty cycle, ensuring accurate voltage regulation. An isolator may also be included for safety and noise reduction, separating the input and output stages.

4. Features of computer power supply.

Here are the main features of a computer power supply:

a) Wattage (Power Output)

- **Wattage Rating:** Indicates the total power the PSU can supply to the components. Common wattages range from 300W to 1200W or more, depending on the needs of the system.

b) Efficiency Rating

- **80 PLUS Certification:** A PSU's efficiency is how much of the AC power it converts to usable DC power. An 80 PLUS certification guarantees a minimum efficiency of 80%, with higher levels (Bronze, Silver, Gold, Platinum, Titanium) offering even better efficiency.

c) Form Factor

- **ATX, SFX, TFX, etc.:** Specifies the physical dimensions and mounting configurations of the PSU, such as ATX (the most common for desktop PCs), SFX for smaller builds, and TFX for compact designs.

d) Connectors

- **24-Pin ATX Connector:** Powers the motherboard.
- **4+4 Pin CPU Connector:** Powers the CPU.
- **PCIe Connectors (6-pin/8-pin):** Powers high-end graphics cards.
- **SATA Connectors:** Powers storage devices like SSDs and HDDs.
- **Molex Connectors:** Older standard used for powering fans, optical drives, etc.

e) Modular Design

- **Non-Modular:** All cables are permanently attached to the PSU.
- **Semi-Modular:** Some cables are attached, while others can be added/removed.
- **Fully Modular:** All cables can be attached or detached based on the system's needs, offering better cable management.

f) Cooling System

- **Fan Size and Type:** Most PSUs come with a fan to dissipate heat. The fan size can vary (e.g., 120mm, 140mm), with larger fans generally being quieter.
- **Fanless Design:** Some PSUs are fanless for silent operation, typically found in lower wattage or highly efficient models.

g) Protection Features

- **Overvoltage Protection (OVP):** Prevents too much voltage from damaging components.
- **Overcurrent Protection (OCP):** Prevents excessive current from damaging components.
- **Short Circuit Protection (SCP):** Protects the PSU and other components if a short circuit occurs.
- **Over Power Protection (OPP):** Shuts down the PSU if too much power is drawn.
- **Over Temperature Protection (OTP):** Shuts down the PSU if it overheats.

h) Input Voltage Range

- **Universal Input:** Many PSUs can automatically adjust to different input voltages (100V-240V), making them suitable for use in different regions without requiring manual adjustment.

i) Active Power Factor Correction (PFC)

- **PFC** improves the efficiency of the power supply by reducing wasted energy and minimizing interference with other electronic devices.

j) RGB Lighting

- Some modern PSUs feature RGB lighting, which can be controlled for aesthetic purposes, especially in gaming PCs.

k) Expansion and upgrade

- Expansion and upgrades to a computer power system typically involve increasing its capacity or capabilities to meet growing demands or accommodate new hardware.

l) Compatibility

- Compatibility in a computer power system refers to the ability of the power supply unit (PSU) to work effectively with the other components in the system.

m) Environmental impact

- The environmental impact of a computer power supply includes energy inefficiency, leading to increased electricity consumption and carbon emissions, along with resource-intensive manufacturing and hazardous e-waste disposal. Inefficient models generate excess heat, requiring more cooling and energy. Opting for energy-efficient and recyclable power supplies helps mitigate these effects.

n) Safety margin

- The safety margin of a computer power supply refers to the extra capacity built into the unit to handle peak power demands without overloading. It ensures reliable performance, preventing overheating, voltage fluctuations, and hardware damage. Choosing a power supply with an adequate safety margin enhances system stability and longevity.

o) Budget and cost consideration

- When considering the budget and cost of a computer power supply, it's important to balance price with efficiency and reliability. Cheaper power supplies may lack durability and energy-saving features, leading to higher long-term electricity costs and potential hardware damage. Investing in a quality, energy-efficient model can save money in the long run through reduced energy consumption and extended system lifespan.

Each of these features can vary depending on the specific PSU model and its intended use (e.g., for basic desktop systems, gaming rigs, or servers).

4. Advantages and disadvantages of computer power supply.

a) Advantages of computer power supply

1. Reliable Power Conversion

- A PSU efficiently converts high-voltage AC from the wall socket to low-voltage DC power, which is safe for the computer's internal components (CPU, GPU, motherboard, etc.).

2. Component Protection

- Many PSUs come with safety features like **Overvoltage Protection (OVP)**, **Overcurrent Protection (OCP)**, **Short Circuit Protection (SCP)**, and **Over Temperature Protection (OTP)** to safeguard the components from damage due to power surges, overheating, or short circuits.

3. Modular Cabling (for modular and semi-modular PSUs)

- **Modular PSUs** allow users to only connect the cables they need, improving airflow and reducing cable clutter. This feature is especially useful in custom PC builds where neat cable management is desired.

4. High Efficiency

- Modern PSUs, especially those with **80 PLUS certification**, are highly efficient, meaning they waste less power as heat and deliver more energy to the system, lowering electricity costs and reducing environmental impact.

5. Wide Range of Wattage

- PSUs come in various wattage options, from 300W to over 1200W, ensuring that systems with different power needs (from basic office PCs to high-end gaming rigs) can be adequately powered.

6. Customizability

- Some high-end PSUs offer **RGB lighting** and are compatible with software controls for gaming setups, adding aesthetic value to custom PC builds.

7. Universal Voltage Support

- Many PSUs feature **Active PFC (Power Factor Correction)** and can automatically adjust to different input voltages (100V-240V), making them versatile for use worldwide without manual switching.

b) Disadvantages of computer power supply

1. Cost (for higher-end models)

- High-efficiency PSUs, especially those with **80 PLUS Gold, Platinum, or Titanium certifications**, tend to be more expensive. Similarly, **modular** and **RGB** models typically cost more than non-modular and basic units.

2. Size and Form Factor Limitations

- Larger, higher-wattage PSUs (like **ATX models**) may not fit in smaller PC cases. Specialized builds (e.g., compact or mini PCs) may require **SFX** or other small form factor PSUs, limiting options and potentially increasing cost.

3. Efficiency Variability

- Lower-end or non-certified PSUs may have poor energy efficiency, wasting more electricity as heat, which not only increases energy costs but also increases system heat, requiring better cooling.

4. Potential for Noise

- Some PSUs, particularly those with smaller fans or cheaper designs, can generate significant noise, especially under heavy loads. This can be bothersome in quiet environments.

5. Complexity of Selection

- Selecting the correct PSU for a system can be challenging, especially for new builders. Choosing a PSU with inadequate wattage or missing connectors can lead to system instability or even component failure.

6. Limited Upgradability

- If a PSU has insufficient wattage or connectors, upgrading components (like adding more powerful GPUs or additional hard drives) may require a PSU replacement, adding to the overall cost.

7. Energy Loss at Low Loads

- Some PSUs are not very efficient at low loads, leading to wasted energy and heat production when the computer is not fully utilizing the PSU's capacity.

8. Cable Clutter (for non-modular PSUs)

- **Non-modular PSUs** come with all cables permanently attached, which can result in unused cables cluttering the inside of the case, reducing airflow and making cable management difficult.

5. Working principle of computer power system

A computer power system, often referred to as a Power Supply Unit (PSU), is responsible for converting the incoming AC (alternating current) power from the electrical outlet into the DC (direct current) voltages required by various components within the computer.

a) Power conversion

- **Transformer:** This component reduces the high voltage from the electrical outlet to a lower voltage suitable for the computer's components.
- **Rectifier:** Converts the AC voltage into a DC voltage. This is typically done using diodes.
- **Filter:** Smooths out the pulsating DC voltage produced by the rectifier, creating a more stable DC output.
- **Voltage Regulators:** These circuits ensure that the output voltage remains constant, even as the load (demand for power) changes. This is crucial to prevent damage to computer components.

b) Power distribution

- **Multiple Output Voltages:** A PSU typically provides multiple DC voltages, each tailored to the specific requirements of different components. These voltages might include:
 - +3.3V: Used for low-power components like memory and input/output devices.
 - +5V: Used for various components, including the motherboard and some peripherals.
 - +12V: Used for high-power components like the CPU and graphics card.
 - -12V: Used for specific components, such as serial ports.

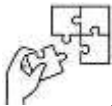


Points to Remember

- An Uninterruptible Power Supply (UPS) is a device that provides backup power to a system when the primary power source fails or experiences significant voltage fluctuations.

- Power management refers to the techniques and technologies used to optimize the use of electrical power in systems and devices.
- Efficiency in electrical and power systems is the ratio of useful power output to the total power input.
- Ripple voltage is the residual periodic variation of the DC voltage within a power supply.
- Ripple current is the alternating component of current in a DC power supply or circuit, often caused by the same factors.
- Harmonic distortion refers to the deviation of a voltage or current waveform from its ideal sinusoidal shape due to the presence of harmonics
- A Battery Management System (BMS) is an electronic system that manages a rechargeable battery.
- A Power Converter is a device that converts electrical energy from one form to another.
- **Types of computer power supply**
Computer power supplies can be classified based on several factors, including their form factor, efficiency rating, output wattage, and type of power regulation.
- Features of computer power supply are:
 - ✓ Wattage
 - ✓ Efficiency ratings
 - ✓ Form factor
 - ✓ Cooling system
 - ✓ Modular design
 - ✓ Protection features
 - ✓ Power factor correction
 - ✓ Input voltage range
 - ✓ RGB lighting
- Advantages of computer power supply are:
 - ✓ Reliable Power Conversion
 - ✓ Component Protection
 - ✓ Modular Cabling (for modular and semi-modular PSUs)
 - ✓ High Efficiency
 - ✓ Wide Range of Wattage
 - ✓ Customizability
 - ✓ Universal Voltage Support
- Disadvantages of computer power supply are:
 - ✓ Cost (for higher-end models)
 - ✓ Size and Form Factor Limitations
 - ✓ Efficiency Variability
 - ✓ Potential for Noise
 - ✓ Complexity of Selection

- ✓ Limited Upgradability
- ✓ Energy Loss at Low Loads
- ✓ Cable Clutter (for non-modular PSUs)
- Working principle of computer power system
A computer power system, often referred to as a Power Supply Unit (PSU), is responsible for converting the incoming AC (alternating current) power from the electrical outlet into the DC (direct current) voltages required and distribute it to various components within the computer.



Application of learning 1.1.

A friend needs a new gaming PC for his children at home because the older one is consuming too much power and lately it produced smoke, the friend asked for the considerations for a new power supply replacement. As the technician, advise your friend about the features of a good computer power supply.



Indicative content 1.2: Analysing Computer Power System Requirement



Duration: 5 hrs



Theoretical Activity 1.2.1: Identification of computer system parts' power consumption

Tasks:

- 1: Answer the following questions related to description of computer power system.
 - i. Identify computer system parts power consumption
 - ii. Discuss the following computer power system mode:
 - a) Idle mode.
 - b) Sleep mode.
 - c) Maximum load power rating mode
- 2: Write the findings/answers on papers or flipchart.
- 3: Present your findings to the class.
- 4: Ask questions for more clarifications.
- 5: Read the key reading 1.2.1 from your manual to get more clarifications.



Key readings 1.2.1:

Analysing computer system power requirement

1. Identification of computer system parts' power consumption

a) Computer system parts

Understanding the power consumption of individual computer components is crucial for optimizing energy efficiency and system performance. Here's a breakdown of common parts:

- Central Processing Unit (CPU)
- Graphics Processing Unit (GPU)
- Motherboard
- RAM
- Storage
- Power Supply
- Cooling System

b) Computer system parts' power consumption under idle mode

Power consumption of different computer system parts in idle mode involves monitoring each component's power usage when the system is not under load. Here's how you can estimate the power consumption of key parts under idle conditions:

1. CPU (Central Processing Unit): The CPU typically consumes less power when idle due to power-saving features such as reducing clock speed and voltage.

- **Identification:**

- Use tools like **HWMonitor**, **CoreTemp**, or **Intel Power Gadget** to monitor CPU power consumption.
- Check the **idle power rating** in the CPU's datasheet. Modern CPUs typically consume between **5W to 20W** in idle mode.

2. GPU (Graphics Processing Unit): A dedicated GPU also lowers its power consumption significantly when idle, often turning off most of its power-hungry cores.

- **Identification:**

- Use tools like **GPU-Z** or **MSI Afterburner** to monitor GPU power consumption in real-time.
- Idle power consumption for high-end GPUs can range from **10W to 30W**, while low-power integrated GPUs (iGPUs) can consume less than 5W.

3. RAM (Random Access Memory): RAM modules consume a constant amount of power, even when idle. The power consumption is primarily related to keeping data accessible.

- **Identification:**

- Most RAM modules consume around **2W to 5W** per module under idle conditions.
- You can estimate this based on the number of RAM sticks and their voltage (e.g., 1.35V or 1.5V) from their technical specifications.

4. Motherboard and Chipset: The motherboard, along with its chipset, voltage regulators, and various controllers, consumes power continuously even when idle.

- **Identification:**

- The power consumption of a motherboard and chipset is usually estimated to be around **20W to 40W**, depending on the complexity and features.
- Some monitoring software like **HWMonitor** may give an overall estimate of motherboard power.

5. Storage Devices (HDD/SSD): Hard drives and SSDs typically have low power consumption when idle, especially SSDs due to the lack of moving parts.

- **Identification:**

- **SSDs** typically consume between **0.5W to 2W** when idle.
- **HDDs** consume more, typically around **4W to 6W** due to motor activity even in idle states.
- Use tools like **CrystalDiskInfo** to monitor storage device power status.

6. Power Supply Unit (PSU): The PSU itself consumes a small amount of power (around 1-5W) in idle mode due to inefficiencies, even if the system is not drawing much power.

- **Identification:**

- You can measure the **total system power draw** from the wall using a power meter (such as a **Kill-A-Watt** device) and subtract the power usage of other components to estimate PSU overhead during idle.

7. Fans and Cooling System: Case fans, CPU coolers, and other cooling components typically run at reduced speeds when the system is idle, lowering their power consumption.

- **Identification:**

- Fans generally consume between **1W to 5W** each depending on their size and speed.
- Use motherboard monitoring software (such as **Fan Xpert** or the BIOS settings) to check fan speed and voltage to estimate power draw.

8. Peripherals and External Devices: Devices like keyboards, mice, and external USB devices consume power, but typically a minimal amount when idle.

- **Identification:**

- **USB devices** may draw between **0.5W to 2W**.
- The total power consumption of peripherals can be observed by using a power meter connected to the entire system.

c) Identification of computer system parts' power consumption under sleep mode

Power consumption of different parts of a computer system while in **sleep mode** involves understanding how each component behaves during low-power states. Here's an estimation of power consumption for the main parts of a system in sleep mode:

Steps to Identify Power Consumption in Sleep Mode:

1. Motherboard: Supplies minimal power to maintain essential processes, like monitoring the power button and enabling wake-up events (keyboard/mouse inputs).

- **Power Consumption:** Typically consumes between **1-5 watts**.

2. CPU (Central Processing Unit): The CPU enters a low-power state (C-states), where it reduces or halts execution. The lowest states consume minimal power while maintaining memory integrity.

- **Power Consumption:** Typically, **<1 watt** in deep sleep modes.

3. RAM (Random Access Memory): The RAM stays partially powered to retain data, especially in S3 sleep states (Suspend-to-RAM).

- **Power Consumption:** Varies depending on the RAM type and capacity. Typically consumes around **1-3 watts**.
- **4. Power Supply Unit (PSU):** Supplies minimal power to components like the motherboard, RAM, and USB devices. The PSU's efficiency also affects overall power draw in low-power states.
- **Power Consumption:** PSUs consume **1-10 watts**, depending on efficiency and the components still drawing power.
- **5. Storage Devices (HDD/SSD):** Hard drives (HDDs) spin down, and solid-state drives (SSDs) enter a low-power state, reducing power draw significantly.
- **Power Consumption:**
 - HDDs: Typically consume **0.5-2 watts**.
 - SSDs: Typically consume less than **1 watt**.
- **6. Graphics Processing Unit (GPU):** The GPU enters a low-power state and stops processing visual tasks.
- **Power Consumption:** Modern GPUs use very little power, often **<1 watt** in sleep mode.
- **7. USB and Peripherals:** Certain peripherals, like USB devices, may still draw power to allow the system to wake from USB inputs.
- **Power Consumption:** USB devices may consume **0.5-2 watts** depending on the device.
- **8. Display Monitor:** Monitors usually enter standby mode, using very little power or turning off entirely.
- **Power Consumption:** Generally, **0.5-2 watts** in standby mode.

Summary of Typical Power Consumption in Sleep Mode:

Component	Power Consumption (Watts)
Motherboard	1-5 watts
CPU	<1 watt
RAM	1-3 watts
PSU (overhead)	1-10 watts
HDD/SSD	0.5-2 watts (HDD) / <1 watt (SSD)
GPU	<1 watt
USB Devices/Peripherals	0.5-2 watts (varies)
Monitor	0.5-2 watts (standby)

d) computer system parts' power consumption under maximum power rating mode

Computer system's power consumption under **maximum power rating mode**, it's important to understand how each component contributes to the overall power draw, especially under full load.

Below is an outline of key parts and their typical power consumption when operating at maximum capacity.

1. Central Processing Unit (CPU): The CPU is one of the most power-hungry components, especially under heavy workloads like gaming, video editing, or scientific simulations.

- **Typical Power Consumption:**

- Mainstream desktop CPUs consume **65–150 watts** under maximum load.
- High-end or overclocked CPUs can draw **150–300 watts**, particularly in enthusiast or workstation systems.

2. Graphics Processing Unit (GPU): The GPU is often the single largest power consumer in a system, particularly during gaming, 3D rendering, or machine learning tasks.

- **Typical Power Consumption:**

- Entry-level GPUs may consume **50–150 watts**.
- Mid-range GPUs consume **150–250 watts** under full load.
- High-end GPUs like NVIDIA's RTX 4080 or AMD's RX 7900 XTX can draw **250–450 watts** or more under extreme conditions.

3. Motherboard: The motherboard itself consumes some power to support its chipset, ports (USB, Ethernet, etc.), and other integrated components.

- **Typical Power Consumption:**

- Most motherboards consume around **30–80 watts** depending on the chipset and the number of connected peripherals.

4. Memory (RAM): RAM power consumption increases with both the amount and the speed (frequency) of the memory modules.

- **Typical Power Consumption:**

- A single stick of DDR4 or DDR5 memory typically consumes **3–5 watts**.
- In systems with multiple memory modules (e.g., 32 GB or 64 GB setups), total power consumption could reach **20–40 watts**.

5. Storage Devices: Power consumption of storage devices depends on the type (SSD vs. HDD) and the number of devices installed.

- **Typical Power Consumption:**

- **HDDs (3.5"):** Consume around **6–10 watts** during read/write operations.
- **SSDs (2.5" SATA):** Use **2–5 watts** under load, making them more power-efficient.
- **NVMe SSDs:** Typically consume **4–7 watts** under full load.

6. Power Supply Unit (PSU): The PSU itself draws some power to operate and loses some energy due to inefficiency (heat generation). The **80 PLUS efficiency rating** gives a general indication of how efficiently the PSU converts AC to DC power.

- **Typical Power Consumption:**
 - A **500-watt system** could draw **10–20%** more from the wall depending on PSU efficiency (e.g., drawing 550–600 watts for a 500W output at 80% efficiency).
- 7. Cooling Fans and Liquid Coolers:** Cooling systems, including case fans and CPU/GPU coolers, use power proportional to their size, speed, and number.
- **Typical Power Consumption:**
 - Standard case fans consume about **2–6 watts** each.
 - All-in-one (AIO) liquid coolers with pumps and multiple fans can consume **10–20 watts**.
- 8. Peripherals (Monitors, USB Devices):** External devices connected via USB ports, like keyboards, mice, external hard drives, or RGB lighting, also draw power from the system.
- **Typical Power Consumption:**
 - USB peripherals (keyboard, mouse) consume about **1–5 watts** each.
 - **External hard drives** consume **5–10 watts**.
 - **RGB lighting** and USB-powered gadgets can add **2–10 watts** depending on their usage.
- 9. Optical Drives (If Present):** Optical drives (DVD, Blu-ray) are less common today, but they consume power during read/write operations.
- **Typical Power Consumption:**
 - Optical drives typically use **15–25 watts** during operation, though idle consumption is lower.



Practical Activity 1.2.2: Drawing a computer power system architecture



Task:

1: With referring to key readings 1.2.2 you are requested to perform the following task:

As technician in computer system architecture, using drawing material you are requested to draw computer power system architecture block diagram.

2: Present the work done to trainer.

3: Ask question if any for clarification.



Key readings 1.2.2

Drawing a computer power system architecture

1. Understand power flow

In a computer power system, the power flow generally follows a hierarchical structure, starting from the AC power source and ending at the individual components. Here's a breakdown:

i. AC Power Input:

The system starts with AC power from a wall outlet. This power is typically 110-120V or 220-240V, depending on the region.

ii. Power Supply Unit (PSU)

The PSU is the primary component responsible for converting AC power to DC power.

iii. Distribution:

The PSU distributes the DC power to various components within the computer using cables and connectors. These connectors are typically modular, allowing for flexibility in connecting components.

2. Draw lines to components

In a computer power system architecture, drawing lines to components refers to the process of visually representing the connections and power flow between various elements within the system. This is crucial for understanding how electrical energy is distributed from the power source through different components, ensuring that each part receives the appropriate voltage and current necessary for operation.

Steps to Draw Power Flow and Lines to Components in Computer Power System:

Step 1: Gather Required Tools: Use drawing materials pen and paper.

Step 2: Identify Key Components, these includes:

- a) Power Source: This could be an AC mains supply or a battery.
- b) Power Supply Unit (PSU): Converts AC to DC and regulates voltage levels.
- c) Motherboard: The main circuit board that distributes power.
- d) Central Processing Unit (CPU): The primary processor that requires power.
- e) Random Access Memory (RAM): Temporary storage that also needs power.
- f) Peripheral Devices: Such as hard drives (HDD/SSD), graphics cards (GPU), and other add-ons.

Step 3: Understand Power Flow.

- a) The power source provides electrical energy.
- b) The PSU receives this energy, converting it from AC to DC while regulating voltage levels suitable for various components (+3.3V, +5V, +12V).
- c) The PSU distributes the converted power through connectors on the motherboard.

Step 4: Create a Basic Layout

Start with a rectangle labelled “Power Source” at the top of your diagram.

Below it, draw another rectangle labelled “Power Supply Unit (PSU)” connected by an arrow indicating the flow of electricity.

Step 5: Add the Motherboard

Draw a rectangle for the “Motherboard” below the PSU.

Connect the PSU to the motherboard with an arrow indicating power distribution.

Step 6: Include Major Components Connected to Motherboard

From the motherboard, draw arrows leading to:

A rectangle for “CPU”

A rectangle for “RAM”

A rectangle for “GPU”

A rectangle for “HDD/SSD”

Step 7: Draw Lines to Components

When drawing this architecture diagrammatically:

- a) Start with a rectangle representing the Power Source, labeled accordingly.
- b) Draw an arrow from the Power Source to another rectangle representing

the Power Supply Unit (PSU) indicating the flow of electricity.

c) From the PSU, draw multiple arrows leading to:

- A rectangle for the Motherboard, indicating that it distributes power further.
- Additional rectangles for each type of peripheral device (like HDDs, SSDs, GPUs) connected via lines showing their respective connections.

Each line should be labelled with its function (e.g., “12V”, “5V”, “3.3V”) to indicate what type of voltage is being supplied.

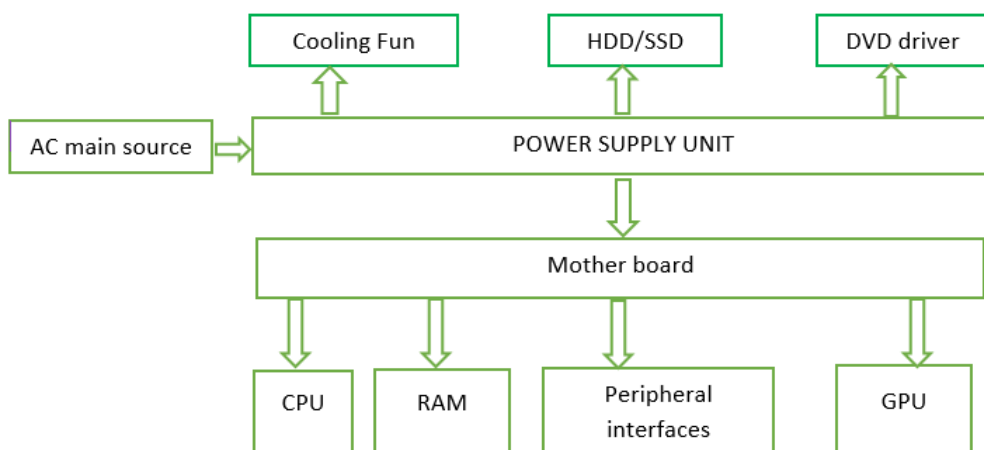
Step 8: Label Your Diagram

Next to each line connecting components, label them with their respective voltage levels:

For example, label lines from PSU to CPU as “+12V”, from PSU to RAM as “+5V”, etc.

Ensure all components are clearly labelled and that arrows indicate directionality of power flow.

The following is a block diagram of a computer power system





Theoretical Activity 1.2.3: Description of features for analysing PSU

Tasks:

1: Describe the following features applied in analysis of power supply unit

- i. Efficiency consideration
- ii. Expansion and upgrades
- iii. compatibility
- iv. Form factor
- v. Environmental impact
- vi. Safety margins
- vii. Budget and cost considerations

2: Write a detailed report for the above analysis.

3: Present the report to trainer.

4: Read the key reading 1.2.3 from trainee's manual to get more clarifications.



Key readings 1.2.3

Analysis of Power Supply Unit (PSU) features

1. Efficiency consideration

Its efficiency directly impacts energy consumption, heat generation, and overall system performance.

Factors Affecting PSU Efficiency:

✓ Efficiency Rating:

- Expressed as a percentage, it indicates how much of the input power is converted into usable output power.
- Higher ratings (e.g., 80+ Bronze, 80+ Silver, 80+ Gold) denote better efficiency.

✓ Load:

The PSU's efficiency can vary depending on the load it's handling.

Many PSUs achieve their peak efficiency at around 50% load.

✓ Internal Components:

The quality of components like capacitors, transformers, and rectifiers significantly impacts efficiency.

Higher-quality components often lead to better efficiency.

✓ Design and Topology:

The PSU's design and topology (e.g., single-rail vs. multi-rail) can influence efficiency.

- ✓ **Cooling:**
Adequate cooling helps prevent overheating, which can reduce efficiency and lifespan.
- ✓ **Age:**
Over time, components can degrade, leading to reduced efficiency.

2. Expansion and upgrade

When considering upgrading or expanding your computer system, the power supply unit (PSU) is often an overlooked but crucial component. An underpowered PSU can limit your system's performance, cause instability, and even damage components.

Reasons for Upgrading or Expanding a PSU:

- ✓ **Adding Components:** If you plan to add components such as a high-end graphics card, multiple hard drives, or a liquid cooling system, you'll need a PSU that can deliver enough power to support the increased demand.
- ✓ **Overclocking:** Overclocking components requires more power, so a PSU with a higher wattage rating is essential.
- ✓ **Future-Proofing:** A more powerful PSU can provide a buffer for potential future upgrades, allowing you to add components without having to replace the PSU immediately.
- ✓ **Improving Efficiency:** Upgrading to a more efficient PSU can reduce energy consumption and heat generation.

3. Verify compatibility

When selecting a power supply unit (PSU) for your computer, compatibility is crucial to ensure proper functionality and prevent damage. Here are the key factors to consider:

- **Wattage Rating:**
 - ✓ **System Requirements:** Determine the total power consumption of your components (CPU, GPU, motherboard, storage devices, etc.).
 - ✓ **PSU Wattage:** Ensure the PSU's wattage rating is significantly higher than your system's total power consumption to allow for headroom and future upgrades.
- **Connectors:**
 - ✓ **Required Connections:** Verify that the PSU has the necessary connectors (ATX 24-pin, CPU power, PCIe power, SATA power, etc.) to power your components.
 - ✓ **Cable Length:** Ensure the cables are long enough to reach your components within the case.

4. Form factor

- **Case Compatibility:** Check if the PSU's form factor (ATX, Micro-ATX, Mini-ITX) matches your computer case's dimensions and mounting points.

5. Environmental impact

Computer power supply units (PSUs) play a crucial role in powering our electronic devices, but they also have environmental implications. Here's a breakdown of the primary environmental impacts:

- **Electronic Waste:**

- ✓ **Disposal:** When PSUs reach the end of their lifespan, they become electronic waste, which can contribute to pollution if not disposed of properly.
- ✓ **Hazardous Materials:** Some PSUs contain hazardous materials like lead and mercury, which can contaminate the environment if not recycled responsibly

- **Heat Generation:**

- ✓ **Energy Loss:** Inefficient PSUs can generate excessive heat, which can lead to energy loss and increased cooling requirements.
- ✓ **Climate Change:** Increased cooling demands can contribute to higher energy consumption and greenhouse gas emissions.

- **Mitigating Environmental Impacts:**

- ✓ **Energy-Efficient Models:** Choose PSUs with high efficiency ratings (e.g., 80+ Gold, 80+ Platinum) to reduce energy consumption.
- ✓ **Proper Disposal:** Recycle old PSUs responsibly to minimize electronic waste and prevent hazardous materials from entering the environment.
- ✓ **Standby Mode:** Turn off or unplug devices when not in use to reduce standby power consumption.
- ✓ **Energy-Efficient Cooling:** Use efficient cooling solutions to minimize energy consumption for cooling purposes.
- ✓ **Support Sustainable Manufacturing:** Choose brands that prioritize sustainable manufacturing practices and minimize environmental impacts.

6. Safety margin

A power supply unit (PSU) is a critical component in a computer system, responsible for delivering power to all internal components. Ensuring a sufficient safety margin in a PSU is essential to prevent system instability, component damage, and potential safety hazards.

Factors Affecting Safety Margins:

- **Wattage Rating:**

- **Overload Protection:** A PSU with a higher wattage rating than your system's total power consumption provides a safety margin to handle unexpected spikes or future upgrades.

- **Overcurrent Protection:** Built-in overcurrent protection circuits within the PSU help prevent damage to the PSU and connected components in case of excessive current draw.
- **Efficiency Rating:**
 - **Heat Generation:** A higher efficiency rating (e.g., 80+ Gold, 80+ Platinum) reduces heat generation, which can improve reliability and safety.
 - **Power Conversion:** Efficient PSUs convert more input power into usable output power, reducing the risk of power fluctuations or voltage drops.
- **Certification:**
 - **Safety Standards:** Certifications like 80+ Platinum and ATX12V ensure compliance with industry safety standards, providing additional assurance of safety.
- **Protection Features:**
 - **Overvoltage Protection:** This feature prevents excessive voltage from reaching connected components, protecting them from damage.
 - **Short-Circuit Protection:** This feature automatically shuts down the PSU if a short circuit occurs, preventing electrical damage.
 - **Over temperature Protection:** This feature shuts down the PSU if it reaches excessive temperatures, preventing overheating and potential fires.
- **Quality Components:**
 - **Reliability:** Using high-quality components can improve the PSU's overall reliability and reduce the risk of failures.
- **Cable Management:**
 - **Heat Dissipation:** Proper cable management helps improve airflow within the case, preventing overheating and improving safety.

7. Budget and cost considerations

When developing a computer power supply unit (PSU), budget and cost are important factors to consider. The price of a PSU can vary widely depending on its wattage, efficiency rating, features, and brand. Here's a breakdown of the key factors influencing PSU costs:

- **Wattage:**
 - **Power Requirements:** Higher wattage PSUs generally cost more as they are designed to handle more demanding components.
 - **Future Upgrades:** Consider your future upgrade plans to avoid having to replace the PSU later.
- **Efficiency Rating:**
 - **Energy Savings:** PSUs with higher efficiency ratings (e.g., 80+ Gold, 80+ Platinum) tend to be more expensive but can save you money on electricity bills in the long run.

- **Environmental Impact:** Higher efficiency ratings also contribute to a smaller carbon footprint.
- **Features:**
 - **Modular Cables:** Modular cables can add to the cost, but they offer more flexibility and can improve airflow within your case.
 - **Additional Features:** Features like active PFC, Japanese capacitors, or RGB lighting can also increase the price.
- **Brand:**
 - **Reputation:** Well-known brands often command higher prices due to their reputation for quality and reliability.
 - **Warranty:** Consider the warranty length and coverage offered by different brands.



Points to Remember

- **The parts of a computer that consume power include the following components:**
 - ✓ Central Processing Unit (CPU)
 - ✓ Graphics Processing Unit (GPU)
 - ✓ Motherboard
 - ✓ Power Supply Unit (PSU)
 - ✓ Memory (RAM)
 - ✓ Storage Devices (HDDs and SSDs)
 - ✓ Cooling Systems (Fans, Liquid Cooling)
 - ✓ Optical Drives (CD/DVD/Blu-ray)
 - ✓ Peripherals Connected via USB
 - ✓ Network Interface Cards (NICs)
 - ✓ Monitors (External)
- **Computer power modes:**
 - ✓ **Idle mode** in a computer refers to a state where the system is powered on but is not actively performing any significant tasks or running resource-intensive programs. In this mode, the computer's components (CPU, GPU, RAM, etc.)

operate at minimal activity levels, conserving power while remaining ready to quickly resume full operation when needed.

✓ **Sleep mode** in a computer is a low-power state that allows the system to conserve energy while still maintaining its operational state. In sleep mode, the computer stops most of its activity, but it stores the current state (open applications, files, and settings) in **RAM** so that the user can quickly resume where they left off without rebooting the system.

✓ **Maximum load power rating** in a computer refers to the highest amount of electrical power that the system or a specific component can safely consume or draw from the power supply under full workload conditions. It indicates the total power requirement that a computer can demand when all components (such as the CPU, GPU, storage, and peripherals) are running at or near their maximum capacity.

- **Steps to Draw Power Flow and Lines to Components in Computer Power System:**

Step 1: Gather Required Tools.

Step 2: Identify Key Components of computer power system.

Step 3: Understand Power Flow.

Step 4: Create a Basic Layout.

Step 5: Add the Motherboard.

Step 6: Include Major Components Connected to Motherboard.

Step 7: Draw Lines to Components.

Step 8: Label Your Diagram.

- **Key features to consider during analysis of power supply unit:**

- ✓ **Efficiency consideration**

Efficiency directly impacts energy consumption, heat generation, and overall computer system performance.

- ✓ **Expansion and upgrade**

When considering upgrading or expanding your computer system, the power supply unit (PSU) is often an overlooked but crucial component. An underpowered PSU can limit your system's performance, cause instability, and even damage components.

✓ **Compatibility**

When selecting a power supply unit (PSU) for your computer, compatibility is crucial to ensure proper functionality and prevent damage. Here are the key factors to consider:

✓ **Form factor**

Case Compatibility: Check if the PSU's form factor (ATX, Micro-ATX, Mini-ITX) matches your computer case's dimensions and mounting points.

✓ **Environmental impact**

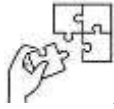
Computer power supply units (PSUs) play a crucial role in powering our electronic devices, but they also have environmental implications. Here's a breakdown of the primary environmental impacts: electronic waste, heat generation, power consumption.

✓ **Safety margin**

Power supply unit (PSU) is a critical component in a computer system, responsible for delivering power to all internal components. Ensuring a sufficient safety margin in a PSU is essential to prevent system instability, component damage, and potential safety hazards.

✓ **Budget and cost considerations**

When developing a computer power supply unit (PSU), budget and cost are important factors to consider. The price of a PSU can vary widely depending on its wattage, efficiency rating, features, and brand. Here's a breakdown of the key factors influencing PSU costs.



Application of learning 1.2.

Your school recently purchased a high-performance desktop computer for the technical lab, designed to support heavy computational tasks such as 3D rendering, machine learning simulations, and gaming development. However, after a few days of use, the computer starts shutting down randomly when running resource-intensive software.

Task:

You need to analyse the computer's power requirements and determine whether the current Power Supply Unit (PSU) is sufficient to support all the components.



Indicative content 1.3: Calculation of System Power Consumption



Duration: 5 hrs



Practical Activity 1.3.1: Calculating System Power consumption



Task:

1: You are requested to perform the following task:

As technician in computer system architecture, you are requested to go in computer lab and pick any computer to calculate its components power consumption, efficiency, total harmonic distortion and power fluctuations.

2: Present the work done to trainer.

3: Ask question if any for clarification.

4: Read the key reading 1.3.1 from your manual to get more clarifications.



Key readings 1.3.1.:

Calculating System Power consumption

To calculate system power consumption and related factors such as component power consumption, efficiency, total harmonic distortion (THD), and power fluctuations, here's a structured approach:

1. Components Power Consumption:

Each component in the system consumes power. The total power consumption of the system is the sum of the individual power consumptions of each component.

Formula:

$$P = V \times I$$

Where:

$P_{\text{component}}$ = Power consumption of the component (Watts)

V = Voltage across the component (Volts)

I = Current flowing through the component (Amps)

Steps:

-  Identify the voltage (V) and current (I) for each component.
-  Multiply voltage and current to get the power consumption of each component.

- ✚ Sum the power consumption of all components to get the total system power consumption.

Example:

If you have a CPU that consumes 12V and draws 5A, and a GPU that consumes 12V and draws 7A:

$$P_{CPU} = 12V \times 5A = 60W$$

$$P_{GPU} = 12V \times 7A = 84W$$

$$P_{Total} = 60W + 84W = 144W$$

2. System Efficiency:

Efficiency is the ratio of useful power output to total power input, expressed as a percentage.

Formula:

$$\eta = \left(\frac{P_{output}}{P_{input}} \right) \times 100$$

Where:

η = Efficiency (%)

P_{output} = Power delivered to the load (Watts)

P_{input} = Power consumed from the power source (Watts)

Steps:

- ✚ Measure or calculate the power output to the load and the power consumed from the source.
- ✚ Divide the output power by the input power and multiply by 100 to get efficiency.

Example:

If a power supply provides 144W output and consumes 160W from the mains:

$$\eta = \left(\frac{144W}{160W} \right) \times 100 = 90\%$$

3. Total Harmonic Distortion (THD):

THD is a measure of how much harmonic distortion is present in a signal. In power systems, it typically refers to distortion in the voltage or current waveform.

Formula for Voltage THD:

$$THD_V = \frac{\sqrt{V_2^2 + V_3^2 + \dots + V_n^2}}{V_1} \times 100$$

Where:

V_1 = Fundamental voltage (first harmonic)

$V_2, V_3, \dots, V_n$ = Harmonic voltages

Formula for Current THD:

$$THD_I = \frac{\sqrt{I_2^2 + I_3^2 + \dots + I_n^2}}{I_1} \times 100$$

Where:

I₁ = Fundamental current

I₂, I₃, ..., I_n = Harmonic currents

Steps:

-  Use an oscilloscope or power quality analyzer to measure the harmonic voltages or currents.
-  Apply the THD formula to calculate the percentage distortion.

Example:

If the fundamental current (I₁) is 10A, and the second and third harmonic currents are 1A and 0.5A respectively:

$$THD_I = \frac{\sqrt{1A^2 + 0.5A^2}}{10A} \times 100 = \frac{\sqrt{1 + 0.25}}{10} \times 100 = 10.6\%$$

4. Power Fluctuations:

Power fluctuations refer to variations in power delivered to the system over time. These fluctuations can arise due to changes in voltage, current, or load conditions.

Formula:

Power fluctuation can be measured by observing the variance or range in power consumption over time.

$$P_{\text{fluctuation}} = P_{\text{max}} - P_{\text{min}}$$

Where:

P_{max} = Maximum power recorded

P_{min} = Minimum power recorded

Steps:

-  Use a power meter or energy analyzer to monitor power consumption over time.
-  Identify the maximum and minimum power values and subtract to find the fluctuation.

Example:

If the power supply fluctuates between 140W and 160W:

$$P(\text{fluctuation}) = 160W - 140W = 20W$$



Points to Remember

- **Power Consumption:** The amount of electrical energy used by a device or system over a certain period, typically measured in watts (W) or kilowatt-hours (kWh).
- **Power Efficiency:** The ratio of useful power output to the total power input, expressed as a percentage. High efficiency means less energy is wasted.
- **Total Harmonic Distortion (THD):** A measure of the distortion in an electrical signal caused by harmonics (undesired frequencies). Lower THD indicates a cleaner and more stable signal, important for ensuring high-quality power and minimizing interference.
- **Power Fluctuation:** Variations in voltage, current, or frequency in an electrical system, which can affect device performance and longevity if they occur frequently or are severe.

Steps to go through while calculating total system power consumption of a computer:

- List the Power Ratings of Components
- Use the Power Consumption Formula

$$P_{\text{component}} = V_{\text{component}} \times I_{\text{component}}$$

- Sum the Power Consumption

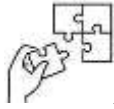
$$P_t = P_1 + P_2 + \dots + P_n$$

- Power Supply Efficiency Factor

$$\eta = \left(\frac{P_{\text{output}}}{P_{\text{input}}} \right) \times 100$$

- Total Harmonic Distortion (THD)

$$\text{THD}_V = \frac{\sqrt{V_2^2 + V_3^2 + \dots + V_n^2}}{V_1} \times 100$$



Application of learning 1.3.

As technician you are given a small DC power supply system designed to run multiple components, such as an LED light, a cooling fan, and a small motor. The system is powered by a DC source, and each component has a power rating. The system is supplied of 12V and current of 0.1A, 0.5A and 1A flow through the system respectively, also it is connected to a multi-meter and an oscilloscope to measure current, voltage, and any distortions in the waveform.

- a) You are asked to calculate the power consumption for each component.
- b) If the power supply provides 25W, and the total power consumed by the components is 19.2W, calculate the system efficiency.



Indicative content 1.4: Selection of Tools, Materials and Equipment



Duration: 2 hrs



Theoretical Activity 1.4.1: Description of tools, material and equipment

Tasks:

- 1: Answer the following questions related to description of computer power system.
 - i. What do you understand by the following terms?
 - a) Tool
 - b) Material
 - c) Equipment
 - ii. Give at least four examples for each term
- 2: Write the findings/answers on papers or flipchart.
- 3: Present your findings to the class.
- 4: Ask questions for more clarifications.
- 5: Read the key reading 1.4.1 from your manual to get more clarifications.



Key readings 1.4.1

Selecting tools, materials and equipment

Developing a computer power system involves a variety of tools, materials, and equipment. Here's a categorized list of what you might need:

1) Tools

A tool is a handheld or mechanical device used to carry out a specific function or task, typically aiding in the construction, maintenance, or repair of systems or products. Tools are often portable and designed to make manual work easier.

These are handheld or software tools used to assemble, test, or manage the system.

The following are some examples of tools:

a) Assembly and Maintenance Tools:

- Soldering Iron and Solder
- Wire Strippers and Cutters
- Screwdrivers
- Pliers and Tweezers
- Cable Ties and Organizers

b) Testing and Debugging Tools:

- Multimeter
- Oscilloscope
- Power Analyzers
- Test Load
- Thermal Cameras

c) Software Tools:

- Power Management Software
- Simulation Tools (MATLAB, LTSpice, Proteus)

2) Materials

A material is a substance or component that is used in the production, construction, or manufacturing of goods. It is typically consumed or used up in the process and forms part of the final product.

These are consumable or physical components used in the building or operation of the system.

The following are some examples of materials used in computer power system:

a) Power Supply Components:

- Power Supply Unit (PSU)
- Voltage Regulators (e.g., LM317, LM7805)
- Capacitors
- Inductors
- Diodes (e.g., Bridge Rectifiers)
- Heat Sinks
- Transformers
- Fuses and Circuit Breakers

b) Control and Monitoring Components:

- Relays
- Power Transistors

c) Energy Storage:

- Batteries (e.g., Li-ion, Lead Acid)
- Supercapacitors

d) Cables and Connectors:

- Power Cables
- Connectors (Molex, ATX, SATA)
- Breadboard and Jumper Wires

e) Protection and Insulating Materials:

- Surge Protectors
- Grounding Equipment
- ESD (Electrostatic Discharge) Protection
- Insulating Materials

3) Equipment

Equipment refers to larger, more complex machines, devices, or installations used for performing specific functions or tasks, often in industrial or professional settings. Equipment is typically not portable and involves machinery or electronic devices that assist in completing tasks.

These are larger, more permanent devices or machines that assist in the creation or operation of the system.

The following are examples of some equipment used computer power system development:

a) **Power Supply Components:**

- Uninterruptible Power Supply (UPS)

b) **Testing Equipment:**

- Circuit Testers
- Power Analyzers

This classification helps distinguish between the various roles and functions of each item in the development of a computer power system.



Practical Activity 1.4.2: Selecting tools, material and equipment



Task:

- 1: Referring to key reading 1.4.2 you are requested to go into workshop and perform the following activities:
- 2: Select tools, materials and equipment from work place needed for computer power system development.
- 3: Prepare a written report of tools, materials and equipment needed for computer power supply development.
- 4: Present the work done to trainer.



Key readings 1.4.2

Selecting tools, materials and equipment

Below are the steps involved in selecting tools, materials, and equipment:

Step1. Define the Project Requirements: Start by identifying the specific goals of the project and the tasks that need to be performed. This will help in determining the types of tools, materials, and equipment required. Determine the scale and complexity of the project (e.g., small-scale installation vs. large industrial project). This will influence the type and capacity of the tools and equipment.

Example: For a solar water pumping system, the main objective is to ensure efficient water pumping using solar energy.

Step2. Identify Key Specifications: Define the technical requirements for the materials, tools, and equipment (e.g., voltage, capacity, efficiency).and also ensure that the selected items meet industry standards and certifications (e.g., CE, ISO, or IEC for electrical components).

Step3. Research Available Options: Investigate available options by consulting manufacturers, suppliers, or technical manuals to see what tools, materials, and equipment are available.

Look for customer reviews, product feedback, or testimonials that can give insights into the reliability and performance of specific items.

Step4. Determine Compatibility: Ensure that the tools, materials, and equipment are compatible with each other and with existing infrastructure or systems and opt for equipment and materials that adhere to common standards, allowing easier integration and future upgrades.

Step5. Evaluate Cost and Budget Constraints: Establish a budget for the tools, materials, and equipment based on the project's financial plan and perform a cost-benefit analysis to ensure the selected items offer the best value for the money, balancing cost with performance, durability, and longevity.

Step6. Prioritize Safety and Compliance: Choose tools and equipment that meet relevant safety regulations to ensure safe operation during and after installation and prioritize tools and equipment that reduce the risk of injury or accidents during use.

Step7. Assess Durability and Longevity: Select materials that can withstand the environmental conditions in which they will be used (e.g., UV exposure, water contact, temperature fluctuations) and choose tools and equipment that are low-maintenance or come with easily available spare parts for long-term reliability.

Step8. Final Selection: Based on the evaluation of cost, safety, durability, and performance, make the final selection of tools, materials, and equipment.

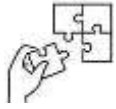


Points to Remember

- A **tool** is a handheld or mechanical device used to carry out a specific function or task, typically aiding in the construction, maintenance, or repair of systems or products. Tools are often portable and designed to make manual work easier.
Examples: Screwdriver, hammer, soldering iron, pliers, wire cutters.
- A **material** is a substance or component that is used in the production, construction, or manufacturing of goods. It is typically consumed or used up in the process and forms part of the final product.
Examples: Copper wire, steel, plastic, diodes, resistors, capacitors, solder.
- **Equipment** refers to larger, more complex machines, devices, or installations used for performing specific functions or tasks, often in industrial or professional settings. Equipment is typically not portable and involves machinery or electronic devices that assist in completing tasks.
Examples: Oscilloscopes, power supplies, computers, CNC machines, 3D printers.

The steps involved in selecting tools, materials, and equipment:

- Step1.** Start by identifying the specific goals of the project and the tasks that need to be performed.
- Step2.** Define the technical requirements for the materials, tools, and equipment (e.g., voltage, capacity, efficiency).
- Step3.** Investigate available options by consulting manufacturers, suppliers, or technical manuals to see what tools, materials, and equipment are available.
- Step4.** Ensure that the tools, materials, and equipment are compatible with each other.
- Step5.** Establish a budget for the tools, materials, and equipment based on the project's financial plan.
- Step6.** Choose tools and equipment that meet relevant safety regulations to ensure safe operation during.
- Step7.** Select materials that can withstand the environmental conditions in which they will be used.
- Step8.** Make the final selection of tools, materials, and equipment based on the evaluation of cost, safety, durability, and performance,



Application of learning 1.4.

A computer refurbishment company need to set up a working environment for computer power system development.

As a skilled computer technician you are required to select and setup tools, material and equipment to accomplish tasks in this company.



Learning outcome 1 end assessment

Theoretical assessment

A. Multiple choice questions: Encircle the letter with the correct answer.

1. What is the main purpose of power management in computers?

- A. To increase the processing power of the CPU
- B. To enhance the computer's graphics performance
- C. To reduce energy consumption when the system is idle
- D. To permanently shut down the computer when not in use.

2. Which of the following is a common feature of power management in modern operating systems?

- A. Overclocking the CPU
- B. Sleep and Hibernate modes
- C. Increasing monitor brightness at night
- D. Preventing the system from shutting down automatically

3. Which of the following power supplies provides battery backup to protect against power outages?

- A. Uninterruptible Power Supply (UPS)
- B. Switching Mode Power Supply (SMPS)
- C. Direct Current (DC) power supply
- D. External Power Adapter

4. Which factor determines the total amount of power a power supply can deliver to a computer's components?

- A. Form factor
- B. Wattage rating
- C. Connector type
- D. Cooling system

5. What is the significance of the 80 PLUS certification for power supplies?
- A. It guarantees the power supply has over 80 connectors
 - B. It ensures the power supply has a minimum efficiency of 80% under typical loads
 - C. It signifies the power supply can operate at 80°C without overheating
 - D. It ensures the power supply supports multiple form factors
6. Which type of connector is typically used to power modern motherboards?
- A. Molex connector
 - B. SATA connector
 - C. 24-pin ATX connector
 - D. 6-pin PCIe connector
7. Which of the following best describes the "ATX" feature of a power supply?
- A. It refers to the wattage of the power supply
 - B. It indicates the form factor and physical dimensions of the power supply
 - C. It describes the cooling system used by the power supply
 - D. It specifies the efficiency rating of the power supply
8. What is the purpose of having a cooling system, such as a fan, in a computer power supply?
- A. To improve the power efficiency
 - B. To cool the processor directly
 - C. To prevent the power supply from overheating and damaging internal components
 - D. To reduce the noise generated by the power supply
9. What is the primary function of a computer power supply's power conversion process?
- A. To convert DC voltage to AC voltage for internal components.
 - B. To regulate the speed of data transmission between components.
 - C. To convert incoming AC voltage to DC voltage for computer components.
 - D. To store power for backup during a power outage.

10. How is power distributed to various computer components from the power supply unit (PSU)?

- A. Through multiple voltage rails that deliver different voltages to specific components
- B. Using a single high-voltage rail that powers all components equally
- C. By converting DC power back to AC for each individual component
- D. Through an external transformer that distributes power directly to the motherboard

B. Match each term in Column A with the correct description from Column B.

Column A	Column B
1. Wattage	A. Converts AC power from the outlet to DC power for internal components.
2. Efficiency Rating (80 PLUS)	B. Ensures efficient power usage with a minimum of 80% under typical loads.
3. Power Conversion	C. The total amount of power the PSU can deliver to a system's components.
4. Power Distribution	D. A 24-pin connector that supplies power to the motherboard.
5. Power fluctuation	E. Standardized size and shape of the PSU to fit into most PC cases.
6. ATX Form Factor	

Practical assessment

You are an IT technician tasked with building a custom desktop computer for a client who needs it for high-performance tasks such as gaming, video editing, and 3D rendering. The client has provided you with a list of components, and you must ensure that the power supply you choose can adequately handle the total power consumption of the system while leaving some headroom for future upgrades. You also need to consider the efficiency of the power supply to avoid energy waste.



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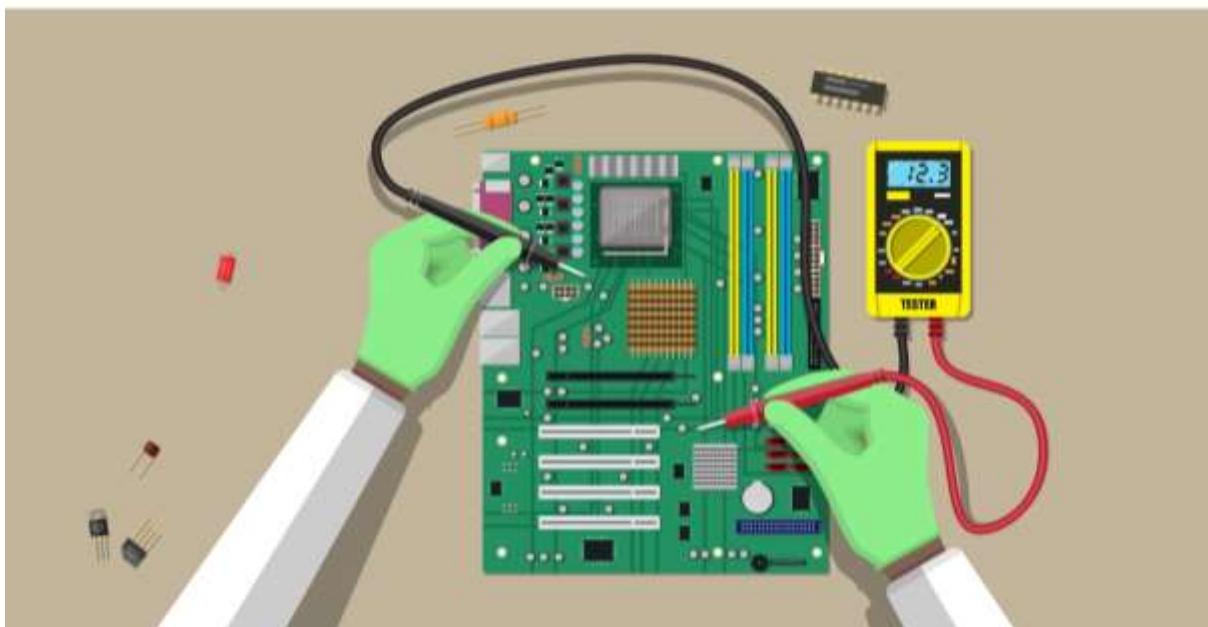
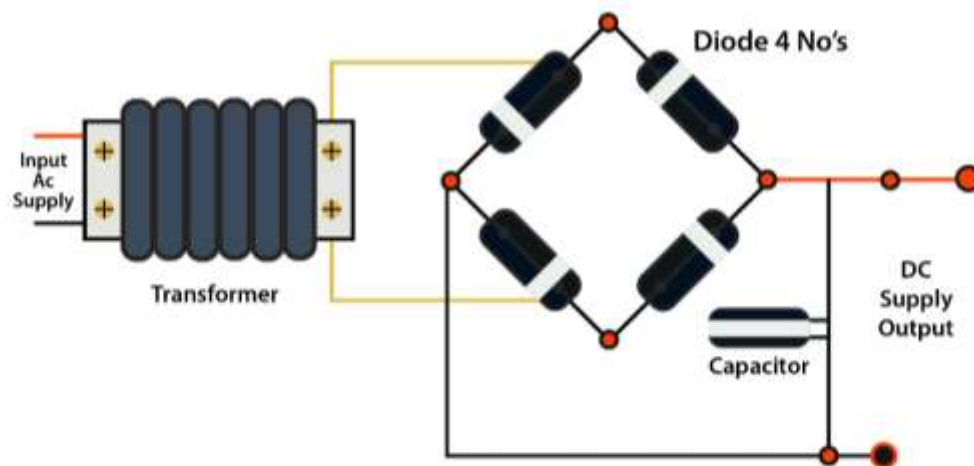
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END

Learning Outcome 2: Implement Power Electronic Converters



Indicative contents

2.1 Design power electronic converters

2.2 Assembling power electronic converter components

2.3 Testing power electronic converters

Key Competencies for Learning Outcome 2: Implement Power Electronic Converters

Knowledge	Skills	Attitudes
• Description of power semiconductor devices • Description of power electronics converter	• Installing of simulation software • Drawing of power electronics converters circuit diagrams • Simulating the power converter • Generating power converter PCB design • Assembling power electronic converter components • Testing power electronic converters	• Willing to explore new ideas • Taking responsibility • Having attention to details • Being organized • Being patient • Being a time manager



Duration: 35 hrs

Learning outcome 2 objectives:



By the end of the learning outcome, the trainees will be able to:

1. Describe clearly power semiconductor devices based on their types and application.
2. Describe correctly power electronics converters based on their types and application.
3. Install properly simulation software for designing power converters in computer.
4. Draw correctly power electronic converters using simulation software.
5. Simulate correctly power electronic converters using simulation software.
6. Generate properly power converter PCB using simulation software.

7. Assemble properly power electronic converter components according to their types.
8. Test accurately power electronic converters of computer power system according to testing methodology.



Resources

Equipment	Tools	Materials
• Personal Computer • PPEs • Oscilloscope • Multi-meter • Function generator • SMD rework station • Drawing table • DC Power supply • ESD workstation • Office UPS	• Screwdrivers • Needle-nose pliers • Universal plier • Tweezers • Flashlight or headlamp • Cable testers • Wire cutter/striper • Pliers • Soldering iron • De-soldering pump • Glue gun • Circuit Simulation and drawing software (NI Multisim, Proteus, Autodesk Eagle, Easy EDA, matlab, CAD) • Drawing accessories	• Power Semiconductor devices • Soldering materials • Power cables • Glue stick • Drawing materials (papers and pencil) • Electronic Passive Components • Isopropyl alcohol and cotton swabs • Electrical tapes and zip ties • Spare screws • Labelling materials • Connectors • Foam cleaner • Battery/cells • Jumper wires • Breadboard • PCB



Indicative content 2.1: Design Power Electronic Converters



Duration: 20 hrs



Theoretical Activity 2.1.1: Description of power semiconductor devices

Tasks:

- 1: You are requested to answer the following questions related to description of power semiconductor devices.
 - i. What do you understand by power semiconductor devices?
 - ii. Identify the symbols of power semiconductor components?
 - iii. Describe the working principles of power semiconductor devices.
 - iv. Identify the power rating of semiconductor devices.
 - v. Identify the applications of power semiconductor devices.
- 2: Write your findings/answers on papers or flipchart.
- 3: Present your findings to the class.
- 4: Ask questions for more clarifications.
- 5: Read the key reading 2.1.1 from your manual to get more clarifications.



Key readings 2.1.1.:

Description of power semiconductor devices

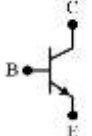
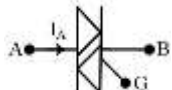
1. Definition

A **power semiconductor device** is a type of semiconductor device designed to handle and control high levels of electrical power. These devices function as switches or rectifiers, efficiently converting and managing electrical energy in power systems.

Power semiconductor devices are specifically engineered for high voltage and current applications, making them essential in power conversion and power supply systems.

2. Symbols of power semiconductor components

Power diode	
Silicon Controlled Rectifier (SCR)	
Metal-Oxide-Semiconductor Field-Effect Transistor (MOSFET)	
Insulated Gate Bipolar Transistor (IGBT)	

Bipolar Junction Transistor (BJT)	
Gate Turn-Off Thyristor (GTO)	
Triode for Alternating Current (TRIAC)	

3. Working principles

Power semiconductor devices are essential for controlling and converting electrical energy in various power electronics systems. Their working principle primarily revolves around their ability to act as switches or amplifiers, enabling the efficient control of high voltages and currents.

Below is a detailed explanation of the working principles of key power semiconductor devices:

3.1 Power Diodes

- Working Principle:** A power diode allows current to flow in one direction (forward-biased) and blocks current in the reverse direction (reverse-biased). When diode is **Forward Bias** the anode is more positive than the cathode, the diode conducts. The electric field at the p-n junction decreases, allowing charge carriers (electrons and holes) to flow, resulting in current conduction. When diode is in **Reverse Bias** the cathode is more positive than the anode, the depletion region widens, preventing current flow. A small reverse leakage current may flow, but the diode essentially blocks current.

3.2 Thyristors (Silicon Controlled Rectifiers - SCR)

- Working Principle:** A thyristor is a four-layer, three-terminal device (anode, cathode, gate) that functions as a switch, allowing current to flow between the anode and cathode when triggered by a gate pulse. Once triggered, it latches into the conducting state, continuing to allow current flow even after the gate pulse is removed, as long as the current remains above the holding threshold. The thyristor turns off only when the current drops below this level, such as when the AC cycle crosses zero. Triggering occurs when a small current applied to the gate forward-biases the inner junction, initiating conduction.

3.3 Power MOSFET (Metal-Oxide-Semiconductor Field-Effect Transistor)

- Working Principle:** A power MOSFET is a voltage-controlled device used for high-speed switching. It operates by applying a voltage to the gate, which creates an

electric field that either forms (enhancement mode) or depletes (depletion mode) a conductive channel between the source and drain, allowing or stopping current flow. Its low gate capacitance enables rapid switching, making it suitable for high-frequency applications.

3.4 Insulated Gate Bipolar Transistor (IGBT)

- **Working Principle:** An Insulated Gate Bipolar Transistor (IGBT) combines the high-speed switching and high efficiency of a MOSFET with the high-current handling and low on-state voltage drop of a bipolar junction transistor (BJT). It is voltage-controlled, where a small gate voltage regulates a larger current flow between the collector and emitter, similar to how a MOSFET operates. When turned on, the IGBT exhibits low conduction losses, making it well-suited for high-power applications due to its efficiency in managing large currents with minimal power dissipation.

3.5 Bipolar Junction Transistor (BJT)

- **Working Principle:** A Bipolar Junction Transistor (BJT) is a current-controlled device with three terminals: base, collector, and emitter. It operates by using a small base current to control a much larger current flow between the collector and emitter, making it effective for amplifying input signals. In addition to amplification, BJTs can function as switches—when in saturation mode, they allow current to flow freely between the collector and emitter, while in cut-off mode, no current flows, effectively turning the switch off.

3.6 Gate Turn-Off Thyristor (GTO)

- **Working Principle:** A Gate Turn-Off Thyristor (GTO) is similar to a conventional thyristor but with the added ability to be turned off by applying a negative gate current. It operates by using a small positive gate pulse to trigger conduction, allowing current to flow between the anode and cathode. Unlike regular thyristors, which require external circuitry to turn off, the GTO can be turned off by applying a negative gate pulse, which extracts charge carriers and interrupts the conductive state. This feature makes it more versatile in controlling high-power circuits.

3.7 Triac

- **Working Principle:** A Triac is a bidirectional device that operates similarly to a thyristor but can conduct current in both directions, making it suitable for controlling AC power. It allows switching of AC loads by conducting during both the positive and negative half-cycles of an AC signal. A small gate current triggers the Triac, and once it is activated, it remains on until the current flowing through it drops below a certain holding value, enabling efficient control of AC power in applications like dimming lights or speed control in motors.

4. Power rating

The **power rating** of power semiconductor devices is a critical parameter that defines the maximum amount of electrical power the device can safely handle without failure. It is influenced by factors such as voltage, current, and heat dissipation capabilities. Here's a breakdown of power ratings for different semiconductor devices:

4.1 Diodes:

- **Power Rating:** Typically ranges from a few watts (for small signal diodes) to several kilowatts (for high-power rectifiers).
- **Key Factors:** Forward current, reverse voltage, and junction temperature.

4.2 Transistors (BJTs, MOSFETs, IGBTs):

- **Power Rating:**
 - **BJT:** From milliwatts to several hundred watts.
 - **MOSFET:** Can range from less than a watt for small MOSFETs to tens or hundreds of kilowatts for power MOSFETs.
 - **IGBT:** Used in high-power applications, with ratings up to several megawatts.
- **Key Factors:** Collector-emitter voltage, collector current, junction temperature, and thermal resistance.

4.3 Thyristors (SCRs, Triacs):

- **Power Rating:** Can handle up to hundreds of kilowatts, commonly used in high-power AC and DC systems.
- **Key Factors:** Forward current, reverse blocking voltage, and thermal characteristics.

4.4 IGBTs (Insulated Gate Bipolar Transistors):

- **Power Rating:** Commonly used in high-power switching applications with power ratings from kilowatts to megawatts.
- **Key Factors:** Gate voltage, switching frequency, current handling capacity, and thermal management.

4.5 Power Modules (combinations of diodes, IGBTs, MOSFETs):

- **Power Rating:** Designed for industrial applications, these can handle from several kilowatts to megawatts.
- **Key Factors:** System voltage, current, and temperature control mechanisms.

5. Applications of power semiconductor devices

Power semiconductor devices are widely used in applications that involve switching, controlling, and converting electrical energy at high efficiency. Below are the primary applications of power semiconductor devices:

- **Rectifiers (AC-DC Converters):** Used in power supplies to convert alternating current (AC) to direct current (DC) for powering various electronic equipment like computers, TVs, and industrial systems.
- **Inverters (DC-AC Converters):** Power semiconductor devices are used to convert DC power (from batteries or renewable energy sources) to AC for powering AC loads or for grid integration.
- **DC-DC Converters:** Widely used in portable electronics devices, electric vehicles, and renewable energy systems to step up or step down DC voltage levels.



Theoretical Activity 2.1.2: Description of power electronic converter



Tasks:

- 1: You are requested to describe by defining and identifying components of each of the following types of power converters:
 - i. AC to DC converter
 - ii. DC to AC converter
 - iii. DC to DC converter
 - iv. AC to AC converter
- 2: Write the findings/answers on papers or flipchart.
- 3: Present your findings to the class.
- 4: Ask questions for more clarifications.
- 5: Read the key reading 2.1.2 from your manual to get more clarifications.



Key readings 2.1.2.:

Description of power converter

1. AC to DC converter

Definition: An AC to DC converter, also known as a rectifier, is an electrical device used to convert alternating current (AC), into direct current (DC). These converters are commonly used in power supplies for electronic devices, battery charging systems, and any application that requires stable DC power from an AC source.

Main Components of an AC to DC Converter:

- i. **Transformer:** Used to step up or step down the AC voltage to a desired level before conversion.
- ii. **Rectifier:** This is the primary component responsible for converting AC to DC. It uses diodes to allow current to flow in only one direction.

There are different types of rectifiers:

- **Half-Wave Rectifier:** Uses a single diode to convert one half of the AC waveform into DC.
 - **Full-Wave Rectifier:** Uses multiple diodes to convert both halves of the AC waveform into DC.
 - **Bridge Rectifier:** A more efficient type of full-wave rectifier using four diodes to allow full utilization of the AC waveform.
- iii. **Filter (Capacitors or Inductors):** After rectification, the output is typically a pulsating DC. A filter is used to smooth the pulsations and provide a more stable DC output by reducing ripple.
 - iv. **Voltage Regulator:** This component stabilizes the DC output voltage to a constant level, which is necessary for sensitive electronic devices.

2. DC to AC converter

A DC to AC converter, commonly known as an **inverter**, is a device that converts direct current (DC), into alternating current (AC). Inverters are widely used in applications where devices need to operate on AC power, but the available power source is DC, such as in solar power systems, battery-powered devices, and uninterruptible power supplies (UPS).



Components of a DC to AC Converter:

- i. **DC Input:** The input source is typically a DC power supply, such as a battery, solar panel, or other DC sources.
- ii. **Oscillator:** The oscillator generates a high-frequency switching signal, which is necessary to convert the DC input into a form suitable for AC generation. The switching frequency depends on the type of inverter and the application.
- iii. **Switching Devices: Transistors, MOSFETs, or IGBTs** (Insulated Gate Bipolar Transistors) are used to control the switching of the DC input. These devices rapidly switch the DC input on and off to simulate the alternating waveform.
- iv. **Transformer:** In some designs, a transformer is used to step up the output voltage to the required AC voltage level (e.g., from 12V DC to 220V or 110V AC).
- v. **Filtering Circuit:** Capacitors and inductors are used to smooth out the sharp pulses generated during the switching process, resulting in a more refined and pure AC output.
- vi. **Output:** The output is an alternating current that can be used to power AC devices. Depending on the design, the output can be either a **modified sine wave**, **pure sine wave**, or **square wave**.

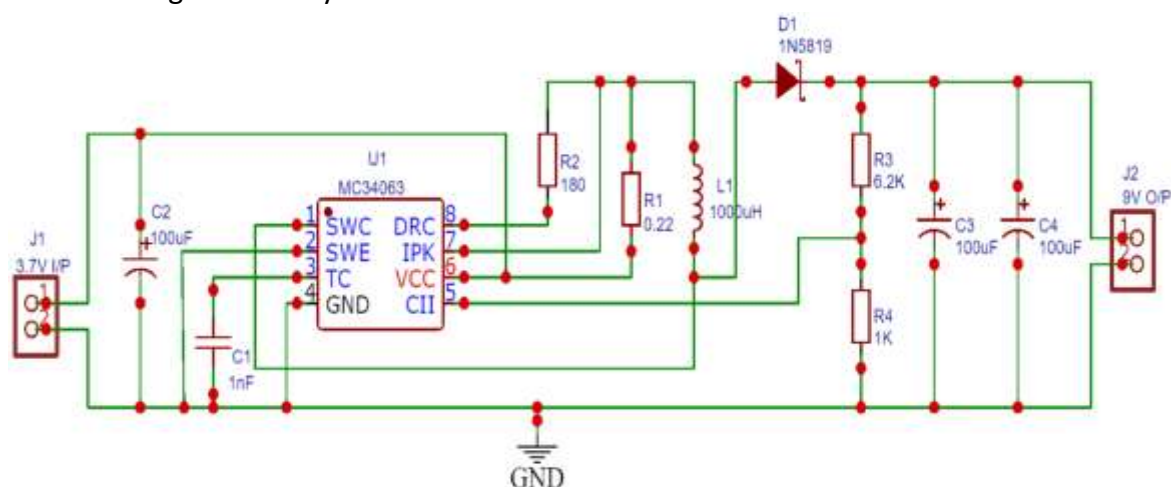
3. DC to DC converter

A DC to DC converter is an electronic device that changes the level of direct current (DC) voltage. It consists of several key components that work together to regulate or change the voltage efficiently:

Main Components of a DC to DC Converter:

- i. **Switching Element (Transistor):** A transistor (e.g., MOSFET or IGBT) is used to rapidly switch the input DC voltage on and off, which helps in regulating the output voltage. This switching is controlled by a pulse-width modulation (PWM) signal.
- ii. **Inductor:** The inductor stores energy when the switch is on and releases it when the switch is off, helping in voltage conversion. It smooths out the current and minimizes voltage ripples.
- iii. **Diode:** The diode provides a path for current when the switching element is off, allowing the stored energy in the inductor to be delivered to the load. It ensures unidirectional current flow.
- iv. **Capacitor:** Capacitors filter and smooth out the voltage at the output, reducing any fluctuations and providing a stable DC voltage.
- v. **Control Circuit:** The control circuit (often using a PWM controller) regulates the switching element, adjusting the duty cycle to maintain a constant output voltage regardless of input variations or load changes.

These components allow the converter to step up (boost), step down (buck), or invert the DC voltage efficiently.



Types of DC to DC converters are:

- **Buck converter** is a type of DC-DC power converter that steps down (reduces) the input voltage to a lower output voltage while maintaining efficient energy transfer.
- **Boost converter** is a type of DC-DC power converter that steps up (increases) the input voltage to a higher output voltage. It is commonly used when the available supply voltage is lower than what is required for a specific application or load.

4. AC to AC converter

An **AC to AC converter** is an electronic device that changes the voltage level or frequency of alternating current (AC) without converting it to direct current (DC). Here are the main components commonly found in AC to AC converters:

Main Components of an AC to AC Converter:

1. Transformer:

- ✓ **Function:** Used to step up (increase) or step down (decrease) the AC voltage. It consists of primary and secondary coils.
- ✓ **Operation:** Based on electromagnetic induction, where alternating current in one coil induces a voltage in another coil.

2. Switching Devices:

- ✓ **Types:** Includes thyristors, triacs, or IGBTs (Insulated Gate Bipolar Transistors).
- ✓ **Function:** Control the flow of AC power by turning it on and off at specific points in the AC waveform to regulate voltage or frequency.

3. Control Circuit:

- ✓ **Function:** Monitors and adjusts the operation of the switching devices to achieve the desired output voltage or frequency.
- ✓ **Components:** May include microcontrollers, feedback loops, and phase-locked loops (PLLs) for precise control.

4. Output Filter:

- ✓ **Function:** Reduces harmonics and smooths the output waveform, ensuring a cleaner AC signal is delivered to the load.
- ✓ **Components:** Typically consists of capacitors and inductors arranged to filter out unwanted frequencies.

5. Rectification Components (if applicable):

- ✓ In some AC to AC converters, rectification components may be used to convert AC to DC momentarily before converting it back to AC at a different frequency.

These components work together to effectively manage AC power conversion, allowing for voltage adjustment and frequency changes as needed for various applications.



Practical Activity 2.1.3: Installing simulation software



Task:

1: With referring to key readings 2.1.3 you are requested to perform the following task:

As technician in computer system architecture, you are requested to go in computer lab to install simulation software in computers.

2: Present the work done to trainer.

3: Ask question if any for clarification.



Key readings 2.1.3

Installation of simulation software

Several simulation software tools are commonly used for designing, analysing, and optimizing computer power systems and power electronics. These tools help engineers model circuits, simulate behaviours, and test designs for various power systems, including power supplies, converters, and distribution systems.

Here are some of the popular ones:

- Easy EDA,
- Proteus,
- Autodesk Eagle,
- NI Multisim,
- Matlab,
- CAD

The installation process for simulation software can vary depending on the specific software package. However, below is a generalized summary of the typical steps involved in installing simulation software.



Typical steps involved in installing simulation software:

- 1) **Download the Software:** Visit the official website of the simulation software provider and locate the download section. Download the installer package compatible with your operating system.
- 2) **Run the Installer:** Locate the downloaded installer file and double-click on it to run the installation process. This might involve an executable file (e.g., .exe for Windows) or a disk image (e.g., .dmg for macOS).

- 3) **Select Installation Location:** During the installation, you may be prompted to choose the directory or folder where you want the simulation software to be installed. The default location is often suggested.
- 4) **Choose Components:** Some simulation software packages allow users to choose specific components or features to install. Select the components you need or go with the default settings.
- 5) **Read and Accept License Agreement:** Review the license agreement or terms of use presented during the installation process. If you agree, check the acceptance box or click the "I agree" button.
- 6) **Configure Settings (if applicable):** Some simulation software may provide configuration options during installation. This may include settings related to user preferences, system integration, or licensing.
- 7) **Install Dependencies (if required):** Depending on the simulation software, there may be dependencies or runtime libraries that need to be installed. The installer may handle this automatically or provide instructions.
- 8) **Wait for Installation to Complete:** Allow the installation process to run. Depending on the size and complexity of the software, this may take some time. Be patient and avoid interrupting the process.
- 9) **Launch the Software:** Once the installation is complete, you may be prompted to launch the simulation software immediately. If not, you can usually find the software in your applications menu or start menu.
- 10) **Check for Updates (Optional):** After installation, it's a good practice to check for software updates. Many simulations tools release periodic updates with bug fixes, improvements, and new features.
- 11) **Activate or Register (if required):** Some simulation software may require activation or registration using a license key. Follow the provided instructions to activate the software if necessary.
- 12) **Verify Installation:** Open the simulation software and ensure that it launches without issues. Verify that all the features are accessible and that the software is functioning correctly.

Example of installing a simulation software:

Proteus software installation steps:

Step 1: Visit the official Labcenter Electronics website

(www.labcenter.com/downloads) or any trusted source to download the Proteus installer.

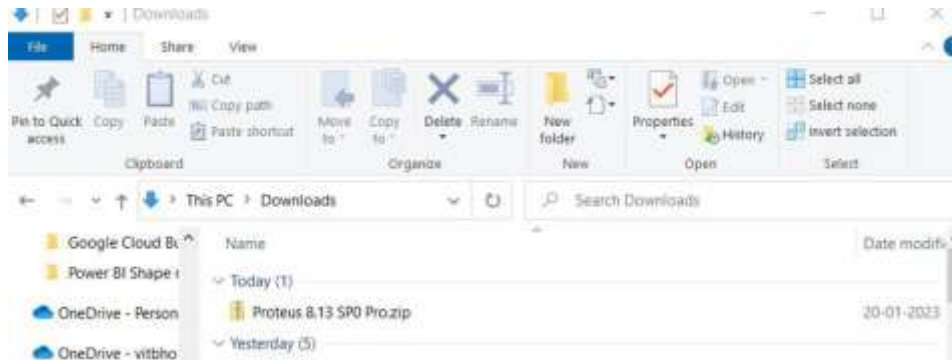
Here we are going to install Proteus 8.13. Let's Click on Download. It will show the following interface now click on Download.



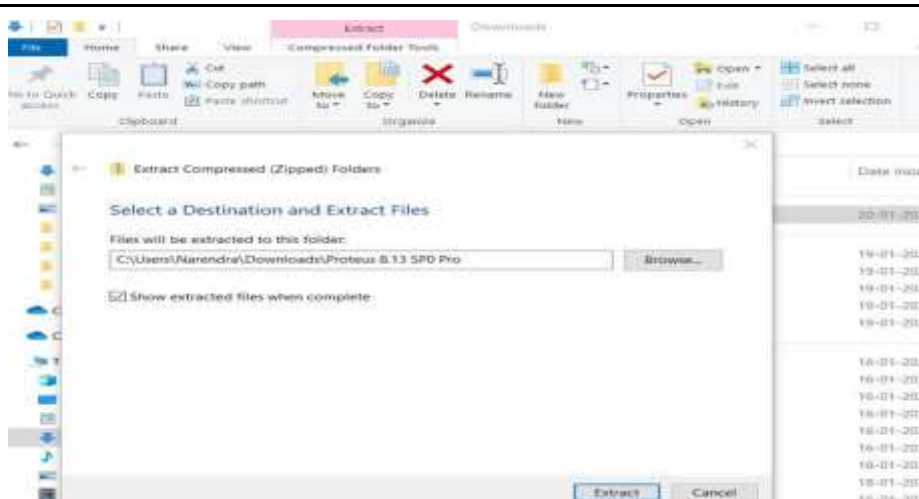
Step 2: It will start to download the zip file. Based on your internet speed it will take some time. Wait until the download process is completed.



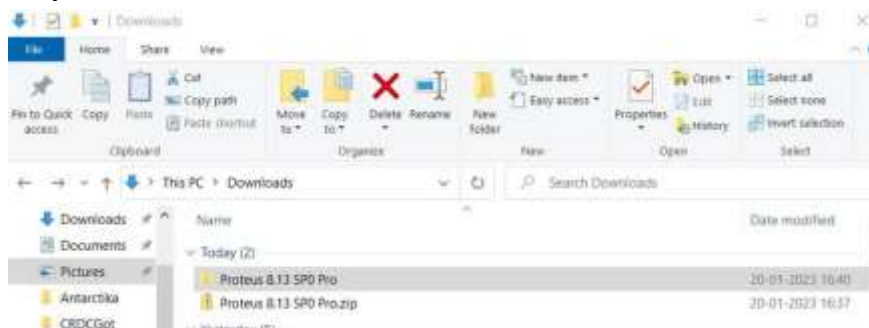
Step 3: When downloading is completed and then going to the download folder it will show the zip file.



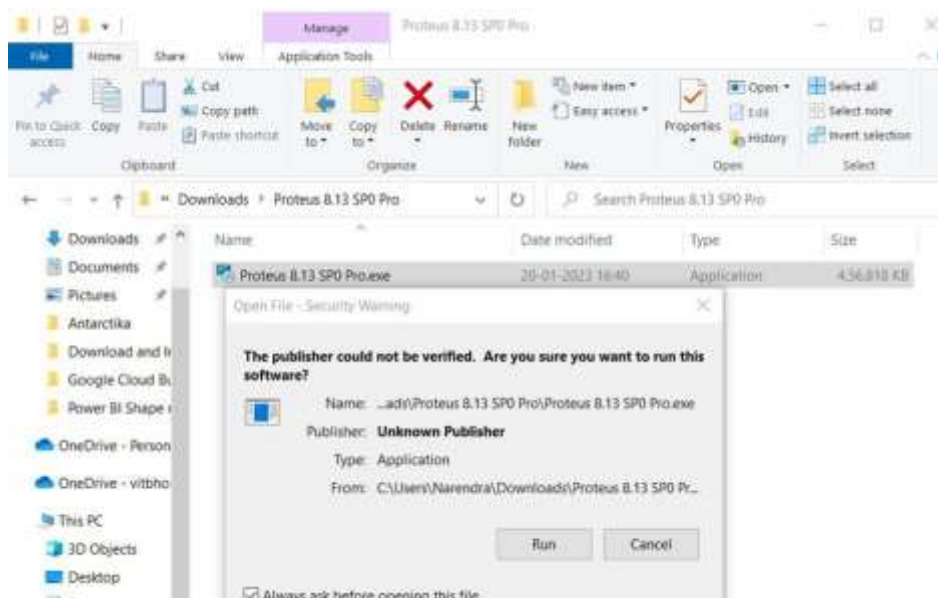
Step 4: Now on right-click the file and extract this using whatever software you have to extract the zip file and select the destination. Here, browse the location where you want to save.



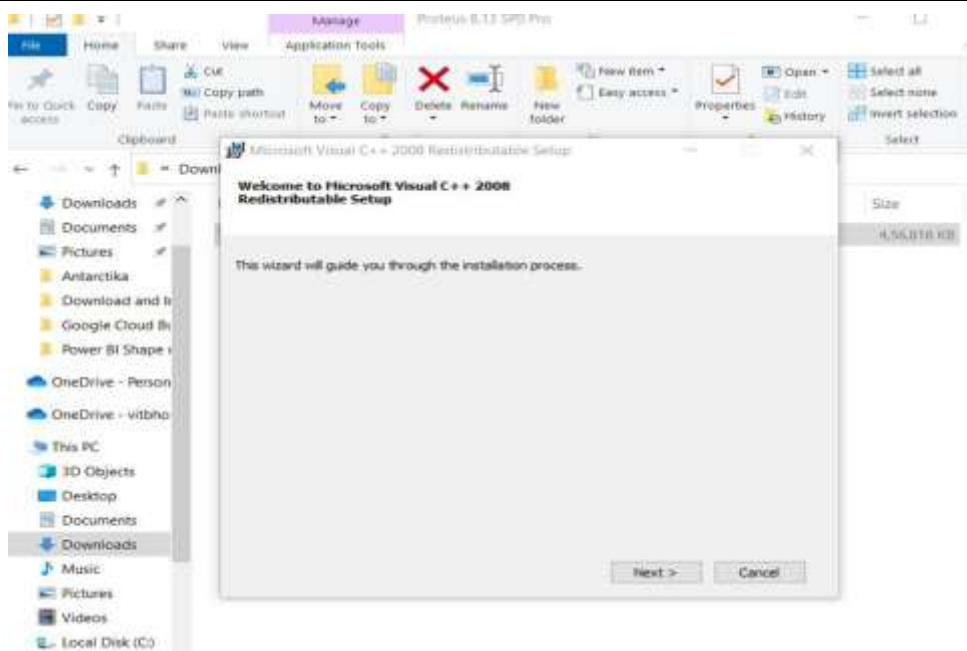
Step 5: Extracted folder will look like this.



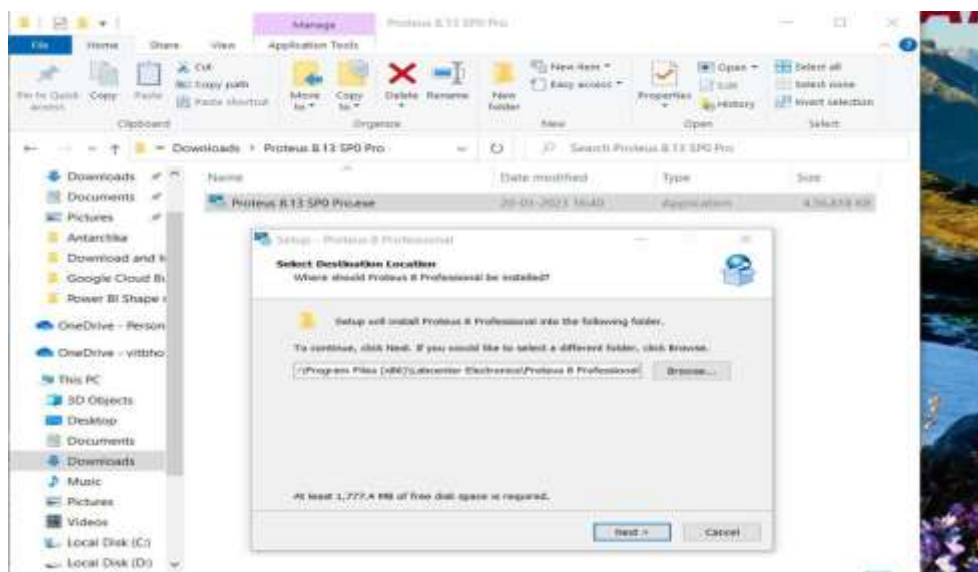
Step 6: Now open Proteus 8.13 SP0 Pro Folder and click on the .exe file. It will ask to run. Click on Run.



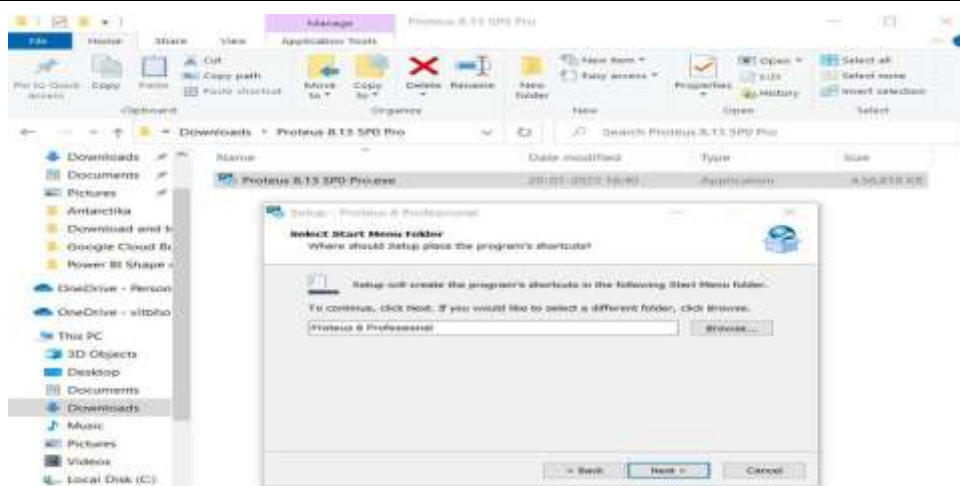
Step 7: In case you don't have an installation wizard then first it will ask the Click an install wizard. Click next and mark check on Agree. Then it will install the wizard, it is an optional process, if you already have this wizard then it will go to the next step directly.



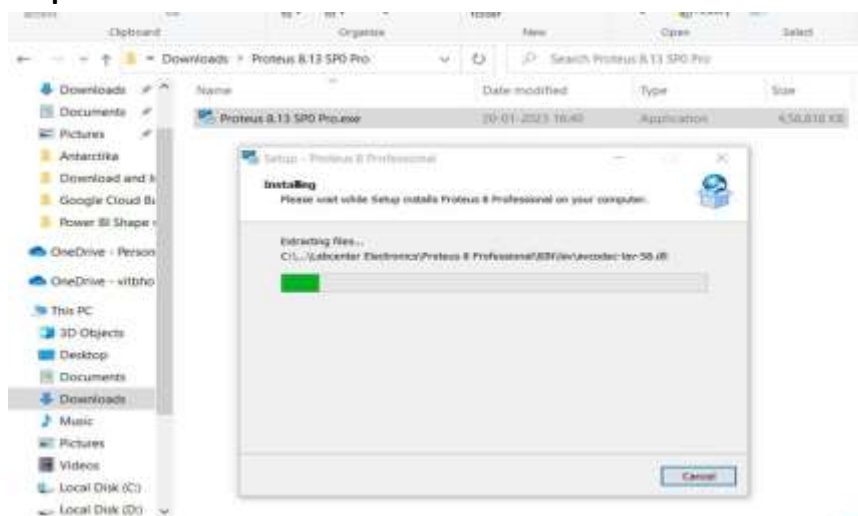
Step 8: Next it will show the following interface and we need to select the location where we want to install this software. Click on Next.



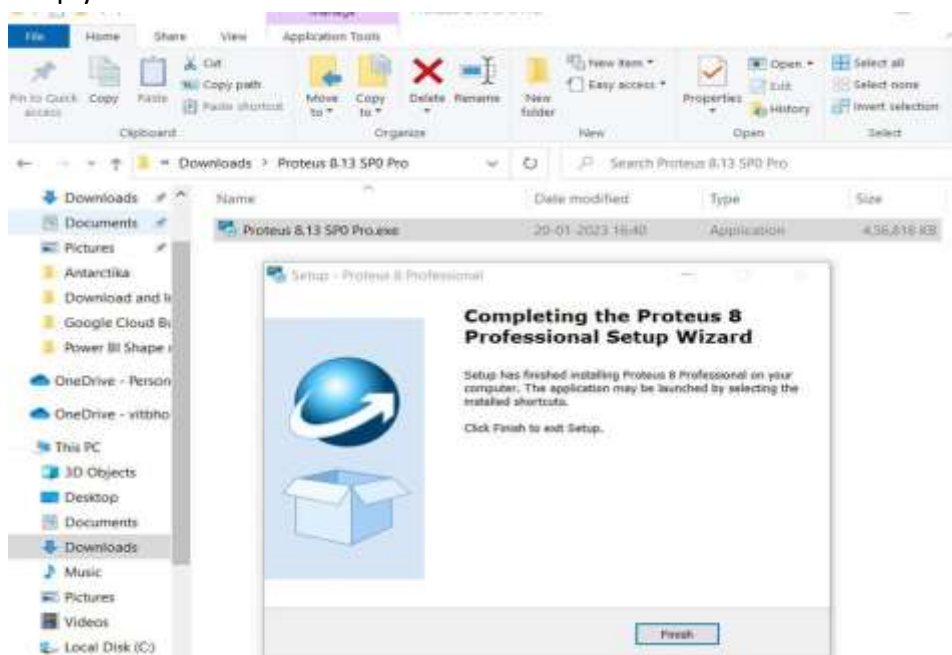
Step 9: Now it will show the following interface. Simply click on Next.



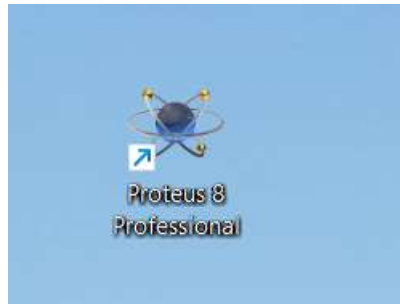
Step 10: Now it will start to install. It will take some time.



Step 11: After Completion of Installation it will show the following interface. Simply Click on Finish. Our Proteus 8.13 is installed in our Windows 10.



Step 12: The desktop icon of this software will show like this. For checking whether it is installed perfectly or not, click on Proteus Professional Icon.



Step 13: It is opening correctly and now we can start to work with this Proteus.



Practical Activity 2.1.4: Drawing power converters circuit diagrams

Task:

1: With referring to key readings 2.1.4 you are requested to perform the following task:
As technician in computer system architecture you are requested to open simulation software and draw the circuit diagram of the power converters below:

- AC to DC converter
- DC to AC converter
- DC to DC converter
- AC to AC converter

2: Present the work done to trainer.

3: Ask question if any for clarification



Key readings 2.1.4

Drawing of power converters circuit diagrams

1. Setting drawing environment in simulation software

Proteus is a design software developed by Labcenter Electronics for electronic circuit simulation, schematic capture and PCB design. Its simplicity and user-friendly design made it popular among electronics hobbyists.

Common interface parts and their functions in Proteus:

Main Toolbar: Contains various icons for quick access to frequently used tools and functions, such as opening, saving, and simulating projects.

Workspace Area: The central area where users design and simulate electronic circuits. It includes the schematic editor (ISIS) and, if applicable, the PCB layout editor (ARES).

Project Explorer: Provides a hierarchical view of the project components, including schematics, PCB layouts, libraries, and source code files. Users can navigate and manage project elements through this panel.

Output Panel: Displays messages, errors, and warnings generated during simulations or compilation. It helps users identify issues in their designs.

Components Toolbar: Contains a library of electronic components that users can drag and drop onto the workspace to create circuits. Components include resistors, capacitors, microcontrollers, sensors, etc.

Component Mode Toolbar: Offers options for placing and editing components on the schematic. Users can switch between placing components, wiring, and other editing modes.

Properties Panel: Displays properties and attributes of selected components or elements. Users can modify component values, set simulation parameters, and customize various properties.

Graphs and Instruments Toolbar: Provides tools for adding simulation probes, oscilloscopes, multimeters, and other measurement instruments to the schematic for real-time data visualisation during simulation.

Control Panel: Includes simulation controls such as play, pause, stop, and step buttons. Users can control the simulation process and analyse the behaviour of their circuits.

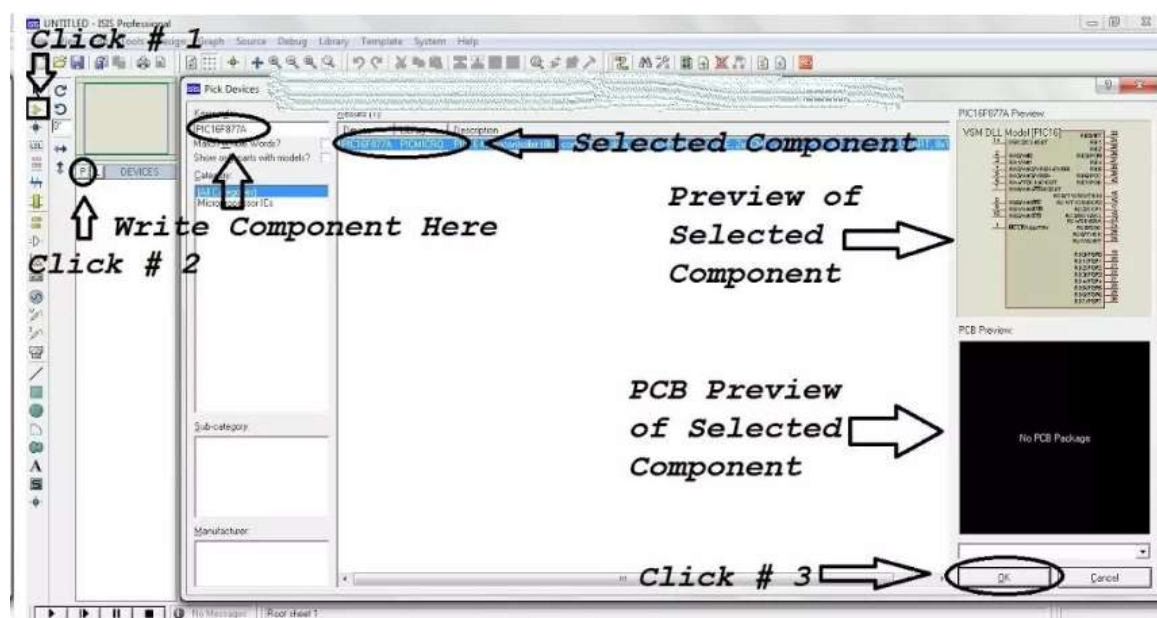
Status Bar: Displays information about the current status of the software, including simulation progress, grid settings, and coordinates of the cursor.

Toolbox: Contains additional tools and functions that users can use during schematic or PCB design. This may include drawing tools, annotation tools, and utilities.

Library Manager: Allows users to manage and customise component libraries. Users can add, remove, or modify existing libraries to expand the available components.

Resource Selector: Enables users to select and switch between different resources, such as schematic sheets, PCB layouts, and 3D views.

3D Viewer (if available): Provides a three-dimensional view of the PCB layout, allowing users to visualise the physical appearance of the designed circuit.

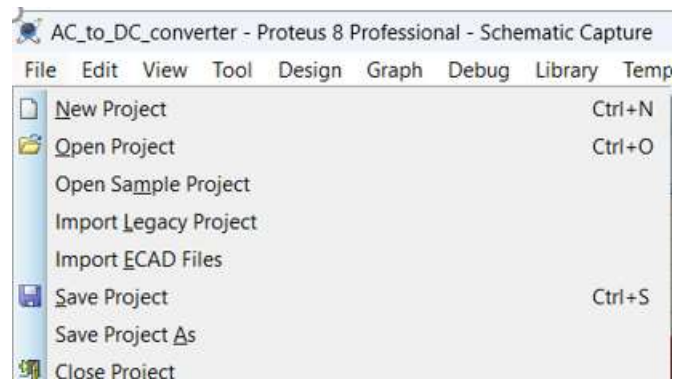


To draw a circuit diagram in Proteus, you can follow these steps:

Step 1: Open Proteus

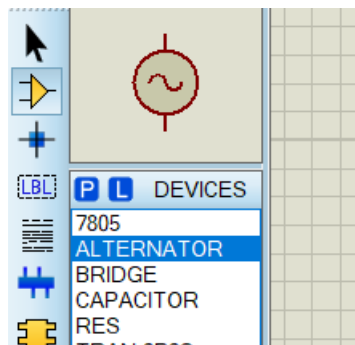
- Launch the Proteus application from your computer.

Step 2: Create a New Project



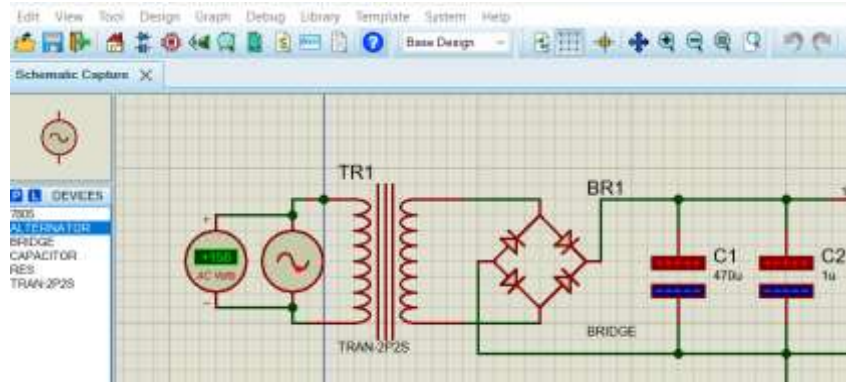
- Go to File → New Project.
- Enter the project name and specify a location for saving your project.
- Follow the wizard to create the project. Choose Create a Schematic from Scratch.

Step 3: Add Components to the Workspace



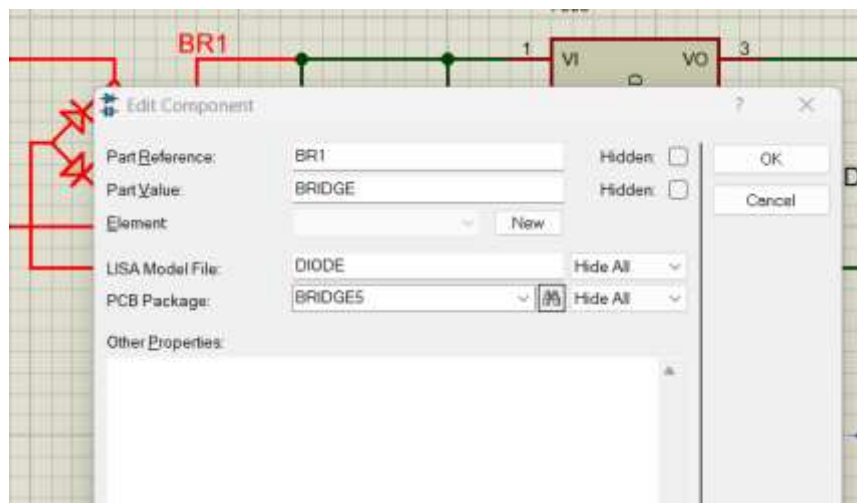
- In the schematic capture window, click on P (Component Mode).
- Search for the components needed for your power converter (e.g., MOSFETs, diodes, capacitors, inductors, power sources, and control ICs).
 - For example, if simulating a Buck Converter, you might need a DC supply, MOSFETs, inductors, and a diode.
- Click on the desired component, and press OK to place it on the workspace.

Step 4: Arrange and Connect Components



- Use your mouse to place each component on the schematic.
- Connect the components using wires by clicking on the pin of one component and dragging it to another.
- Add other circuit elements like resistors, capacitors, or switches as needed.
- Ensure you have a load connected (like a resistor or other loads) to test the output.

Step 5: Configure Components

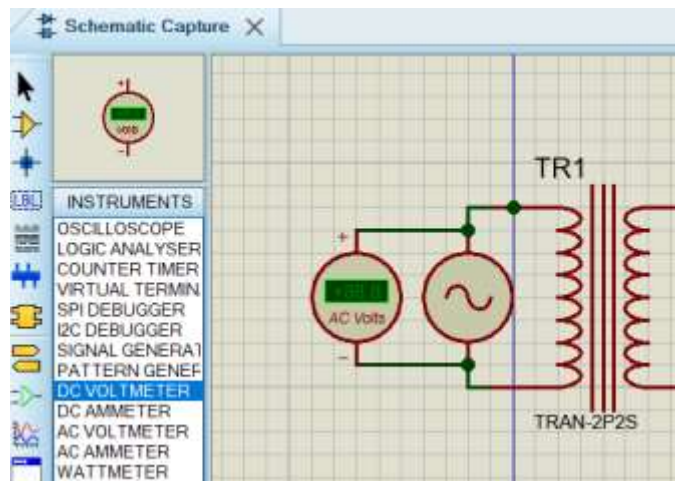


- Double-click on each component to configure its parameters.
 - Set the values for inductors, capacitors, resistors, voltage sources, and any other parameters relevant to your simulation.

Step 6: Add Ground and Power Supply

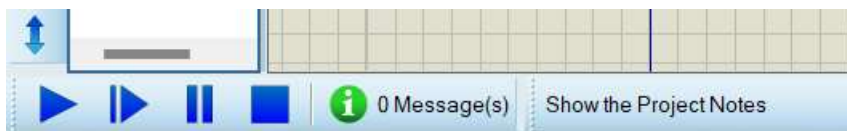
- Don't forget to add a ground to your circuit by selecting Terminal Mode and choosing the Ground symbol.
- If necessary, also add a DC or AC voltage source from the component list (for the input of your converter).

Step 7: Add Probes or Virtual Instruments



- For monitoring voltage or current in specific points of the circuit, you can add virtual instruments like oscilloscopes, voltmeters, or ammeters.
 - Go to Instruments Mode, select the instrument, and place it in your circuit.
 - Wire the probes to the relevant points (e.g., to measure input and output voltages).

Step 8: Run the Simulation



- After completing the circuit, press the Play button at the bottom of the window to run the simulation.
- Observe the behaviour of the power converter and adjust component values as needed.

Step 9: Analyze the Results

- You can view the output waveforms or values on the virtual instruments (oscilloscopes, meters, etc.).
- You can stop the simulation and tweak the circuit if needed, then run it again to see how changes affect the system.

Step 10: Save the Project

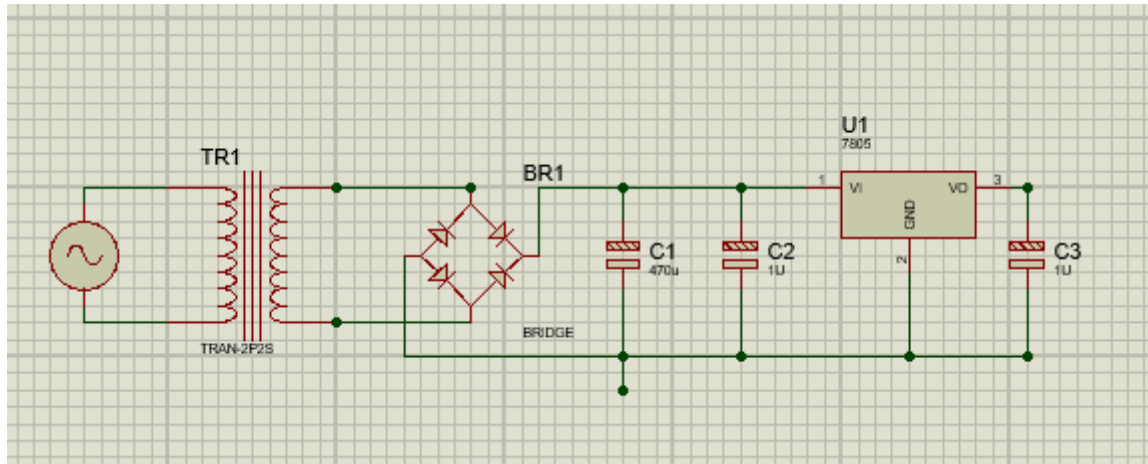
- Once satisfied, save your project by going to File → Save Project.

With these steps, you can simulate different types of power converters such as buck

converters, boost converters, or buck-boost converters in Proteus.

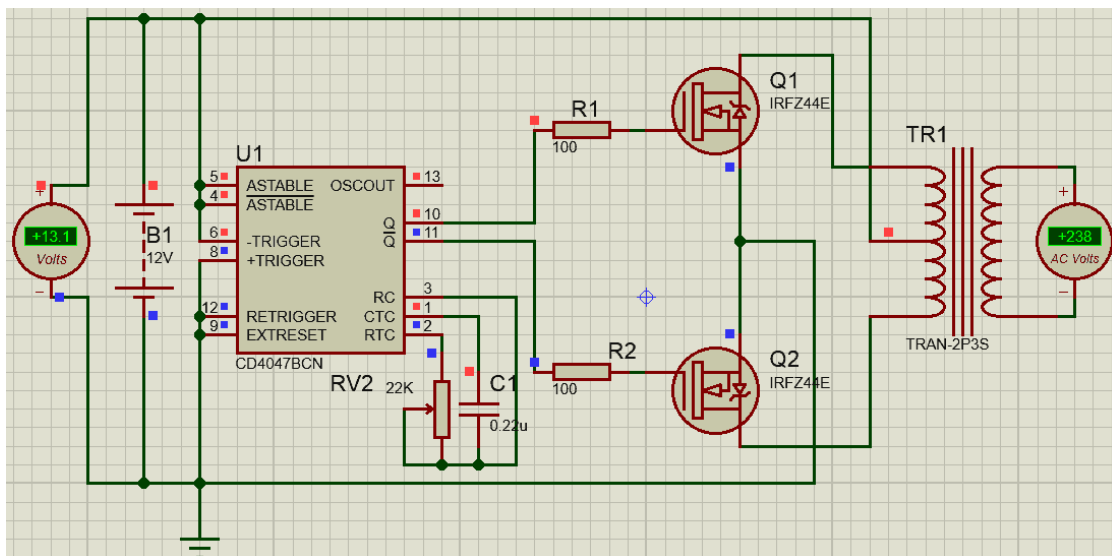
2. Power converters circuit diagrams

2.1 AC to DC converter



The circuit above consists of AC main power source, a step down transformer, a bridge rectifier, two capacitors for filtration and a regulator to provide a steady DC power.

2.2 DC to AC converter



The 12V DC to 220V AC inverter circuit is designed using IC CD4047. The IC CD4047 acts as a switching pulse oscillating device. The n-channel power MOSFET IRFZ44n acts as a switch. The 12-0-12V secondary transformer inversely used as a Step-up transformer from converting low AC to High AC.

Components Required

The following components are required for making this Inverter project.

1. IC CD4047
2. IRFZ44 Power MOSFET – 2

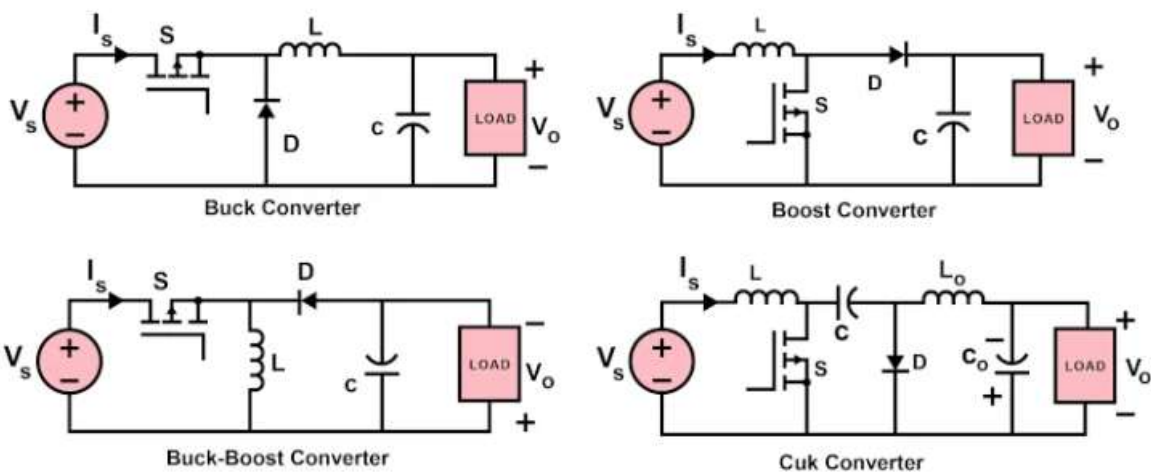
3. 12-0-12/1A Secondary Transformer
4. 22K Ω Variable Resistor
5. 100 Ω /10W Resistors – 2
6. 0.22 μ F Capacitors
7. 12V Sealed Lead Acid Battery

2.3 DC to DC converter

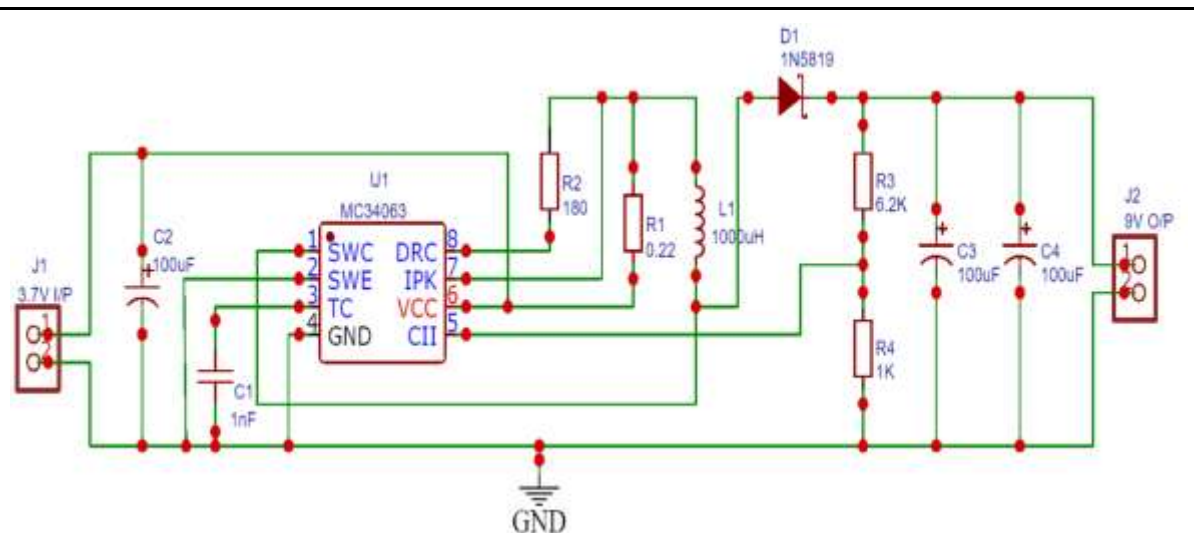
A DC to DC power converter is an electronic device that changes a direct current (DC) input from one voltage level to another, either stepping it up (boost) or stepping it down (buck). These converters are essential in applications requiring voltage regulation or transformation in portable devices, power supplies, and renewable energy systems.

Include the following types:

- **Buck Converter:** Steps down the voltage.
- **Boost Converter:** Steps up the voltage.
- **Buck-Boost Converter:** Steps up or down the voltage depending on the input.
- **Cuk Converter:** Provides an inverted output voltage.



The following is a **3.7V to 9V Boost Converter Circuit** using **MC34063** DC-to-DC converter IC.

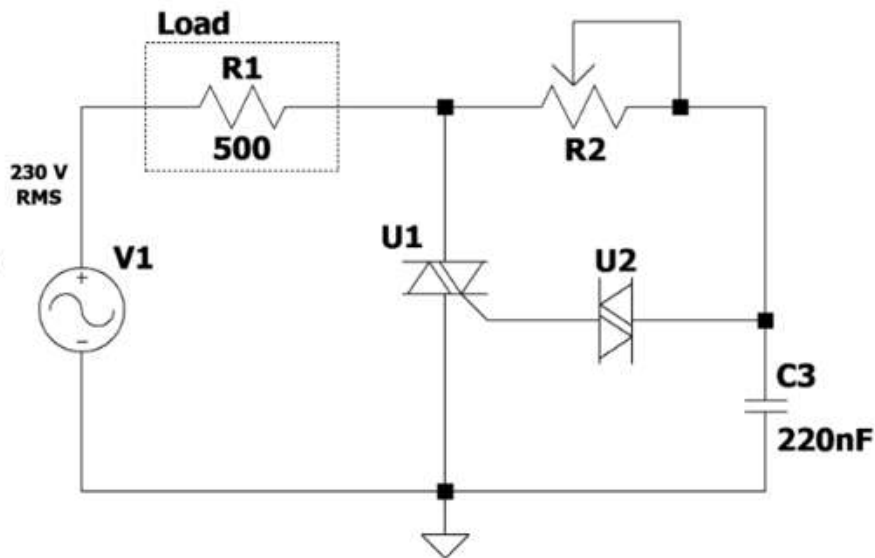


The list of components that we need for designing 3.7V to 9V Boost Converter Circuit is given below:

S.N.	Components Name	Description
1	MC34063	DC-DC Converter IC
2	1N5819	Schottky Diode
3	100uH	Inductor
4	100uF	Capacitor
5	1nF	Capacitor
6	6.2K	Resistor
7	1K	Resistor
8	180 Ohm	Resistor
9	0.22 ohm	Resistor
10	3.7V Battery	3.7V Lithium-Ion/Lithium Polymer Battery

2.4 AC to AC converter

AC-AC conversion, also known as frequency conversion, voltage conversion, or phase conversion, converts one alternating voltage (AC) to another at a different frequency or voltage. This process is used in many applications, including powering electric motors, speed regulation, and voltage modulation.



As seen in the circuit diagram above, the basic circuit consists of a Triac, a Diac, a potentiometer, a resistor, and a capacitor.



Practical Activity 2.1.5: Simulating power converters



Task:

- 1: With referring to key readings 2.1.5 you are requested to perform the following task:
As technician in computer system architecture, you are requested to go in computer lab, open simulation software and simulate the following power converters:
 - a) AC to DC converter
 - b) DC to AC converter
 - c) DC to DC converter
 - d) AC to AC converter
- 2: Present the work done to trainer.
- 3: Ask question if any for clarification



Key readings 2.1.5

Simulating the power converter

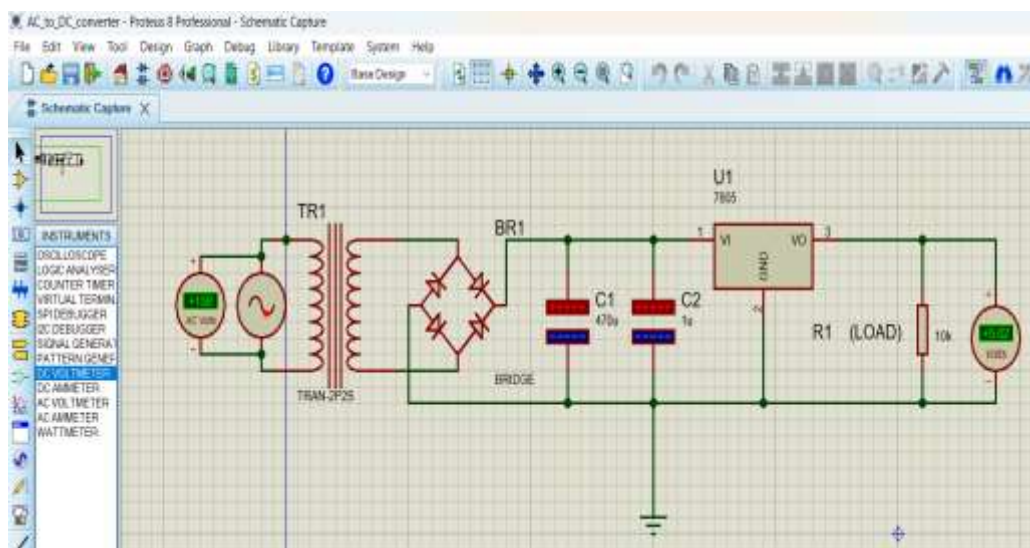
To simulate an electronic power converter circuit (such as a DC-DC converter, AC-DC rectifier, DC-AC inverter and AC to AC), follow these key steps:

- Define the Objective of the Simulation
- Select Simulation Software
- Model the Power Converter Circuit
- Configure the Circuit Topology
- Set Simulation Parameters
- Run the Simulation
- Analyse the Results
- Optimize the Design

1. Simulating AC to DC converter

To simulate an AC to DC converter in any simulation software, you set up a basic circuit that includes components such as a transformer, a bridge rectifier, a capacitor for filtering, and a load. Proteus provides a visual environment for electronics simulation and is particularly useful for designing circuits and embedded systems.

Below is a simulation of AC to DC converter using Proteus:

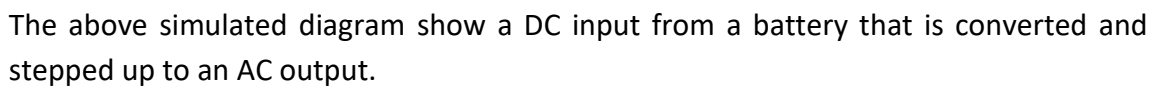


The above simulated diagram show AC input that is converted and regulated to DC at the output.

2. Simulating DC to AC converter

To simulate a DC to AC converter (inverter) in Proteus, you will need to design a basic inverter circuit that converts a DC input (e.g., from a battery) into an AC output. A simple way to achieve this is by using components like MOSFETs or transistors as switches, a PWM controller such as Timer (CD4047BCN) for signal generation, and an optional

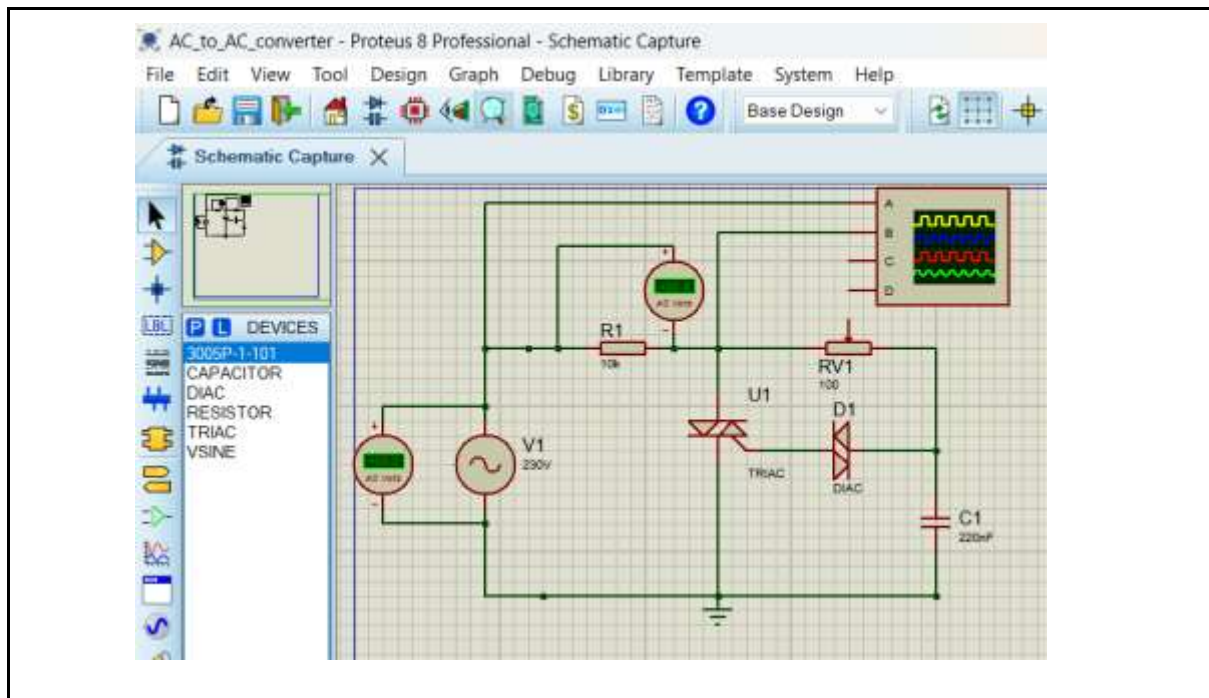
Here is a simple DC to AC simulation using Proteus:



Below is a simple simulation of DC to DC converter using a DC to DC converter IC



To simulate a simple AC to AC converter (such as an AC voltage controller), you will need an AC power source, a Triac to control the voltage, a Diac for triggering the Triac, two resistors and a capacitors for the timing circuit that sets the firing angle, a load (like a resistor) to represent the output, and an oscilloscope to observe the input and output AC waveforms. These components will allow you to adjust the amplitude of the output AC voltage effectively.



Practical Activity 2.1.6: Generating power converter PCB Design



Task:

- 1: With referring to key readings 2.1.6 you are requested to perform the following task:
As technician in computer system architecture, you are requested go in computer lab, open a project in simulation software and generate a PCB design of power converter.
- 2: Present the work done to trainer.
- 3: Ask question if any for clarification



Key readings 2.1.6

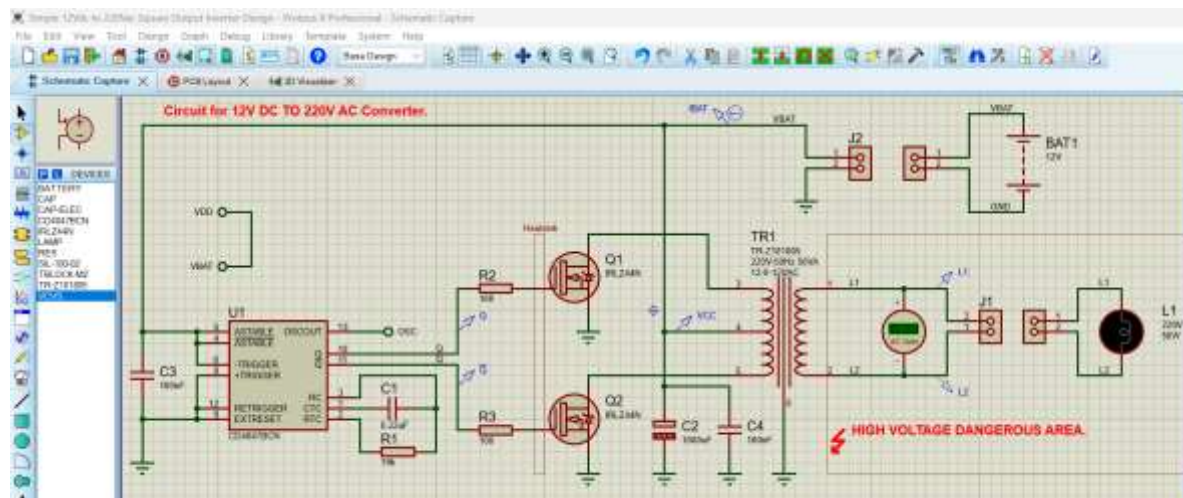
Generating power converter PCB Design

Designing a DC to AC power converter PCB involves several stages, from understanding the circuit to generating the final PCB layout. Here's a step-by-step guide to creating a DC to AC power converter PCB design using Proteus or any PCB design tool:

1. Create a Schematic in Proteus

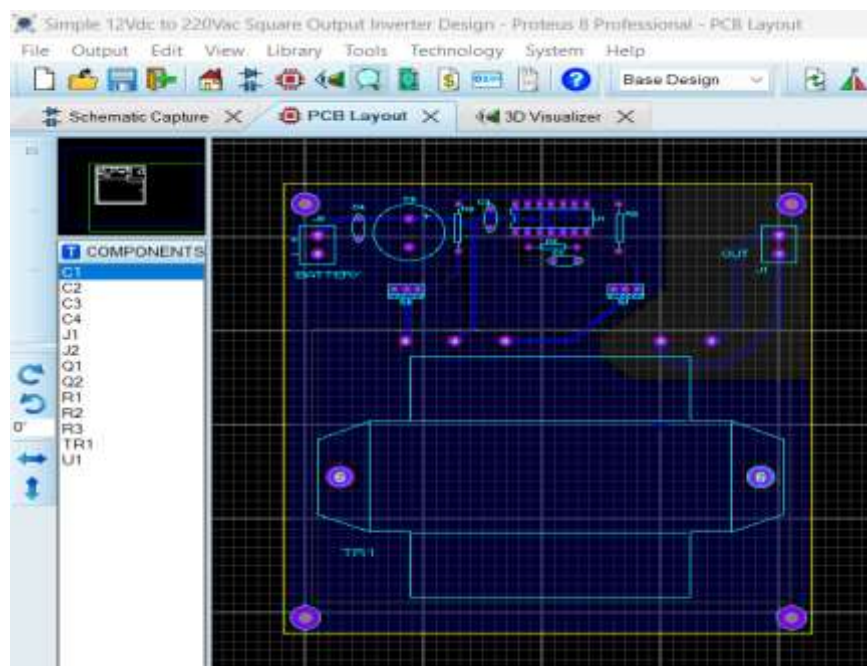
- Open a New Project: Create a new project and select Schematic Capture.
- Place Components: Select and place the components based on your DC to AC circuit design.
- Connect Components: Use the wire tool to connect the components as per

your circuit diagram.



2. Assign PCB Footprints

- Assign Footprints: For each component in the schematic, assign a corresponding PCB footprint that matches its physical size and package type.
- Check Component Footprints: Ensure that high-power components like MOSFETs, transformers, and capacitors have sufficient space for heat dissipation and current handling.



3. Design the PCB Layout

- Define Board Shape and Size: Set the dimensions of your PCB according to the design.

- Place Components: Arrange the components on the PCB, considering the following:
 - High-power components should have adequate heat dissipation (MOSFETs, transformers).
 - Separate low-power and control circuits from high-power switching circuits.
 - Group components by their functions (e.g., the MOSFETs and their gate drivers should be placed close together).



By following these steps, you can design a DC to AC power converter PCB, ensuring correct component placement, routing, and thermal management for efficient operation.



Points to Remember

- **Common types of power semiconductor devices:**
 - **Diodes:** is a two terminal electronic components that allow current to flow in one direction only.
 - **Transistors:** is a three-terminal semiconductor device used to amplify or switch electronic signals and power.
 - **Thyristors:** is a four-layer, three-terminal semiconductor device used to control and switch high voltage and current.

- **IGBTs (Insulated Gate Bipolar Transistors):** is a power semiconductor device that combines the fast switching of a MOSFET with the high current and voltage-handling capabilities of a bipolar transistor.
- **MOSFETs (Metal-Oxide-Semiconductor Field-Effect Transistors):** is a voltage-controlled semiconductor device used for switching and amplifying electronic signals.
- **Power electronic converters** are devices that convert electrical energy from one form to another using power semiconductor devices. They are widely used in modern electrical systems to control and manage the flow of electrical energy efficiently. There are four main types of power electronic converters:
 - **AC-DC Converters (Rectifiers):** Converts alternating current (AC) to direct current (DC).
 - **DC-DC Converters (buck and boost converter):** Converts one DC voltage level to another DC voltage level (either step-up or step-down).
 - **DC-AC Converters (Inverters):** Converts direct current (DC) into alternating current (AC).
 - **AC-AC Converters:** Converts AC from one frequency and/or voltage level to another.
- **Simulation software** for power electronic converters allows engineers to model, analyse, and optimize converter designs before actual implementation. These tools help in understanding the converter's behaviour, switching characteristics, thermal performance, and efficiency under various operating conditions. Examples of Circuit Simulation and drawing software are NI Multisim, Proteus, Autodesk Eagle, Easy EDA, matlab, CAD. You may try many software to explore different features.
- Steps which apply when you're drawing power converter:
 - i) Understand the Requirements
 - Determine input/output specifications: Know the input voltage, output voltage, and current requirements.
 - Select the type of converter
 - ii) Choose Components
 - iii) Draw the Power Stages
 - iv) Draw the Control Stages
 - v) Label Components
- To simulate an electronic power converter circuit (such as a DC-DC converter, AC-DC rectifier, DC-AC inverter and AC to AC), follow these key steps:
 - i. Define the Objective of the Simulation

- ii. Select Simulation Software
 - iii. Model the Power Converter Circuit
 - iv. Configure the Circuit Topology
 - v. Set Simulation Parameters
 - vi. Run the Simulation
 - vii. Analyse the Results
 - viii. Optimize the Design
- Step-by-step guide to generate PCB of power converter using any PCB design tool:
 - i. Create a Schematic
 - ii. Assign PCB Footprints
 - iii. Generate PCB Layout



Application of learning 2.1.

ABC is a computer manufacturing company; it has a large command of designing electronic power converts.

As one of the designers in that company you are requested to design an AC to DC converter by drawing, simulating, and generating PCB of AC to DC converter and this converter will convert standard 230V AC from the wall socket into the required 12V DC voltages for different parts of the computer.



Indicative content 2.2: Assembling Power Electronic Converter Components



Duration: 10 hrs



Practical Activity 2.2.1: Assembling power electronic converter components



Task:

- 1: With referring to key readings 2.2 you are requested to perform the following task:
As technician in computer system architecture you are requested go in workshop to assemble power electronic converter components.
- 2: Present the work done to trainer.
- 3: Ask question if any for clarification



Key readings 2.2.1

Assembling power electronic converter components

1. Placing components into PCB

Fixing components on a printed circuit board (PCB) requires precision and care to ensure the components are correctly installed without damaging the board. Here are some essential tips:

i. Preparation

- Clean the PCB: Ensure the board is free from dust, debris, or old solder using a soft brush or an isopropyl alcohol solution.
- Use the right tools: Ensure you have appropriate tools like a temperature-controlled soldering iron, tweezers, solder, flux, desoldering braid/pump, and magnifying glass.

ii. Component Placement

- Double-check polarity: For polarized components (capacitors, diodes, etc.), ensure correct polarity before soldering.
- Fit snugly: Insert through-hole components so they sit flush with the PCB. For surface-mount devices (SMDs), align them carefully with the pads.
- Use flux: Flux helps the solder flow better and adheres well to the component and the PCB pads, creating a solid connection.

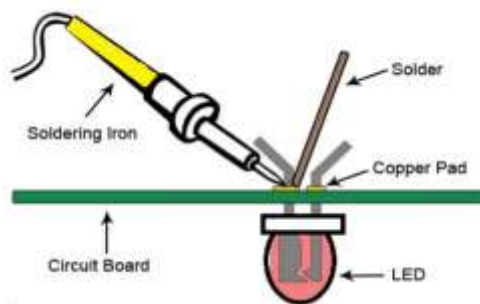
2.2.2 Soldering the components

Soldering is an essential skill for assembling and repairing electronics. Proper soldering

techniques ensure strong, reliable electrical connections. Here are various soldering techniques and best practices:

Through-Hole Soldering

- **Step 1:** Insert the component lead through the hole in the PCB.
- **Step 2:** Heat the joint by applying the soldering iron to both the lead and the pad simultaneously.
- **Step 3:** After heating for about 1-2 seconds, introduce the solder to the joint (not directly to the iron).
- **Step 4:** Allow the solder to flow and form a concave shape around the joint, then remove the soldering iron and let the joint cool naturally.



2.2.3 Wiring and connection of components

Wiring and connecting components in electronics requires careful attention to detail for proper circuit functionality and reliability.

It starts with selecting the appropriate wires based on current and voltage ratings, ensuring that each connection is solid and free from short circuits. For through-hole components, wires are inserted into the PCB holes and soldered on the opposite side, while surface-mount components use pads.

Proper handling of polarized components like diodes, capacitors, and transistors is crucial to avoid damage or malfunction. Twisting and tinning wires before soldering improves connection strength, and heat shrink tubing can be used to insulate exposed joints.



Points to Remember

- Placing components on PCB:
 - **Correct Orientation:** Ensure that polarized components like diodes, electrolytic capacitors, and ICs are oriented correctly according to the circuit design and markings on the PCB.

- **Component Spacing:** Leave enough space between heat-generating components, such as transistors and voltage regulators, to allow for heat dissipation and avoid thermal issues.
- **Soldering components on PCB:**
 - **Clean the Solder Pads:** Ensure that the PCB pads are clean and free of oxidation or contaminants to allow the solder to flow properly, ensuring strong connections.
 - **Use the Right Temperature:** Use a soldering iron at the correct temperature (typically around 350°C for most components) to avoid damaging components and ensure clean, reliable solder joints.
- **Wiring and connecting components:**
 - **Secure Wiring:** Ensure wires are neatly routed and securely fastened to the PCB or connectors, using cable ties or clips to prevent loose or tangled wiring, which can cause stress on solder joints or short circuits.



Application of learning 2.2.

You are a technician tasked with assembling a DC power converter for a small electronic device. The device requires a stable 12V DC power supply, but the available input is 220V AC from a mains power source.

Using a PCB design provided by your engineering team, you must assemble a working AC to DC power converter.



Indicative content 2.3: Testing Power Electronic Converters



Duration: 5 hrs



Practical Activity 2.3.1: Testing power electronic converters



Task:

1: With referring to key readings 2.2 you are requested to perform the following task:

As technician in computer system architecture, you are requested go in workshop to test power electronic converter.

2: Present the work done to trainer.

3: Ask question if any for clarification



Key readings 2.3.1

Testing power electronic converters

2.3.1 Continuity testing

A continuity test in the context of power electronic converters is a diagnostic procedure used to check for proper electrical connections between components. It verifies that there is a continuous electrical path between two points in the circuit. This test ensures that there are no open circuits, broken connections, or unintended high-resistance paths, which could affect the converter's performance.

Steps for Performing a Continuity Test:

1. **Power Down the System:** Make sure the converter is disconnected from the power supply to avoid accidents or damage to the testing equipment.
2. **Set the Multimeter to Continuity Mode:** Most multimeters have a dedicated continuity setting.
3. **Test the Multimeter:** Touch the probes together; a beep should indicate proper function.
4. **Test the Component or Circuit:** Place the multimeter probes across the two points you want to test. If the multimeter beeps, it indicates that the path is

continuous (low resistance). No beep or a high-resistance reading suggests an open circuit or a bad connection.

2.3.2 Measure voltage and current

A measurement test in a power electronic converter involves assessing various electrical parameters to ensure that the converter is functioning as intended. These tests are critical for verifying the performance, efficiency, and safety of the converter and typically involve measuring quantities such as voltage, current, power, and switching characteristics.

Key Measurement Tests in Power Electronic Converters:

1. Voltage Measurement:

- ✓ **Input Voltage:** Ensures that the correct voltage is supplied to the converter. This is essential for the proper functioning of the power electronics components.
- ✓ **Output Voltage:** Verifies that the converter produces the expected output voltage, whether it's AC or DC. For example, a DC-DC converter should provide the correct regulated output voltage.
- ✓ **Ripple Voltage:** Measures the AC component superimposed on the DC output voltage, which is crucial for ensuring low noise and stable output.

Equipment: Digital multimeter (DMM), oscilloscope.

2. Current Measurement:

- ✓ **Input Current:** Ensures that the converter draws the correct amount of current, which indicates proper power consumption.
- ✓ **Output Current:** Verifies that the load is receiving the expected amount of current. Too much current may indicate an overload or short circuit, while too little may suggest an underperforming converter.
- ✓ **Switching Current (for switches like MOSFETs/IGBTs):** Used to monitor the current through power switches, which is important for analyzing their performance and ensuring they don't exceed their rated limits.

Equipment: Clamp meter, DMM, current probes (for oscilloscopes).

Procedure for Performing Measurement Tests:

1. **Prepare the Converter:** Ensure the converter is in a safe and stable operating condition, connected to the correct input source and load.

2. **Select the Appropriate Test Points:** Identify the points in the circuit where measurements need to be taken (input/output, across switches, etc.).
3. **Choose the Right Equipment:** Use calibrated and appropriate instruments based on the type of measurement (voltage, current, temperature, etc.).
4. **Record Data:** Take measurements under various operating conditions (e.g., different load levels, temperatures, and switching frequencies).
5. **Analyze the Results:** Compare the measured values against design specifications or expected performance criteria.

2.3.3 Analyse output waveform

Analysing the output waveform of a power electronic converter is essential to ensure that it operates as expected and meets the desired performance specifications. Here's how to effectively analyse the output waveform of a power electronic converter:

Steps to Analyze the Output Waveform:

1. Select the Appropriate Equipment:

To capture and analyze the waveform, you need suitable tools:

- **Oscilloscope:** The primary tool for observing the shape, amplitude, and frequency of the waveform.
- **Digital Multimeter (DMM):** For basic measurements like RMS voltage, current, etc.

2. Configure the Oscilloscope:

- **Set Time and Voltage Scales:** Choose appropriate time base (time/division) and voltage scale (volts/division) based on the expected waveform characteristics.
- **Trigger Settings:** Set the trigger mode to capture the waveform reliably

3. Capture the Output Waveform:

- **Connect the Oscilloscope Probes:** Ensure proper probe grounding to avoid noise and interference in the measurement.
- **Observe the Waveform:** Take note of the shape (sine wave, square wave, sawtooth, or complex pulse wave) and any anomalies such as ripples, spikes, or distortions.

4. Analyse the Waveform Characteristics:

- **Amplitude:**

- ✓ **Peak Voltage (V_{peak}):** The maximum voltage value from the waveform.
- ✓ **Root Mean Square (RMS) Voltage:** This is particularly important for AC waveforms to assess the equivalent DC voltage. For AC inverters, the RMS voltage should match the desired output level.
- **Frequency:** Measure the output frequency to ensure it matches the design specifications.
- **Ripple: Voltage Ripple:** For DC converters, check the magnitude of the ripple voltage (small AC component superimposed on the DC output). Excessive ripple can affect the performance of sensitive electronics.



Points to Remember

- **Continuity testing** is a process used to check whether there is an uninterrupted path (or complete circuit) for current to flow in an electrical system or component. It helps identify open circuits, short circuits, or loose connections in wires, cables, fuses, or electronic components.
- **Measurement testing** refers to the process of determining and verifying various electrical, physical, or performance parameters in systems, components, or devices using specific measurement instruments. It involves quantifying properties like voltage, current, resistance, power, frequency, to ensure the system or device meets its design specifications and operates within expected tolerances.



Application of learning 2.3

You are an electronics technician responsible for testing the functionality and performance of a power electronic converter. The converter under test could be a DC-DC step-down (buck) converter, a DC-AC inverter, or an AC-DC rectifier.

Your goal is to ensure that the converter meets the design specifications and operates safely and reliably under different conditions.



Learning outcome 2 end assessment

Theoretical assessment

A. Multiple-choice questions on power electronic converters:

Question 1:

Which of the following is a type of DC-DC converter?

- A) Inverter
- B) Buck Converter
- C) Rectifier
- D) Transformer

Question 2:

What is the main purpose of an inverter in power electronics?

- A) To convert DC to AC
- B) To convert AC to DC
- C) To step down voltage
- D) To step up voltage

Question 3:

In a full-bridge rectifier, what is the output waveform for an AC input?

- A) Pure sine wave
- B) Pulsating DC
- C) Pure DC
- D) Square wave

Question 4:

Which of the following converters is used to step up DC voltage?

- A) Buck Converter
- B) Boost Converter
- C) Inverter

D) AC-DC Converter

Question 5:

What is the role of a PWM (Pulse Width Modulation) technique in converters?

- A) To reduce harmonic distortion
- B) To regulate output voltage
- C) To increase current flow
- D) To convert AC to DC

B. Match each converter component with its respective description

Column A: Converter Component	Column B: Function/Description
1. Inductor	A. Smooths the output voltage by reducing ripple
2. Diode	B.
3. Resistor	C. Switches on and off to control energy flow
4. Capacitor	D. Ensures current flows in one direction
5. Switching Device (e.g., MOSFET)	E. Stores energy in a magnetic field
6. Controller (PWM Control)	F. Regulates the switching to achieve desired voltage level

Practical assessment

XYZ Company is dedicated to manufacturing of computer accessories. As a technician in computer system architecture, your task is to design, assemble and test a power converter for the desktop computer.

The power converter will take the incoming AC mains voltage and convert it to regulated DC voltages to power various computer components.



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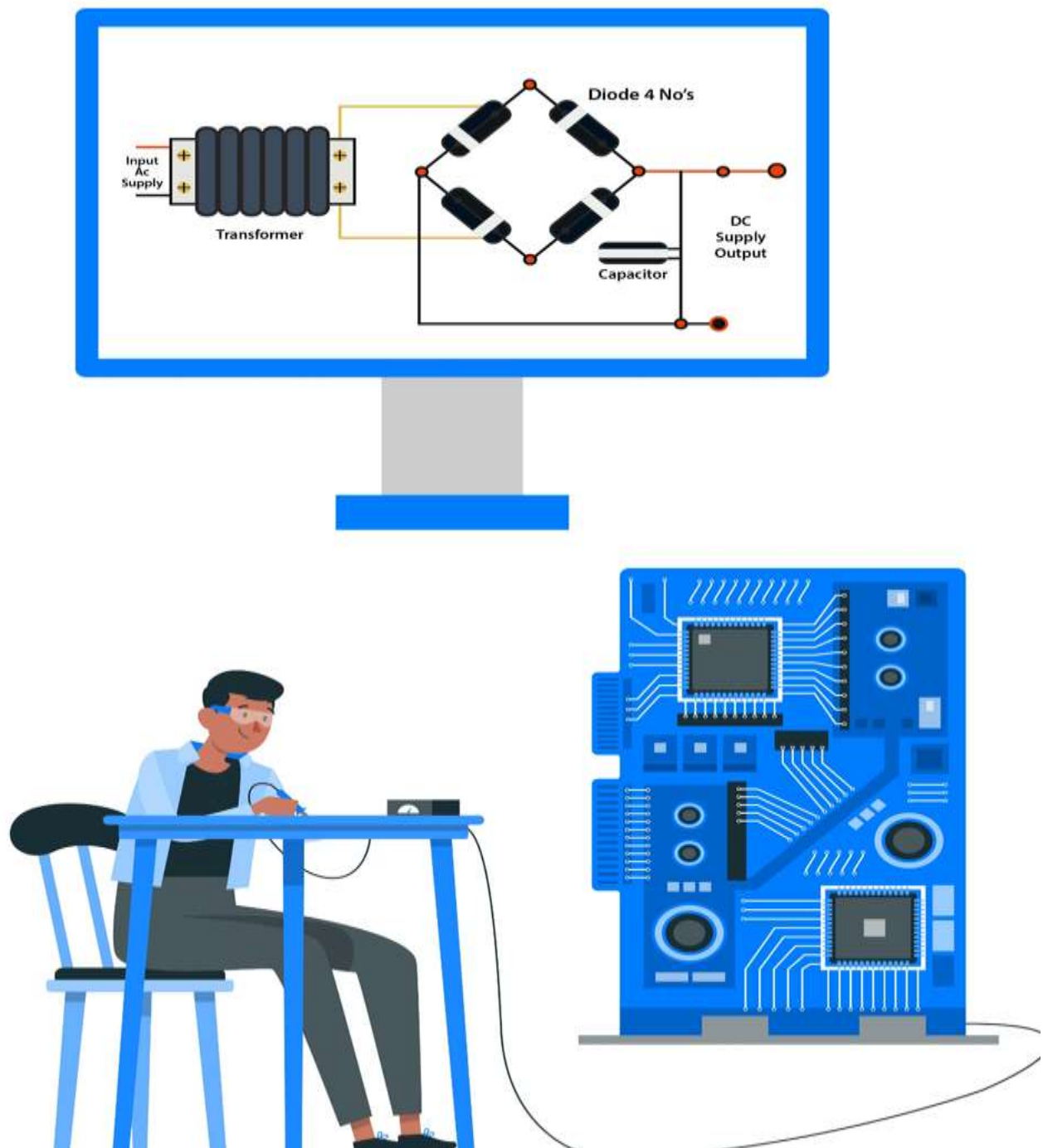
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END

Learning Outcome 3: Implement a Power Supply Unit (PSU) of Computer System



Indicative contents

3.1 Designing computer power distribution system

3.2 Assembling power supply unit

3.3 Testing computer supply unit

Key Competencies for Learning Outcome 3: Implement a Power Supply Unit (PSU) of a Computer System

Knowledge	Skills	Attitudes
• Interpretation of voltage regulation concepts • Description of a power supply types • Description of electrical quantities	• Designing a power distribution scheme • Drawing of schematic diagram of power supply unit • Simulating the power supply unit circuit diagrams • Assembling power supply unit • Testing computer supply unit	• Being responsible • Having attention to details • Being rapid • Being organized • Willing to explore new ideas



Duration: 20 hrs

Learning outcome 3 objectives:



By the end of the learning outcome, the trainees will be able to:

1. Interpret correctly the voltage regulation in accordance with the power supply unit.
2. Design correctly the computer power distribution system according to system power requirements
3. Describe appropriately the power supply types in accordance with the power supply unit.
4. Draw correctly the schematic diagram of power supply unit in line with Computer power supply system.
5. Simulate systematically the power supply unit circuit diagrams in line with Computer power supply system.
6. Generate correctly power supply PCB in line with Computer power supply system.
7. Assemble properly the power supply unit in line with Computer power distribution system design.
8. Describe correctly testing concepts in line with Computer power distribution system design.
9. Inspect appropriately computer power supply unit in line with Computer power supply system.
10. Measure correctly input and output voltage and current of power supply unit in line with Computer power supply system.
11. Test systematically the power supply unit concepts in line with Computer power supply system.



Resources

Equipment	Tools	Materials
• Personal Computer,	• Screwdrivers	• Labelling materials

• PPEs, • Oscilloscope, • Multi-meter, • ESD Workbench, • SMD rework station,	• Needle-nose pliers • Universal plier • Tweezers • Soldering iron • Desoldering pump • Glue gun • Circuit Simulation and drawing software	• Connectors • Power Semiconductor devices • Soldering materials • Power cables • Glue stick • PCB • Electronic Passive Components • Spare screws • Isopropyl alcohol and cotton swabs
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Indicative content 3.1: Designing Computer Power Distribution System



Duration: 4 hrs



Theoretical Activity 3.1.1: Interpretation of voltage regulation concepts.

Tasks:

- 1: Interpret the following questions related to the voltage regulation in power supply.
 - i. What do you understand by the following terms?
 - a) Voltage divider circuit
 - b) Voltage rails
 - c) Automatic voltage regulator
 - d) Voltage regulator module
 - e) Power distribution Module
- 2: Write the findings/answers on papers or flipchart.
- 3: Present your findings to the class.
- 4: Ask questions for more clarifications, if any.
- 5: Read the key reading 3.1.1 from your manual to get more clarifications.



Key readings 3.1.1.:

Interpretation of Voltage Regulation concepts in Computer Power Supply Unit

1. Introduction

Voltage regulation ensures that the output voltage of a power supply remains constant despite variations in load or input voltage. It is typically expressed as a percentage and helps protect electronic devices from damage due to voltage fluctuations.

For example, a power supply with good voltage regulation will keep the output at 5V, whether the input voltage varies (line regulation) or the current drawn by the load changes (load regulation).

2. Voltage Divider Circuit

A voltage divider circuit is a simple electrical circuit that converts a higher voltage into a lower voltage. It consists of two resistors connected in series across a voltage source. The output voltage is taken from the junction of the two resistors. The formula for the output voltage (V_{out}) can be expressed as:

$$V_{out} = V_{in} \times \left(\frac{R_2}{R_1 + R_2} \right)$$

Where:

V_{in} is the input voltage,

R_1 is the resistor connected to the input voltage, and

R_2 is the resistor connected to ground.

In computer power supplies, this principle can be used for reference voltages or to create specific lower voltages needed for various components.

3. Voltage Rails

Voltage rails refer to the specific output voltages provided by a power supply unit (PSU). Each rail corresponds to a different voltage level required by various components within a computer system. Common voltage rails include +3.3V, +5V, and +12V. These rails are essential because different components like CPUs, GPUs, and storage devices require different operating voltages to function correctly. The stability and regulation of these rails are crucial for reliable operation; fluctuations can lead to system instability or hardware damage.

4. Automatic Voltage Regulator (AVR)

An Automatic Voltage Regulator (AVR) is an electronic device that automatically maintains a constant voltage level. In computer power supplies, AVRs are used to ensure that the output voltage remains stable despite variations in input voltage or load conditions. They work by monitoring the output voltage and making adjustments as necessary through feedback mechanisms. This regulation helps protect sensitive electronic components from overvoltage or under voltage conditions, which can cause malfunction or damage.

AVRs are commonly used in generators, power distribution systems, and large electrical equipment. They ensure that devices receive stable voltage regardless of input fluctuations, which is critical in places with unstable power supply conditions.

How it works:

- Senses output voltage.
- Compares it to a reference.
- Automatically adjusts the internal control circuitry to stabilize the voltage.

5. Voltage Regulator Module (VRM)

A Voltage Regulator Module (VRM) is a specialized circuit designed to provide precise control of output voltages for specific components such as CPUs and GPUs within computers. VRMs convert higher input voltages down to lower levels while maintaining high efficiency and low ripple noise. They typically consist of multiple phases that distribute current load evenly across several inductors and capacitors, improving thermal performance and reliability. VRMs are critical in modern computing systems where power demands are high and efficiency is paramount.

A **Voltage Regulator Module (VRM)** is a component used in motherboards and other electronic systems to regulate and supply a precise voltage to microprocessors and other high-demand components.

In modern computer systems:

- VRMs are crucial for stepping down the voltage from the power supply (e.g., 12V) to the specific voltage required by the CPU or GPU (e.g., 1.2V–1.5V).
- They dynamically adjust the output voltage based on the power demand of the processor, ensuring stable performance.

A VRM is made of power MOSFETs, inductors, and capacitors and is responsible for protecting sensitive electronics from voltage fluctuations.

6. Power distribution scheme

A **power distribution scheme** in a **computer power system** refers to the way electrical power is delivered and managed across various components of the computer. It ensures

that each component (e.g., motherboard, processor, memory, storage drives, and peripheral devices) receives the right voltage and current required for optimal operation.



Practical Activity 3.1.2: Designing a power distribution scheme



Task:

1. Referring to the instruction from your trainer, you are requested to perform the given task below. The task should be done individually.

As electronician, you are asked to go in the electronic workshop and design a power distribution scheme for a computer system, understanding how different components of the system require various power levels and how power is distributed efficiently.

2. Read key reading 3.1.2 then perform the activity.

3. Present your work to the trainer and whole class

4. Ask questions if any for clarification

5. Perform the task provided in application of learning 3.2



Key readings 3.1.2

Designing a power distribution scheme

Designing a power distribution scheme in a computer system's power supply involves a series of careful considerations and steps. The aim is to ensure stable and efficient power delivery to all components.

Steps of Designing power distribution scheme

Here's a step-by-step outline:

Step1. Understand Power Requirements: Analyse the power requirements of each component in the system (e.g., CPU, GPU, RAM, storage, motherboard) including Voltage and Current Requirements and Power Ratings.

Step2. Select the Power Supply Unit (PSU): Based on the power requirements of each component, calculate the total power the PSU needs to provide. Also consider the Efficiency Ratings and Form Factor Compatibility.

Step3. Define Power Distribution Buses: Design power buses to distribute the correct voltage to the corresponding components.

Step4. Protection Mechanisms: Incorporate protective mechanisms to safeguard the system in case of faults or surges by designing the system with necessary fuses and circuit breakers to prevent damage to components.

Step5. Regulation and Conversion: Include VRMs to ensure a steady voltage supply even under varying loads and convert the main power rail voltages to other necessary levels where needed.

Step6. Power Connectors and Cabling: Ensure the proper use of power connectors and design the layout for proper cable management to ensure clean power flow and reduce heat build-up in the case.

Step7. Cooling Considerations: Ensure the PSU and other high-power components have adequate cooling (e.g., fans or heat sinks) to prevent overheating.

Step8. Testing and Validation: After assembling the power distribution system, perform stress tests to ensure the PSU and power rails can handle maximum loads and use the voltage monitoring tools to check for stability and efficiency under different operating conditions.

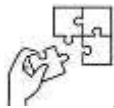
Step9. Compliance and Safety Standards: Design the system to comply with efficiency standards like Energy Star, reducing overall power consumption and heat output.



Points to Remember

- **Description of voltage regulation concepts in Power supply system**
 - ✓ A voltage divider circuit is a simple electrical circuit that converts a higher voltage into a lower voltage.
 - ✓ Voltage rails refer to the specific output voltages provided by a power supply unit (PSU). Each rail corresponds to a different voltage level required by various components within a computer system.
 - ✓ An Automatic Voltage Regulator (AVR) is an electronic device that automatically maintains a constant voltage level.
 - ✓ A Voltage Regulator Module (VRM) is a specialized circuit designed to provide precise control of output voltages for specific components such as CPUs and GPUs within computers.

- ✓ A power distribution scheme in a computer power system refers to the way electrical power is delivered and managed across various components of the computer.
- **Steps of Designing power distribution scheme**
 - Step1.** Understand Power Requirements
 - Step2.** Select the Power Supply Unit (PSU)
 - Step3.** Define Power Distribution Buses
 - Step4.** Incorporate protective mechanisms to safeguard the system in case of faults
 - Step5.** Include VRMs to ensure a steady voltage supply even under varying loads
 - Step6.** Ensure the proper use of power connectors
 - Step7.** Ensure the PSU and other high-power components have adequate cooling to prevent overheating.
 - Step8.** Tests to ensure the PSU and power rails can handle maximum loads and use the voltage monitoring tools to check for stability and efficiency under different operating conditions.
 - Step9.** Design the system to comply with efficiency standards



Application of learning 3.1.

You are a system designer tasked with creating a custom high-performance workstation for a company specializing in 3D rendering, AI computations, and video editing.

The workstation will have multiple high-end graphics cards (GPUs), large storage arrays (SSDs and HDDs), and a powerful multi-core processor (CPU).

You are requested to design the power distribution scheme for the system to ensure reliable performance.



Indicative content 3.2: Assembling Power Supply Unit



Duration: 9 hrs



Theoretical Activity 3.2.1: Description of a power supply types



Tasks:

1: Describe the following questions related to power supply types

- i. Describe the following types of power supply:
 - a) Linear power supply
 - b) Switched Mode power supply (SMPS)
 - c) Uninterruptable power supply (UPS)
 - d) Programmable Power supply
 - e) Variable Power Supply
 - f) Circuit Simulation
- ii. What do you understand by Circuit Simulation?

2: Write the findings/answers on papers or flipchart.

3: Present your findings to the class.

4: Ask questions for more clarifications, if any.

5: Read the key reading 3.1.1 from your manual to get more clarifications.



Key readings 3.2.1.:

Description of a power supply types

1. Linear power supply

a) Definition

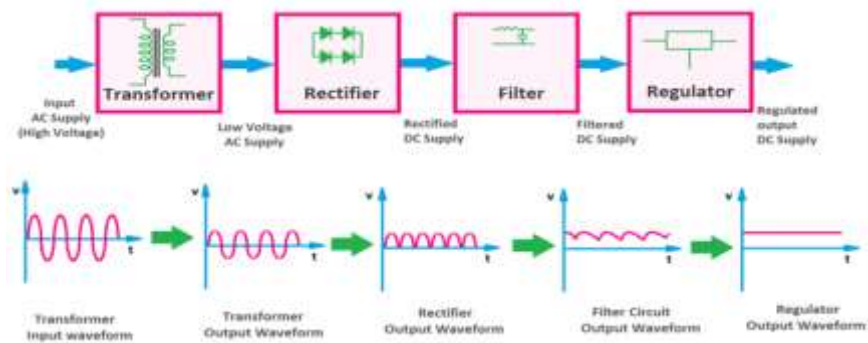
A linear power supply converts AC voltage to a stable DC voltage. It uses a transformer with an iron core to step down the AC voltage. Then, a rectifier converts AC to DC. Finally, a filter smooths the DC voltage, and a regulator maintains a constant output. Linear power supplies are known for their reliability and low noise, making them ideal for sensitive electronic equipment.

b) Types of Linear Power Supply

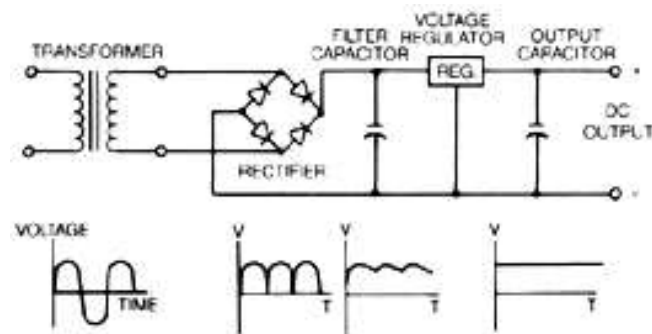
There are typically two types of linear power supplies:

- ✧ Unregulated Linear
- ✧ Regulated Linear
- ✓ **Regulated Linear Power Supply**

Voltage regulation in regulated power supplies refers to maintaining the voltage at the desired level. This is particularly suitable for sensitive connected appliances as regulated voltage provides a smooth, steady voltage supply.



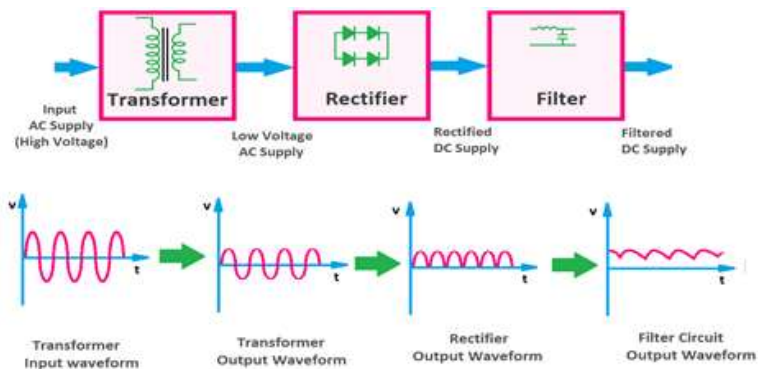
The process of producing regulated voltage includes a series of sub-functions. First, the AC supply is stepped down to the desired output level, and then the reduced AC voltage is transformed into a positive waveform by a diode bridge rectifying circuit. A filter circuit comprising a parallel-connected capacitor turns the positive waveform into a rippled-DC voltage, and finally a regulator circuit eliminates the ripples in the DC to deliver stable voltage to the connected load.



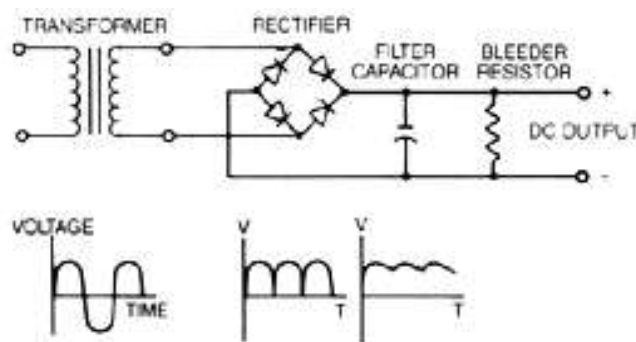
Regulated linear power supplies are ideal for almost all types of electronic devices, as these power supplies ensure smooth supply of voltage.

✓ **Unregulated Linear Power Supply**

As mentioned earlier and as the name suggests, regulation of voltage isn't involved in an unregulated linear power supply. Apart from that, the basic functions of an unregulated power supply are similar to regulated linear power supplies.



The AC input voltage, just like a regulated power supply, is initially processed and then transformed into rippled DC voltage through a filter capacitor. But since the unregulated power supplies don't have a voltage regulator circuit, the ripples in the DC output voltage cannot be done away with. This leads to voltage sags, which are often reflected in the output.



c) Regulated and Unregulated Power Supply: How Are They Different?

The basic difference between a regulated and an unregulated power supply is the presence or absence of voltage regulation. In the case of regulated linear, the voltage is regulated, which means any change in the input voltage isn't reflected in the output. On the contrary, unregulated power supplies do not have voltage regulation and for this reason, voltage is not regulated at the output.

d) Applications

- **Regulated power supply** are Application areas of a regulated power supply include TVs, computers, and other highly sensitive electronic devices where voltage variations can be fatal.

- **Unregulated linear power supply** are used in electrical devices like LED lamps and DC motors that are not so sensitive to voltage fluctuations can use unregulated power supplies.

2. Switched-Mode Power Supply (SMPS)

a) Definition

A Switched Mode Power Supply (SMPS) is an electronic power supply that efficiently converts electrical power using a switching regulator. Unlike traditional linear power supplies, which dissipate excess power as heat, an SMPS rapidly switches the input voltage on and off at high frequencies to regulate the output voltage, making it highly efficient.

SMPS are often smaller than and not as heavy as normal power supplies. This makes them perfect for tiny devices that you can take with you. They can also adjust to slight changes in the electricity that comes in, so they can deal with different voltages. This makes them more flexible than regular power supplies that only work with one specific voltage.

b) Key Features of SMPS:

- ✓ **Efficiency:** SMPS are much more efficient than linear power supplies because they minimize energy loss. The switching action reduces heat generation, which allows for smaller, more compact designs.
- ✓ **High-Frequency Operation:** SMPS operates at high switching frequencies (usually in the range of 20 kHz to several MHz), which allows for smaller components, such as transformers and inductors, contributing to the compact size of SMPS.
- ✓ **Versatility:** SMPS can work with a wide range of input voltages and can provide multiple regulated outputs at different voltages.

c) Types of SMPS:

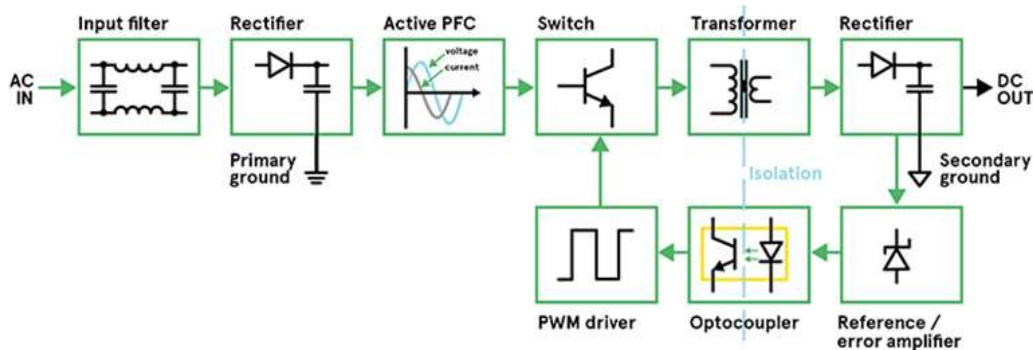
- ✓ **AC-DC Converters:** Converts AC input to a regulated DC output. Commonly used in computer power supplies.
- ✓ **DC-DC Converters:** Converts one level of DC voltage to another level, such as stepping up or stepping down the voltage.
- ✓ **DC-AC Inverters:** Converts DC input to AC output, often used in uninterruptible power supplies (UPS).

d) Working principle

An SMPS works by turning a quick, pulsing direct current (DC) into a slower, steady direct current (DC). This conversion process is accomplished through several stages, including the input stage, switching stage, output stage, and control circuit.

- **Input Stage:** The AC (or DC) input is first rectified and filtered to produce a high DC voltage.

- **Switching Stage:** A high-frequency switch (typically a MOSFET) turns the power on and off rapidly, controlling the flow of current through the transformer or inductor.
- **Output Stage:** The output is then rectified and filtered to remove high-frequency noise, providing a stable DC output.
- **Feedback Control:** The output is constantly monitored and fed back to the control circuit, which adjusts the switching duty cycle to maintain a stable output voltage under varying loads.



e) Applications of SMPS:

- Power supplies for computers, TVs, and mobile chargers
- Industrial equipment that requires efficient, high-power conversion
- LED drivers and lighting systems
- Telecommunications equipment

f) Selecting an AC-DC switched-mode power supply

When selecting an AC/DC switched-mode power supply, in addition to the required electrical specifications such as:

- ✧ input and output voltage,
- ✧ power handling, and
- ✧ operating efficiency,

3. Uninterruptible Power Supply (UPS)

a) Definition

An Uninterruptible Power Supply (UPS) is an electrical device that provides emergency power to connected equipment in case of a primary power failure. It ensures continuous power delivery for a limited duration, allowing users to safely shut down equipment or switch to an alternative power source, minimizing data loss, hardware damage, or operational downtime.

b) Major parts of a UPS:

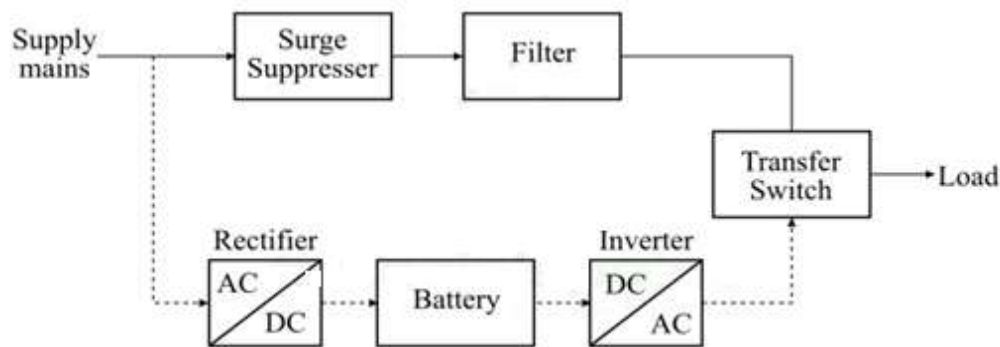
1. **Battery:** The core of a UPS, it stores electrical energy and delivers it when the main power source fails. Most UPS systems use rechargeable lead-acid or lithium-ion batteries.
2. **Rectifier/Charger:** This component converts incoming AC (Alternating Current) power to DC (Direct Current) to charge the battery. It also powers the inverter when the main power is present.
3. **Inverter:** When the main power supply fails, the inverter converts the stored DC power from the battery back into AC power to supply connected devices.
4. **Transfer Switch:** The transfer switch automatically switches between main power and battery power. When a power failure is detected, it shifts to battery power, ensuring no interruption to connected devices.
5. **Surge Protector:** Many UPS units also include a surge protection feature, safeguarding connected equipment from voltage spikes caused by power surges or lightning strikes.
6. **Control and Monitoring Circuit:** Modern UPS systems have monitoring circuits to display battery status, load levels, and alerts. Some are integrated with network monitoring tools to manage power remotely.

c) Types of UPS Systems:

The uninterrupted power supply (UPS) systems are broadly classified into the following three types:

- ✧ **Off-Line UPS:** In an off-line UPS system, the ac power from the supply mains is first converted into DC power using a rectifier circuit and then stored in the battery connected to the output port of the rectifier.

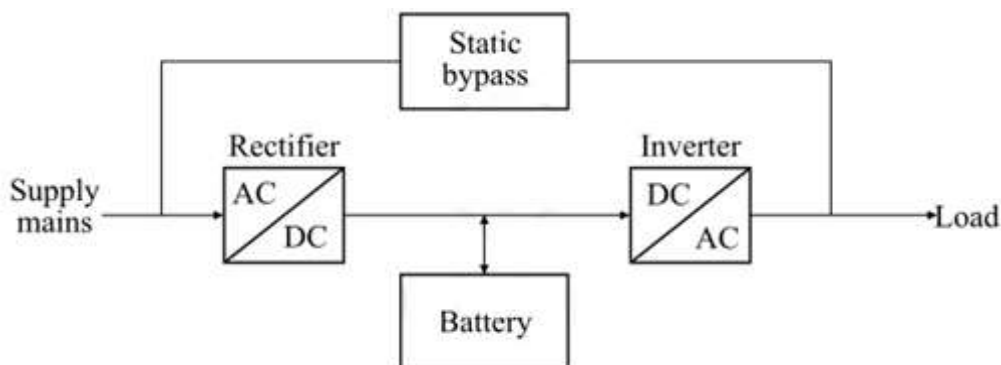
When, there is any power failure that occurs in the supply mains, the dc power of the battery is converted into ac power using a power inverter and is then transferred to the load connected to the ups system. The off-line ups systems provide surge protection in addition to the emergency power. These are the least expensive UPS systems available in the market.



Standby (Offline) UPS provides basic protection and is ideal for smaller loads. The inverter kicks in only when the main power source fails, making it less expensive but offering limited protection.

- ✧ **On-Line UPS:** In this system, the rectifier circuit receives ac power from the supply mains and directly drives the inverter circuit. Thus, it involves two simultaneous conversion processes and is hence referred to as a Double Conversion UPS System.

In the case of supply mains power failure, the rectifier circuit becomes inactive and the DC power stored in the battery is converted into AC by the inverter and supplied to the load. Once the mains power is restored, the rectifier circuit again starts charging the battery. A current limiting mechanism is also provided in the system to protect the battery from overheating. Although, on-line ups systems are slightly expensive due to their design and components used.

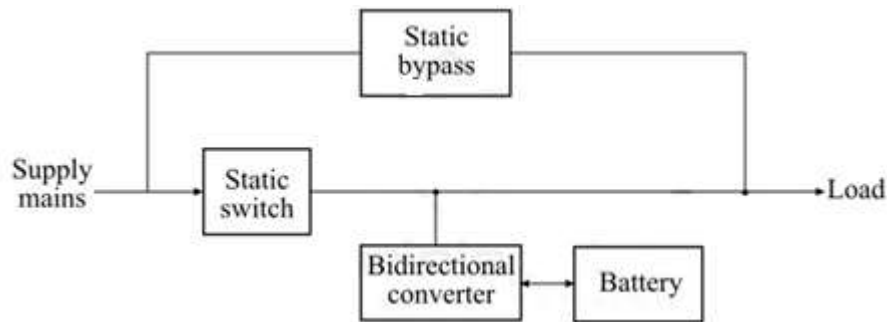


Line-Interactive UPS: Offers intermediate protection by regulating minor fluctuations in voltage (e.g., brownouts or overvoltages) without switching to battery power, extending battery life.

- ✧ **Line Interactive UPS:** Under normal operating conditions, the load is supplied with AC power through a parallel connection of the AC source and the bidirectional converter.

The converter charges the battery, and also provides AC power to the load in the situation of breakage of supply mains. In this type of UPS system, a bypass is also

provided which transfers the load to a bypass AC input when the UPS fails to function. The line-interactive UPS systems are well suited for medium to high power rating loads.



d) Functions and Features:

- ✧ **Power Continuity:** The UPS keeps essential equipment (e.g., servers, medical devices, networking gear) powered during an outage, preventing data loss, system crashes, and hardware failure.
- ✧ **Voltage Regulation:** Some UPS systems regulate voltage, smoothing out spikes and sags in power without switching to battery power, ensuring a steady and safe voltage supply to sensitive equipment.
- ✧ **Data Loss Prevention:** For systems like computers and servers, a UPS provides enough time for users to save work, close applications, and safely shut down in an organized manner during power outages.
- ✧ **Network Connectivity:** Advanced UPS units are equipped with network interfaces, allowing users to monitor power status, receive notifications, and perform automatic shutdowns through remote control.

e) Role of UPS

When there is any failure in the main power supply from the utility, the UPS supplies emergency power to the load for a short duration of time. This is the primary function of a UPS. Modern UPSs can also provide protection against electrical faults such as short-circuit, voltage fluctuations, low voltage, instability of mains frequency, etc.

f) Applications

These days, UPS systems are very common in almost all electrical and electronic systems. Following are some of the important applications of different types of UPS systems:

- ✧ UPS systems are used in computer data centres as Servers, data centers, and network equipment rely on UPS systems for power backup to maintain continuous operations and prevent data corruption during outages.
- ✧ UPS systems are used in industrial control and monitoring systems. Manufacturing plants use UPS systems to maintain control systems, preventing production stoppages and equipment damage due to power loss.

- ✧ UPS systems are also used in telecommunication systems.
- ✧ They are also used in hospitals, banks, insurance offices, and other commercial applications for backup power, etc.
- ✧ **Medical Equipment:** In hospitals, UPS systems are used for critical medical devices like ventilators, ensuring they keep running during power interruptions.

4. Programmable power supply

a) Definition

A programmable power supply is a type of power supply unit (PSU) that allows the user to control and adjust its output parameters (such as voltage, current, or power) via external inputs. These adjustments can be made through various interfaces, using a keypad or rotary switch from the front panel of the power supply, computer software, or communication protocols like USB, Ethernet, or GPIB (General Purpose Interface Bus). This capability enables the power supply to adapt to a wide range of applications.

b) Key Features:

- ✧ **Adjustable Output:** The output voltage and current can be precisely controlled according to the requirements of the connected device or system. This is useful for testing and prototyping.
- ✧ **Automation:** Programmable power supplies can be integrated into automated testing systems where different voltages or currents are required at different stages of the test process. This helps in performing tasks without manual intervention.
- ✧ **Remote Control:** These power supplies often support remote control through software or interfaces, allowing users to control them from a computer or programmable logic controller (PLC).
- ✧ **Protection Features:** Programmable power supplies often come with advanced protection features such as overvoltage protection (OVP), overcurrent protection (OCP), and short-circuit protection to safeguard both the power supply and the connected devices.
- ✧ **Memory Functions:** Some programmable power supplies can store user-defined settings for different applications, enabling quick switching between preset configurations.

c) Types of programmable power supply

You can find programmable power supplies in various form factors including:

- ✧ **DC programmable power** supplies are designed to deliver stable voltage.

A DC programmable power supply operates in two primary modes:

- ✚ **Constant Voltage (CV):** The output voltage is set by the user's configurations, whereas the output current is determined by the load's resistance.
- ✚ **Constant Current (CC):** The Constant Current (CC) mode primarily serves as a safety feature but can also be used for other purposes. In this mode, the current remains fixed according to the user's current limit setting. If the power supply operates in Constant Voltage (CV) mode and the current exceeds the set limit, it will automatically switch to CC mode. It will revert to CV mode once the load current drops below the specified limit.
- ✧ **AC Programmable Power Supply:** An AC programmable power supply provides alternating current to test equipment by simulating grid disturbances, such as interruptions, harmonics, and surges, which could potentially damage a Device Under Test (DUT).
- ✧ **Benchtop Programmable Power Supply:** Benchtop programmable power supplies are light enough to be placed on a supported flat surface such as a workbench.

Benchtop programmable power supplies are essential tools for electrical engineers and circuit designers, used to test and measure various electrical devices, including circuit boards and other electronic components.

- ✧ **Chassis-mounted Programmable Power Supply:** Chassis-mounted programmable power supplies are installed into a chassis or frame and come in both linear and switching models. These supplies convert AC current to DC current based on the application's requirements. The chassis can take various forms, with controls typically located on the front panel of the chassis.
- ✧ **Rack mount Programmable Power Supply:** Rack-mounted programmable power supplies are designed to be installed in a support rack, where multiple units can be stacked together.
- ✧ **Digital Programmable Power Supply:** Digital programmable power supplies offer precise control and monitoring via a digital interface and encoder knob.
- ✧ **Analog Programmable Power Supply:** Analog programmable power supplies utilize knobs and dials for adjusting output parameters. Although they are generally less precise than digital models, they are valued for their simplicity and suitability for specific applications.

e) Applications:

- ✓ **Product Testing:** Used in labs and manufacturing to test electronic devices by simulating different power conditions.
- ✓ **Research and Development (R&D):** Ideal for R&D environments where different voltages or currents are needed for experimenting with circuit designs.
- ✓ **Automated Test Equipment (ATE):** Integrated into automated systems for testing large quantities of products under different conditions.
- ✓ **Battery Testing:** Programmable power supplies are often used for charging and discharging batteries at controlled rates to analyze their performance.

5. Variable power supply

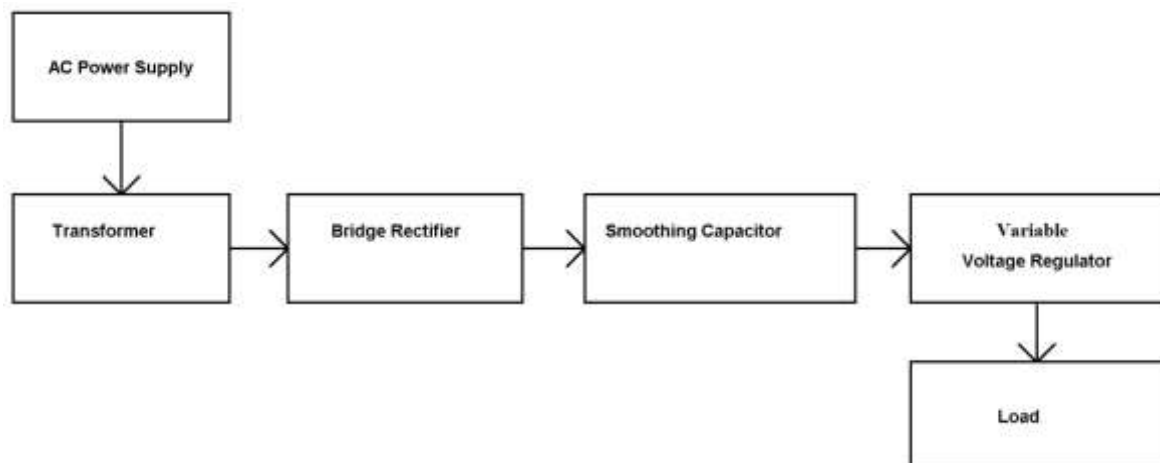
a) Definition

A variable regulated power supply, also called a variable bench power supply, is one where you can continuously adjust the output voltage to your requirements.

The variable power supply circuit is equipped with an adjustable voltage regulator to adjust the output according to the output. An adjustable voltage regulator has line regulation and load regulation.

A variable power supply is a device that allows you to adjust its output voltage and, in some cases, the current as well. This adaptability is essential for applications that require different voltages for testing or operational purposes. The adjustments are typically made using a potentiometer or a similar control, which lets users fine-tune the voltage output according to their specific needs.

The flexibility of a variable power supply makes it invaluable in laboratories, electronics manufacturing, and repair workshops. By simply turning a knob or setting a control, users can modify the output to match the exact requirements of the device they are working with, ensuring optimal performance and protection against voltage-sensitive components. This ability to customize output helps prevent damage and extends the lifespan of electronic devices.



b) Working principle

The working principle of a variable power supply revolves around converting and regulating input power to provide a stable, adjustable output.

Here's how it works:

- ✧ **AC to DC Conversion:** The power supply typically takes an AC input and converts it to DC using rectifiers. This is the first step in providing a usable DC output.
- ✧ **Voltage Regulation:** The rectified DC voltage is then regulated using either linear or switching regulators. Linear regulators are simpler and provide low noise output, whereas switching regulators are more efficient and suitable for high power applications.
- ✧ **Adjustment Mechanism:** The adjustable feature is achieved through variable resistors or digital control circuits. In analog designs, potentiometers are commonly used, while digital designs employ microcontrollers or digital signal processors for precise control.

This combination of conversion, and regulation ensures that the variable power supply can provide a wide range of output settings with high precision and stability.

c) Applications

Variable power supplies are used across a wide array of applications due to their versatility and precision.

Here are some common applications:

- ✧ **Electronics Development and Testing:** In research and development labs, engineers use variable power supplies to test and prototype electronic circuits. The ability to adjust voltage and current helps in analyzing circuit behavior under different conditions.
- ✧ **Education:** In educational institutions, variable power supplies are used in laboratories to teach students about electronics and electrical engineering principles.
- ✧ **Repair and Maintenance:** Technicians use variable power supplies to diagnose and repair electronic devices. The adjustable output helps in testing different components and circuits.
- ✧ **Battery Charging:** Variable power supplies can be used to charge batteries, allowing precise control over the charging voltage and current to optimize battery life and performance.
- ✧ **Industrial Automation:** In industrial settings, these power supplies are used to power and test various automation components and systems.

These applications demonstrate the wide-ranging utility of variable power supplies in both educational and professional settings.

5. Circuit simulation

Simulating a circuit involves creating a detailed circuit schematic and using the software's simulation tools to analyse its behaviour under various conditions.

Proteus software provides a comprehensive environment for designing and testing electronic circuits, including Linear power supply, SMPS, variable, etc



Practical Activity 3.2.2: Drawing of schematic diagram of power supply unit



Task:

1: Referring to key reading 3.2.2 and the instruction from your trainer, you are requested to perform the given task below. The task should be done individually.

As technician, you are asked to go in the Computer Lab to draw the schematic diagram of a basic DC power supply unit, which converts AC mains power into regulated DC voltage, utilizing standard electronic components using Proteus.

2: Present your work to the trainer

3: Ask questions if any for clarification

4: Perform the task provided in application of learning 3.2



Key readings 3.2.2

Drawing of schematic diagram of power supply unit

Steps of Drawing of schematic diagram of power supply unit using Proteus

Here is a step-by-step guide for drawing a schematic diagram of a **power supply unit (PSU)** in **Proteus** software:

Step 1: Define Power Supply Specifications: Before you start with the schematic, determine the **Input Voltage** and **Output Voltage**.

Step 2: Open Proteus Software: Start the Proteus software then Go to **File > New Project**, name your project and select the directory for saving.

Step 3: Select and Add Components: Go to **Library > Pick Devices** or click the **P** button on the toolbar. Search for components using their names or descriptions then after **Place Components on Schematic** by clicking on the schematic workspace to place each component.

Step 4: Connect Components: Select the **wire tool** (blue dot or pencil icon) from the toolbar to connect the components.

Step 5: Add Labels and Power Sources: Use the **Label Tool** to add text labels for components like Vout, Vin, GND, and other important nodes.

Step 6: Annotate and Finalize Schematic: Use the **Annotate** tool to automatically assign identifiers (e.g., R1, C1, D1) to all components.

Step 7: Save the Schematic: Save it by going to **File > Save** or clicking the save icon



Practical Activity 3.2.3: Simulating the power supply unit circuit diagrams



Task:

1: Referring to the key reading 3.2.3 and instruction from your trainer, you are requested to perform the given task below. The task should be done individually.

As technician, you are asked to go in the Computer Lab to design and simulate a 12V DC regulated power supply using the following components:

- a) Step-down transformer (220V AC to 15V AC).
- b) Bridge rectifier (using four diodes or an integrated bridge rectifier).
- c) Smoothing capacitor (2200 μ F to filter the rectified signal).
- d) 7812 voltage regulator (to provide a stable 12V DC output).
- e) Load resistor (to simulate a typical load).

2: Present your work to whole class.

3: Perform the task provided in application of learning 3.3



Key readings 3.2.3

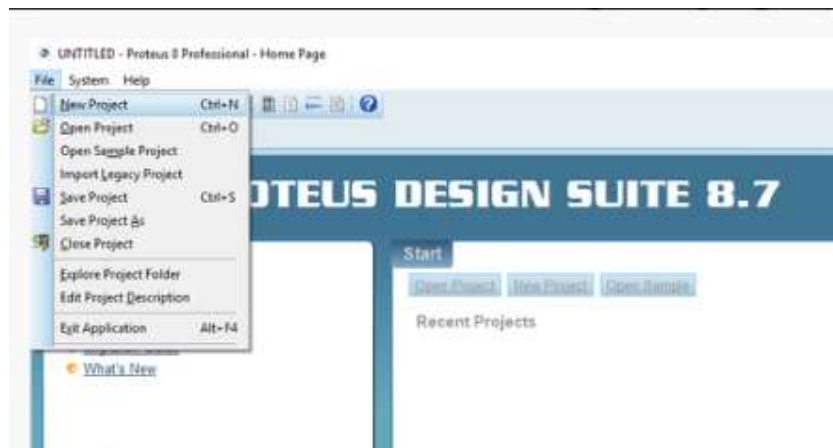
Simulating the power supply unit circuit diagrams

1. Steps of simulating Linear power supply using Proteus

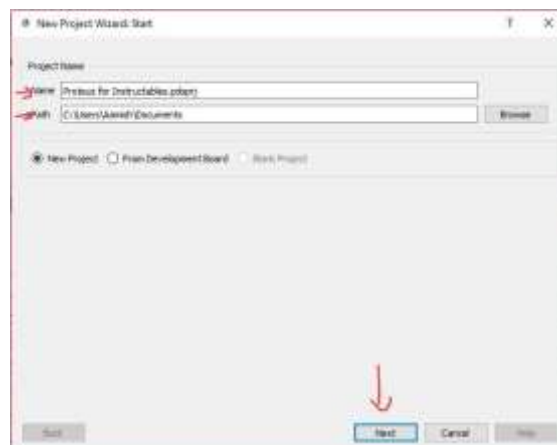
Below are the steps to design and simulate a linear power supply in Proteus.

Step 1: Define the Power Supply Requirements such as Input Voltage, Output Voltage and Components.

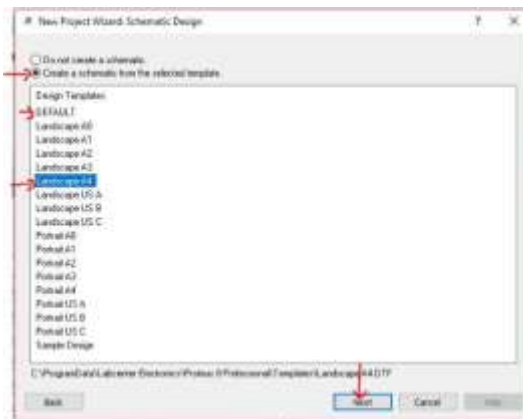
Step 2: Open Proteus and Create a New Project: Launch Proteus and Go to **File > New Project** then name your project and specify the directory to save it.



There you can change the Name and Path of the project after setting click Next.



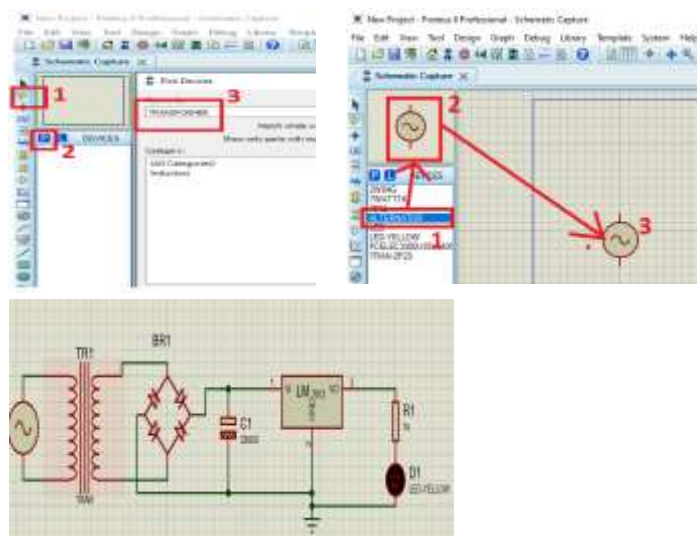
Now you need to Mark Create a schematic from the selected template then you can either select Default or you can select any templates size normally we select landscape A4,



By selecting the Landscape A4 and click Next, you will have a schematic window as shown in the picture below.

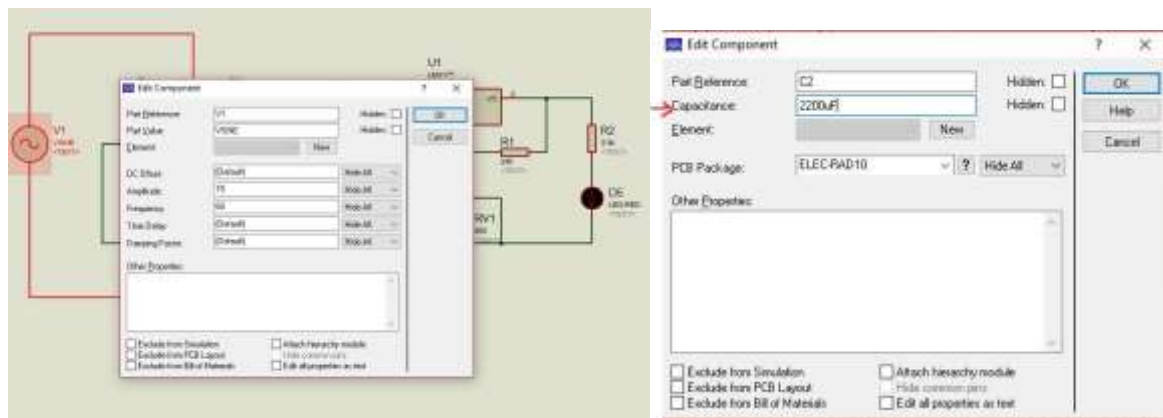


Step 3: Select and Add Components: Click on library then **Pick Devices** button (or press **P**) to search for and add components then after selecting the components, click on the schematic editor and arrange them logically, ensuring correct orientation.

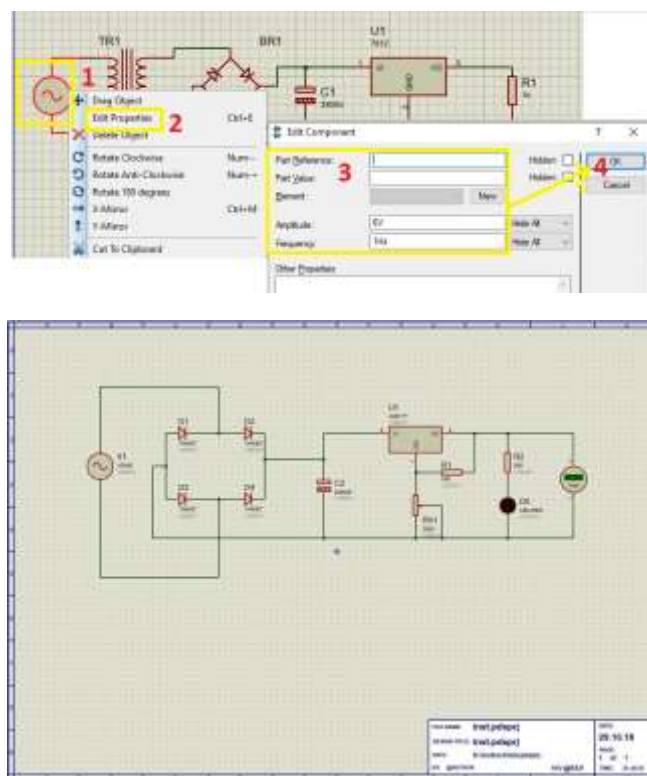




You can also edit the component's parameters by right clicking on it and click on Edit parameters.

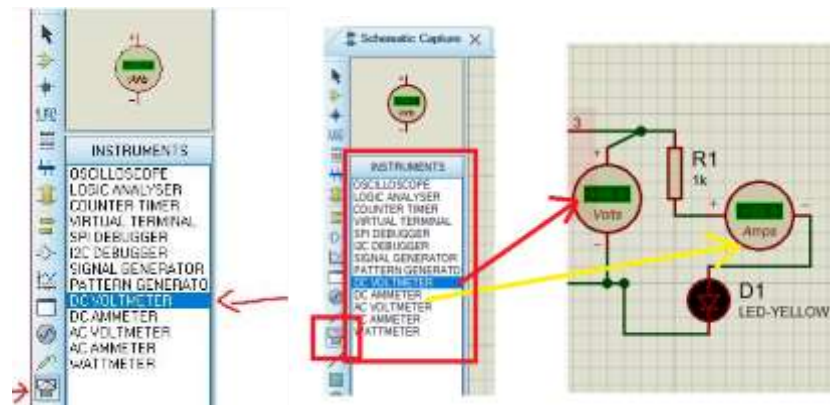


Step 4: Build the Schematic: Connect Components Using the Wire Tool and Label Connections for easy identification.



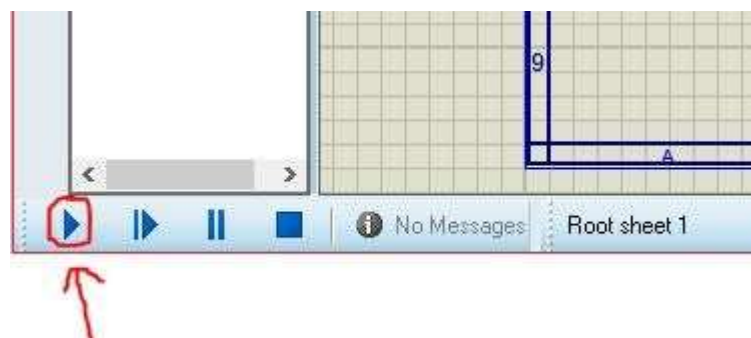
Step 5: Set Up Simulation Parameters: Add **Voltage Probes** or **Virtual Instruments** (e.g., oscilloscope or voltmeter) to measure AC input and DC output. (Probes can be placed at the output of the transformer, the rectifier, the filter capacitor, and after the voltage regulator to observe the waveform changes at each stage), then set the **Simulation Mode** to "Analog" for accurate power supply simulation.

To add a voltmeter click on the **meter** icon and select **DC VOLTMETER** then click where ever you want to place it connect it by wires(lines) if necessary

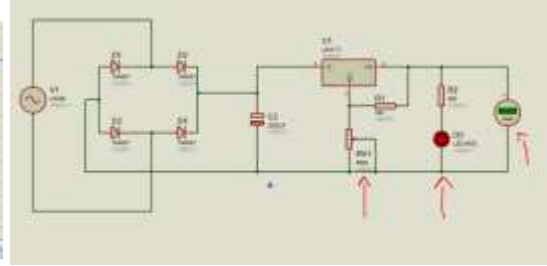
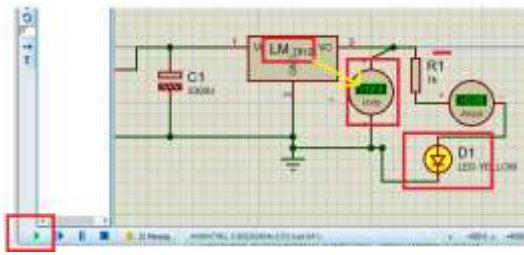


Step 6: Run the Simulation: Click the **Run Simulation** button (green play icon) and the circuit will begin simulating, and you can observe the waveforms or voltage readings from the probes and instruments.

To check the circuit click on the play icon on bottom left of the proteus.



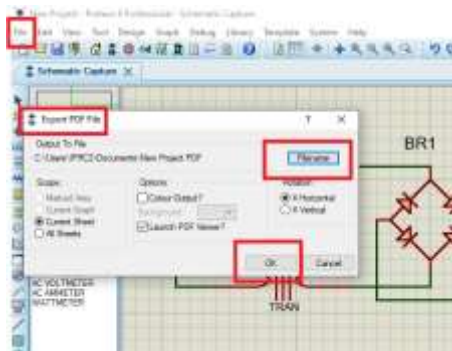
Step 7: Analyse Simulation Results: Check if the output voltage matches the expected value. If the output voltage is not as expected, recheck the transformer configuration, rectifier connections, and regulator wiring.



Step 8: Save and export the Simulation: Once the design works as expected, save your project by going to **File > Save**.



You can export the schematic as an image or a PDF for sharing or documentation purposes by going to **File > Export**.



2. Steps of simulating Switched-Mode Power Supply (SMPS) using Proteus

Here's a step-by-step guide to simulating an SMPS in Proteus:

Step1. Install and Open Proteus: Make sure you have a licensed version of Proteus installed on your computer then launch the Proteus software.

Step2. Create a New Project: Go to **File > New Project** to start a new project the follow the wizard to set up the project, including naming it and selecting a directory to save it.

Step3. Design the SMPS Circuit: Use the **P (Pick Devices)** button or **Library > Pick Devices** to search for and place components and position the components on the schematic sheet, use the **Wiring** tool to connect the components as per your SMPS design and **configure Components** by double-clicking on each component to set its parameters.

Step4. Configure Simulation Settings: Go to Simulation > Edit Simulation Profile to define simulation parameters then choose the type of simulation you want to perform, such as Transient Analysis, AC Analysis, or DC Sweep.

Step5. Run the Simulation: Click on the Play button or go to Simulation > Run to start the simulation then observe the simulation results in the Virtual Oscilloscope or other measurement tools provided by Proteus.

Step6. Analyze the Results: Observe the output voltage and current waveforms to ensure they match the expected values.

Step7. Document and Save: Save the schematic and simulation results regularly to avoid losing progress by clicking File > Save to save your project. And you can generate reports and export simulation results for documentation purposes.

3. Steps of simulating Uninterruptible Power Supply (UPS) using Proteus

Here's a step-by-step guide to simulate a UPS in Proteus:

Step1. Install and Open Proteus: Make sure Proteus is installed on your computer then open the Proteus software.

Step2. Create a New Project: Go to File > New Project to create a new project and follow the setup wizard to name the project and choose a directory.

Step3. Design the UPS Circuit: Use the P (Pick Devices) button or Library > Pick Devices to search for and place components. Then position the components on the schematic sheet according to the UPS design and connect the components using the Wiring tool. Also double-click each component to set its parameters.

Step4. Configure Simulation Settings: Go to Simulation > Edit Simulation Profile to define simulation parameters and choose the type of simulation, such as Transient Analysis for time-based behavior or AC Analysis for frequency response.

Step5. Run the Simulation: Click on the Play button or go to Simulation > Run to begin the simulation and observe the UPS operation through the Virtual Oscilloscope or other measurement tools provided by Proteus.

Step6. Analyze the Results: Verify that the output voltage remains stable and within the expected range during normal operation and battery backup modes. Ensure that the UPS switches correctly between mains power and battery power without interruption and analyze how the UPS manages battery charging and discharging, including the charging rate and cutoff thresholds.

Step7. Document and Save: Save the schematic and simulation results regularly to prevent data loss by clicking on File > Save to save your project.

4. Steps of simulating Programmable power supply using Proteus

Simulating a **Programmable Power Supply** using **Proteus** involves designing a circuit schematic that includes components to control and program the output voltage and current.

Here's a step-by-step guide to simulating a programmable power supply in Proteus:

Step1. Install and Open Proteus: Verify that Proteus is installed on your computer then open the Proteus software.

Step2. Create a New Project: Go to File > New Project to create a new project and follow the setup wizard to name your project and select a directory for saving it.

Step3. Design the Programmable Power Supply Circuit: Use the P (Pick Devices) button or go to Library > Pick Devices to search for and place components then position the components on the schematic sheet according to your design and after use the Wiring tool to connect components according to the circuit design also set parameters of components such as resistor values, capacitor values, and control settings.

Step4. Configure Simulation Settings: Go to Simulation > Edit Simulation Profile to define the type of simulation and parameters and choose Transient Analysis to observe time-based behavior or DC Sweep for varying input voltages.

Step5. Run the Simulation: Click on the Play button or go to Simulation > Run to begin the simulation and monitor the simulation results using virtual instruments by using virtual instruments like Oscilloscope, Multimeter, or Virtual Analyzer to measure output voltage, current, and control signals.

Step6. Analyze the Results: Ensure that the output voltage and current are adjustable and meet the desired specifications based on the control signals or programming settings then analyze the performance of the programmable power supply, including response time, accuracy, and stability.

Step7. Document and Save: Save the schematic and simulation results regularly to avoid data loss by clicking on File > Save to save your project then generate reports and export simulation results for documentation purposes.

5. Steps of simulating Variable power supply using Proteus

Here's a step-by-step simulating Variable power supply using Proteus:

Step1. Install and Open Proteus: Make sure that Proteus is installed on your computer then Open the Proteus software.

Step2. Create a New Project: Go to File > New Project to create a new project and follow the setup wizard to name your project and choose a directory for saving it.

Step3. Design the Variable Power Supply Circuit: Use the P (Pick Devices) button or Library > Pick Devices to search for and place components then position the components on the schematic sheet according to your design then after connect the components according to the circuit design and configure their parameters.

Step4. Set Up Simulation Profile and Define Simulation Parameters: Go to Simulation > Edit Simulation Profile to define the type of simulation and parameters then after insert voltage sources to simulate input power and control signals.

Step5. Run the Simulation: Click on the Play button or go to Simulation > Run to begin the simulation then observe the circuit behavior using virtual instruments like Oscilloscope, Multimeter, or Virtual Analyzer to measure the output voltage, current, and adjustment settings.

Step6. Analyze the Results: Ensure that the output voltage is adjustable over the desired range and that it matches the specifications set by the potentiometer or control signal.

Step7. Optimize Design: Make necessary adjustments to improve the performance of the variable power supply. This might involve changing component values, improving stability, or enhancing the control mechanism.

Step8. Save Your Work and Generate Reports: Save the schematic and simulation results regularly to avoid data loss by File > Save to save your project and also generate and export reports of your simulation results for documentation purposes.



Practical Activity 3.2.4: Generating a power supply PCB



Task:

1: Referring to the instruction from your trainer, you are requested to perform the given task below. The task should be done individually.

As technician, you are asked to go in the Computer Lab to generate the PCB layout of a regulated DC power supply that converts 220V AC input into a stable 5V DC output, suitable for powering microcontrollers or other low-power electronic circuits using Proteus software.

3: Present your work to whole class.

4: Read key reading 3.2.4 and perform the task provided in application of learning 3.2.4



Key readings 3.2.4

Generating a power supply PCB

Steps of Generating a power supply PCB using Proteus

Generating a **PCB (Printed Circuit Board)** for a power supply using **Proteus** involves several key steps, from designing the schematic to creating and verifying the PCB layout.

Here's a step-by-step guide to help you through the process:

Step1. Design the Schematic by creating a New Project, designing the schematic and after simulate the circuit.

As you know we can not place **VSINE** or Transformer on the PCB board we need to add a block which is called as **TBLOCK-I2** it is a two terminal connector at which we will connect our output terminals of the transformer similarly for the voltmeter we are going to put the same terminal block.

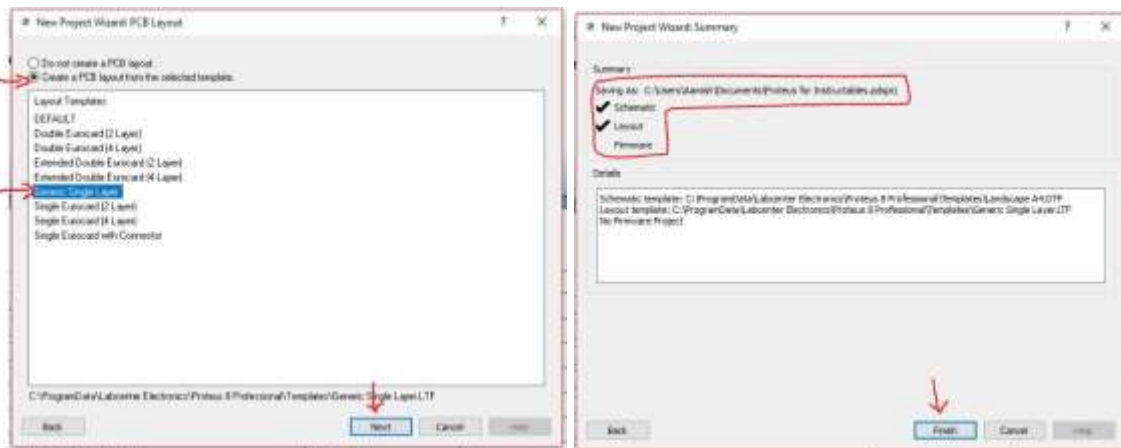
For this purpose we will replace VSINE and voltmeter with terminal block (TBLOCK)

Now only remaining component is LED who's PCB footprint is missing so we need to add the PCB footprint of LED for that purpose double click the LED click on the question mark a new window named packages would appear type LED and double-click LED now and the PCB package you can see there would be written LED at the end.

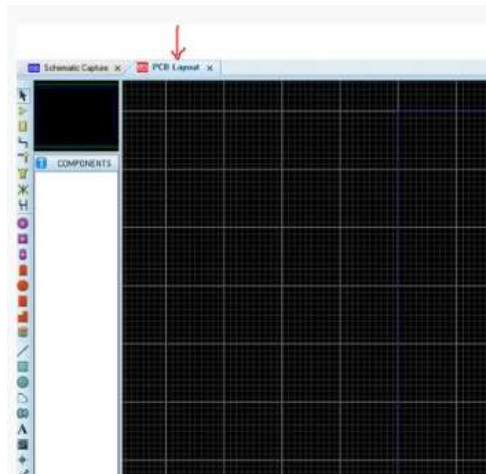
You can rename the terminal blocks by input and output or whatever you like

Step2. Create the PCB Layout: Once the schematic is complete and tested, convert it to PCB layout by going to Tools > Update PCB Layout and choose a PCB layout based on your schematic.

Also while launching the the Proteus software, you can choose create a PCB layout from the selected remplate and select Generic Single Layer.



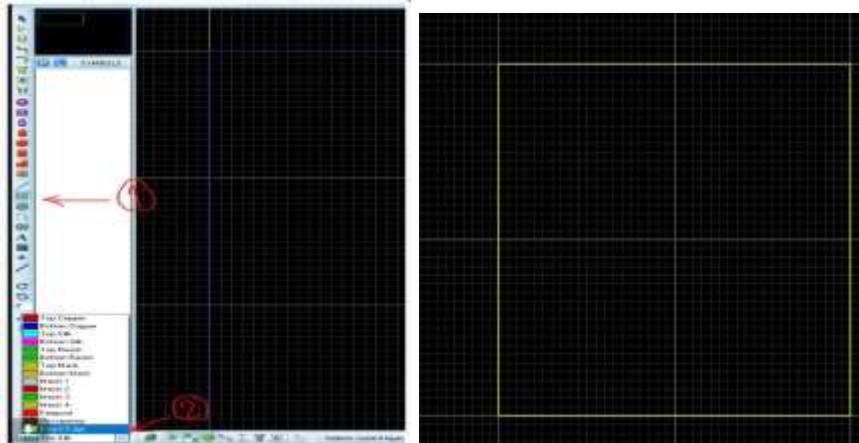
1. Click on the PCB layout tab then zoom in to upper left corner as shown in pictures this block is **1x1 cm** in area



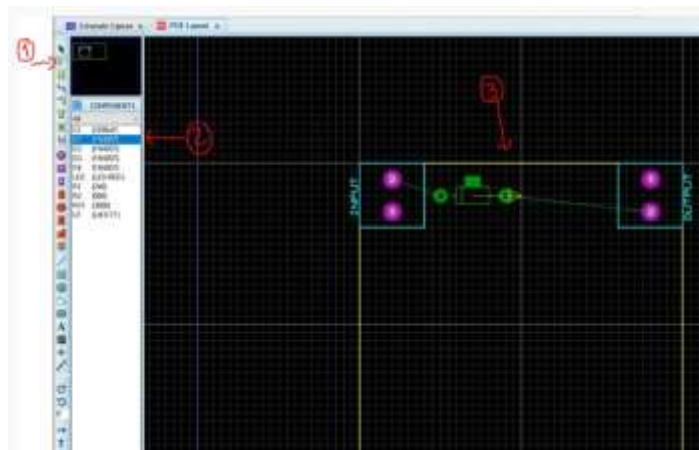
2. Now from the left toolbar click on the **Square** button and from the bottom toolbar select **Board Edge**.



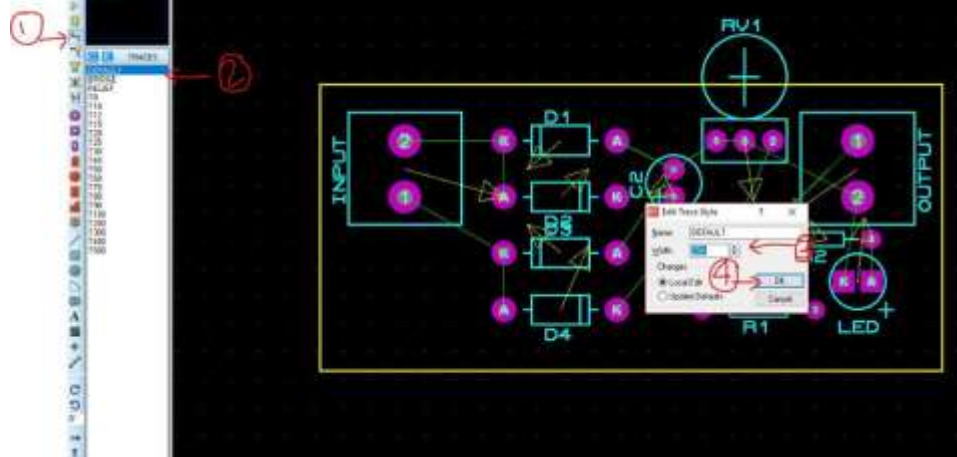
3. Draw a rectangle of any size by clicking anywhere inside the blue area of the PCB window, as you can see I have drawn a 2 x 2 cm² block. This is the size of your PCB you can change it by hovering your mouse pointer at any corner off this Block. As I have also changed it after completing the design.



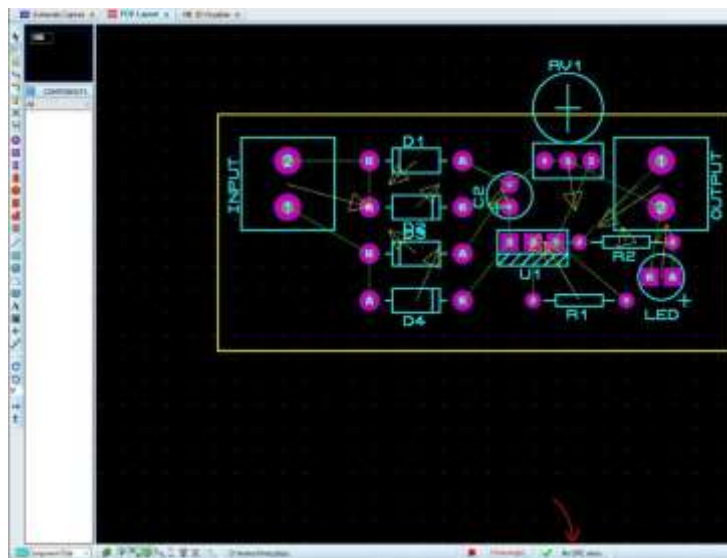
4. Now click on the small diode icon which indicate that component lists. Select the component and then click inside the yellow Square to place it



5. When you would be placing a component you will see green lines which indicates the connection of one point to the other point and the Yellow arrow head which indicates that this component is to join with this terminal of the other component. Place the components wherever you like keep in mind that you choose simple and easy path which is indicated by the green lines. After placing all the components you can adjust the yellow rectangle



6. Make sure there are no DRC errors on the bottom status bar, DRC error appears when there is an overlapping of a terminal with other terminals or with board edge.



Step3. Define PCB Settings: Define the board dimensions, layer stack, and other settings using Design > Board Shape to set the dimensions of the PCB. Then set the number of layers and other properties via Design > Layer Stack Manager.

Step4. Place Components: Arrange the components on the PCB layout by dragging them from the Components list then position components to minimize trace lengths and avoid clutter.

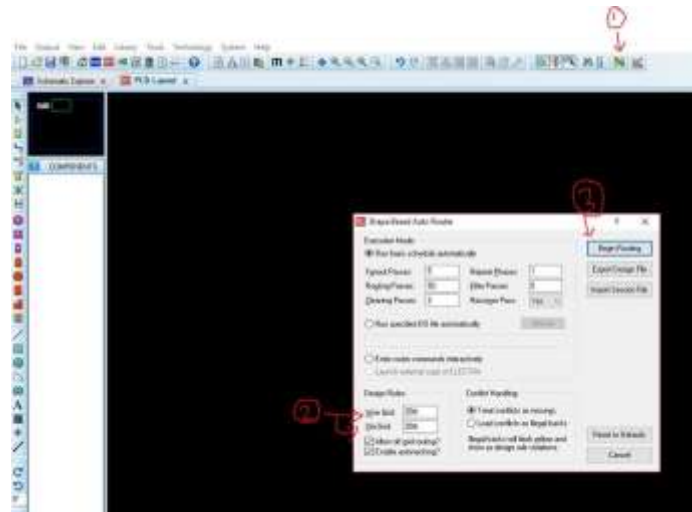
Step5: Route Traces: Use the Route tool to connect the components according to the schematic and ensure proper routing of signal traces, power lines, and ground planes.

Auto-Routing

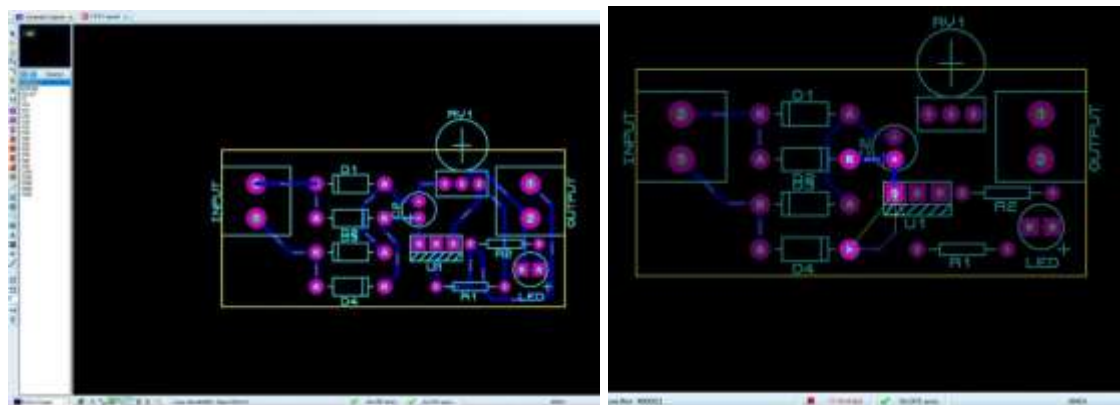
1. Click on the **routing button** from left toolbar as described in picture then double

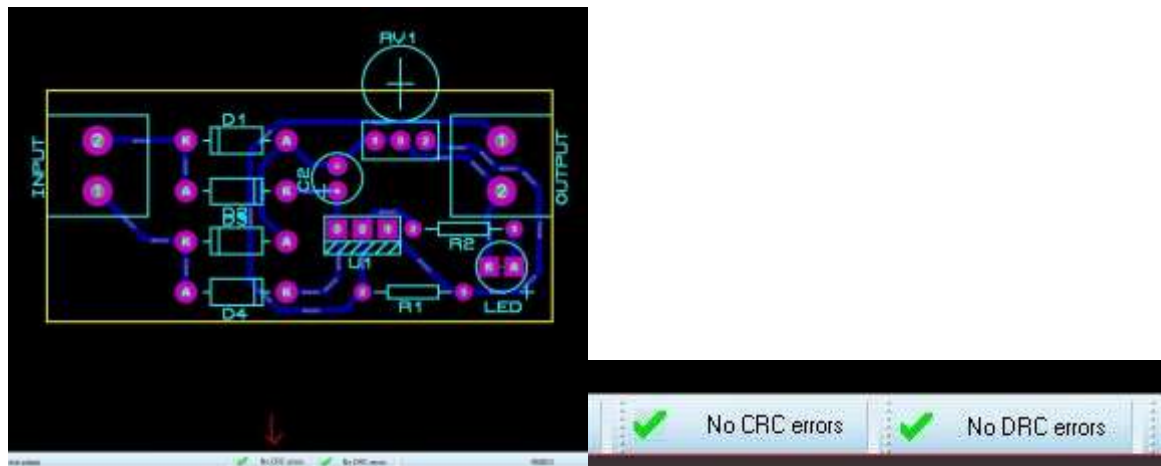
click **default**, change the default **width** to **20th** or **25th** (this is the **thickness** of your PCB routing) and click **OK**

1. Now click on the second last icon (**Auto-Router**) and check the **Wire-Grid** and **Via-Grid** it should be **more than 15** other wise soldering would be difficult in close connections. then click **Begin Routing**



2. After routing make sure there are no **CRC errors**, in case there is an error there would be a connection missing press **CTRL+Z** and re-arrange that component and start routing again.





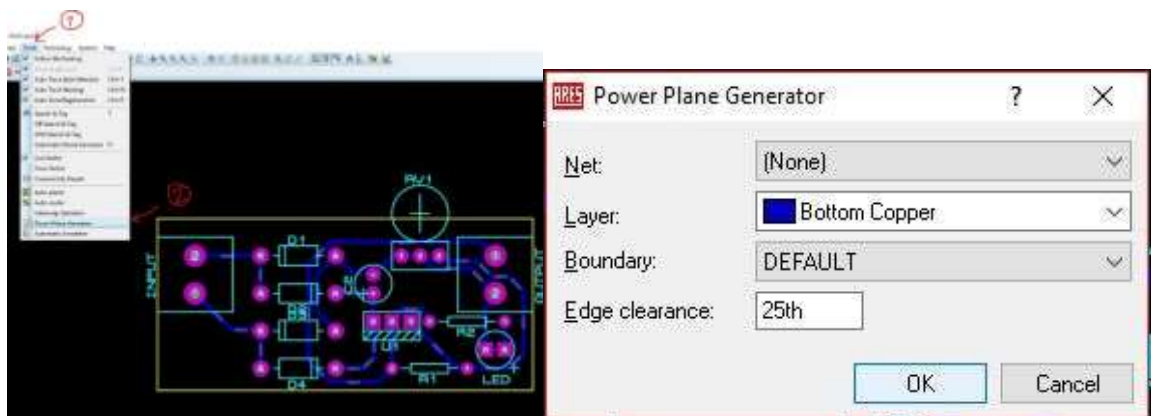
Manual-Routing

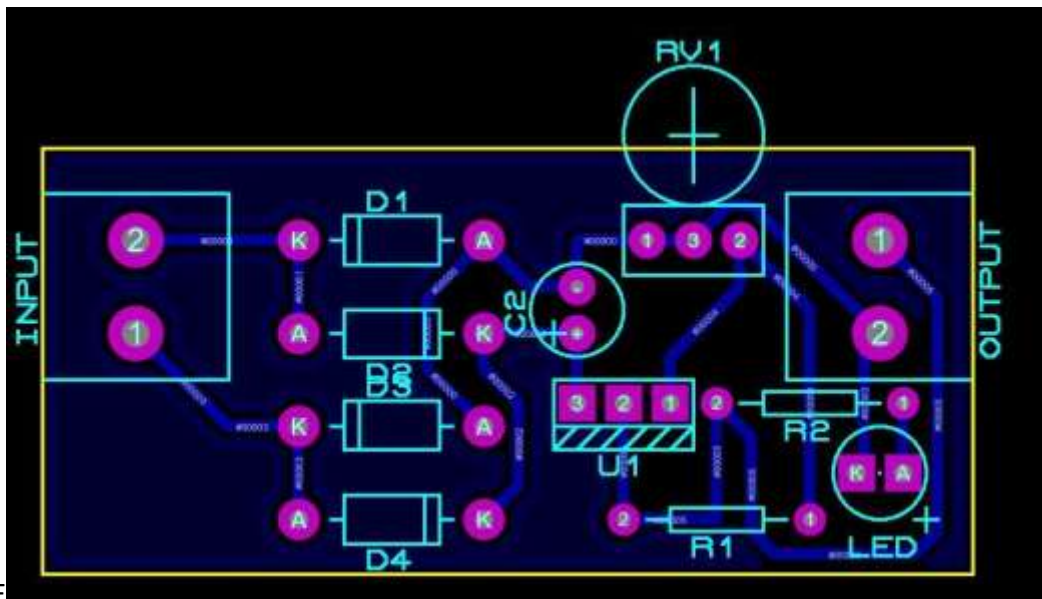
1. Click on the **routing** button from left toolbar as described in picture then double click **default**, change the **default width to 20th or 25th** (this is the thickness of your PCB routing) and click **OK**
2. Click on any terminal the point that terminal wants to connect would be highlighted just click on that highlighted terminal same as we did while making the schematic. Make all the connections you would notice that whenever you are making the connection the CRC errors would be reduced after all the connections there would be no CRC error.

If you want to make your PCB design more professional click on tools then the second last item name power plan generator Just click ok and see what happens with your PCB design

Step6: Add PCB Features: Insert necessary PCB features such as:

- **Pads:** For mounting components.
- **Silkscreen:** To mark component outlines and labels.
- **Copper Pour:** For ground planes or power distribution.

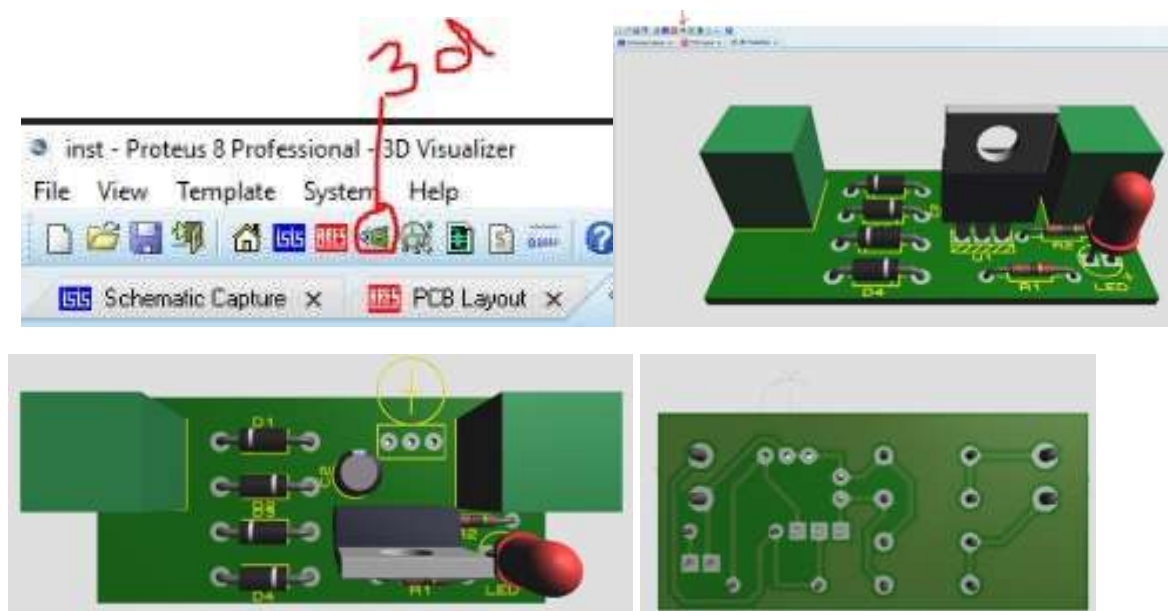




Step7. Verify and Finalize the PCB Design: Verify that there are no electrical issues with the layout and ensure that all connections are correct and there are no unconnected nets.

Step8. Review the PCB Design: Double-check component placements, trace routing, and board layout and make sure there are no errors or design issues before proceeding to fabrication.

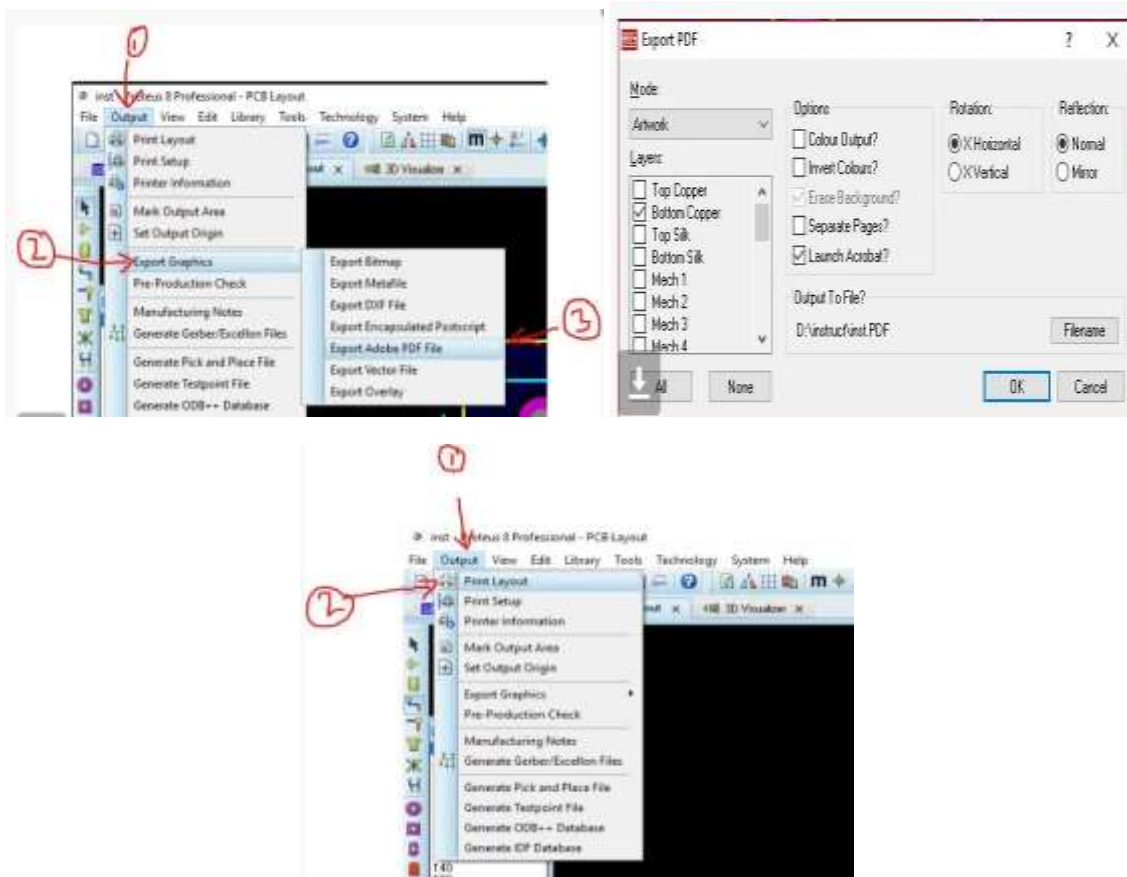
Proteus offer to students a 3D visualisation tool you can see your circuit in 3D which is quite awesome.



Step9. Generate Gerber Files: Go to File > Export > Gerber Files to generate the files required for PCB fabrication.

Step10. Printing PCB: For printing the PCB layout there are two methods

1. Go to **output** click on **export graphics** then click on **export Adobe PDF file** in this method you will save the PCB layout as a PDF file and dialog box will open from this box only select the **bottom copper** and the **Board Edge** and un-check everything else and click **OK**. You need to check the **actual size** printing option while printing from a adobe or any other PDF viewer.
2. Go to **output** click on **print layout**, a dialog box will appear only check **bottom copper** and **board Edge** adjust the PCB as shown in the print preview you can print it anywhere on the page and click **OK**.



Practical Activity 3.2.5: Assembling power supply unit



Task:

- 1: Referring to the instruction from your trainer, you are requested to perform the given task below. The task should be done individually.

As technician, you are asked to go in the electronic workshop to assemble a power supply unit that converts 230V AC mains into a regulated 12V DC output.

2: Present your work to whole class.

3: Read key reading 3.2.5 and perform the task provided in application of learning 3.5



Key readings 3.2.5

Assembling power supply unit

Steps of Assembling power supply unit

Step1. Gather Tools and Components: This includes tools such as soldering iron, solder, screwdrivers, multimeter, tweezers, etc and components such as PCB, connector terminals, enclosure, heat sink, electronic components, etc

Step2. Prepare the PCB: This involves checking for any visible defects or damage on the PCB and ensure that the board is clean and free of dust or debris. Also compare the PCB layout with the schematic to ensure that all components are correctly placed.

Step3. Assemble the Components: Insert components into their respective holes or slots starting from smaller to larger, then heat the soldering iron and apply solder to the component leads and PCB pads. After soldering, use wire cutters to trim any excess component leads that extend beyond the PCB.

Step4. Inspect the Assembly: Ensure that all components are properly aligned and that there are no physical defects, then use a multimeter to check for continuity between relevant points to ensure that there are no open circuits or incorrect connections.

Step5. Test the Assembly: Connect the PSU to an appropriate power source and verify that the output voltages are within the specified range and check for stability. Then monitor the PSU's performance under load conditions, checking for correct voltage regulation, stability, and any signs of overheating or instability.

Step6. Check for Proper Operation: Verify that the PSU can handle the expected load and that it operates correctly within its specifications.

Step7. Finalize Assembly: Place the assembled PCB into an enclosure or casing to protect it from environmental factors and mechanical damage. Then label the PSU's input and output connections for easy identification and safe operation and conduct a final set of tests to ensure that the PSU functions correctly and meets all design specifications.

Step8. Documentation and Safety: Keep records of the assembly process, including any issues encountered and how they were resolved also follow all safety guidelines when working with electrical components and power supplies.



Points to Remember

- **Description of a power supply types**
 - ✓ A linear power supply converts AC voltage to a stable DC voltage.
 - ✧ There are typically two types of linear power supply as Unregulated Linear and Regulated Linear
 - ✓ A Switched Mode Power Supply (SMPS) is an electronic power supply that efficiently converts electrical power using a switching regulator.
 - ✧ Types of SMPS are: AC-DC Converters, DC-DC Converters, and DC-AC Inverters.
 - ✓ An Uninterruptible Power Supply (UPS) is an electrical device used to provide emergency electrical power to different electrical loads in the case of a main power supply failure.
 - ✓ A programmable power supply is a type of power supply unit (PSU) that allows the user to control and adjust its output parameters (such as voltage, current, or power) via external inputs.
 - ✓ A variable regulated power supply, also called a variable bench power supply, is one where you can continuously adjust the output voltage to your requirements.
- **Steps of Drawing of schematic diagram of power supply unit using Proteus**

To **draw of schematic diagram of power supply unit** in Proteus, begin by determining the **Input Voltage** and **Output Voltage** for your project. Next, launch the Proteus software and navigate to **File > New Project**; here, you will name your project and select a directory for saving it. After setting up the project, proceed to select and add the necessary components for your schematic. Utilize the wire tool, represented by a blue dot or pencil icon in the toolbar, to connect these components effectively. To ensure all components are properly identified, use the Annotate tool to automatically assign

identifiers. Finally, save your work by going to **File > Save** or by clicking the save icon to preserve your schematic design.

- **Steps of simulating linear power supply using Proteus**

To simulate your linear power supply project in Proteus, first define the **Power Supply Requirements**, including the **Input Voltage**, **Output Voltage**, and the necessary components. Next, launch the Proteus software and navigate to **File > New Project**, where you will name your project and specify the directory for saving it. Once your project is set up, click on the library and select the **Pick Devices** button (or press P) to search for and add the required components. After selecting the components, connect them using the **Wire Tool** and label the connections for easy identification. To measure AC input and DC output, add voltage probes or virtual instruments to your schematic. When everything is in place, click the **Run Simulation** button (represented by a green play icon) to start simulating your circuit. After running the simulation, analyze the results carefully. Finally, remember to save and export your simulation to preserve your work for future reference.

- **Simulating an SMPS in Proteus involves:**

To design and simulate a switched-mode power supply (SMPS) circuit, begin by **starting a new project** where you will add and configure the necessary components for the SMPS. Next, **define the type of simulation** you wish to conduct, along with the input sources that will drive your circuit. This is crucial for ensuring that the simulation accurately reflects real-world conditions. After setting up the simulation parameters, **observe and analyze the performance** of the SMPS circuit through the simulation tools available. Finally, based on the results obtained from your simulations, **make adjustments to optimize performance** and verify that the circuit meets your specifications.

- **Simulating a UPS in Proteus involves:**

To simulate an uninterruptible power supply (UPS) circuit, begin by **designing the circuit** and configuring the necessary components to ensure optimal functionality. Once the design is in place, **define the type of simulation** you will conduct, along with the input sources that will be used to test the circuit's performance. After setting up these parameters, **observe the UPS performance** during the simulation and make any adjustments as needed to enhance its efficiency and reliability. Finally, if any issues arise

during this phase, take the time to **address those problems** and improve the overall design to meet your specifications.

- **Simulating a programmable power supply in Proteus involves:**

Simulating a programmable power supply in Proteus involves several key steps. First, you need to **design the circuit and configure the components** appropriately to ensure that they function as intended. Once the circuit is set up, the next step is to **define the type of simulation** you wish to run and specify the input sources that will be used during the simulation. After these preparations, proceed to **run and analyze the simulation** to observe how the circuit performs under various conditions. Finally, if any issues arise during this analysis, it's important to **address these problems and improve the design** accordingly.

- **Simulating a variable power supply in Proteus involves:**

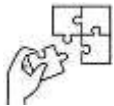
To simulate a variable power supply in Proteus, first, you need to **design the circuit** and configure the necessary components, laying the groundwork for your simulation. Next, it's important to **define the type of simulation** you wish to conduct, along with the input sources that will power your circuit. Once these parameters are set, you can proceed to **run and analyze the simulation**, observing how the circuit performs under various conditions. Finally, if any issues arise during this analysis, you should **address these problems and refine your design** accordingly to enhance performance.

- **Generating a PCB for a power supply in Proteus involves:**

Generating a printed circuit board (PCB) for a power supply in Proteus involves several key steps. First, you need to **create and test the circuit design**, ensuring that all components are correctly configured and functioning as intended. Once the circuit is validated, the next step is to **convert the schematic to a PCB layout**; this includes placing the components, routing the traces, and adding any necessary features to enhance the design. Following the layout creation, it's essential to **perform design checks** to ensure everything is in order before generating Gerber files, which are crucial for fabrication. Finally, after preparing the files for manufacturing, you will **assemble the PCB** and verify its functionality to confirm that it operates as expected.

- **Assembling a power supply unit involves:**

Assembling a power supply unit (PSU) involves several important steps to ensure a successful build. First, making sure you have everything required for the assembly process. Next, prepare the printed circuit board (PCB) by inspecting and verifying both the PCB and its components to ensure they meet specifications. Once everything is in order, **assemble the components** by carefully placing and soldering them onto the PCB. After assembly, it's crucial to **inspect and test** the unit; this includes performing visual inspections, conducting continuity checks, and executing load testing to confirm functionality. Following these tests, **finalize the assembly** by enclosing the PSU, labelling all connections clearly, and conducting final tests to ensure everything operates correctly.



Application of learning 3.2.

You are an Electronics Technician working for a startup company that is developing a smart home automation hub. This hub will control various IoT devices like lights, thermostats, and security cameras. The hub must operate on a reliable and efficient power supply to ensure continuous operation. It will convert 230V AC mains power to 5V DC with a maximum current output of 2A to power the hub's microcontroller and communication modules.

Your supervisor tasked you with the following tasks:

- i. Designing and drawing a 05V DC regulated power supply circuit, using Proteus, for an embedded system project.
- ii. Before physically building the power supply unit (PSU), you are tasked by simulating the design to verify its performance, efficiency, and safety.
- iii. Create a printed circuit board (PCB) for this power supply circuit,
- iv. Assemble a 05V DC power supply unit to fit the system



Indicative content 3.3: Testing Computer Power Supply Unit



Duration: 7 hrs



Theoretical Activity 3.3.1: Description of testing concepts

Tasks:

1: Answer the following questions related to description of power supply testing concepts.

i. What do you understand by the following terms?

- a) Voltage
- b) Current
- c) Transient response
- d) Ripple
- e) Short circuit
- f) Electromagnetic Interference
- g) Electromagnetic Compatibility

2: Write the findings/answers on papers or flipchart.

3: Present your findings to the class.

4: Ask questions for more clarifications, if any.

5: Read the key reading 3.3.1 from your manual to get more clarifications.



Key readings 3.3.1.:

Description of testing concepts

1. Voltage: is the electrical potential difference between two points in a circuit. It's the force that drives the flow of current through a circuit.

It's measured in volts (V) by the instrument called Voltmeter connected in parallel with the component under test. Testing voltage helps ensure that a circuit or component is operating at the correct voltage levels.

2. Current: is the flow of electric charge through a conductor or circuit, measured in amperes (A) by the instrument called Ammeter connected in series with the component under test.

Testing current helps to determine if the correct amount of current is flowing through a component or circuit, which can be critical for proper operation and safety.

3. Transient Response: It refers to how a circuit reacts to sudden changes in voltage or current, such as switching on or off. It's crucial in assessing how quickly and effectively a circuit returns to its steady-state operation after a disturbance. Testing transient response helps to evaluate the stability and performance of circuits, especially in dynamic conditions.

Transient response testing in a computer power supply evaluates how well the supply responds to sudden changes in load conditions, particularly in terms of voltage stability. This is important to ensure that the power supply can handle rapid changes in the power demand from components like the CPU and GPU without causing system instability.

4. Ripple: is the small, unwanted AC component superimposed on the DC output of a power supply or regulator. It is typically caused by incomplete smoothing of the output voltage. Testing for ripple is important in power supplies to ensure that the DC output is stable and free from excessive fluctuations that could affect circuit performance.

Measuring ripple in a computer power supply involves detecting the small, high-frequency variations superimposed on the DC output voltage. Ripple is an important parameter as it indicates the quality of the power supply's filtering and regulation. Excessive ripple can affect the stability and performance of electronic components.

5. Short Circuit: A short circuit occurs when there is an unintended path of low resistance that bypasses the intended circuit path. This can cause excessive current flow, potentially leading to overheating or damage. Testing for short circuits involves checking for unexpected connections between circuit points that should be isolated.

6. EMI/EMC: EMI (Electromagnetic Interference) refers to unwanted electromagnetic signals that can disrupt the operation of electronic devices. **EMC (Electromagnetic Compatibility)** is the ability of a device to operate correctly in its electromagnetic environment without causing or being affected by EMI. Testing for EMI and EMC involves ensuring that devices are shielded or designed to minimize interference and comply with regulatory standards for electromagnetic emissions and immunity.

- **EMI (Electromagnetic Interference):** This refers to unwanted electromagnetic emissions generated by electronic devices, such as a PSU, which can interfere with the performance of nearby equipment.
- **EMC (Electromagnetic Compatibility):** This ensures that a device, like a PSU, can operate properly in its electromagnetic environment without causing or being affected by interference.

Testing the Electromagnetic Interference (EMI) and Electromagnetic Compatibility (EMC) of a computer power supply unit (PSU) is crucial to ensure it complies with industry standards and does not disrupt other devices or systems



Practical Activity 3.3.2: Visual inspecting of computer power supply unit

Task:

1: Referring to the instruction from your trainer, you are requested to perform the given task below. The task should be done individually.

As technician, you are asked to go in the electronic workshop to conduct a visual inspection of a computer power supply unit to identify potential issues and the components involved.

2: Present your work to whole class.

3: Perform the task provided in application of learning 3.3



Key readings 3.3.2

Visual inspection of computer power supply unit

1. Steps of visual inspection of computer power supply unit

The visual inspection of a computer power supply unit (PSU) is an essential part of diagnosing potential hardware issues or ensuring that the PSU is functioning properly.

Here are the steps for performing a thorough visual inspection:

Step1. Turn off the Computer and Unplug the Power Supply: Before inspecting the PSU, ensure that the computer is turned off and unplugged from the wall socket to prevent electrical hazards and ensure no power is connected.

Step2. Inspect the External Casing: Look for any dents, cracks, or physical damage to the PSU's external casing.

Step3. Inspect the Power Cable: Ensure the power cable is intact with no visible cuts, fraying, or loose connections. A damaged power cable can cause improper power delivery or pose a safety risk.

Step4. Check for Bulging or Leaking Capacitors: If possible, remove the cover and check inside the PSU to look at the capacitors for any signs of bulging, leaking, or physical distortion. Damaged capacitors are a common cause of PSU failure.

Step5. Look for Burn Marks or Discoloration: Check the circuit board and components for any burn marks, soot, or discoloration, which can indicate overheating or electrical shorts.

Step6. Examine Wiring and Connectors: Check that connectors are securely attached and not bent, missing, or loose. Loose connectors can cause poor power delivery.

Step7. Inspect for Dust and Debris: Dust accumulation can cause the PSU to overheat. Make sure to clean any dust from the vents, fan, and around the PSU. A can of compressed air can be used to blow dust out of hard-to-reach areas.

Step8. Smell for Burnt Odors: A burnt smell from the PSU could indicate overheating or internal component failure. If you detect such an odor, the PSU may be damaged and unsafe to use.

Step9. Check the PSU Label and Ratings: Ensure the PSU label is intact and clearly visible. The label should provide information about the PSU's wattage, voltage ratings, and certification

Step10. Ensure Proper Grounding: Inspect the PSU's ground connection (if visible) to ensure it's secure and free of corrosion because improper grounding can lead to electrical hazards or malfunction.

Step11. Look for Signs of Liquid Damage: Check for any signs of liquid spills or corrosion, which could indicate water damage or exposure to moisture.



Practical Activity 3.3.3: Measuring input voltage and current of a power supply unit



Task:

1: Referring to the instruction from your trainer, you are requested to perform the given task below. The task should be done individually.

As technician, you are asked to go in the electronic workshop to measure the input voltage and current of a computer power supply unit, and relate these measurements to the overall performance and safety of the PSU.

2: Present your work to whole class.

3: Read key reading 3.3.3 and perform the task provided in application of learning 3.3



Key readings 3.3.3

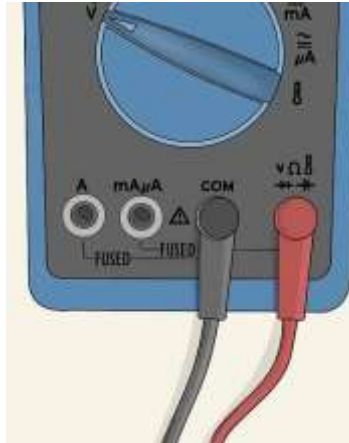
Measuring input voltage and current of a power supply unit

1. Steps to measure input Voltage

Step 1: Set the Multimeter to Voltage Mode: Turn the dial on the multimeter to the DC Voltage (V) setting, as input voltage in most circuits is DC (e.g., power supplies). If you are working with AC, set the multimeter to the AC Voltage (V~) setting and Select a range that is higher than the expected voltage to avoid overloading the meter. If unsure, start with the highest setting and adjust accordingly.



Step 2: Connect the Probes: Red probe goes to the positive input voltage terminal, Black probe goes to the ground (negative) terminal and place the probes directly across the points you want to measure the voltage difference between.



Step 3: Turn on the Circuit: Power on the circuit to measure the input voltage under operating conditions.

Step 4: Read the Measurement: The voltage will be displayed on the multimeter screen and if the value exceeds the multimeter's range, adjust the range setting or select a higher-capacity multimeter.



Step 5: Turn Off the Circuit: After recording the measurement, turn off the circuit and disconnect the probes.



2. Steps measure input Current in a power supply

Step1. Turn Off the Power: Power off the circuit or device where you will measure the current to ensure safety while connecting the multimeter.

Step2. Set the Multimeter to the Current Measurement Mode: If you're measuring direct

current (**A–**) (e.g., from batteries or DC circuits), set the multimeter to DC current and if you're measuring alternating current (**A~**) (e.g., from AC mains), set it to AC current.

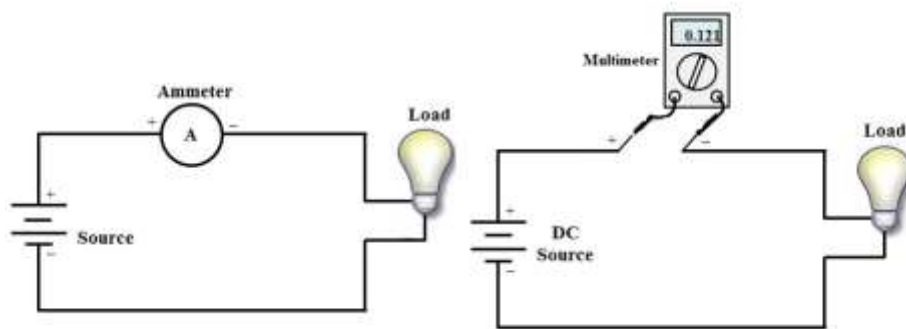
- ✧ Choose the appropriate current range on the multimeter. If unsure, select the highest range to prevent overloading the multimeter.

Step3. Switch Probe Connections: Insert the black probe into the "COM" port (common ground) and insert the red probe into the port labeled "mA" or "A" for current for low current measurements (typically below 200mA), while you have to insert the red probe into the high-current port labeled "10A" or similar, depending on your multimeter for higher current (usually above 200mA),

Step4. Break the Circuit: To measure current, you must connect the multimeter **in series** with the circuit. This means you need to disconnect or break the circuit at the point where you want to measure the current flow.

Step5. Connect the Probes in Series: Connect the black probe to the part of the circuit that leads to the negative (or ground) side and connect the red probe to the positive side where the current will flow.

For AC measurements, the orientation of the probes doesn't matter since the current alternates.



Step6. Turn On the Power: Once the multimeter is connected in series, turn the circuit or device back on to allow current to flow through the multimeter.

Step7. Read the Display: The current will be displayed on the multimeter's screen. Be sure to note the units (mA, A, etc.) and ensure that the value is within the expected range for your circuit.

Step8. Turn Off the Power and Disconnect the Multimeter: After measuring the current, turn off the power to the circuit, then disconnect the multimeter probes.

- ✧ Switch the red probe back to the voltage/resistance port if you're finished measuring current.

Step9. Turn Off the Multimeter: Turn off the multimeter to conserve battery life.



Practical Activity 3.3.4: Measuring output voltage and current



Task:

1. Referring to the instruction from your trainer, you are requested to read the given task below. The task should be done individually.

As technician, you are asked to go in the electronic workshop to measure the output voltage and current of a computer power supply unit to understand its performance characteristics and ensure it meets specifications.

2. Read key reading 3.3.4 to understand the procedures, then perform the activity.

3. Present your work to whole class.

4. Perform the task provided in application of learning 3.3



Key readings 3.3.4

Output voltage and current measurement

Measuring the output voltage and current in a computer power supply involves using appropriate tools and following safety guidelines. Here are the steps you can follow:

1. Steps for output Voltage Measurement:

Step1: Power Off the Computer: Before performing any measurement, ensure that the computer is turned off and unplugged to avoid any electric shocks.

Step2: Locate the Power Supply Connectors: Identify the connectors coming from the power supply and use a power supply pinout diagram to identify which pins carry the different voltage rails and ground.

Step3: Set Multimeter to Voltage Mode: Set your digital multimeter to measure DC voltage. Ensure the multimeter probes are properly connected: the red probe to the VΩ port and the black probe to the COM port.

Step4: Turn On the Computer: Power on the computer to start the power supply.

Step5: Measure the Output Voltage: Carefully place the black probe on a ground pin (black wire) and the red probe on the pin carrying the voltage you want to measure then record the voltage readings for different rails. Ensure the readings are within the expected tolerance levels.

2. Steps for output current Measurement:

Step1: Set Multimeter to Current Mode: Turn off the computer and switch your digital multimeter to the appropriate current range then ensure the red probe is in the correct port for current measurement.

Step2: Disconnect Load: Disconnect the load (e.g., motherboard, drives) from the power supply connector where you plan to measure the current.

Step3: Insert Multimeter in Series: To measure current, the multimeter must be placed in series with the load and connect one end of the multimeter probe to the output pin (and the other probe to the wire going to the load).

Step4: Turn On the Power Supply: Power on the computer or the power supply then the current will flow through the multimeter.

Step5: Check Current Reading: Record the current drawn by the load at different voltage rails.

Step6: Turn Off and Reconnect: After taking measurements, turn off the power supply, disconnect the multimeter, and reconnect the load.



Practical Activity 3.3.5: Testing transient response of a computer power supply



Task:

1. Referring to the instruction from your trainer, you are requested to read the given task below. The task should be done individually.

As technician, you are asked to go in the electronic workshop to test the behavior of a computer power supply unit (PSU) under sudden load changes and evaluate its transient response without causing instability in the output voltage.

2. Read key reading 3.3.5, then perform the activity.

3. Present your work to whole class.

4. Perform the task provided in application of learning 3.3



Key readings 3.3.5

Testing transient response

Steps of Transient response testing

Here are the detailed steps for performing transient response testing in a computer power supply:

Step 1: Understand the Purpose of Transient Response Testing: Measure how quickly the PSU stabilizes the output voltage after a load change and identify if there are significant voltage overshoots, undershoots, or oscillations during transient conditions.

Step 2: Prepare the Required Equipment: These are Electronic Load, Oscilloscope, Multimeter and the device being tested (The power supply unit).

Step 3: Select Test Parameters: Set the input voltage and choose the critical voltage rails to test, as they are used to power important components like the CPU, GPU, and motherboard.

Step 4: Connect the Test Equipment: Attach the output rails of the PSU to the electronic load>>connect the oscilloscope to the voltage rail you want to monitor and perform then attach the current probes to measure the load current dynamically during the test.

Step 5: Define the Transient Load Profile: Program the **Electronic Load** to switch between different loads quickly.

Step 6: Run the Transient Test: Provide AC power to the PSU and ensure it operates normally under stable load conditions. Then **initiate the Transient Load Test** by using the electronic load to create abrupt changes in load then using **the Oscilloscope**, observe how the voltage rail behaves during load changes.

Step 7: Analyze the Oscilloscope Waveforms: Look for key indicators of PSU performance such as: Peak-to-Peak Voltage, Voltage Ripple and Stabilization Time.

Step 8: Repeat the Test for Other Voltage Rails: Use the same transient load profile for each rail to assess the PSU's performance under various conditions.

Step 9: Document the Results: Document the amount of overshoot, undershoot, and recovery time for each rail during different load changes and after **compare with Specifications** to ensure that the PSU performs within the acceptable limits specified by the manufacturer or industry standards.

Step 10: Conclude with a Final Report: Summarize the performance of the PSU in terms of transient response, highlighting any issues that might need correction and after recommend whether the PSU is suitable for use in systems that experience frequent or significant load variations, such as gaming PCs or workstations.



Practical Activity 3.3.6: Measuring ripple in a computer power supply



Task:

1: Referring to the instruction from your trainer, you are requested to read the given task below. The task should be done individually.

As technician, you are asked to go in the Electronic Workshop tasked with testing the quality of a computer's power supply by measuring the ripple voltage on the output rails.

2: Read key reading 3.3.6, then perform the activity.

3: Present your work to whole class.

4: Perform the task provided in application of learning 3.3



Key readings 3.3.6

Measuring ripple in a computer power supply

Steps of Measuring ripple in a computer power supply

Here are the steps to measure ripple in a computer power supply:

Step 1: Understand the purpose of Ripple Measurement: The small AC voltage fluctuations present on the DC output, typically measured in millivolts (mV) or microvolts (μV).

Step 2: Prepare the Equipment such as a digital oscilloscope is preferred for capturing high-frequency ripple with adequate resolution, a high-frequency probes to minimize loading effects on the circuit and the unit under test (UUT) from which you want to measure ripple.

Step 3: Connect the Equipment: Set Up the Oscilloscope by choosing an oscilloscope channel with appropriate bandwidth and ensure that the oscilloscope probe is properly calibrated and then connect the probe tip to the output terminal of the power supply where you want to measure ripple.

Step 4: Set Up the Oscilloscope: Set the time base to capture a sufficient number of ripple cycles>>adjust the voltage scale (vertical scale) to ensure that the ripple signal is visible **and** set the oscilloscope to trigger on the DC voltage level of the ripple to capture stable waveforms. Use edge triggering for a stable display.

Step 5: Measure Ripple: Observe the waveform on the oscilloscope display. The ripple will appear as a small AC component superimposed on the DC voltage. This will help you to determine the peak-to-peak voltage of the ripple and the frequency of the ripple.

Step 6: Analyze the Results: Compare the measured ripple voltage with the specifications of the power supply to ensure it is within acceptable limits and ensure that the ripple frequency corresponds to the expected switching frequency or harmonics.

Step 7: Document the Results: Record the measurements and compare with Specifications to ensure that the ripple levels are within the manufacturer's specifications or industry standards then after prepare a report summarizing the ripple measurements, including any observations or issues identified during the testing.



Practical Activity 3.3.7: Protecting overload and short circuit



Task:

1. Referring to the instruction from your trainer, you are requested to perform the given task below. The task should be done individually.

As technician, you are asked to go in the Electronic Workshop to design and implement protective circuits to safeguard a computer power supply from overloads and short circuits.

2. Read key reading 3.3.7, then perform the activity.

3. Present your work to whole class.

4. To get more clarification. perform the task provided in application of learning 3.3



Key readings 3.3.7

Protecting overload and short circuit

1. Steps of protecting overload in a computer power supply

Here's a step-by-step guide to protect against overload in a circuit:

Step1. Understand the Circuit's Power Requirements: Calculate or look up the current and voltage requirements of the components in the circuit. Ensure that the circuit's power supply provides the necessary voltage and current within safe operating limits.

Step2. Add an appropriate fuse: Place a fuse in series with the circuit's power input. The fuse should be rated slightly higher than the maximum expected current draw but lower than what would cause damage.

Step3. Implement a Circuit Breaker: For circuits requiring continuous operation or easy resetting after an overload, use a circuit breaker instead of a fuse.

Step4. Use Current-Limiting Resistors: In simpler circuits, placing resistors in series with certain components or inputs can limit the maximum current that flows through those sections of the circuit.

Step5. Install Thermal Protection: Thermal protection devices such as thermal fuses or Positive Temperature Coefficient (PTC) thermistors can help protect the circuit from overcurrent by increasing their resistance when the temperature rises due to overload.

Step6. Use a Current Limiter IC: Integrated circuits (ICs) specifically designed to limit current can protect more advanced circuits from overload. These devices monitor the current flow and shut down or limit the output when the current exceeds a set threshold.

Step7. Use Overcurrent Protection (OCP) Circuits: Overcurrent protection circuits can monitor current flow and automatically cut power or limit current when an overload condition is detected.

Step8. Implement a Power Management IC: For complex systems, power management ICs can regulate current and voltage levels, offering overload protection along with other features like undervoltage lockout (UVLO) and overvoltage protection (OVP).

Step9. Use a Zener Diode for Voltage Clamping: In some circuits, overload can happen due to excessive voltage. A Zener diode can be used in reverse bias to clamp the voltage to a safe level.

Step10. Employ a Crowbar Circuit for Short-Circuit Protection: This is a protective circuit that uses a thyristor (like an SCR) to short the power supply when excessive voltage or current is detected, essentially "crowbarring" the power supply and preventing further damage to the circuit..

Step11. Design with a Soft Start Mechanism: These circuits gradually increase the voltage and current when powering up a system, avoiding the sudden inrush current that can cause

overload. They are particularly useful in circuits with capacitors, motors, or inductive loads that draw high initial currents.

Step12. Use a Power Supply with Built-in Protection: When designing a circuit, using a regulated power supply that has built-in overload protection (like foldback current limiting or shutdown features) can prevent damage to both the power supply and the circuit itself.

Step13. Regular Maintenance and Testing: Over time, components can degrade, increasing the likelihood of overloads. Regularly inspect and test the circuit, especially in industrial or high-power applications, to ensure that all protection mechanisms are functioning properly.

2. Steps of testing short circuit in a computer power supply

Testing for a short circuit in a general electronic circuit is a critical troubleshooting step. Here's a step-by-step guide to help you safely test for a short circuit in a circuit:

Step1. Power Off and Disconnect the Circuit: Before testing for a short circuit, ensure that the circuit is completely disconnected from any power sources to avoid the risk of electric shock or damaging components.

Step2. Visual Inspection: Check the circuit board and components for visible signs of damage, such as burnt marks, melted components, or damaged wires. These are often indicators of a short circuit and also look for any solder bridges that could be shorting adjacent pins or traces on the PCB (Printed Circuit Board). Poor soldering can often cause accidental shorts.

Step3. Set the Multimeter to Continuity: If your multimeter has a continuity setting, set it to this mode. In continuity mode, the multimeter will beep when there is a direct electrical path between two points, which is useful for finding short circuits.

Step4. Test for Continuity Across the Circuit: Put one probe on a known ground (like the ground rail of the circuit) and use the other probe to check various points in the circuit and check between power and ground.

Step5. Isolate Sections of the Circuit: If the circuit is complex, try isolating different sections to narrow down where the short may be occurring and also test components individually by desoldering it from the circuit and test it with your multimeter.

Step6. Check the Power Rails: If your circuit uses voltage regulators, check the input and output for shorts to ground then after check the entire power rail of the circuit, which connects the power supply to various components, for any shorts to ground.



Practical Activity 3.3.8: Testing temperature in computer power supply unit



Task:

1: Referring to the instruction from your trainer, you are requested to read the given task below. The task should be done individually.

As technician, you are asked to go in the Electronic Workshop to measure and analyze the temperature of various components within a computer power supply unit (PSU) to ensure safe and optimal operation.

2: Read key reading 3.3.8, then perform the activity.

3: Present your work to whole class.

4: Perform the task provided in application of learning 3.3



Key readings 3.3.8

Testing temperature in computer power supply unit

Steps of testing temperature in computer power supply unit

Here's a step-by-step guide on how to safely test and monitor the temperature of a computer PSU:

Step 1. Safety Precautions: Before testing, make sure the computer is powered off and unplugged from the wall socket to prevent electric shock and allow it to cool before testing, as it may be hot.

Step 2. Use an Infrared Thermometer or Thermal Camera (Non-Contact Method)

This method allows you to safely check the PSU temperature without opening the unit.

- **Turn on the computer:** Plug the power supply back in and turn the computer on.
- **Aim the infrared thermometer:** Point the infrared thermometer at the PSU's exhaust vent (where hot air exits). This will give you a general reading of the air temperature being expelled, indicating the PSU's internal heat.
- **Check temperature readings:** Take note of the temperature. For most PSUs, exhaust air should be between 30°C and 50°C (86°F to 122°F) under normal conditions.

Higher temperatures may indicate that the PSU is under stress or not properly ventilated.

Step 3. Use a Temperature Probe (Contact Method)

For more precise readings, you can use a thermocouple or temperature probe. This method is more advanced but provides better accuracy for internal temperatures.

- **Turn off and unplug the PSU:** Make sure the PSU is completely disconnected before opening it.
- **Open the PSU casing:** If you are experienced and comfortable working with electronics, you can carefully unscrew the PSU casing to access internal components.
- **Place the temperature probe:** Attach the temperature probe to a key internal component, such as a heat sink, transformer, or power MOSFET. These areas typically generate the most heat.
- **Reassemble the PSU:** Once the probe is securely attached, close the PSU (if possible) and reconnect it to the power supply.
- **Turn on the computer:** Power on the computer and monitor the temperature readings from the probe.
- **Monitor over time:** Observe the temperature as the computer runs under normal and load conditions. PSUs typically operate best between 40°C and 60°C (104°F to 140°F). Consistently high temperatures (above 70°C or 158°F) may indicate poor airflow, dust buildup, or an overloaded PSU.

Step 4. Use PSU Monitoring Software

Many high-end power supplies come with built-in temperature sensors and monitoring software that can provide real-time temperature data. If your PSU supports this feature, follow these steps:

- **Install the PSU manufacturer's software:** Some PSUs come with proprietary software (e.g., Corsair Link, EVGA Precision, etc.) that allows you to monitor various parameters, including temperature.
- **Check the temperature readings:** Open the software and look for temperature data under the PSU monitoring section.
 - Some software can provide the temperature of internal components like the fan or power regulation modules.

- Compare the readings with the normal operating range (40°C to 60°C is typical).

Step 5. Ensure Proper Ventilation

- **Check PSU ventilation:** Ensure the PSU has adequate airflow and that there are no obstructions blocking the fan or ventilation grills.
- **Clean dust buildup:** Dust accumulation can block airflow and cause the PSU to overheat. Use compressed air to clean out the PSU exhaust and fan area.

Step 6. Compare to Manufacturer Specifications

- **Look up the PSU's rated operating temperature:** Different PSUs have different operating temperature ranges. Review the manufacturer's manual or product specifications for guidance on normal and maximum temperature ratings.



Practical Activity 3.3.9: Testing EMI/EMC in computer power supply unit



Task:

1. Referring to the instruction from your trainer, you are requested to read the given task below. The task should be done individually.

As technician, you are asked to go in the Electronic Workshop tasked with testing the electromagnetic interference (EMI) levels and ensuring electromagnetic compatibility (EMC) of a newly designed computer power supply unit (PSU). Your goal is to verify that the PSU complies with regulatory standards and does not cause interference to other nearby electronic devices.

2. Read key reading 3.2.3, then perform the activity.

3. Present your work to whole class.

4. Perform the task provided in application of learning 3.3



Key readings 3.3.9

Testing EMI/EMC in computer power supply unit

Steps of testing EMI/EMC in computer power supply unit

Here's a step-by-step guide on how EMI/EMC testing is performed for a PSU:

Step1. Prepare tools and Equipment Required such as spectrum Analyzer: Measures electromagnetic emissions over a wide frequency range, EMC Software, Oscilloscope, etc

Step2. Prepare the Test Setup: Place the PSU in the testing environment and connect the LISN between the PSU and the power supply source.

Step3. Conduct Conducted Emissions Testing: Connect the PSU's input power line to the LISN, which provides a stable environment for measuring the conducted emissions and **turn on the EMI receiver/spectrum analyser then after observe the results.**

Step4. Conduct Immunity Testing: To check the PSU's ability to operate correctly when exposed to external electromagnetic disturbances. This is performed by the following methods:

- ✓ **Place the PSU in the test setup:** For radiated immunity, place the PSU in an anechoic chamber with antennas generating high-power electromagnetic fields.
- ✓ **Apply disturbances:** Introduce RF interference in various frequency bands and monitor the PSU's performance. For conducted immunity, inject noise through the power lines and observe the PSU's behavior.
- ✓ **Analyze functionality:** Ensure the PSU continues to function without interruption or significant performance degradation.

Step5. Check for Compliance: Once testing is complete, compare the results with the relevant EMC regulations and standards.

Step6. Document Results: After all tests are complete, document the results with detailed information about the test setup, procedures followed, and whether the PSU passed or failed the tests.



Practical Activity 3.3.10: Testing safety in computer power supply unit



Task:

1: Referring to the instruction from your trainer, you are requested to read the given task below. The task should be done individually.

As technician, you are asked to go in the Electronic Workshop to test the safety features of a computer power supply unit (PSU) to ensure that it operates correctly and safely

within specified limits. You will identify and assess key safety mechanisms such as overload protection, overvoltage protection, short circuit protection, and proper grounding.

2: Read key reading 3.3.10, then perform the activity.

3: Present your work to whole class.

4: Perform the task provided in application of learning 3.3



Key readings 3.3.10

Testing safety in computer power supply unit

Here's a step-by-step guide on how to perform safety testing for a PSU:

Step1. Gather Necessary Testing Equipment such as multimeter, Insulation Resistance Tester (Megger), Oscilloscope, automatic Test Equipment (ATE), etc

Step2. Visual Inspection and Physical Safety Check by inspecting the PSU for any visible damage, like cracks, loose parts, exposed wires, or burnt components, then ensure that the power connectors and cables are in good condition, also make sure the cooling fan is operational and free of obstructions.

Step3. Ground Continuity Test: Place one probe on the PSU's ground (the metal chassis) and the other on the ground pin of the power cord, the resistance should be very low, usually less than 0.1 ohms, to ensure proper grounding.

Step4. Insulation Resistance Test (Megger Test): Ensure the unit is powered down and disconnected from any load the use the insulation resistance tester set to the appropriate voltage (typically 500V or 1kV, depending on the PSU's rating).

Step5. Overload Protection Test: Connect the PSU to a load tester that can simulate different power demands.

Step6. Short-Circuit Protection Test: Create a short circuit between the PSU's output terminals using a controlled test setup and check if the PSU should shut down immediately upon detecting the short circuit.

Step7. Overvoltage Protection Test: Use a load tester to push the PSU to an overvoltage condition by manipulating the load and input voltage and **observe if the** PSU should shut down when the output voltage exceeds the specified limit.

Step8. Temperature and Thermal Protection Test: Place thermal sensors or use a thermal imaging camera to monitor key components of the PSU (such as the transformer, heat sinks, and capacitors) then check that the temperature remains within safe operating limits.

Step9. Functional Testing: Use an oscilloscope to check for voltage ripples, noise, and power fluctuations and measure the PSU's efficiency (input power vs. output power) to ensure it operates within the specified efficiency ratings.



Points to Remember

- **Description of power supply testing concepts**
 - ✓ **Voltage** is the electrical potential difference between two points in a circuit.
 - ✓ **Current** is the flow of electric charge through a conductor or circuit,
 - ✓ **Transient Response** refers to how a circuit reacts to sudden changes in voltage or current, such as switching on or off.
 - ✓ **Ripple** is the small, unwanted AC component superimposed on the DC output of a power supply or regulator.
 - ✓ **A short circuit** occurs when there is an unintended path of low resistance that bypasses the intended circuit path.
 - ✓ **EMI** (Electromagnetic Interference) refers to unwanted electromagnetic emissions generated by electronic devices, such as a PSU, which can interfere with the performance of nearby equipment.
 - ✓ **EMC** (Electromagnetic Compatibility) ensures that a device, like a PSU, can operate properly in its electromagnetic environment without causing or being affected by interference.

- **Steps of Visual Inspection of Computer Power Supply Unit**

To conduct a thorough visual inspection of a computer power supply unit (PSU), follow a systematic approach. Begin by **inspecting the exterior** of the PSU, looking for any visible signs of damage. Next, **check for burn marks** as these can indicate overheating or electrical shorts. Move on to **inspect the capacitors** on the PSU circuit board for any signs of leaking. Additionally, **assess the condition of the cables** that can lead to short circuits. It's also important to **evaluate the integrity of the connectors** and **the power cord** itself, as a

damaged cord poses a significant safety hazard. Finally, ensure that the **ground pin** on the power cord is intact and undamaged.

- **To measure input voltage with a multimeter, follow these steps:**

To measure input voltage with a multimeter, begin by ensuring that the multimeter is powered on. Next, set the multimeter to the appropriate voltage mode: choose DC voltage (V–) if measuring input voltage from dc sources or select AC voltage (V~) for mains power or other alternating current sources. It is advisable to select an appropriate voltage range; if uncertain, start with the highest range to prevent damaging the multimeter. After configuring the settings, insert the black probe into the "COM" port and the red probe into the port labeled "VΩ" or "V" for voltage measurement.

Connect the probes to the circuit or device by placing the black probe on the negative terminal or ground of the power source and the red probe on the positive terminal. Ensure that the **probes are connected in parallel** to the input terminals of the device or circuit for an accurate voltage measurement. Once connected, read the voltage displayed on the multimeter, taking note of the units (volts, millivolts, etc.). Finally, after completing your measurement, disconnect the probes and turn off the multimeter to preserve battery life and ensure safety.

- **To measure current with a multimeter, follow these steps:**

To measure current with a multimeter, begin by **setting the multimeter to current mode** by turning the dial to the appropriate setting for either DC or AC current, depending on the type of current you are measuring. Next, **open the circuit**; unlike voltage measurements, current must be measured by inserting the multimeter in series with the circuit. Proceed to **connect the multimeter in series**, ensuring that the red probe is placed at the point where the current enters the circuit and the black probe is positioned at the point where the current continues to flow after passing through. After making these connections, **turn on the circuit** to restore power, allowing current to flow through both the multimeter and the circuit, which will enable the multimeter to measure and display the amount of current on its screen. Once you have observed and noted the **measurement**, ensure that the current value is within expected limits. Finally, after completing your measurement, **turn off the circuit**, remove the multimeter, and reconnect everything to restore the circuit to its original state.

- **Steps of testing transient response of a computer power supply**

First, it's important to **understand the purpose of transient response testing**, which is to assess how well a PSU can handle sudden changes in load. Next, you should **prepare the required equipment**, including oscilloscopes, electronic loads, and any necessary connectors. Once your equipment is ready, **select the test parameters** that will guide your testing process and **connect the test equipment** properly to ensure reliable measurements. After setting up the load, **provide AC power to the PSU** and ensure it operates normally under stable load conditions before proceeding. Once stable operation is confirmed, **analyze the oscilloscope waveforms** to evaluate the transient response characteristics of the PSU. To obtain a comprehensive understanding, **repeat the test for other voltage rails**, as different outputs may respond differently to load changes. After completing all tests, it's crucial to **document the results** meticulously for future reference.

- **Here are the steps to measure ripple in a computer power supply:**

First, understand the purpose of ripple measurement, which is to assess the small AC voltage component present on a DC output. Next, prepare the necessary equipment, such as a digital oscilloscope capable of capturing high-frequency ripple with adequate resolution. Then, set up the oscilloscope by adjusting the time base to capture a sufficient number of ripple cycles and adjusting the voltage scale (vertical scale) to make the ripple signal visible. After setup, observe the waveform on the oscilloscope display, where the ripple will appear as a small AC component superimposed on the DC voltage.

Once measured, compare the ripple voltage with the power supply's specifications to ensure it meets the required standards. Finally, record the measurements and compare them with the manufacturer's specifications to confirm that the ripple levels are within acceptable limits.

- **Protecting your circuit from overload involves a combination of preventive design steps and protective components:**

To ensure proper protection in electrical circuits, start by using fuses or circuit breakers as a simple and effective method for safeguarding against overcurrent. Next, incorporate current-limiting resistors or PTC thermistors to regulate current flow and prevent damage to components. For more advanced control, implement overcurrent protection (OCP) circuits, current limiter ICs, or power management ICs, which provide precise regulation. Zener diodes and crowbar circuits can be added for voltage clamping and short-circuit protection. Lastly,

consider using a power supply that includes built-in overload protection, providing an extra layer of safety to the overall system.

- **Steps of testing temperature in computer power supply unit**

Before conducting any testing on your computer, it is essential to ensure that the device is powered off and unplugged from the wall socket to prevent electric shock. Once the computer is safely prepared, you can proceed to use one of several temperature measurement methods, such as an infrared thermometer or thermal camera for a non-contact approach, a temperature probe for a contact method, or PSU monitoring software to gather temperature data. Additionally, it is crucial to ensure proper ventilation during this process, as adequate airflow helps maintain optimal operating temperatures. Finally, after measuring the temperatures, compare your findings to the manufacturer specifications to determine if the components are functioning within the recommended temperature ranges.

- **Steps of testing EMI/EMC in computer power supply unit**

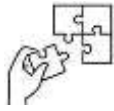
To carry out effective testing, begin by **preparing the tools and equipment required** for the process, ensuring you have everything necessary for accurate measurements. Once your tools are ready, proceed to **prepare the test setup**, arranging the equipment according to the specifications needed for the tests. Next, conduct **conducted emissions testing** to evaluate the electromagnetic interference generated by your device. Following this, perform **immunity testing** to assess how well the device can withstand external electromagnetic disturbances. After completing these tests, it's essential to **check for compliance** with relevant standards to ensure that your device meets regulatory requirements. Finally, make sure to **document the results** thoroughly, as this will provide a comprehensive record of the testing process and outcomes, which is crucial for any future assessments or audits.

- **Testing the safety of a PSU involves multiple steps:**

To ensure the safety and reliability of a power supply unit (PSU), several critical testing steps must be undertaken. First, **ground continuity** should be verified to ensure that the PSU is properly grounded, which is essential for preventing electrical shock hazards. Next, it is important to check **insulation resistance** to confirm that the insulation between live and ground circuits is intact, safeguarding against potential short circuits.

Additionally, performing **overload, short-circuit, and overvoltage tests** is crucial for verifying the PSU's protection mechanisms. Temperature monitoring should also be implemented to

confirm that thermal protection systems are functioning correctly and preventing overheating. Finally, conducting **EMC/EMI and functional testing** will ensure the PSU's stability and compliance with safety standards, confirming that it operates effectively without causing electromagnetic interference.



Application of learning 3.3.

You are an electronics technician in a computer repair shop. The goal of the company is to design a power supply unit and repair the faulty computer power supplies of their clients.

- a) You receive a faulty computer power supply, and you are tasked by the following:
 - i. Perform a visual inspection of a computer PSU to diagnose potential issues before repairing it.
 - ii. Measure input voltage and current in a computer power supply unit (PSU) for diagnostic and performance evaluation purposes.
 - iii. Measure the output voltage and current from various rails of the PSU to ensure they meet the specified standards.
- b) Your supervisor asks you to design a 9Vdc new power supply unit to supply an robotics system for an automated manufacturing plant and after perform the following tasks:
 - i. Test transient response of a computer power supply unit (PSU) to determine if it meets the necessary performance standards for reliable operation.
 - ii. Measure the ripple voltage in the output of a computer power supply to determine if it meets the necessary performance standards for reliable operation.
 - iii. Perform an overload protection mechanism to prevent damage to both the power supply and the electronic system during a power surge or excessive current draw.
 - iv. Test temperature of the PSU under varying loads to identify overheating issues and assess whether cooling mechanisms are functioning effectively.
 - v. Test electromagnetic emissions to avoid causing interference with other electronic devices



Learning outcome 3 end assessment

Theoretical assessment

Instruction: Attempt all provided questions

Section A: Multiple Questions: Encircle the letter of the correct answer.

1. What is the primary purpose of simulating a computer power supply in Proteus?

- a) To build a physical prototype
- b) To test and analyze the circuit behavior before hardware implementation
- c) To create a schematic design for presentation purposes only
- d) To train students on hardware soldering techniques

2. Which component is commonly used to convert AC mains voltage to DC in a computer power supply simulation in Proteus?

- a) Diode bridge
- b) Resistor
- c) Transformer
- d) Inductor

3. When simulating the power supply, which component would you typically use to filter the ripples from the rectified DC voltage?

- a) Capacitor
- b) Transistor
- c) Resistor
- d) Diode

4. Which Proteus tool or feature is used to measure the output voltage of the power supply in the simulation?

- a) Oscilloscope
- b) Multimeter
- c) Ammeter
- d) Logic analyzer

5. In a computer power supply simulation, what could cause the output voltage to fluctuate or drop unexpectedly?

- a) Incorrect diode placement
- b) Insufficient load applied
- c) Lack of ground connection
- d) All of the above

6. Which type of regulator is commonly used in Proteus simulations to maintain a stable 5V output for a computer power supply?

- a) LM317
- b) LM7805
- c) LM555
- d) 2N2222

7. In Proteus, which of the following indicates that your simulated power supply circuit is functioning correctly?

- a) Constant current at input
- b) Stable DC voltage output at the correct levels
- c) Increased heat dissipation in components
- d) Flickering of voltage values on the display

8. Which test should be performed in Proteus to ensure that the computer power supply can handle different loads?

- a) Ripple test
- b) Load regulation test
- c) Line regulation test
- d) Short circuit test

9. What is the function of a capacitor placed after the rectifier in a computer power supply circuit?

- a. To reduce the AC component
- b. To regulate voltage
- c. To increase the current

d. To filter out noise

10. In Proteus, which library contains most of the components required for designing a computer power supply circuit?

- a. Active Components Library
- b. Transformer Library
- c. Power Library
- d. Electromechanical Library

11. Which software module in Proteus allows you to simulate the behavior of power supply circuits?

- A) ISIS
- B) ARES
- C) PSpice
- D) LTspice

Section B: Matching Question

12. Match the statements in Column A with their corresponding significance in column B

Answer	ColumnA	ColumnB
1=.....	1) It's role is filter and reduce the ripple in the DC output of a in a computer power supply circuit	a) Transformer
2=.....	2) It's important to protect the circuit from overcurrent conditions in the computer power supply simulation.	b) Multimeter
3=.....	3) It is used stabilize the output voltage in a computer power supply circuit.	c) Smoothing capacitor
4=.....	4) Proteus tool or feature is used to measure the output voltage of the power supply in the simulation	d) Fuse
		e) The voltage regulator

Practical assessment

You are an electronics engineer working for a company that designs high-performance custom gaming PCs. The company's latest project requires a Power Supply Unit (PSU) that can support high-end components such as powerful graphics cards, multi-core processors, and advanced cooling systems. Additionally, the PSU must be energy-efficient, modular, and capable of supporting overclocking features while ensuring stable power delivery across all components.

Your task is to design and implement a PSU for the system. This includes selecting components, performing necessary calculations, building the circuit, and testing the PSU to ensure it meets the necessary power demands and safety standards.



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Learning Outcome 4: Implement Power Management System



Indicative contents

4.1. Designing computer power management system architecture

4.2. Designing power backup system components

4.3. Integration of power backup system components

4.4. Maintaining power backup

4.5. Estimation of power system cost

Key Competencies for Learning Outcome 4: Implement Power Management System

Knowledge	Skills	Attitudes
• Description of computer power management system • Description of battery management system • Description of power backup • Description of cost estimation • Description of bill of quantity	• Selecting tools and technologies • Developing computer power management plan • Calculating power backup size • Building battery charging system • Installation of electrical power backup • Integrating power backup to power supply unit • Establishing routine maintenance procedures	• Being patient • Willingness to explore new ideas • Demonstrate team spirit while working with others. • Having attention to details. • Being rapid • Being able to take responsibility. • Being organized. • Establish resilience in your daily activities.

	• Designing bill of quantity • Applying cost estimation	
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Duration: 30 hrs

Learning outcome 4 objectives:



By the end of the learning outcome, the trainees will be able to:

1. Assess and gather effectively the requirements computer power management system architecture
2. Identify correctly computer power management techniques based on computer power management system architecture
3. Describe correctly battery management system based on computer power management system architecture
4. Design correctly computer power management system plan based on power management system architecture
5. Describe correctly power backup based on power backup system components
6. Identify effectively backup power sources based on power backup system
7. Calculate efficiently power backup size based on Total power consumption by system
8. Describe properly battery charging controller based on backup system components
9. Design correctly battery charging controller based on backup system components
10. Describe properly component placement compatibility and placement assessment based on backup system components
11. Install correctly electrical power backup to power supply unit based on backup system components

12. Apply effectively communication interface based on backup system
13. Integrate correctly power backup system to power supply unit in line with power backup system
14. Establish Routine Monitoring and maintenance Procedures in line with power backup system
15. Estimate accurately the cost of power system in line with power backup system



Resources

Equipment	Tools	Materials
• Personal Computer • PPEs • Multi-meter • Function generator • SMD rework station • DC Power supply • Office UPS	• Screwdrivers • Needle-nose pliers • Universal plier • Tweezers • Flashlight or headlamp • Cable testers • ESD Tools applicator • Wire cutter/stripping Pliers • Soldering iron • Desoldering pump • Glue gun • Circuit Simulation and drawing software (NI Multisim, Proteus, Autodesk Eagle, Easy EDA, matlab, CAD) • Drawing accessories	•Soldering wire •Jumper wires •Breadboard •PCB •Power cables •Desoldering wick •Drawing materials (papers and pencil) •Electronic Passive Components •Wires (2.5mm², 1.5mm²,4mm², and 6mm²) •Insulator tapes •Water proof tapes •Connection boxes •Screws •Electrical pipes •Cat 6e •RJ 45 connectors



Indicative content 4.1: Designing Computer Power Management System Architecture



Duration: 6 hrs



Theoretical Activity 4.1.1: Description of Assessment and requirements gathering

Tasks:

- 1: Answer the following questions related to the assessment and requirements gathering in computer management system.
 - i. Describe power requirements of a computer power management system.
 - ii. Discuss about usage pattern of a computer power management system.
 - iii. Identify critical components of computer power consumption.
 - iv. List the steps of documenting your findings on computer power system requirements.
- 2: Write the findings/answers on papers or flipchart.
- 3: Present your findings to the class.
- 4: Ask questions for more clarifications.
- 5: Read the key reading 4.1.1 from your manual to get more clarifications.



Key readings 4.1.1.:

Description of Assessment and requirements gathering

1. Evaluate Power Requirements

1.1 What are the power requirements?

Power requirement refers to the specific amount of computing power, in terms of either microprocessors and their speed or the peak watts from the computing power supply, needed for a particular project or task in the field of computer science.

The power requirements of a computer power system depend on the components of the system and the workload it handles.

1.2 Evaluation of Power Requirements in a Computer Power Management System

Evaluating the power requirements of a **computer power management system** involves determining how efficiently the system uses and distributes power, identifying the power

needs of each component, and ensuring that the system has enough capacity for optimal performance under varying loads.

The evaluation of power requirements in a computer power management system depends on several key factors:

- a) **Hardware Configuration:** The type and specifications of hardware components significantly influence power consumption. More powerful CPUs and GPUs consume more energy, especially during intensive tasks.

For instance, high-performance graphics cards can substantially increase electricity usage during gaming or graphic design activities.

- b) **Software Efficiency:** The efficiency of the operating system and applications plays a crucial role in energy consumption. Efficient software can minimize the processing demands on hardware, thereby reducing power usage. Conversely, resource-intensive applications can lead to increased energy consumption due to higher CPU and memory usage.

- c) **Usage Patterns:** Understanding how and when computers are used is vital for evaluating power needs. Many systems remain active during idle times, leading to unnecessary energy consumption.

For example, studies indicate that computers are often left on overnight or during long breaks, which contributes to excessive energy use.

- d) **Power Management Settings:** The configuration of power management features within the operating system can optimize energy consumption. Features like sleep modes, scheduled shutdowns, and dynamic voltage scaling help reduce power draw when systems are not in use.

- e) **Environmental Factors:** External conditions such as temperature and humidity can also affect power requirements. Higher temperatures may necessitate increased cooling efforts, which in turn raises overall energy consumption.

- f) **Component Interactions:** The interplay between different hardware components—such as the CPU, GPU, RAM, and storage devices—can impact overall power efficiency.

For example, if a high-performance CPU is paired with an inefficient power supply unit (PSU), it may not operate optimally, leading to wasted energy.

1.3 Factors Influencing Energy Consumption

Several factors affect the overall energy consumption of computer systems:

- **Hardware Efficiency:** The efficiency ratings of components (e.g., CPUs and PSUs) can greatly influence power usage. Higher efficiency ratings (like 80 Plus) indicate better performance with less wasted energy.
- **User Behavior:** How users interact with their computers (e.g., leaving systems on overnight) contributes to unnecessary power consumption.
- **Operating System and Software:** The efficiency of the operating system and applications plays a role in managing power usage effectively.

2. Analysis of Power usage pattern.

Power usage patterns in a computer power supply system refer to the variations in electricity consumption based on different operational states and tasks performed by the computer.

Analyzing the power usage pattern of a computer system involves understanding how power is consumed across different components and under various conditions. The pattern can vary based on workload, system configuration, and user behavior.

Understanding power usage patterns in computers is essential for optimizing energy efficiency and reducing costs.

Here is an analysis of the power usage pattern for a typical computer system:

2.1. Idle State

When a computer is powered on but not under any significant load (e.g., only basic background tasks like system processes or minimal applications are running), the power usage is at its lowest.

- **CPU:** Minimal usage, typically operating at low clock speeds to conserve energy (around 10W–30W depending on the model).

- **GPU:** Very low power draw (usually less than 10W), especially if it switches to idle or low-power states when not used for rendering.
- **Storage (HDD/SSD):** Minimal activity, usually around 2W–5W.
- **RAM:** Memory is used to hold basic system processes, consuming around 3W–5W.
- **Total Power Consumption:** 50W–100W for an average desktop, even lower for laptops due to better power efficiency mechanisms.

2.2. Moderate Load (Browsing, Streaming, Office Applications)

In this state, the computer is running typical everyday tasks such as web browsing, video streaming, and word processing. These tasks don't require the full processing power of the CPU or GPU.

- **CPU:** Power consumption increases to moderate levels (around 20W–50W), depending on the number of active applications and whether they are single-threaded or multi-threaded.
- **GPU:** Still low power usage unless a GPU-accelerated task like video playback is occurring (10W–30W).
- **Storage:** Occasional access to storage, but consumption remains relatively low (around 2W–10W).
- **RAM:** More memory is active, consuming around 5W–10W.
- **Total Power Consumption:** 100W–200W, depending on the system configuration.

2.3. High Load (Gaming, Video Editing, 3D Rendering)

High-performance tasks such as gaming, video editing, and 3D rendering demand significant computational power.

- **CPU:** The CPU may run at or near full load, consuming between 60W–150W depending on the number of cores and frequency.
- **GPU:** This component can reach its maximum power draw, especially during intense gaming or rendering (between 150W–350W for high-end GPUs).
- **Storage:** More frequent read/write activity, but consumption typically remains below 15W for modern SSDs and HDDs.

- **RAM:** More memory access, especially for rendering or large-scale data tasks (10W–15W).
- **Cooling Fans:** Increased workload requires more cooling, increasing fan speeds and power usage (5W–15W).
- **Total Power Consumption:** 300W–600W for gaming desktops, higher for systems with multiple GPUs or overclocked components.

2.4. Peak Power Usage (Boot-up and Intensive Tasks)

During boot-up or when launching heavy software (such as a game or editing software), there may be a temporary spike in power consumption as all components briefly run at high capacity.

- **CPU:** High spike in power for a short period as the system initializes (100W+).
- **GPU:** Briefly powers up to load graphical interfaces or initialization screens (up to 300W depending on the GPU).
- **Storage:** Increased access as system files are loaded from disk, especially if using traditional HDDs (up to 20W).
- **Cooling System:** Some fans may spin faster to accommodate higher heat output during boot (up to 10W–20W).
- **Total Power Consumption:** May momentarily spike to 500W–700W depending on the system.

2.5. Sleep/Hibernation Mode

In these power-saving states, the system consumes very little power. Most components are either shut down or in low-power mode.

- **CPU:** Minimal power draw, sometimes as low as 1W.
- **GPU:** Effectively zero power consumption unless there is a wake-on-GPU feature.
- **Storage:** Inactive, using near-zero power.
- **Total Power Consumption:** Typically 2W–10W depending on the system and peripherals.

2.6. Power Usage in Laptops vs. Desktops

- **Laptops:** Optimized for energy efficiency. Modern laptops adjust power consumption dynamically depending on usage. They may consume between 10W (idle) to 100W (heavy load), which is significantly less than desktops.
- **Desktops:** Tend to consume more power due to larger components, more powerful GPUs, and higher-performing CPUs. The consumption range can be from 50W (idle) to 600W or more (under heavy load).

3. Identify Critical Components

Identifying the critical components that contribute to energy consumption in computer power systems is essential for optimizing efficiency and reducing operational costs.

Here are the primary components:

3.1. Central Processing Unit (CPU)

- **Power Consumption:** The CPU is one of the most power-hungry components, typically consuming between 50 to 150 watts depending on its architecture and workload.
- **Impact:** The power usage can vary significantly based on the tasks being performed, with higher consumption during intensive processing tasks like gaming or data analysis

3.2. Graphics Processing Unit (GPU)

- **Power Consumption:** GPUs can consume a substantial amount of power, often ranging from 20 watts for low-end models to over 350 watts for high-performance gaming cards. In gaming PCs, a high-quality GPU can use up to 250 watts alone.
- **Impact:** The GPU's power draw is particularly pronounced during graphics-intensive tasks such as gaming or video rendering, where both the CPU and GPU operate at peak performance levels.

3.3. Power Supply Unit (PSU)

- **Efficiency:** The PSU's efficiency plays a crucial role in overall energy consumption. Higher efficiency ratings (e.g., 80 Plus Gold or Platinum) indicate better performance in converting AC to DC power, reducing wasted energy and heat generation.

- **Impact:** An inadequate or inefficient PSU can limit performance and lead to increased energy costs due to wasted power.

3.4. Memory (RAM)

- **Power Consumption:** RAM typically consumes between 2 to 5 watts per module and does not fluctuate significantly with load. However, having more RAM modules increases overall consumption.
- **Impact:** While RAM is not the largest consumer of power, its cumulative effect in systems with multiple modules can contribute to total energy usage.

3.5. Storage Devices

- **Hard Disk Drives (HDDs):** HDDs consume about 5 to 15 watts while spinning and during read/write operations.
- **Solid-State Drives (SSDs):** SSDs generally consume less power than HDDs, using energy only during active read/write processes. This makes SSDs a more energy-efficient option for storage.

3.6. Motherboard and Other Components

- **Power Consumption:** The motherboard itself consumes approximately 20 to 40 watts, along with other peripherals like cooling fans and network cards which add to the total system draw.
- **Impact:** Although individually these components consume less power compared to CPUs and GPUs, their collective impact can be significant in larger systems.

Summary Table of Power Consumption

The following table summarizes the estimated power consumption of critical components under various states:

Component	Idle (W)	Network (W)	Gaming (W)	Streaming (W)	3D Editing (W)
CPU	5 - 15	15 – 30	Up to 200	Up to 100	Up to 150
GPU	<10	<20	Up to 350	Up to 150	Up to 250
Motherboard	25 - 40	30 – 50	45 - 80	40 – 70	40 – 70

RAM	<5	<5	<10	<5	<10
Storage (HDD/SSD)	<3	<3	<5	<3	<5

4. Document Findings and Requirements

After evaluating the system's power needs and identifying critical components, document the key findings and establish the requirements for optimizing power consumption. The goal is to create a comprehensive report that outlines current power usage, areas for improvement, and potential solutions for more efficient power management.

4.1 Documentation Should Include:

- **Power usage profile:** Summarize power consumption across different workloads (idle, active, peak usage).
- **Component analysis:** Detail the power consumption of critical components and identify those that contribute most to overall power usage.
- **Usage patterns:** Highlight key trends and usage scenarios that lead to high energy consumption (e.g., during idle time, heavy multitasking).
- **Recommendations for improvement:** Suggest power-saving strategies, such as enabling power management features, upgrading components, or adjusting usage habits.
- **Requirements for optimization:** Identify the technical and operational requirements for implementing effective power management solutions.

4.2. Recommendations for Reducing Energy Consumption

To optimize energy efficiency in computer systems, consider the following strategies:

- **Utilize Power Management Features:** Enable sleep modes and other built-in power-saving settings.
- **Choose Energy-Efficient Components:** Select hardware with high-efficiency ratings to minimize energy waste.
- **Regularly Update Software:** Keep operating systems and applications updated to benefit from improvements in energy management.

- **Monitor Usage Patterns:** Use tools to track energy consumption and identify opportunities for reduction.
- **Educate Users:** Promote awareness about energy-saving practices among users to encourage responsible usage.

To ensure accurate assessment and effective planning for power supply needs:

- Identify Current Power Supply:** Determine the specifications of the existing power supply.
- Check Motherboard Requirements:** Verify the motherboard's power requirements and compatibility with potential upgrades.
- Check Other Components' Requirements:** Assess the power needs of all critical components including CPU, GPU, and peripherals.
- Calculate Total Power Consumption:** Sum the power requirements of all components to understand total system demand.
- Compare Power Supply Options:** Evaluate different power supply units based on efficiency ratings and compatibility with existing hardware.
- Document Power Supply Upgrade:** Record details of any upgrades made to the power supply system.
- Consider Additional Factors:** Document other considerations such as cooling requirements and future expansion plans which may impact power needs.



Theoretical Activity 4.1.2: Identification of computer power management techniques and power saving strategies

Tasks:

- 1: Answer the following questions related to the identification of computer power management techniques.
 - i. What do you understand by computer power management?
 - ii. Describe the Processor power states
 - iii. Discuss the software optimization
 - iv. Identify the computer power saving strategies
 - v. Identify the power management policies
- 2: Write the findings/answers on papers or flipchart.
- 3: Present your findings to the class.

4: Ask questions for more clarifications.

5: Read the key reading 4.1.2 from your manual to get more clarifications.



Key readings 4.1.2.:

Identification of computer power management techniques

1. Computer power management

1.1 Definition

- **Power management:** is how power is efficiently directed to different components of a system. Power management is especially important for portable devices that rely on battery power. By reducing power to components that are not being used, a good power management system can double or triple the lifetime of a battery.

Computer power management (CPM) refers to the strategies and technologies used to control and optimize the energy consumption of computers and related devices. This practice is essential for enhancing energy efficiency, reducing operational costs, and promoting sustainability.

1.2 Key Aspects of Computer Power Management

- **Energy Conservation:** CPM aims to minimize power usage by transitioning computers into low-energy states when they are not in active use. For instance, systems can automatically enter sleep mode after a set period of inactivity, significantly reducing energy consumption from approximately 35-70 watts to as low as 2 watts
- **Automated Control:** Power management systems often include automated features such as scheduled shutdowns, wake-ups, and sleep modes. These features allow organizations to customize power settings based on specific operational needs, ensuring that computers are only using power when necessary.

- **Software and Hardware Integration:** Effective power management requires collaboration between software, hardware, and the operating system. Device drivers play a crucial role in managing power for individual components, allowing for intelligent decisions regarding power usage across the entire system.
- **User Behavior Monitoring:** Understanding user activity is vital for optimizing power management. Many computers remain powered on during idle times due to user habits or software processes that inhibit low-power states. Addressing these behaviors can lead to substantial energy savings.
- **Environmental Impact:** By implementing effective CPM strategies, organizations can significantly reduce their carbon footprint. The Environmental Protection Agency estimates that if all office desktops had power management activated, it could save billions in energy costs annually.

1.3 Benefits of Computer Power Management

- **Cost Savings:** Reducing energy consumption directly translates into lower electricity bills. Organizations can save an average of \$36 per PC per year through effective power management practices.
- **Extended Hardware Lifespan:** By minimizing the operational load on computers, CPM can help extend the lifespan of hardware components, reducing maintenance and replacement costs.
- **Improved Efficiency:** Automated power management allows for better resource allocation and operational efficiency, enabling IT departments to focus on more critical tasks rather than manual power management.

2. Sleep modes

2.1 Definition

Sleep mode is a power saving state that stops all actions on the computer. Any open documents and applications are moved to the system memory (RAM) and the computer goes to a low-power state.

2.2 System Sleeping States

Power state	ACPI state	Description
Working	S0	The system is fully usable. Hardware components that aren't in use can save power by entering a lower power state.
Sleep (Modern Standby)	S0 low-power idle	The system can very quickly switch from a low-power state to high-power state in response to hardware and network events. Systems that support Modern Standby do not use S1-S3.
Sleep	S1 S2 S3	The system appears to be off. The amount of power consumed in states S1-S3 is less than S0 and more than S4. S3 consumes less power than S2, and S2 consumes less power than S1. Systems typically support one of these three states, not all three. In states S1-S3, volatile memory is kept refreshed to maintain the system state. Some components remain powered so the computer can wake from input from the keyboard, LAN, or a USB device. Hybrid sleep, used on desktops, is where a system uses a hibernation file with S1-S3. The hibernation file saves the system state in case the system loses power while in sleep. Note: SoC systems that support Modern Standby don't use S1-S3.
Hibernate	S4	The system appears to be off. Power consumption is reduced to the lowest level. The system saves the contents of volatile memory to a hibernation file to preserve system state. Some

		components remain powered so the computer can wake from input from the keyboard, LAN, or a USB device. Fast startup is where the user is logged off before the hibernation file is created. This allows for a smaller hibernation file, more appropriate for systems with less storage capabilities.
Soft off	S5	The system appears to be off. This state is comprised of a full shutdown and boot cycle.
Mechanical off	G3	The system is completely off and consumes no power. The system returns to the working state only after a full reboot.

3.Processor power states

3.1 Definition

Processor power states are mechanisms that allow CPUs to manage their energy consumption efficiently by adjusting operational parameters based on workload and activity levels.

Processor power management involves various states that help optimize energy consumption during different operational conditions.

These states are categorized into **P-states** (performance states) and **C-states** (idle states).

3.2. P-States (Performance States)

P-states allow the processor to adjust its frequency and voltage dynamically based on workload demands. This helps in reducing power consumption without significantly impacting performance.

The states are defined as follows:

- **P0:** Maximum performance state, where the CPU operates at its highest frequency and voltage.

- **P1, P2,...Pn:** Successively lower performance states where the CPU operates at reduced frequencies and voltages. The exact number of P-states varies by processor model.

Initially, the operating system would set these states by writing values into special processor registers. However, modern architectures like Intel's Skylake allow the CPU itself to manage P-states more autonomously through technologies such as Speed Shift.

3.3. C-States (Idle States)

C-states are designed to reduce power consumption when the CPU is not executing instructions. They represent various levels of inactivity where different parts of the processor can be powered down.

The various C-states include:

- **C0:** Active state where the CPU is executing instructions. P-states are applicable here.
- **C1:** Auto Halt state, where the CPU is halted but can return to C0 instantly.
- **C1E:** Enhanced version of C1, with low frequency and voltage.
- **C2:** Temporary state before deeper sleep states; some memory paths remain active.
- **C3:** Deeper sleep state where caches are flushed, and clocks are turned off.
- **C6:** Core states saved before shutdown; PLL (Phase-Locked Loop) is off.
- **C7:** Similar to C6 but with additional cache flushing.
- **C8:** Deeper state where even more components are powered down.

Higher C-states typically indicate a deeper level of sleep, resulting in lower power consumption but longer wake-up times

3.4 ACPI (Advanced Configuration and Power Interface)

ACPI (Advanced Configuration and Power Interface) is a standard used to manage power consumption and configuration settings in computers. It provides a common interface for

operating systems and hardware components to communicate and coordinate power-related functions. ACPI is designed to improve energy efficiency, system responsiveness, and overall user experience.

- **QuickStart technology:** Extends battery life by entering a low-power state during the briefest pauses in user activity, such as between key strokes. Instantly returns to full-power state when prompted.
- **Dynamic Cache Sizing:** power savings mechanism flushes system memory dynamically, based on demand or during periods of inactivity. Power savings occur as the cache ways are turned off once the data has been saved in memory. Once the Dynamic Cache Sizing feature flushes the entire cache data to memory, the processor transitions to Intel® Enhanced Deep Sleep. This allows the processor to lower voltage below the Deep Sleep minimum voltage for enhanced power savings and/or efficiencies.

4. Software optimization

4.1 Definition

Software optimization refers to the process of improving the efficiency and performance of software applications, particularly in terms of resource usage, speed, and energy consumption. It involves making adjustments to code, algorithms, and system configurations to achieve better performance while minimizing resource requirements.

Program optimization, code optimization, or software optimization is the process of modifying a software system to make some aspect of it work more efficiently or use fewer resources.

Configure operating system settings to enable power-saving features like screen dimming, hard drive spin-down, and automatic sleep.

4.2 Function

Software optimisation manage the following functions:

4.2.1 Processor Power States:

- **Dynamic Frequency Scaling:** Software can dynamically adjust the processor's clock speed and voltage based on workload, reducing power consumption during periods of low activity.
- **C-State Management:** Software can intelligently manage the processor's power states (C0-C6) to minimize power consumption while maintaining performance.

4.2.2 Memory Management:

- **Page Replacement Algorithms:** Efficient page replacement algorithms can minimize memory access, reducing the power consumption of memory controllers and cache.
- **Memory Compression:** Software can compress memory data to reduce the overall memory footprint and, consequently, the power required to access it.

4.2.3 I/O Device Management:

- **Power Management Policies:** Software can implement power management policies for I/O devices, such as hard drives and network interfaces, to reduce their power consumption during idle periods.
- **Device Scheduling:** Optimizing the scheduling of I/O operations can reduce the overall power consumption of I/O subsystems.

4.2.4 Operating System Optimization:

- **Background Task Management:** Software can prioritize and manage background tasks to minimize their impact on power consumption, especially during idle periods.
- **Power-Saving Features:** Operating systems can provide features like screen dimming, hard drive spin-down, and automatic sleep to reduce power consumption.

4.2.5 Application Optimization:

- **Energy-Efficient Algorithms:** Developers can design algorithms and data structures that are more energy-efficient, especially for computationally intensive tasks.

- **Idle Detection and Power Management:** Applications can implement mechanisms to detect periods of inactivity and automatically reduce their power consumption.

4.2.6 Cloud-Based Optimization:

- **Virtualization and Consolidation:** Cloud platforms can consolidate multiple virtual machines onto fewer physical servers, improving resource utilization and reducing power consumption.
- **Workload Balancing:** Software can distribute workloads across multiple servers to optimize resource utilization and minimize power consumption.

5. Identify Power Saving Strategies.

5.1 Definition

Power saving strategies are essential for optimizing energy consumption in computer systems, enhancing efficiency, and reducing operational costs. These strategies can be categorized into hardware-based and software-based approaches.

5.2 Dynamic Voltage and Frequency Scaling (DVFS)

Dynamic Voltage and Frequency Scaling (DVFS) is a power management technique that optimizes energy consumption in computing systems by dynamically adjusting the voltage and frequency of processors and other components based on workload requirements.

a. Importance

The primary goal of DVFS is to balance computational capacity with power efficiency. By adapting the voltage and frequency according to performance needs, DVFS ensures that systems operate efficiently under varying workloads, thereby reducing overall energy consumption while maintaining performance levels.

b. Basic Concepts

- **Voltage Scaling (Undervolting):** This involves reducing the supply voltage to the processor. Lowering voltage leads to significant power savings due to the quadratic relationship between voltage and active power consumption.
- **Frequency Scaling (Underclocking):** This refers to lowering the clock frequency of the processor, which also reduces dynamic power consumption. By decreasing frequency, the processor consumes less power during less demanding tasks.
- **Overvolting:** In contrast, increasing the voltage can enhance performance for demanding tasks, allowing the processor to operate at higher frequencies.

c. DVFS Techniques

- **Dynamic Voltage Scaling (DVS):** A simpler form of DVFS that scales down only the voltage based on performance requirements.
- **Adaptive Voltage and Frequency Scaling (AVFS):** An extension of DVFS that utilizes closed-loop feedback mechanisms to compensate for variations in temperature, process conditions, and IR drop. AVFS provides active monitoring and adjustments for better power reduction but requires more complex control systems.

d. Implementation Challenges

Implementing DVFS involves several challenges:

- **Power Domains:** DVFS requires distinct power domains where different voltage levels can be applied. Consistent definitions across these domains are crucial for effective management.
- **Operating Modes:** Each power domain can operate in various modes, each with specific supply voltages and operating frequencies.
- **EDA Tools Optimization:** Efficiently optimizing different voltage-frequency combinations can be complex and may require longer design times but offers substantial power savings.

e. Benefits

- DVFS can improve dynamic power consumption by **40-70%** and reduce leakage by **2X to 3X**.
- It enhances battery life in portable devices and lowers thermal dissipation, which is critical for maintaining optimal operating conditions.
- The technique also contributes to reducing a device's carbon footprint by lowering overall energy usage.

f. Applications

DVFS is widely used across various platforms:

- It is implemented in mobile devices, laptops, desktops, servers, and data centers to optimize performance while conserving energy.
- In virtual environments like VMware vSphere, DVFS helps reduce energy consumption by dynamically adjusting CPU states based on resource demands.

Below are some effective power-saving strategies:

a. Hardware-Based Strategies

- **Low-Power Components:** Utilize energy-efficient hardware components, such as low-power processors, energy-efficient memory modules, and high-efficiency power supply units. These components are designed to minimize energy consumption while maintaining performance.
- **Dynamic Voltage and Frequency Scaling (DVFS):** Implement DVFS techniques that adjust the voltage and frequency of the processor based on workload demands. This allows the CPU to operate at lower power levels during less intensive tasks, thereby saving energy without sacrificing performance.
- **Power Gating:** Use power gating techniques to selectively turn off power to unused components within the system. This helps reduce leakage power and overall energy consumption when certain parts of the hardware are not in use.

- **Efficient Cooling Solutions:** Invest in advanced cooling technologies that require less power, such as variable-speed fans or liquid cooling systems. Efficient cooling reduces the need for excessive energy consumption to maintain optimal operating temperatures.

b. Software-Based Strategies

- **Power Management Policies:** Implement operating system-level power management policies that control when and how devices enter low-power states. Features like sleep modes, hibernation, and automatic shutdowns can significantly reduce energy usage during idle periods.
- **Screen Dimming and Timeout Settings:** Configure monitors to dim or turn off after a period of inactivity. This simple adjustment can lead to substantial energy savings, especially in office environments where computers may remain idle for extended periods.
- **Background Process Management:** Optimize background processes to minimize resource usage when the system is not actively being used. This includes managing startup programs and ensuring that non-essential applications do not consume resources unnecessarily.
- **Intelligent Task Scheduling:** Use intelligent scheduling algorithms that prioritize tasks based on their resource requirements and timing. By scheduling intensive tasks during off-peak hours or when other resources are available, overall power consumption can be reduced.

c. Centralized Management Tools

- **Power Management Software:** Utilize centralized power management solutions that allow IT administrators to monitor and control power settings across all devices within an organization. This enables uniform application of power-saving measures and facilitates automated actions like scheduled shutdowns or wake-ups.

d. User Education and Awareness

- **Promote Energy-Saving Practices:** Educate users about the importance of energy conservation and encourage practices such as shutting down computers when not in use, utilizing sleep modes, and being mindful of resource-intensive applications.

6. Smart power Management Policies.

Smart power management policies are essential components of computer power management systems designed to optimize energy consumption while maintaining system performance. These policies leverage both hardware and software techniques to intelligently manage power usage based on current demands and usage patterns. Here's an overview of the basics of smart power management policies:

6.1. Definition and Purpose

Smart power management policies refer to a set of rules and algorithms that govern how a computer system manages its power consumption. The primary objectives include:

- **Reducing Energy Consumption:** Minimize unnecessary power usage during idle times or low-demand periods.
- **Extending Device Longevity:** Reduce wear on components by managing heat generation and power cycles.
- **Enhancing Performance:** Ensure that the system remains responsive by balancing power savings with performance needs.

6.2. Types of Smart Power Management Policies

- **Dynamic Frequency Scaling:** Adjusts the processor's clock speed based on workload demands. When the workload decreases, the system can lower the frequency, thereby saving power.
- **Sleep and Hibernate Modes:** Automatically transitions systems into low-power states after a period of inactivity. Sleep mode keeps the session in memory for quick wake-up, while hibernate saves the session to disk and powers down completely.

- **Task Scheduling Optimization:** Implements intelligent scheduling algorithms to prioritize high-demand tasks while reducing resource allocation for less critical processes during low activity periods.
- **Adaptive Power Management:** Utilizes real-time data to adjust power settings dynamically based on current usage patterns, such as temperature, workload, and user behavior.

6.3. Implementation Mechanisms

- **Operating System Integration:** Modern operating systems incorporate built-in power management features that allow for seamless implementation of smart policies. This includes APIs for developers to create applications that can leverage these features.
- **Hardware Support:** Many processors come with built-in support for various power states (P-states and C-states) that can be managed through these policies. For instance, Intel processors use Speed Shift Technology to optimize P-state transitions.
- **Monitoring Tools:** Software tools monitor system performance and energy consumption, providing feedback that helps refine power management strategies over time.

6.4. Benefits

- **Cost Savings:** By reducing energy consumption, organizations can significantly lower electricity bills.
- **Environmental Impact:** Decreased energy usage contributes to reduced carbon emissions, supporting sustainability initiatives.
- **User Experience Improvement:** Systems that efficiently manage power can provide better performance during active use while ensuring quick recovery from low-power states.

6.5. Challenges

Implementing smart power management policies can present challenges, including:

- **Complexity of Implementation:** Developing effective algorithms that balance performance and energy savings requires sophisticated programming and testing.
- **User Acceptance:** Users may need to adapt to changes in how their devices behave regarding performance and responsiveness during low-power states.



Theoretical Activity 4.1.3: Description of battery management system



Tasks:

1: Answer the following questions related to the description of battery management system.

- i. What do you understand by battery?
- ii. What do you understand by battery management system?
- iii. Identify the types of battery used in computer power system
- iv. How do you manage charging and discharging cycles of battery?
- v. Describe the term optimized charging efficiency.

2: Write the findings/answers on papers or flipchart.

3: Present your findings to the class.

4: Ask questions for more clarifications.

5: Read the key reading 4.1.3. from your manual to get more clarifications.



Key readings 4.1.3.:

Description of battery management system

1. Battery

1.1 Definition and Function

A **battery** is a device that stores electrical energy in chemical form and converts it into electrical energy when needed. It consists of one or more electrochemical cells that generate voltage through chemical reactions.

Batteries are widely used in various applications, including consumer electronics (like smartphones and laptops), electric vehicles, and renewable energy systems, providing a portable power source that enables devices to operate without a direct connection to a power supply.

Batteries are characterized by their capacity (measured in ampere-hours or milliampere-hours), voltage, energy density, and cycle life, which indicates how many charge-discharge cycles they can undergo before significant capacity loss occurs.

Here are some key points about batteries:

- **Electrochemical Cells:** A battery is made up of electrochemical cells that undergo chemical reactions to generate electricity. When discharging, these reactions produce electrical energy, which can be used to power devices.
- **Rechargeable vs. Non-Rechargeable:** Batteries can be classified as primary (non-rechargeable) or secondary (rechargeable). Primary batteries, like alkaline batteries, can only be used once, while secondary batteries, such as lithium-ion and lead-acid batteries, can be recharged and reused multiple times.

1.2 Types of Batteries

- **Lead-Acid Batteries:** Commonly used in vehicles, these are one of the earliest types of rechargeable batteries. They have a high power-to-weight ratio and are cost-effective for automotive applications.
- **Lithium-Ion Batteries:** Widely used in portable electronics and electric vehicles, lithium-ion batteries are known for their high energy capacity, low self-discharge rate, and absence of memory effect.

1.3 Applications

Batteries are essential for powering a wide range of devices, including:

- **Consumer Electronics:** Such as smartphones, laptops, and cameras.
- **Automobiles:** Providing the necessary power to start engines and operate electrical systems.

- **Renewable Energy Systems:** Storing energy generated from sources like solar panels for later use.

1.4 Importance

Batteries play a crucial role in modern technology by enabling portable power solutions and supporting the transition to renewable energy sources. Their ability to store and supply energy efficiently makes them vital for both everyday consumer products and larger-scale applications like electric vehicles and grid storage systems.

2. Battery Management System (BMS)

2.1 Definition

A Battery Management System (BMS) is an electronic system that manages and monitors rechargeable battery packs or cells to ensure their safe and efficient operation.

It ensures safe operation by regulating charging and discharging processes, monitoring critical parameters such as voltage, current, temperature, and state of charge (SoC). The BMS also protects against overcharging, over-discharging, and thermal runaway, thereby enhancing the battery's lifespan and performance.

The BMS plays a crucial role in various applications, including electric vehicles, renewable energy storage systems, and portable electronic devices.

2.2 Functions of a Battery Management System

- Monitoring:** The BMS continuously monitors critical parameters such as voltage, current, temperature, and state of charge (SOC) of individual battery cells or modules. This real-time data collection is essential for ensuring safe operation and optimal performance.
- Protection:** The BMS protects the battery from operating outside its safe limits. It prevents overcharging, over-discharging, and excessive heating, which can lead to battery damage or failure. By managing these conditions, the BMS helps prolong the lifespan of the battery.
- Balancing:** To maximize the battery's capacity and efficiency, the BMS balances the charge levels across individual cells within a battery pack. This ensures that all cells operate at similar states of charge, preventing localized overcharging or undercharging.
- State Estimation:** The BMS calculates various states of the battery, including:

- ✧ **State of Charge (SOC):** Indicates how much energy is left in the battery relative to its total capacity.
 - ✧ **State of Health (SOH):** Reflects the overall condition and capacity of the battery compared to its original specifications.
- e) **Communication:** The BMS communicates with external systems or devices to relay information about the battery's status, including alerts for any faults or operational issues. This communication is vital for integrated energy management systems.
- f) **Thermal Management:** The BMS monitors the temperature of the battery cells and can control cooling systems to prevent overheating, which is critical for maintaining safety and performance.

2.3 Types of Battery Management Systems

- a) **Centralized BMS:** A single controller oversees all cells in a battery pack. This approach is cost-effective but may struggle with scalability in larger systems due to wiring complexity.
- b) **Modular BMS:** Multiple smaller units monitor individual modules within a battery pack. This design allows for easier management but can be more complex and expensive.
- c) **Distributed BMS:** Each cell has its own management unit connected through a communication bus to a central controller. While this system offers high accuracy and flexibility, it can also be more costly to implement.

2.4 Importance of a Battery Management System

The BMS is critical for ensuring that batteries operate safely and efficiently in various applications. It enhances performance by optimizing charging and discharging cycles, extending battery life, and preventing hazardous conditions that could lead to fires or explosions.

2.5 BMS Components and their functions:

A Battery Management System (BMS) consists of several key components, each serving specific functions to manage the battery effectively.

Here are the primary components of a BMS and their roles:

- a) **Microcontroller (MCU):** Acts as the central processing unit of the BMS, orchestrating all operations. It continuously collects data from various sensors, processes this information, and makes critical decisions regarding cell balancing, thermal regulation, and communication with external devices.
- b) **Cell Monitoring Circuit:** Monitors individual cells' voltage, current, and temperature to detect discrepancies early. This ensures that all cells operate within safe limits and helps in balancing the charge among cells.
- c) **Balancing Circuit:** Ensures that all cells within a battery pack maintain similar states of charge (SoC). It redistributes energy among cells to prevent overcharging or undercharging, which can lead to capacity loss or reduced performance. Balancing can be achieved through passive (dissipating excess energy) or active methods (moving charge from higher voltage cells to lower voltage ones).
- d) **Thermal Management System:** Monitors and regulates the temperature of the battery cells to keep them within safe operating ranges. It can activate cooling or heating systems as needed to prevent overheating or excessive cooling, thereby preserving battery life.
- e) **Communication Interfaces:** Enables data exchange between the BMS and external systems using protocols like CAN bus or UART. This communication allows for real-time monitoring and control, including receiving charge/discharge commands and reporting diagnostic information.
- f) **Pre-Charge Circuit:** Manages inrush current during charging to prevent component failure. It ensures that the initial charging current does not exceed safe levels when connecting to a power source.
- g) **Insulation Monitoring Device (IMD):** Detects unintentional leakage between hot conductors and ground, enhancing safety by preventing potential electrical hazards.
- h) **Real-Time Clock (RTC):** Keeps track of time for scheduling tasks related to battery management, such as logging data and executing timed operations.
- i) **Current Sensing and Control:** Current Sensing and Control mechanisms monitor and regulate charge and discharge currents for optimal battery usage. Adding

current sensors can measure the flow of electric charge, providing essential data for managing the charging and discharging processes.

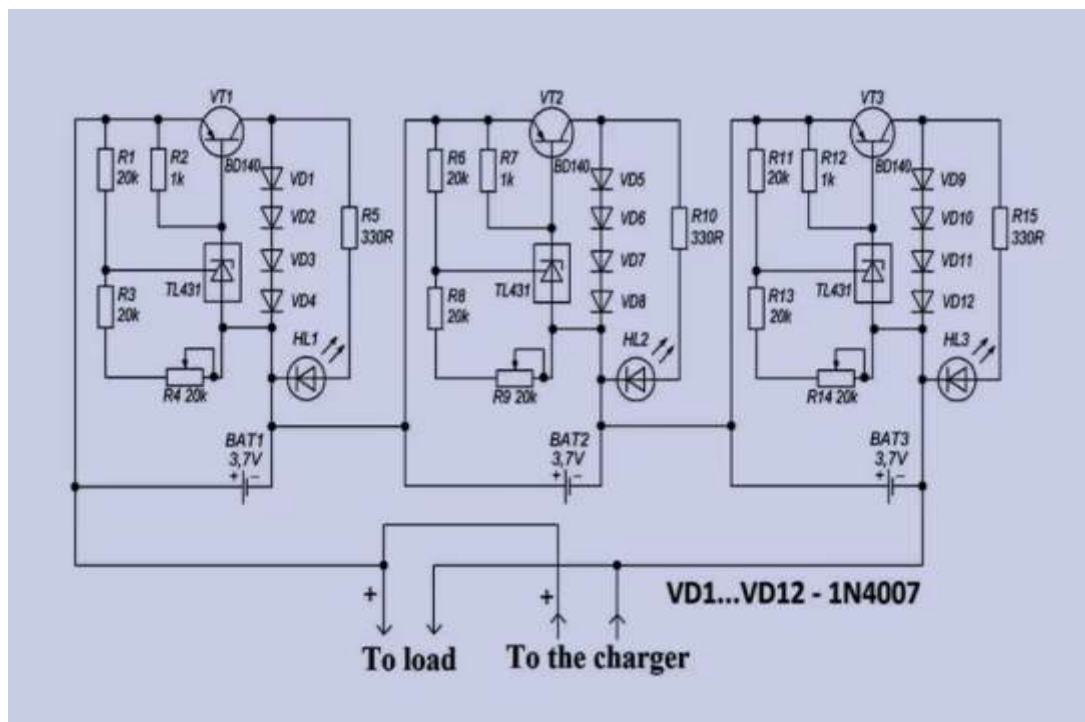
- j) **Protection Circuits:** They may include voltage limiters and disconnect switches. Also, over-discharging can damage cells and reduce battery lifespan. Protection circuits implement mechanisms to disconnect the load when the voltage drops to a critical level.

- **Overview of BMS Circuit Diagram Symbols and Notations**

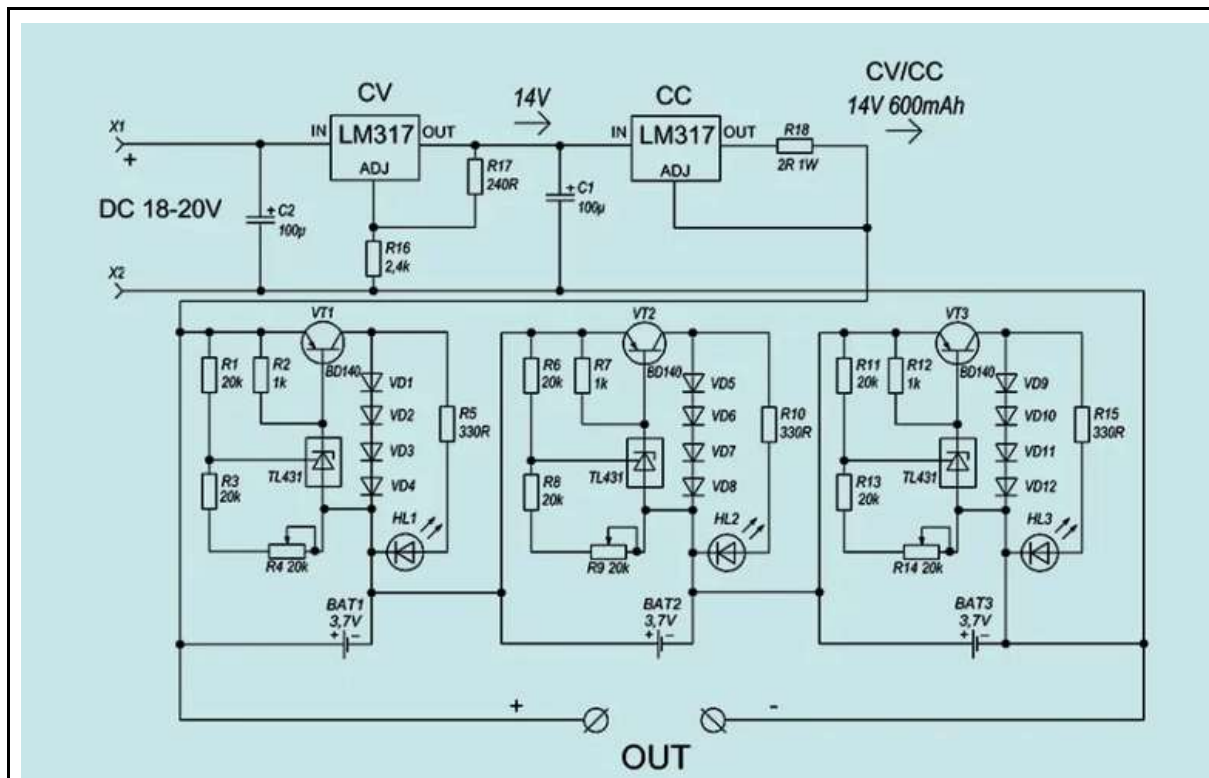
BMS circuit diagrams use standardized symbols and notations to represent various components, ensuring clear communication and understanding.

-Common Symbols: Symbols such as resistors, capacitors, and specific icons for BMU, voltage balancing, temperature sensors, and other components are universally recognized in BMS circuit diagrams.

-Standardized Representation: Following a standardized representation helps engineers and technicians accurately interpret and implement the circuit diagram



- **A BMS Circuit Diagram with Adjustable regulator**



The circuit diagram appears to be a **battery charging circuit** designed for charging multiple lithium-ion (Li-ion) cells, likely configured in parallel.

Here's a breakdown of the key sections and components:

Power Input Section consists of **DC 18-20V** input at terminals X1 and X2 and the input is filtered by two capacitors (**C2** and **C1**, both **100μF**) to smooth out the voltage supply.

Voltage Regulation Section consists of **LM317 voltage regulators** (**CV** and **CC**) are used for **constant voltage (CV)** and **constant current (CC)** regulation. Where the LM317 marked as **CV** is responsible for regulating the output to a **fixed voltage (14V)**. The resistors **R16 (2.4kΩ)** and **R17 (240Ω)** set the output voltage and the LM317 marked as **CC** works in constant current mode. Resistor **R18 (2Ω, 1W)** limits the current to 600mA. This ensures a constant current supply to the charging section.

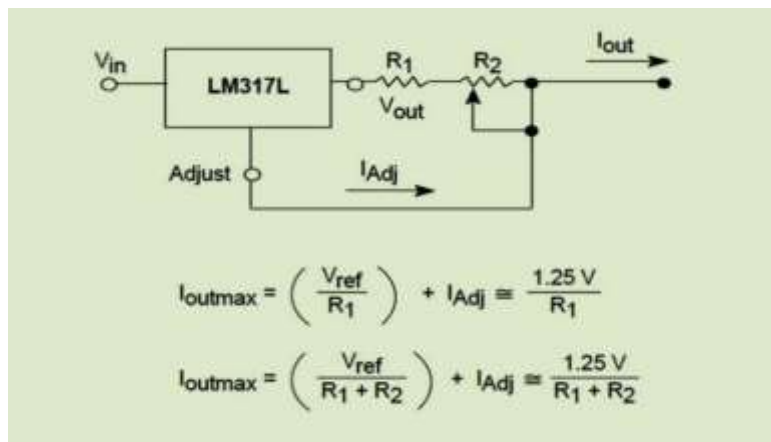
Battery Charging Section consists of there are **three parallel charging circuits**, each corresponding to a battery (**BAT1, BAT2, BAT3**, rated **3.7V**), **VT1, VT2, VT3** are **BD140 PNP** transistors controlling the charging process, **TL431** components are **adjustable voltage references** used to regulate the voltage across the batteries. They ensure the battery charging voltage does not exceed safe limits, **Resistors R1, R6, R11 (1kΩ each)** and **R2, R7, R12 (20kΩ each)** form voltage dividers to control the charging voltage, **VD1-VD12** are likely

diodes used for current flow regulation and protection and **HL1, HL2, HL3** seem to be **indicator LEDs** to show the charging status for each battery.

Operation:

- The input voltage (18-20V) is regulated by the LM317s to provide a stable 14V output with 600mA current limiting.
- Each battery (BAT1, BAT2, BAT3) is connected to its own charging circuit, controlled by BD140 transistors and regulated by TL431.
- The diodes and transistors ensure that the batteries are charged safely with the correct voltage and current.

This circuit is a **multi-cell lithium-ion battery charger** with **constant voltage (CV)** and **constant current (CC)** control, capable of charging three 3.7V cells in parallel. The use of LM317 voltage regulators provides stable voltage and current for safe charging, and the transistors ensure proper charge regulation for each battery individually.



The experiments demonstrated that the balancing circuit functions optimally. The charging process reaches completion upon attaining the designated voltage of 4.2 Volts.

3. Types of battery used in computer power system

Computer power systems utilize various types of batteries, each with its unique characteristics and applications.

The most common types include:

3.1. Lithium-Ion (Li-ion) Batteries: Li-ion batteries are the predominant choice for laptops and other portable electronics due to their high energy density, lightweight, and low self-discharge rates.

- **Subtypes:**

- **Lithium Nickel Manganese Cobalt Oxide (NMC):** Commonly used in consumer electronics for its balance of performance and safety.
- **Lithium Iron Phosphate (LiFePO4):** Known for safety and longevity, often used in power backup systems.
- **Lithium Cobalt Oxide (LCO):** Offers high energy density but has a shorter lifespan, suitable for devices requiring compact batteries

3.2. Nickel-Metal Hydride (Ni-MH) Batteries: Ni-MH batteries are a more environmentally friendly alternative to Ni-Cd batteries, providing higher capacity and less susceptibility to memory effects.

- **Applications:** Although less common in modern laptops, they are still used in some devices that require moderate energy density and are less hazardous than older technologies

3.3. Nickel-Cadmium (Ni-Cd) Batteries: Once the standard for portable devices, Ni-Cd batteries have largely fallen out of favor due to their weight, toxicity, and memory effect issues.

- **Current Use:** They are now primarily found in older devices or specific applications where their characteristics are still beneficial

3.4. Lithium Polymer (Li-Po) Batteries: Li-Po batteries use a polymer electrolyte instead of liquid, allowing for thinner and lighter designs. They are increasingly popular in high-end laptops and mobile devices.

- **Advantages:** They provide high energy density and can be shaped to fit various designs, making them ideal for compact electronics

3.5. Lead-Acid Batteries: While not commonly used in laptops, lead-acid batteries are prevalent in larger power systems like UPS (Uninterruptible Power Supplies) due to their robustness and cost-effectiveness.

- **Limitations:** They are heavier and have lower energy density compared to modern alternatives

Summary Table of Battery Types

Battery Type	Energy Density (Wh/kg)	Lifespan (Cycles)	Common Applications
Lithium-Ion	126-190	500-1,000	Laptops, smartphones
Nickel-Metal Hydride	100	300-500	Some laptops, hybrid vehicles
Nickel-Cadmium	45-80	1,000	Older laptops, specific tools
Lithium Polymer	185	300-500	High-end laptops, drones
Lead-Acid	30-50	200-300	UPS systems

4. Monitoring battery health and status

Monitoring the health and status of a battery is crucial for ensuring its reliability, performance, and longevity. This process involves tracking various parameters and metrics that indicate the condition of the battery and its ability to hold and deliver charge effectively.

4.1 Parameters for Monitoring

- **State of Charge (SoC):** Indicates the current charge level relative to the battery's capacity.

SoC represents the current charge level of the battery, expressed as a percentage of its total capacity.

Importance: Monitoring SoC helps determine how much energy is available for use and when the battery needs to be charged.

- **State of Health (SoH):** Reflects the overall condition of the battery, including its ability to hold charge compared to when it was new.

SoH indicates the overall condition of the battery compared to its original specifications, often expressed as a percentage.

Factors Affecting SoH: Capacity fade, internal resistance, and the number of charge-discharge cycles.

Importance: A declining SoH indicates that the battery's performance is deteriorating and may require replacement.

- **Temperature:** Monitoring temperature is crucial as it affects battery performance and lifespan.
- **Voltage and Current:** These parameters help assess charging efficiency and detect potential issues.

Importance: Overvoltage can damage the battery, while undervoltage can lead to a loss of capacity.

- **Cell Balancing:** Ensuring that all cells in the battery pack are at a similar voltage level to prevent overcharging or undercharging.
- **Overcharge and Overdischarge Protection:** Preventing the battery from being charged beyond its maximum capacity or discharged below its minimum voltage.
- **Temperature Monitoring and Management:** Monitoring the temperature of the battery pack and taking appropriate measures to prevent overheating or freezing.

Importance: High temperatures can lead to thermal runaway, while low temperatures can reduce efficiency.

- **Short Circuit Protection:** Detecting and preventing short circuits within the battery pack.

4.2 Monitoring Technologies

4.2.1. Battery Management Systems (BMS)

- **Functionality:** BMS monitors SoC and SoH, ensuring safe operation within specified limits. It prevents overcharging, deep discharging, and overheating.
- **Real-Time Data:** Advanced BMS can provide real-time data on battery parameters, allowing for immediate corrective actions if needed

4.2.2. Impedance Spectroscopy

- **Method:** This technique involves applying an AC current or voltage perturbation to measure the impedance response of the battery at various states of charge. It can diagnose internal degradation due to overcharge or over-discharge conditions.
- **Benefits:** Impedance data helps classify battery health zones, alerting users to potential failures before they occur.

4.2.3. Electrochemical Models

- **Adaptive Observers:** Researchers have developed electrochemical model-based systems that estimate lithium concentration in electrodes and aging parameters in real-time. This approach enhances understanding of non-measurable internal variables critical for battery management.
- **Applications:** This technology is particularly beneficial for electric vehicles and stationary storage systems, providing insights into battery performance over its entire lifecycle.

4.2.4. Dedicated Monitoring Systems

- **BQMS (Battery Health Monitoring System):** Designed for stationary applications, it monitors parameters such as string voltage, cell temperature, and connection resistance. It operates continuously and can provide alerts for maintenance needs.
- **Efftronics Battery Health Monitoring:** This system offers 24/7 monitoring of individual cell voltages, temperatures, and charging/discharging currents, helping identify weak cells before failure

5. Managing charge/discharge cycles

5.1 Charge/discharge cycle

A **charge/discharge cycle** is a complete sequence of charging and discharging a battery. It is a measure of how many times a battery can be recharged and used before its capacity and performance decline significantly.

A charge/discharge cycle refers to the process of charging a battery from a low state of charge to its maximum capacity and then discharging it back to a low state. This cycle is essential for assessing battery health and longevity, particularly for lithium-ion batteries, which can endure approximately 300-500 cycles before significant capacity loss occurs

5.2 The guide of performing a charge/discharge cycle:

Performing a charge/discharge cycle for a battery involves a series of systematic steps that ensure the battery is charged to its maximum capacity and then discharged appropriately.

Below are the detailed steps involved in this process:

a. Prepare the Battery

- **Check Compatibility:** Ensure that the battery is compatible with the charger and the device it will power.
- **Inspect Condition:** Visually inspect the battery for any signs of damage, swelling, or leakage.

b. Charging the Battery

- **Connect to Charger:** Attach the battery to an appropriate charger, ensuring correct polarity (positive to positive, negative to negative).
- **Set Charging Parameters:** If applicable, set parameters such as current and voltage on the charger according to manufacturer specifications.
- **Start Charging:** Initiate the charging process. Monitor the charging status through indicators (LEDs or a display).
- **Charge to Full Capacity:** Allow the battery to charge until it reaches its maximum voltage or capacity (typically indicated by a full charge light or display).

c. Monitor Charging

- **Temperature Monitoring:** Keep an eye on the battery temperature during charging. Excessive heat can indicate problems and may shorten battery life.
- **Voltage Checks:** Ensure that the voltage does not exceed safe limits (e.g., for lithium-ion batteries, typically no more than 4.2V per cell).

d. Discharging the Battery

- **Disconnect from Charger:** Once fully charged, disconnect the battery from the charger.
- **Connect to Load:** Attach the battery to a load (device or circuit) that will utilize its energy.
- **Set Discharge Parameters:** If available, set parameters for controlled discharge (current draw).
- **Gradual Discharge:** Allow the battery to discharge gradually until it reaches a predetermined cutoff voltage (e.g., 20% capacity for lithium-ion batteries).

e. Monitor Discharge

- **Voltage and Current Monitoring:** Continuously monitor voltage and current during discharge to ensure they remain within safe limits.

- **Avoid Deep Discharge:** Do not allow the battery to discharge below its recommended cutoff level, as this can cause damage.

f. Recharge After Discharge

- **Reconnect to Charger:** After reaching the desired discharge level, reconnect the battery to its charger.
- **Repeat Cycle:** Continue this process periodically based on usage patterns and manufacturer recommendations.

g. Analyze Performance

- **Data Collection:** Record data from each charge/discharge cycle, including voltage levels, charging times, and temperatures.
- **Evaluate Capacity Loss:** Over time, compare current performance metrics against baseline values to assess any degradation in capacity or performance.

6. Battery testing and characterization

6.1 Definition

Battery testing and characterization are essential processes for evaluating the performance, safety, and longevity of batteries. Various methods are employed to assess different parameters of battery health.

They involve applying different charge/discharge cycles to the batteries under various conditions and measuring their parameters, such as voltage, current, capacity, energy, power, efficiency, impedance, and temperature. These parameters can provide useful information about the battery's state of charge (SOC), state of health (SOH), and state of function (SOF).

Here's some of the key testing methods and their significance:

6.2 Key Battery Testing Methods

6.2.1. Voltage Measurement: A simple method to gauge the state of charge (SoC) by measuring the open circuit voltage when the battery is at rest.

Limitations: Voltage alone cannot accurately determine the state of health (SoH) of the battery, as it does not account for internal resistance or capacity degradation

6.2.2. Internal Resistance Testing: This involves measuring the internal resistance through techniques like pulse or AC impedance methods.

Importance: High internal resistance can indicate corrosion or mechanical defects, which may signal the end of a battery's life. This test is also known as the ohmic test

6.2.3. Coulomb Counting: This method tracks the flow of charge into and out of the battery to estimate its capacity.

Application: While providing instant readouts of Full Charge Capacity (FCC), it requires periodic calibration through full charge/discharge cycles to correct tracking errors.

6.2.4. Electrochemical Impedance Spectroscopy (EIS): EIS provides a detailed snapshot of the battery's chemical state by analyzing its impedance across a range of frequencies.

Benefits: This method can identify internal degradation and assess overall health without disassembling the battery

6.2.5. Load Bank Testing (Discharge Testing): This comprehensive test evaluates a battery's performance under specific load conditions, determining actual capacity and identifying weak cells.

Procedure: The test typically requires taking batteries offline, which can be a drawback due to downtime

6.2.6. Visual and Mechanical Inspections

Importance: Regular visual inspections help identify physical signs of wear or damage, ensuring that any potential issues are addressed before they lead to failure

6.2.7. Advanced Non-Invasive Techniques: Techniques such as X-ray analysis, MRI, and magnetic sensing are used to examine battery cells without disassembly.

Applications: These methods can provide insights into internal structures and detect issues like lithium plating or electrolyte depletion without altering the battery's state

7. Optimizing charging efficiency

7.1 Introduction

Optimized charging efficiency refers to the effective management of energy transfer during the battery charging process, aiming to maximize the amount of energy stored in the battery while minimizing losses.

This concept is crucial for enhancing battery performance, extending lifespan, and ensuring safety in various applications, particularly in lithium-ion batteries.

7.2 Key Aspects of Optimized Charging Efficiency

a. Definition: Charging efficiency is quantified as the ratio of energy output (energy stored in the battery) to energy input (energy consumed during charging). It is expressed as:

$$\text{Charging Efficiency} = \left(\frac{\text{Output Energy}}{\text{Input Energy}} \right) \times 100\%$$

b. Importance: High charging efficiency is critical for:

- Reducing energy losses during charging.
- Enhancing overall battery utilization.
- Extending the operational range of devices powered by batteries.

c. Methods to Improve Charging Efficiency:

- **Selecting Appropriate Chargers:** Using chargers that match the battery's specifications helps avoid energy losses and protects against damage.
- **Optimizing Charging Algorithms:** Implementing strategies like constant current (CC), constant voltage (CV), or hybrid CC-CV methods can improve both charging speed and efficiency.
- **Temperature Management:** Maintaining optimal temperature during charging prevents degradation and enhances efficiency.
- **Battery Management Systems (BMS):** Advanced BMS can monitor and control charging parameters in real-time, ensuring optimal performance and safety.

d. Charging Techniques:

- **Pulse Charging:** This method uses periodic current pulses, allowing for better electrolyte distribution and higher overall efficiency.
- **Multi-Stage Charging:** Gradually adjusting current and voltage throughout the charging process can optimize performance while minimizing risks of overheating or overcharging.

e. Real-World Applications: Optimized charging efficiency is vital in sectors like electric vehicles, renewable energy storage, and consumer electronics where battery performance directly impacts usability and sustainability.

7.3 Charging Methods

7.3.1. Constant Current (CC) Charging: Charges the battery at a fixed current until a predefined voltage is reached.

Advantages: Simple implementation and effective for many battery types, including NiCd, NiMH, and Li-ion.

This help to monitor the battery closely to prevent overcharging, which can damage the cells and reduce lifespan

7.3.2. Constant Voltage (CV) Charging: Maintains a constant voltage while allowing the current to decrease as the battery approaches full charge.

This help to reduce the risk of overheating and overcharging during the final stages of charging.

7.3.3. Constant Current/Constant Voltage (CC/CV) Charging: Combines CC and CV methods. Initially charges at a constant current until a set voltage is reached, then switches to constant voltage mode.

Advantages: Efficiently charges batteries while minimizing the risk of damage from overcurrent or overvoltage.

It is commonly used for lithium-ion batteries

7.3.4. Pulse Charging: Involves charging with periodic pulses rather than a continuous current.

Advantages: Allows for better distribution of charge within the battery, potentially increasing efficiency and reducing unwanted chemical reactions.

It helps maintain electrolyte uniformity and can improve overall charging rates

7.3.5. Burp Charging: A technique that applies short discharge pulses during rest periods in pulse charging to dislodge gas bubbles on electrodes.

It can enhance charge rates and improve battery longevity by preventing dendrite formation

7.4. The benefits of using optimized charging

Using optimized charging offers several significant benefits, particularly for lithium-ion batteries commonly found in smartphones, laptops, and electric vehicles.

7.4.1. Prolonged Battery Lifespan

- **Reduced Wear and Tear:** Optimized charging minimizes the time a battery spends at full charge, which can reduce stress on the cells and slow down capacity

degradation over time. By avoiding constant exposure to high voltage levels, the overall lifespan of the battery can be extended.

7.4.2. Mitigation of Memory Effect

- **Improved Performance:** For certain battery types, optimized charging helps reduce the memory effect, where batteries lose their maximum energy capacity if not fully discharged regularly. This is particularly beneficial for older battery technologies but also aids in maintaining performance in lithium-ion batteries

7.4.3. Enhanced Charging Efficiency

- **Smart Charging Patterns:** Optimized charging utilizes machine learning to adapt to user habits, allowing devices to charge quickly up to a certain percentage (e.g., 80%) and then slow down as it approaches full capacity. This approach not only saves energy but also enhances efficiency by preventing unnecessary energy loss during charging

7.4.4. Temperature Control

- **Safety and Efficiency:** By regulating the charging speed based on usage patterns and environmental conditions, optimized charging helps maintain an ideal temperature range during the charging process. This reduces the risk of overheating, which can lead to safety hazards and decrease battery efficiency.

7.4.5. Convenience for Users

- **Predictive Charging:** Devices equipped with optimized charging features can predict when they will be unplugged based on user routines. For example, if a phone is plugged in overnight, it will charge to 80% quickly and then hold off on reaching 100% until just before the user typically unplugs it. This ensures that users always have a full battery when needed without unnecessary wear during prolonged charging sessions.

7.4.6. Cost Savings

- **Reduced Energy Costs:** Higher charging efficiency translates to lower energy consumption over time. For commercial applications, this can lead to significant cost savings due to reduced electricity usage during charging cycles.



Practical Activity 4.1.4: Designing computer power management system plan



Task:

1: After learning basic of power management system, you are requested to read the given task below. The task should be done individually.

As electronician, you are asked to go in the Computer Lab and design a plan for a computer power management.

2. Read key reading 4.1.4 then perform the activity.
3. Present your work to the trainer and whole class
4. Ask questions if any for clarification
5. Perform the task provided in application of learning 4.1



Key readings 4.1.4.:

Designing computer power management system plan

Designing a computer power management system plan involves several critical steps to ensure efficiency, effectiveness, and user satisfaction.

Here's a structured approach to developing such a plan:

Step1. Define Objectives and Scope

- **Identify Goals:** Determine the primary objectives of the power management system, such as reducing energy consumption, minimizing costs, or enhancing user convenience.
- **Scope:** Define the scope of the project, including which devices (desktops, laptops, servers) will be managed and the extent of control (e.g., scheduling, remote management).

Step2. Conduct a Needs Assessment

- **User Requirements:** Gather input from stakeholders (IT staff, end-users) to understand their needs and expectations.

- **Current System Evaluation:** Analyze existing power management practices and identify areas for improvement.

Step3. Research and Select Technologies

- **Power Management Software:** Evaluate available software solutions that can automate power management tasks (e.g., ManageEngine, Microsoft Configuration Manager).
- **Hardware Integration:** Consider any necessary hardware components, such as smart plugs or energy monitoring devices, that can enhance the system's capabilities.

Step4. Develop a System Architecture

- **Design Framework:** Create a detailed architectural design that outlines how different components will interact (e.g., software interfaces with hardware).
- **Central Control Unit:** Plan for a central control unit that can manage power settings across all devices.

Step5. Implement Power Schemes

- **Create Power Policies:** Develop customized power schemes based on usage patterns (e.g., sleep mode after inactivity, hibernation settings).
- **Automation Features:** Incorporate automated features that adjust power settings based on time of day or user activity.
- **Step6. Testing and Validation**
- **Pilot Testing:** Run a pilot program with a small group of users to test the system's functionality and gather feedback.
- **Adjustments:** Make necessary adjustments based on pilot results to optimize performance and user experience.

Step7. Deployment

- **Rollout Plan:** Develop a phased rollout plan to implement the system across all targeted devices.
- **User Training:** Provide training sessions for users to familiarize them with new features and benefits.

Step8. Monitor and Optimize

- **Performance Monitoring:** Continuously monitor system performance and energy savings through analytics tools.
- **Feedback Loop:** Establish channels for ongoing user feedback to identify issues or areas for further improvement.

Step9. Maintenance and Support

- **Regular Updates:** Schedule regular updates for software and hardware components to ensure compatibility and security.
- **Technical Support:** Provide accessible technical support for users encountering issues with the power management system.

Step10. Evaluate Impact

- **Review Metrics:** After implementation, assess the impact of the power management system on energy consumption, cost savings, and user satisfaction.
- **Report Findings:** Document findings and share them with stakeholders to demonstrate the value of the system.



Points to Remember

- **Description of Assessment and requirements gathering**

Power requirement refers to the specific amount of computing power, in terms of either microprocessors and their speed or the peak watts from the computing power supply, needed for a particular project or task in the field of computer science.

The evaluation of power requirements in a computer power management system depends on several key factors such as hardware configuration, software efficiency, usage patterns, power management settings, environment factors and components interactions

- **Analysis of Power usage pattern.**

Power usage patterns in a computer power supply system refer to the variations in electricity consumption based on different operational states and tasks performed by the computer.

Analyzing the power usage pattern of a computer system involves understanding how power is consumed across different components and under various conditions. The pattern can vary based on workload, system configuration, and user behavior.

- **Identify Critical Components**

Identifying the critical components that contribute to energy consumption in computer power systems is essential for optimizing efficiency and reducing operational costs.

- **Document Findings and Requirements**

After evaluating the system's power needs and identifying critical components, document the key findings and establish the requirements for optimizing power consumption. The goal is to create a comprehensive report that outlines current power usage, areas for improvement, and potential solutions for more efficient power management.

- **Identification of computer power management techniques**

Power management: is how power is efficiently directed to different components of a system. Power management is especially important for portable devices that rely on battery power.

Sleep mode is a power saving state that stops all actions on the computer. Any open documents and applications are moved to the system memory (RAM) and the computer goes to a low-power state.

- **Processor power states**

Processor power states are mechanisms that allow CPUs to manage their energy consumption efficiently by adjusting operational parameters based on workload and activity levels.

These states are categorized into **P-states** (performance states) and **C-states** (idle states).

- **Software optimization**

Software optimization refers to the process of improving the efficiency and performance of software applications, particularly in terms of resource usage, speed, and energy consumption.

- **Identify Power Saving Strategies.**

Power saving strategies are essential for optimizing energy consumption in computer systems, enhancing efficiency, and reducing operational costs. These strategies can be categorized into hardware-based and software-based approaches.

Dynamic Voltage and Frequency Scaling (DVFS) is a power management technique that optimizes energy consumption in computing systems by dynamically adjusting the voltage and frequency of processors and other components based on workload requirements.

- **Smart power Management Policies.**

Smart power management policies are essential components of computer power management systems designed to optimize energy consumption while maintaining system performance.

- **Description of battery management system**

A **battery** is a device that stores electrical energy in chemical form and converts it into electrical energy when needed. It consists of one or more electrochemical cells that generate voltage through chemical reactions.

Batteries are characterized by their capacity (measured in ampere-hours or milliampere-hours), voltage, energy density, and cycle life, which indicates how many charge-discharge cycles they can undergo before significant capacity loss occurs.

- **Categories of battery**

Rechargeable vs. Non-Rechargeable: Batteries can be classified as primary (non-rechargeable) or secondary (rechargeable). Primary batteries, like alkaline batteries, can

only be used once, while secondary batteries, such as lithium-ion and lead-acid batteries, can be recharged and reused multiple times.

✓ **Types of battery**

Lithium-Ion (Li-ion) Batteries: Li-ion batteries are the predominant choice for laptops and other portable electronics due to their high energy density, lightweight, and low self-discharge rates.

Nickel-Metal Hydride (Ni-MH) Batteries: Ni-MH batteries are a more environmentally friendly alternative to Ni-Cd batteries, providing higher capacity and less susceptibility to memory effects.

Nickel-Cadmium (Ni-Cd) Batteries: Once the standard for portable devices, Ni-Cd batteries have largely fallen out of favor due to their weight, toxicity, and memory effect issues.

Lithium Polymer (Li-Po) Batteries: Li-Po batteries use a polymer electrolyte instead of liquid, allowing for thinner and lighter designs. They are increasingly popular in high-end laptops and mobile devices.

Lead-Acid Batteries: While not commonly used in laptops, lead-acid batteries are prevalent in larger power systems like UPS (Uninterruptible Power Supplies) due to their robustness and cost-effectiveness.

✓ **Battery Management System (BMS)**

A Battery Management System (BMS) is an electronic system that manages and monitors rechargeable battery packs or cells to ensure their safe and efficient operation.

✓ **Types of Battery Management Systems are**

Centralized BMS: A single controller oversees all cells in a battery pack.

Modular BMS: Multiple smaller units monitor individual modules within a battery pack.

Distributed BMS: Each cell has its own management unit connected through a communication bus to a central controller.

A **lithium battery bank** is a collection of lithium-ion batteries connected together to store electrical energy for backup power or renewable energy applications. Lithium battery banks are known for their high energy density, longer life cycles, and improved efficiency compared to traditional lead-acid batteries.

Redundant power supply refers to the inclusion of additional power supply units (PSUs) or backup systems to ensure uninterrupted power delivery in critical applications. The concept of redundancy is crucial in systems where power availability is essential for operation, such as data centers, hospitals, and industrial environments.

✓ **Designing computer power management system plan**

Here's a structured approach to developing such a plan:

- Step1.** Determine the primary objectives of the power management system,
- Step2.** Gather input from stakeholders to understand their needs and expectations.
- Step3.** Evaluate available software solutions that can automate power management tasks
- Step4.** Create a detailed architectural design that outlines how different components will interact
- Step5.** Develop customized power schemes based on usage patterns
- Step6.** Run a pilot program with a small group of users to test the system's functionality and gather feedback.
- Step7.** Develop a phased rollout plan to implement the system across all targeted devices.
- Step10.** After implementation, assess the impact of the power management system on energy consumption, cost savings, and user satisfaction.



Application of learning 4.1.

You are an electronics technician in a computer repair shop. The goal of the company is to repair the faulty computer power supplies and designing new power supplies for their clients.

As part of the company, you have been tasked to design an effective computer power management plan and policies for a new power supply unit to be designed.



Indicative content 4.2: Integration of Power Backup System Components



Duration: 6 hrs



Theoretical Activity 4.2.1: Description of power backup concepts



Tasks:

1: Answer the following questions related to description of computer power backup system.

- i. What do you understand by Power backup?
- ii. Identify the types of power backup system
- iii. Why is protection necessary in power backup system?
- iv. What are the power back up technologies?

2: Write the findings/answers on papers or flipchart.

3: Present your findings to the class.

4: Ask questions for more clarifications.

5: Read the key reading 4.2.1 from your manual to get more clarifications.



Key readings 4.2.1.:

Description of power backup

1. Definition

Power backup refers to the methods and systems used to provide a temporary or uninterrupted power supply during outages or fluctuations in the main power source. Power backup solutions are critical for ensuring that essential equipment, systems, and operations remain functional even when the grid power is unavailable or unstable. They can vary in capacity, duration, and technology, ranging from small portable devices to large-scale industrial solutions. Power backup systems help maintain operational continuity, protect sensitive electronics, and prevent data loss, making them vital for both residential and commercial applications.

2. Overview of Power Backup Systems

Power backup systems automatically switch to battery or generator power when the main power supply fails, preventing disruptions in service. They are crucial for homes, businesses, and facilities that require continuous power for safety, productivity, and data integrity.

3. Types of Power Backup Systems

3.1 Uninterruptible Power Supply (UPS): Provides immediate backup power to critical equipment during outages.

- **Use Cases:** Commonly used for computers, servers, and medical equipment to prevent data loss and damage.

3.2 Battery Backup Systems: Store electricity in batteries for use during outages.

They typically include a battery (often lithium-ion), charger, inverter (to convert DC to AC), and outlets for connecting devices.

- **Applications:** Suitable for homes and small businesses needing short-term power.

3.3 Generator Systems categorized as:

- **Standby Generators:** Automatically start during an outage and provide power through fuel sources like diesel or natural gas.
- **Portable Generators:** Manually operated units that can be moved to provide power where needed.

✧ **Use Cases:** Ideal for larger loads or prolonged outages.

3.4 Solar Backup Systems: Utilize solar panels to charge batteries, providing an eco-friendly backup option.

- **Advantages:** Low operating costs and sustainability; suitable for remote areas with limited grid access.

3.5 Components of Power Backup Systems

- **Energy source:** Store energy for later use; common types include lead-acid and lithium-ion or generators which provide backup power by converting fuel (diesel, gasoline, or natural gas) into electrical energy
- **Inverters (for battery source):** Convert stored DC energy from batteries into AC power for household appliances.

- **Chargers (for battery source):** Manage the charging of batteries from the main power supply or solar panels.
- **Transfer Switches:** Automatically switch between main power and backup power sources.
- **Cooling Systems:** Used in larger systems to dissipate heat generated during operation, ensuring that batteries and other components remain within safe temperature ranges.
- **Monitoring Systems,** this includes:

User Interface: Displays real-time information about the status of the power backup system, including battery charge level, load, and health indicators.

Remote Monitoring Systems: Allow users to track system performance and receive alerts about potential issues from a distance.

3.6 Benefits of Power Backup Systems

- **Continuous Power Supply:** Ensures that essential devices remain operational during outages, preventing disruptions in service.
- **Protection Against Surges:** UPS systems protect against voltage spikes that can damage sensitive electronics.
- **Cost Savings:** Reduces downtime costs for businesses by maintaining operations during outages.
- **Improved Safety:** Critical facilities (e.g., hospitals) can continue functioning without interruption, enhancing safety for patients and staff.
- **Scalability:** Many systems can be expanded by adding more batteries or integrating solar panels as needs grow.

4. Protection in Power Backup Systems.

Protection in power backup systems is essential for several reasons:

a) Safety: To prevent electrical hazards, including short circuits, fires, or explosions that can occur during power fluctuations or failures.

b) Equipment Integrity: Protecting sensitive electronics from voltage spikes, surges, and dips is crucial for preventing damage and ensuring reliable operation.

c) Battery Longevity: Effective management and protection of batteries (e.g., overcharging, over-discharging) can significantly extend their lifespan and performance.

d) Operational Continuity: Ensuring that backup systems operate correctly and reliably during an outage is vital for maintaining business operations and protecting critical processes.

e) Regulatory Compliance: Many industries are subject to regulations that require proper safeguards to protect equipment and personnel from power-related risks.

5. Power Backup Technologies

Power backup technologies refer to the various methods and systems employed to provide backup power and ensure continuity during outages.

Key technologies include:

a) Battery Technologies:

- **Lead-Acid Batteries:** Widely used for UPS systems and backup power due to their affordability and reliability.
- **Lithium-Ion Batteries:** Increasingly popular due to higher energy density, longer lifespan, and faster charging capabilities.
- **Flow Batteries:** Suitable for larger scale storage, offering long-duration backup and scalability.

b) Inverter Technology: Converts direct current (DC) from batteries into alternating current (AC) for use in most household and industrial applications.

c) Generator Technology: Combustion engines (diesel, gasoline, natural gas) that generate electricity for backup power.

d) Flywheel Systems: Utilize kinetic energy stored in a rotating mass to provide immediate backup power.

e) Supercapacitors: Provide rapid bursts of power for short periods, used in conjunction with batteries for enhanced performance.

f) Smart Grid Technology: Integrates renewable energy sources, energy storage, and advanced monitoring to enhance power reliability and efficiency.

g) Microgrid Systems: Localized grids that can operate independently or in conjunction with the main grid, providing backup power and integrating renewable sources.

h) Energy Management Systems: Software solutions that monitor and control energy use, optimizing the operation of backup systems based on demand and available resources.



Theoretical Activity 4.2.2: Identification backup power sources



Tasks:

1: Answer the following questions related to the identification of computer power backup sources.

- i. What do you understand by Power backup sources?
- ii. Identify the types of power backup sources.
- iii. What do you understand by lithium battery bank?
- iv. Discuss about the redundant power supply.

2: Write the findings/answers on papers or flipchart.

3: Present your findings to the class.

4: Ask questions for more clarifications.

5: Read the key reading 4.2.2 from your manual to get more clarifications.



Key readings 4.2.2.:

Identification of backup power sources

1. Introduction

Backup power sources are systems designed to provide uninterrupted electrical power in case of failure of the primary sources of electricity. They allow for continuous power supply at any time, regardless of whether the grid is functioning or not.

Identifying backup power sources is crucial for ensuring continuous operation during power outages.

2. Types of backup power sources

Here are the primary types of backup power sources along with their characteristics and applications:

2.1. Uninterruptible Power Supply (UPS)

A UPS provides immediate backup power using batteries when the main power source fails. It ensures a seamless transition without interrupting the power supply to critical devices.

A UPS is a device that provides immediate backup power to connected equipment when the main power source fails. It is designed to protect hardware such as computers and networking equipment from power interruptions and surges. UPS systems can vary in size and capacity, offering features like surge protection and battery backup. Uninterruptible Power Supply (UPS)

- **Types of UPS:**

- ✓ **Offline/Standby UPS:** Activates during a power failure, suitable for low-power devices.
- ✓ **Line Interactive UPS:** Provides protection against minor fluctuations and can handle moderate loads.
- ✓ **Online UPS:** Always on, providing clean and stable power, ideal for sensitive equipment like servers.

2.2. Lithium Battery Bank

Description: Use batteries to store electricity and automatically switch on during outages to provide power to critical systems.

- **Types of Batteries:**

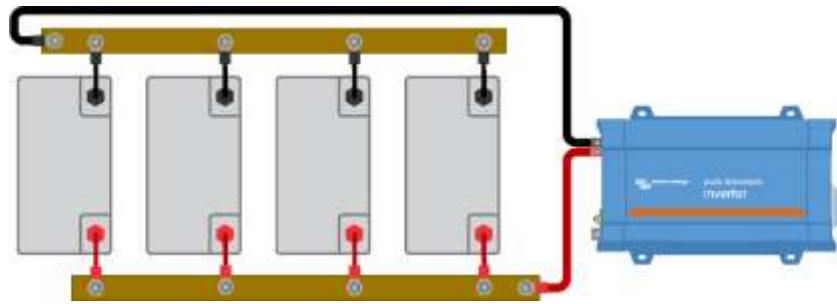
- ✓ **Lead-Acid Batteries:** Cost-effective but shorter lifespan.
- ✓ **Lithium-Ion Batteries:** Higher upfront cost but longer lifespan and better performance.
- ✓ **Applications:** Effective for homes, small businesses, and industries needing short-term backup.

A lithium battery bank consists of multiple lithium-ion batteries configured to store energy for longer durations. These systems are known for their high energy density, efficiency, and longevity compared to traditional lead-acid batteries. They are suitable for larger setups requiring extended backup times.

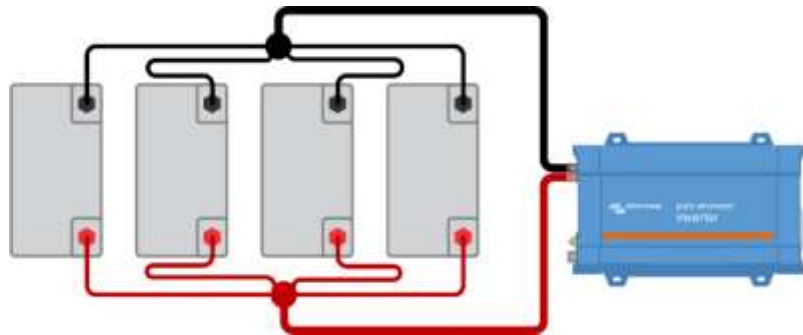
2.2.1. Parallel Battery Bank Connections: In a parallel circuit, components are connected across common points, creating multiple paths for current to flow.

There are four ways to correctly wire a parallel battery bank:

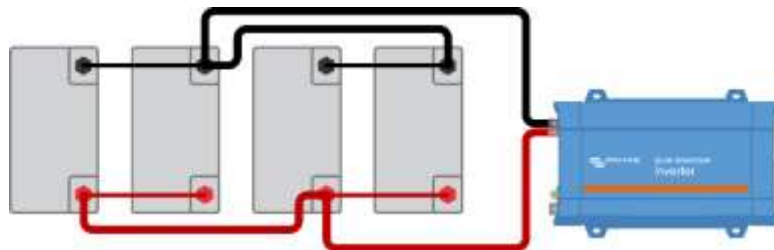
Use busbars.



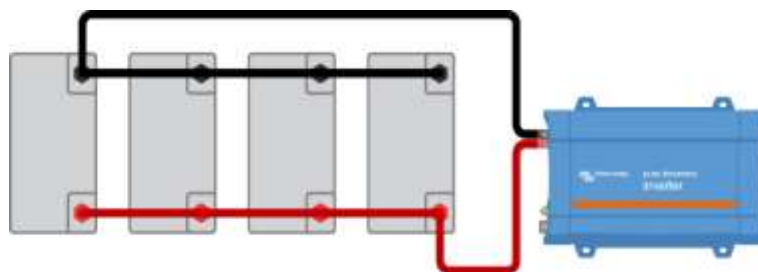
Connect using positive and negative positions, Ensure equal cables



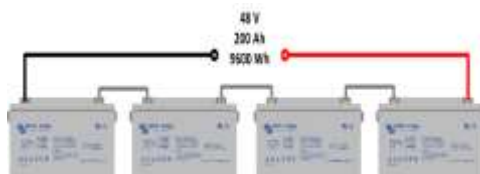
Connect halfway. Ensure all cables have the same thickness.



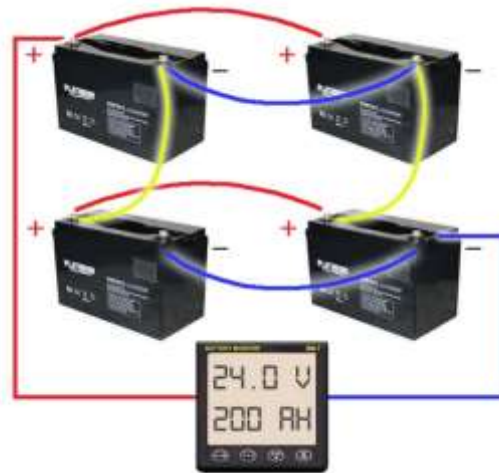
Connect diagonally. Note that while connecting the battery this way is simple and effective; it is not perfect. There may still be slight differences in the individual battery currents.



2.2.2. Battery Bank Connected in Series: A series circuit is a configuration where all components are connected end-to-end in a single path.



Combined Series/Parallel Battery Connections



2.2.3. The difference between battery bank and power bank

Power banks typically have capacities ranging from 2,000mAh to 20,000mAh, while battery packs can have capacities up to 50,000mAh. A larger capacity means the device can charge your smartphone multiple times without needing to be recharged. Power banks tend to be slower than battery packs in terms of charging speed.

2.3. Standby Generators: These are permanently installed units that automatically activate during outages. They can run on various fuels, including diesel, natural gas, or propane.

- **Applications:** Commonly used in homes, businesses, hospitals, and data centers where continuous power is critical.

2.4. Portable Generators: These generators are movable and can be used in various locations. They typically run on gasoline or propane.

- **Applications:** Ideal for outdoor events, construction sites, or as emergency backup at homes during outages.

2.5. Solar Power Backup Systems: These systems store solar energy in batteries for use during outages. They are environmentally friendly and have low maintenance costs.

- **Applications:** Suitable for homes and businesses in sunny regions or remote areas without reliable grid access.

2.6. Diesel Generators: These generators use diesel engines to convert mechanical energy into electrical energy. They are known for their reliability and efficiency.

- **Applications:** Commonly used in industrial settings, hospitals, and remote locations where consistent power is essential.

2.7. Gas Turbines: Operate on natural gas or biogas to generate electricity. They are often used in larger installations due to their scalability.

- **Applications:** Suitable for industrial applications requiring substantial backup power capacity.

3. Redundant Power Supply

3.1 Definition

Redundant power supply refers to the inclusion of additional power supply units (PSUs) or backup systems to ensure uninterrupted power delivery in critical applications. The concept of redundancy is crucial in systems where power availability is essential for operation, such as data centers, hospitals, and industrial environments.

3.2 Key Features of Redundant Power Supply:

3.2.1. Dual Power Supply Units: Systems may be equipped with two or more PSUs, which can operate simultaneously or take over automatically if one fails.

If one PSU fails, the other(s) continue to provide power without interruption.

3.2.2. Hot-Swap Capability: Many redundant power supplies are designed to be hot-swappable, allowing users to replace a failed unit without shutting down the entire system.

This feature minimizes downtime and maintains operational continuity.

3.2.3. Load Balancing: Redundant power supplies can distribute the load between multiple units, enhancing overall system efficiency and reliability.

If one supply is under heavy load, the other can share the burden, preventing overheating and wear.

3.2.4. Monitoring and Alerts: Advanced systems often include monitoring capabilities that track the status of each power supply unit.

Alerts can be generated to notify operators of failures or maintenance needs, enabling proactive management.

3.3 Applications:

a. Data Centers: To ensure servers remain operational even during power outages or equipment failures.

- b. Telecommunications:** To provide continuous service availability for critical communication systems.
- c. Healthcare:** In hospitals, where power outages can jeopardize patient care and safety.



Practical Activity 4.2.3: Calculating power backup size



Task:

1: Referring to the instruction from your trainer, you are requested to perform the given task below. The task should be done individually.

As technician, you are asked to go in the working environment to calculate the power backup size by following given computer system: Desktop Computer: 0.3 kW, Monitor: 0.2 kW, Router: 0.05 kW. If a backup system is rated at 1 kW and is expected to run the system for 4 hours:

- Calculate the power delivered by the backup system
- Provide a summation of power consumed by the system components
- Calculate power backup duration of the backup system

2: Present your work to the trainer and whole class

3: Ask questions if any for clarification

4: Read key reading 4.2.3 from your manual to get more clarifications and perform the task provided in application of learning 4.2



Key readings 4.2.3

Calculating power backup size

Calculating the size of a power backup system involves several steps to ensure that the system can provide sufficient energy to meet the needs during a power outage.

Here's a structured approach to determine the appropriate size of a power backup system:

Step1. Determine the Total Load

- List All Devices:** Identify all the devices and equipment that need to be powered during an outage (e.g., computers, servers, lights, HVAC systems).

- **Power Ratings:** Check the power ratings (in watts) of each device, typically found on the device's label or in the manufacturer's specifications.

To calculate the power delivered by each device in the backup system, use the formula:

Power Delivered = Voltage (V) × Current (I) in Watts (W)

- **Calculate Total Power:** Sum the power ratings of all devices to find the total load:

Total Load (W) = Σ Power of Each Device (W)

Step2. Estimate Usage Duration

- **Determine Required Backup Time:** Decide how long you need the backup power to last (in hours). This duration will influence the size of the battery bank or generator capacity required.

Step3. Calculate Energy Requirements

- **Calculate Total Energy Consumption:** Convert the total load into energy requirements based on the estimated usage duration:

Total Energy (Wh) = Total Load (W) × Usage Duration (h)

Step4. Select the Type of Backup System

- **Choose Backup System Type:** Decide whether you will use batteries, generators, or a combination of both for your power backup solution.
- **Consider System Efficiency:** Account for the efficiency of the chosen system (e.g., UPS systems may have an efficiency rating, typically around 90-95%).

5. Calculate Battery Size (if applicable)

If you are using batteries for backup, follow these additional steps:

- **Adjust for Efficiency:** Account for the efficiency of the battery system. Use a derating factor to calculate the effective capacity:

Adjusted Energy Requirement(Wh) = Total Energy (Wh)/Efficiency

- **Determine Battery Capacity:** Calculate the required battery capacity in amp-hours (Ah):

Battery Capacity(Ah) = Adjusted Energy Requirement(Wh)/ Battery Voltage(V)

- Choose the nominal voltage of the battery bank (commonly 12V, 24V, or 48V).

If using a generator:

- **Generator Sizing:** Ensure the generator can handle the total load, including surge

power needs (start-up power for devices such as refrigerators or air conditioners). A common rule of thumb is to size the generator for 1.2 to 1.5 times the total load to accommodate surge demands.

Step6. Consider Additional Factors

- **Future Expansion:** Consider potential future loads when sizing the backup system to ensure it can accommodate growth.
- **Environmental Factors:** Account for the environment where the backup system will be used, as temperature and humidity can affect battery performance and lifespan.
- **Regulations and Standards:** Check local codes and regulations that may affect sizing requirements or installation standards.

Step7. Finalize and Document Findings: Record all calculations, assumptions, and chosen specifications for future reference and maintenance.

Example:

Given the following given computer system components: Desktop Computer: 0.3 kW, Monitor: 0.2 kW, Router: 0.05 kW. If a backup system is rated at 1 kW and is expected to run the system for 4 hours:

- a) Calculate the power delivered by the backup system
- b) Provide a summation of power consumed by the system components
- c) Calculate power backup duration of the backup system

Solution:

a) Total power consumption by system

To determine the total power consumption of your computer system, sum up the power ratings of all components that will be running during the backup period. This includes:

- **Desktop or Laptop Computer:** Typically ranges from 0.1 kW to 0.5 kW depending on specifications.
- **Monitor:** Usually around 0.1 kW to 0.3 kW.
- **Networking Equipment:** Routers, switches, etc., can add another 0.05 kW to 0.1 kW.
- **Peripherals:** Printers or external drives may add additional power consumption.

Example Calculation

Let's say you have the following components in your computer system:

- **Desktop Computer:** 0.3 kW
- **Monitor:** 0.2 kW
- **Router:** 0.05 kW

b) Total Power Consumption Calculation

Total Power Consumption=Power of Desktop+Power of Monitor+Power of Router

Total Power Consumption=0.3 kW+0.2 kW+0.05 kW=0.55 kW

Total energy Delivered Calculation

Assuming you have a backup system rated at 1 kW and you want it to run for 4 hours:

Total Power Delivered=1 kW×4 hours=4 kWh
Total Power Delivered=1kW×4hours=4kWh

c) Backup Duration Calculation

To find out how long your backup can sustain your computer system:

Backup Duration (hour) = Total Power Delivered divided by the
Total Power Consumption in one hour = 4 kWh/ 0.55 kW ≈ 7.27 hours



Theoretical Activity 4.2.4: Description of battery charging system



Task:

1: Answer the following questions related to the description of computer power backup sources.

- What do you understand by charging method?
- Discuss about charging profile
- What is the charging controller?
- What do you understand by Managing temperature?
- What is the Monitoring and control of batteries?

2: Write the findings/answers on papers or flipchart.

3: Present your findings to the class.

4: Ask questions for more clarifications.

5: Read the key reading 4.2.4. from your manual to get more clarifications.



Key readings 4.2.4.

Description of battery charging system

1. Apply charging method

Developing a proper battery charging method is an essential part of the BMS. The method is based on accurate battery estimations for state of charge (SOC), state of health (SOH) and temperature.

There are four commonly used and popular charging methods:

- Constant current (CC) charging
- Constant-voltage (CV) charging
- Constant-current-constant-voltage (CC-CV) charging
- Multi-stage constant-current (MCC) charging

CC charging is a simple method that uses a small constant current to charge the battery during the whole charging process. CC charging stops when a predefined value is reached. This method is widely used for charging NiCd or NiMH batteries, and Li-ion batteries.

CV charging, which regulates a predefined constant voltage to charge batteries. Its main advantage is that it circumvents over voltages and irreversible side reactions, thus prolonging battery life. Since the voltage is constant, the charging current decreases as the battery charges.

The **CC-CV charging** method uses CC charging in the first charging stage, and when the voltage reaches the maximum safe threshold value, the charging process shifts to the CV charging method. The charging process is complete when the current levels off or when full battery capacity is reached. The charging time is mainly defined by the constant current value (CC mode), while the capacity utilization is predominantly influenced by the constant voltage value (CV mode).

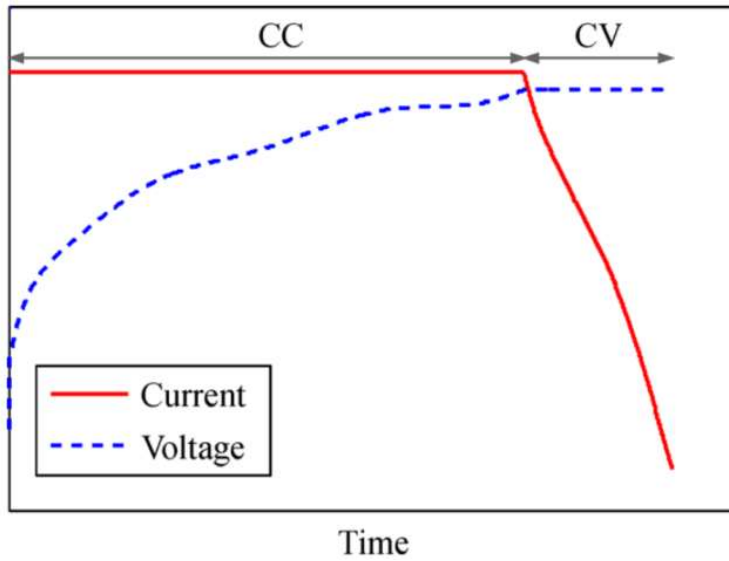


Figure: A graph of the battery current and voltage in CC-CV charging mode.

The **MCC charging** method includes several constant current stages, where the current is gradually decreased as the terminal voltage reaches a default voltage threshold. The charging process continues until the battery reaches the terminal conditions. The MCC method is shown in Figure below.

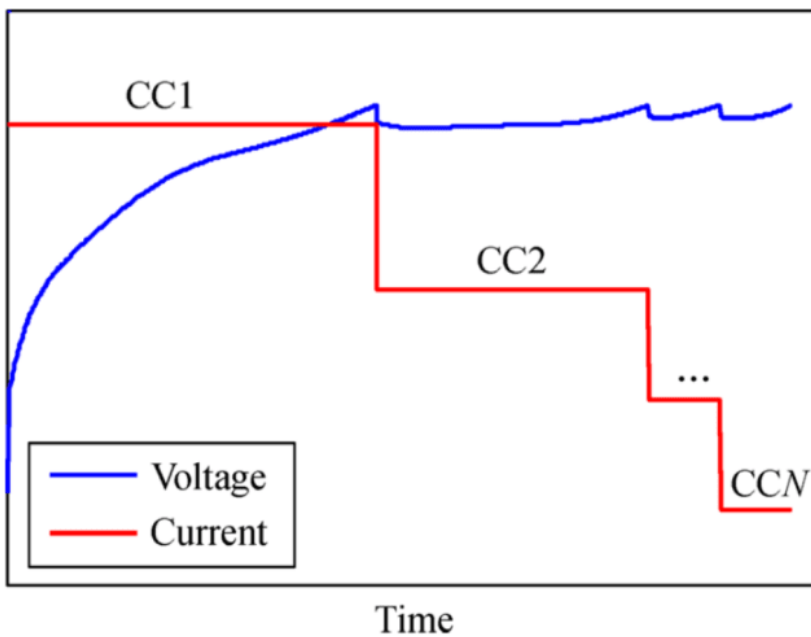


Figure 2: A graph of the battery current and voltage in MCC charging mode.

The MCC method is suitable for charging the following battery types: lead-acid, NiMH, and Li-ion batteries.

Comparison of Charging Methods

Table below summarizes the features of the four traditional charging methods.

Method	Advantages	Disadvantages	Key Parameters
Constant Current (CC)	- Easy to implement	- Capacity utilization is low	- Charging current rate
Constant Voltage (CV)	- Easy to implement	- Causing the lattice collapse of battery	- Charging voltage rate
Constant Current Constant Voltage (CC-CV)	- Capacity utilization is high - Stable terminal voltage	- Challenge to balance charging speed, energy loss, temperature variation	- Charging current rate in CC mode - Charging voltage rate in CV mode
Multi-Stage Constant Current (MCC)	- Easy to implement - Fast charging	- Challenge to balance charging speed, capacity utilization and battery lifetime	- The number of CC stages - Charging current rates for each stage

Table 2. Comparison of the traditional charging methods.

2. Design charging controller

The **charge controller** in a system is to protect the battery from overcharge and over discharge. Any system that has unpredictable loads, user intervention, optimized or undersized battery storage (to minimize initial cost), or any characteristics that would allow excessive battery overcharging or over discharging requires a charge controller and/or low-voltage load disconnect.

3. Testing efficiency and power factor

Testing the efficiency and power factor of battery charger controllers is crucial for ensuring optimal performance and compliance with industry standards. This process involves a series of systematic evaluations that assess how effectively a charger converts input power into usable output while minimizing losses.

3. 1. Efficiency Testing

Efficiency in battery chargers is defined as the ratio of output power to input power, typically expressed as a percentage. The formula used to calculate efficiency (η) is:

$$\eta = \left(\frac{P_{out}}{P_{in}} \right) \times 100$$

Where:

- P_{out} is the output power (in watts).
- P_{in} is the input power (in watts).

To conduct efficiency testing:

- **Setup:** Connect the battery charger to a controlled load, which can be resistive or a combination of batteries and resistors.
- **Measurements:** Measure the input voltage and current using a calibrated power

meter to determine P_{in} . Simultaneously, measure the output voltage and current to calculate P_{out} .

- **Calculations:** Use the above formula to compute efficiency and compare it against manufacturer specifications to ensure it meets acceptable ranges

3. 2. Power Factor Testing

Power factor (PF) is another critical parameter that indicates how effectively electrical power is being converted into useful work output. It is defined as the ratio of real power (P) in kilowatts (kW) to apparent power (S) in kilovolt-amperes (kVA):

$$PF = \frac{P}{S}$$

Where:

- Real Power (P) is the actual power consumed by the circuit.
- Apparent Power (S) is the product of the current and voltage in the circuit.

Procedure for Power Factor Testing

- **Measurement Tools:** Utilize a power quality analyzer capable of measuring both real and apparent power.
- **Testing Conditions:** Conduct tests under various load conditions to assess how PF varies with different charging scenarios.
- **Analysis:** A PF below 0.95 may indicate inefficiencies, potentially leading to increased operational costs due to higher current requirements and penalties from utility providers.

4. Combined Testing Approach

For comprehensive evaluation:

- Perform both efficiency and power factor tests simultaneously under identical conditions.
- Ensure that testing adheres to established protocols, such as those outlined by energy commissions or industry standards, which may specify temperature limits, voltage ranges, and acceptable measurement uncertainties

5. Factors affecting the efficiency of a battery charger controller

The efficiency of a battery charger controller is influenced by several key factors that can significantly affect its performance and longevity. Here are the primary considerations:

5.1. Design and Technology

- **Charger Type:** Different types of chargers, such as **Switch Mode Power Supplies (SMPS)** and **Pulse Width Modulation (PWM)** controllers, exhibit varying efficiencies. SMPS typically offers higher efficiency due to reduced heat

generation, while PWM provides precise control over charging rates but may generate electromagnetic interference.

- **Control Algorithms:** The effectiveness of the charging process is largely determined by the controller's algorithm. Advanced algorithms can optimize energy transfer from the power source to the battery, enhancing overall efficiency.

5.2. Compatibility with Battery Types

- **Battery Chemistry:** The type of battery being charged (e.g., lead-acid, lithium-ion) affects efficiency. For instance, lithium-ion batteries generally perform better in terms of efficiency compared to lead-acid batteries under similar conditions.
- **Voltage Matching:** Ensuring that the charger's voltage matches that of the battery bank is critical. Mismatched voltages can lead to inefficient charging and potential damage.

5.3. Environmental Conditions

- **Temperature Effects:** Temperature plays a significant role in charging efficiency. Higher temperatures can decrease battery voltage and charging rates, particularly for lead-acid batteries, while lower temperatures may reduce overall system efficiency. Some controllers include temperature compensation features to adjust charging parameters based on ambient conditions

5.4. Installation Environment: Factors such as humidity, dust, and exposure to elements can affect the performance and reliability of charge controllers. Proper installation and maintenance are essential for optimal operation

5.5. System Sizing and Configuration

- **Proper Sizing:** The charge controller must be appropriately sized for the system's voltage and current requirements. An undersized or oversized controller can lead to inefficiencies, overcharging, or undercharging.
- **Load Management:** Effective load management features in a charger can prevent deep discharges and maintain battery health, which indirectly contributes to overall efficiency.

5.6. Power Quality Factors

- **Power Factor:** The power factor reflects how effectively the charger uses electricity. A low power factor indicates inefficiencies in energy use, which can lead to increased operational costs and reduced charger performance.

- **Harmonic Distortion:** Nonlinear loads can introduce harmonic distortion into the system, negatively impacting both efficiency and power quality. Chargers designed to minimize these effects will perform better overall

6. Managing temperature

Temperature is a dominant factor affecting battery charging performance. High temperature decreases the life cycle of Li-ion batteries, and charging is not recommended in below freezing conditions. Table below shows the influence of temperature on charging performance for different battery types.

Battery types	Charging performance
Li-Ion	• High temperature speeds up charging but adversely affects the battery lifespan • Charging below freezing is dangerous
Lead-acid	• Higher temperature causes lower V threshold by 3 mV/°C • Charging rate less than 0.3 C below freezing
NiMH, NiCd	• High temperature decreases charging acceptance (70% at 45°C and 45% at 60°C) • Charging rate 0.1 C below freezing • Charging rate 0.3 C between 0 °C and 6 °C

Managing temperature in battery charger controllers is essential to ensure optimal performance, safety, and longevity of the batteries being charged. Effective thermal management systems (TMS) are designed to maintain battery temperatures within a specified range, typically between 20°C and 45°C, as extreme temperatures can adversely affect battery efficiency and lifespan.

6.1 Key Components of Temperature Management

a. Battery Thermal Management System (BTMS)

A BTMS is integral to regulating the temperature of battery packs. It employs various methods to either cool or heat the batteries as needed:

- **Cooling Systems:** These systems use liquid or air cooling methods to dissipate heat generated during charging and discharging. Liquid cooling systems are particularly effective as they can absorb heat more efficiently than air.
- **Heating Systems:** In colder conditions, heating elements may be employed to raise the temperature of the battery pack, facilitating better performance during charging.

b. Sensors and Controllers

Precision temperature sensors play a crucial role in monitoring battery temperatures. These sensors provide real-time data to a thermal management controller, which adjusts

cooling or heating mechanisms accordingly. This dynamic response helps maintain optimal operating conditions for the battery.

7. Monitoring and control of batteries

Monitoring and controlling batteries is essential for ensuring their safety, efficiency, and longevity. Battery Management Systems (BMS) play a crucial role in this process by providing real-time data on battery performance, health, and environmental conditions. Here's an overview of the key aspects involved in battery monitoring and control.

7.1 Key Functions of Battery Management Systems (BMS)

7.1.1. Monitoring

A BMS continuously monitors various parameters to ensure optimal battery operation:

- **Voltage:** Measures total voltage and individual cell voltages to detect imbalances and prevent overcharging or deep discharging.
- **Temperature:** Monitors the temperature of the battery cells to prevent overheating, which can lead to thermal runaway.
- **Current:** Tracks the current flowing in and out of the battery to manage charging and discharging cycles effectively.
- **State of Charge (SoC):** Estimates the remaining charge in the battery, which is critical for determining when to recharge.
- **State of Health (SoH):** Assesses the overall condition of the battery, indicating its ability to hold charge compared to its original capacity.

7.1.2. Control

The BMS not only monitors but also controls various functions:

- **Balancing:** Ensures that all cells within a battery pack are charged evenly, which helps prolong battery life.
- **Protection:** Engages protective measures against overvoltage, undervoltage, overcurrent, and excessive temperatures.
- **Charging Control:** Adjusts charging parameters based on real-time data to optimize charging efficiency and safety.

7.1.3. Communication

Modern BMS solutions often include communication capabilities that allow for:

- **Data Reporting:** Sending performance data to external systems for analysis and decision-making.
- **Remote Monitoring:** Enabling users to monitor battery status from remote locations through IoT connectivity.

7.2 Types of Battery Monitoring Systems

7.2.1. Wired vs. Wireless Systems

- **Wired Systems:** Provide reliable connections in environments with high electromagnetic interference. They are often used in stationary applications like data centers or utility installations.
- **Wireless Systems:** Offer flexibility in installation and are suitable for applications where wiring is impractical. They typically use RF signals for communication.

7.2.2. Specialized Monitoring Solutions

Different systems cater to specific needs:

- **Electrolyte Level Monitoring:** For lead-acid batteries, monitoring electrolyte levels is crucial for performance.
- **Temperature Monitoring Solutions:** Some systems focus specifically on temperature management, essential for lithium-ion batteries used in electric vehicles.

7.3 Benefits of Effective Battery Monitoring and Control

Implementing a robust BMS provides numerous advantages:

- **Increased Reliability:** Continuous monitoring helps identify potential issues before they lead to failures, improving overall system reliability.
- **Cost Savings:** Proactive maintenance enabled by real-time data can reduce operational costs associated with unexpected downtime or battery replacements.
- **Enhanced Safety:** By preventing conditions that could lead to thermal runaway or other hazardous situations, a BMS significantly enhances safety.



Practical Activity 4.2.5: Designing battery charging controller



Task:

1: Referring to the instruction from your trainer, you are requested to perform the given task below. The task should be done individually.

As technician, you are asked to go in Computer Lab design and simulate a battery charging controller to manage the charging process of a 12V rechargeable battery. The controller must ensure safe and efficient charging, including overcharge protection, current regulation, and automatic cutoff once the battery is fully charged.

2: Present your work to whole class.

3: Read key reading 4.2.5 and perform the task provided in application of learning 4.2



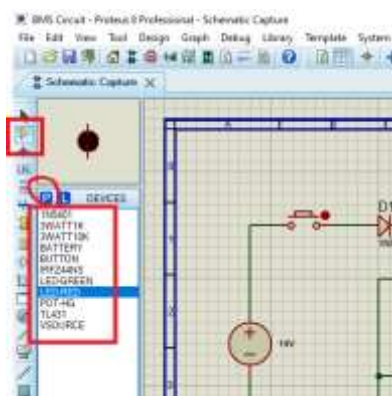
Key readings 4.2.5.

Designing battery charging controller

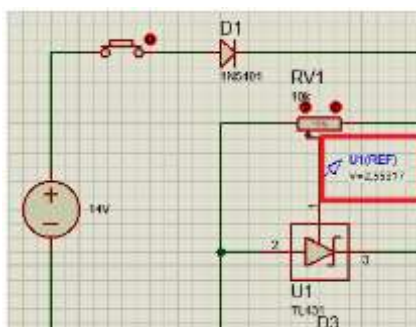
To design and simulate a Battery charger controller using Proteus, follow these steps:

Step1. Start a New Project: Open Proteus and create a new project. Set up your workspace for schematic design.

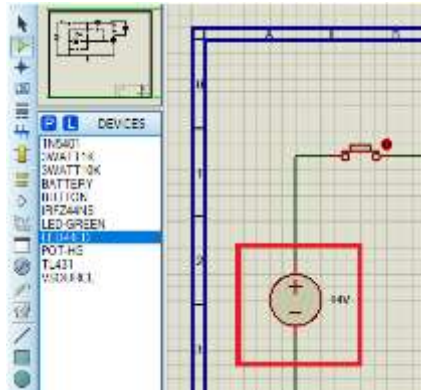
Step2. Add Components: Begin by adding the necessary components to your schematic. This includes the TL431 voltage reference, resistors, transistors (such as BD140), diodes, and any other components required for your specific BMS design.



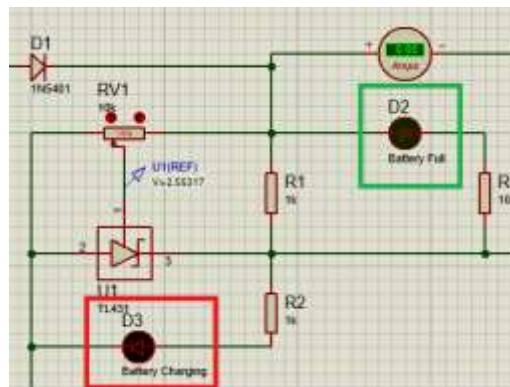
Step3. Configure the TL431: Connect the TL431 according to its typical application circuit. Ensure that you set up the feedback network with resistors to establish the desired reference voltage (usually around 2.5V or adjustable to 4.2V for lithium-ion batteries).



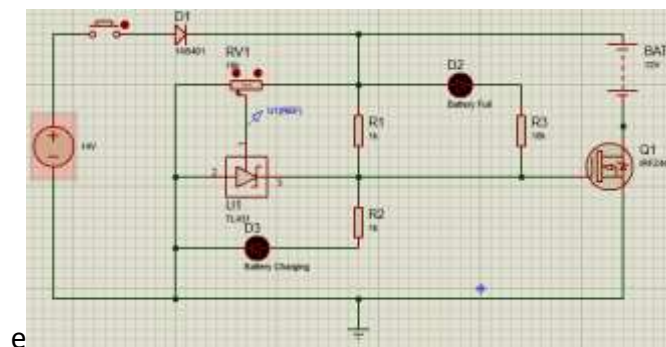
Step3. Connect the Power Supply: Add a DC power source to simulate the battery or power supply input for your circuit. Ensure that the voltage levels are appropriate for your design.



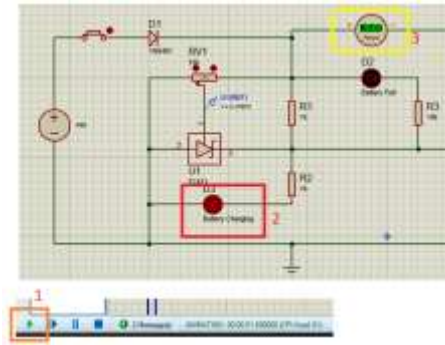
Step4. Set Up Load Balancing: If your BMS includes load balancing features, configure additional components like shunt resistors and indicators (LEDs) to monitor the charging status of each cell.



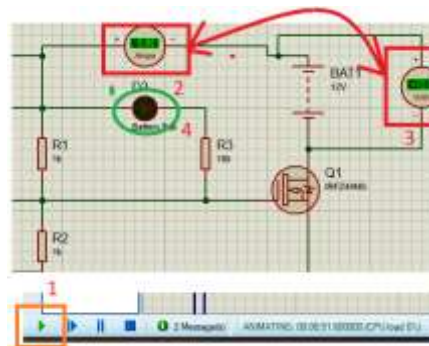
Step5. Wire the Circuit: Carefully wire all components together, ensuring that connections are correct according to your schematic design.



Step6. Simulate the Circuit: Once everything is connected, run the simulation in Proteus. Monitor the behavior of the circuit, particularly how the TL431 regulates voltage and how transistors control charging and discharging processes.

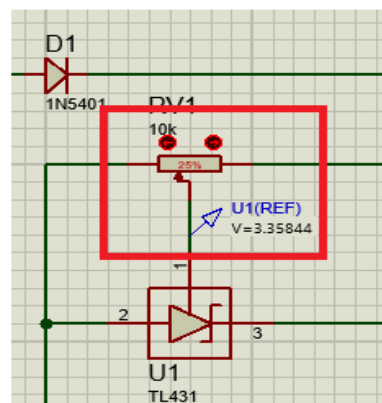


Insert the DC Ammeter (3) in the circuit to indicate if the battery is charging, Press the Simulation Button (1) then the Charging LED is glowing whereas Ammeter is reading, indicating the charging process.



Once the battery is full charged, the Ammeter (2) stop reading and Green LED (4) glows.

Step7. Adjust Parameters: If necessary, adjust component values (like resistor values for voltage division) to achieve optimal performance based on simulation results.



Step8. Finalize Design: After successful simulation and adjustments, finalize your design by saving the project (by clicking on File>Save once the project was saved of Save as when it is the first time to save it) and preparing any documentation needed for further development or physical implementation.



Points to Remember

- **Description of power backup concepts**

Power backup refers to the methods and systems used to provide a temporary or uninterrupted power supply during outages or fluctuations in the main power source. Power backup systems automatically switch to battery or generator power when the main power supply fails, preventing disruptions in service.

They are crucial for homes, businesses, and facilities that require continuous power for safety, productivity, and data integrity. Power backup systems automatically switch to battery or generator power when the main power supply fails, preventing disruptions in service. They are crucial for homes, businesses, and facilities that require continuous power for safety, productivity, and data integrity.

- **Types of Power Backup Systems** includes Uninterruptible Power Supply (UPS), Battery Backup Systems, Generator Systems and Solar Backup Systems

Protection in Power Backup Systems.

Protection in power backup systems is essential for several reasons: Safety, Equipment Integrity, Battery Longevity, Operational Continuity, Regulatory Compliance and Power Backup Technologies

Power backup technologies refer to the various methods and systems employed to provide backup power and ensure continuity during outages.

Key technologies include: Battery Technologies, Inverter Technology, Generator Technology, Flywheel Systems, Supercapacitors, Smart Grid Technology, Microgrid Systems and Energy Management Systems

- **Identification of backup power sources**

- ✓ **Definition**

Backup power sources are systems designed to provide uninterrupted electrical power in case of failure of the primary sources of electricity.

- ✓ **Types of backup power sources**

Here are the primary types of backup power sources along with their characteristics and applications:

- ✧ **Uninterruptible Power Supply (UPS):** A UPS provides immediate backup power using batteries when the main power source fails.
- ✧ **Lithium Battery Bank:** Use batteries to store electricity and automatically switch on during outages to provide power to critical systems.
- ✧ **Standby Generators:** These are permanently installed units that automatically activate during outages.

- ✧ **Portable Generators:** These generators are movable and can be used in various locations.
- ✧ **Solar Power Backup Systems:** These systems store solar energy in batteries for use during outages.
- ✧ **Diesel Generators:** These generators use diesel engines to convert mechanical energy into electrical energy.
- ✧ **Gas Turbines:** Operate on natural gas or biogas to generate electricity.
- ✧ **Redundant Power Supply** refers to the inclusion of additional power supply units (PSUs) or backup systems to ensure uninterrupted power delivery in critical applications.

- **Calculating power backup size**

- ✓ **Definition**

Calculating the size of a power backup system involves several steps to ensure that the system can provide sufficient energy to meet the needs during a power outage.

- ✓ Here's a structured approach to determine the appropriate size of a power backup system:

Start by identifying all the devices and equipment that need to be powered during an outage, then decide how long you need the backup power to last (in hours). Convert the total load into energy requirements based on the estimated usage duration also Decide whether you will use batteries, generators, or a combination of both for your power backup solution. Consider potential future loads when sizing the backup system to ensure it can accommodate growth. And finally, record all calculations, assumptions, and chosen specifications for future reference and maintenance.

- **Description of battery charging system**

- ✧ **Apply charging method:** Developing a proper battery charging method is an essential part of the BMS. The method is based on accurate battery estimations for state of charge (SOC), state of health (SOH) and temperature.
- ✧ **There are four commonly used and popular charging methods:**
- ✧ **CC charging** is a simple method that uses a small constant current to charge the battery during the whole charging process.
- ✧ **CV charging**, which regulates a predefined constant voltage to charge batteries.
- ✧ The **CC-CV charging** method uses CC charging in the first charging stage, and when the voltage reaches the maximum safe threshold value, the charging process shifts to the CV charging method.
- ✧ **Charging controller:** The **charge controller** in a system is to protect the battery from overcharge and over discharge.

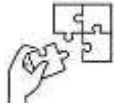
- **Testing efficiency and power factor**

- ✧ Testing the efficiency and power factor of battery charger controllers is crucial for ensuring optimal performance and compliance with industry standards.
- ✧ **Factors affecting the efficiency of a battery charger controller**
- ✧ The efficiency of a battery charger controller is influenced by several key factors that can significantly affect its performance and longevity, as:
 - ✧ Design and Technology, Compatibility with Battery Types, Environmental Conditions, System Sizing and Configuration, Power Quality Factors, Monitoring and control of batteries
- ✧ **Types of Battery Monitoring Systems are:** Wired system, Wireless System and Specialized Monitoring Solutions.
- ✧ **Benefits of Effective Battery Monitoring and Control**
- ✧ Implementing a robust BMS provides numerous advantages as: Increased Reliability, Cost Savings and Enhanced Safety,

- **Designing battery charging controller**

To design a battery charging controller using Proteus, start by opening the software and creating a new project to set up your schematic workspace. Next, add essential components such as the TL431 voltage reference, resistors, transistors, and diodes, tailoring your selection to the specific requirements of your Battery Management System (BMS) design. Configure the TL431 according to its typical application circuit, ensuring that the feedback network with resistors establishes the desired reference voltage, typically around 2.5V or adjustable to 4.2V for lithium-ion batteries.

Incorporate a DC power source to simulate the battery or power supply input, confirming that the voltage levels align with your design specifications. If your BMS includes load balancing features, set up additional components like shunt resistors and indicators (LEDs) to monitor each cell's charging status. Carefully wire all components together according to your schematic, then run the simulation in Proteus to observe circuit behavior, focusing on how the TL431 regulates voltage and how transistors manage charging and discharging processes. Adjust component values as needed based on simulation results to optimize performance, and finally, save your project by clicking on File > Save or Save As for future reference and documentation preparation.



Application of learning 4.2.

The company that code, test and deploy applications needs a reliable power backup solution that ensures uninterrupted operation of all computer systems during power outages. The existing setup lacks sufficient backup capabilities, which can lead to significant delays in project timelines and affect client satisfaction.

You are part time hired by the company to design and install the backup system to facilitate the company continuing its activities even if the main supply fails for a period of 1 hour.



Indicative content 4.3: Integration of Power Backup System Components



Duration: 8 hrs



Theoretical Activity 4.3.1: Description of component placement compatibility and consideration assessment



Tasks:

1: Answer the following questions related to the description of component placement compatibility and consideration assessment.

- i. What do you understand by the following terms?
 - a. Voltage compatibility
 - b. communication protocols
 - c. physical dimensions
 - d. power rating and capacities
 - e. Environmental compatibility
 - f. Space utilization
 - g. cable management
 - h. heat dissipation
 - i. Weight distribution
 - j. Future expansion and scalability

2: Write the findings/answers on papers or flipchart.

3: Present your findings to the class.

4: Ask questions for more clarifications.

5: Read the key reading 4.3.1 from your manual to get more clarifications.



Key readings 4.3.1.:

Description of component placement compatibility and consideration assessment

Introduction

In the context of power backup systems, particularly uninterruptible power supplies (UPS) and battery management systems (BMS), several critical terms are essential for effective design and implementation.

Below is an explanation of each term:

a. Voltage Compatibility: Voltage compatibility refers to the ability of a power backup system to operate within a specified voltage range. This ensures that the UPS or BMS

can accept input from various power sources and deliver appropriate output to connected devices without causing damage or operational issues. For example, a UPS may support input voltages ranging from 120V to 240V AC, accommodating different regional standards.

Input and Output Voltage Matching

Battery Systems: Common battery voltages include 12V, 24V, 36V, and 48V. For instance, a 12V battery system is often used in smaller applications, while 48V systems are preferred for larger installations due to improved efficiency in power transmission.

Power Ratings and Current Draw

Power Requirements: The total power requirements of connected loads must be calculated. For example, if a device requires 1000W at 120V, the current draw can be calculated using the formula: $I=P/V$

Thus, $I=1000W/120V=8.33A$

Adjusting for Voltage Fluctuations: If the input voltage drops to 100V during a brownout, the current draw increases: $I=1000W/100V=10A$

This increase in current necessitates ensuring that the wiring and components can handle higher currents without overheating or causing failures.

Battery Charging Voltages

Batteries typically require a higher voltage for charging than their nominal voltage. For example, a 12V lead-acid battery usually requires a charging voltage of about 13.8V to 14.4V to ensure proper charging. This difference is crucial for maintaining battery health and performance.

b. Communication Protocols: Communication protocols are standardized methods that allow different devices within a power backup system to exchange information. Common protocols include Modbus, CAN bus, and Ethernet-based protocols like EtherCAT. These protocols facilitate monitoring and control functions, enabling the UPS to communicate status updates, battery health data, and operational commands to external systems or devices.

Compatibility Assurance: Ensure that both systems can communicate effectively, allowing for monitoring and management of power conditions during outages.

RS-485 Cables: Often used for communication between the power backup system and monitoring devices or control systems.

Ethernet Cables (Cat5e/Cat6): Used for network connectivity in smart backup systems that require remote monitoring and control.

c. Physical Dimensions: Physical dimensions refer to the size and shape of the power backup system, which can impact installation and integration into existing infrastructure. Understanding these dimensions is crucial for ensuring that the device fits within designated spaces and complies with safety regulations regarding airflow and accessibility.

Space Measurement: Measure the physical dimensions of the power backup system and compare them with the available installation space. This ensures that the system fits well without obstructing other equipment or creating safety hazards.

Installation Considerations: Consider any additional space required for ventilation or access to maintenance points.

Power Cables:

H07 R-NF: A rubber-insulated cable suitable for outdoor use, often used in installations requiring flexibility and resistance to weather conditions.

NYCWY: A multi-core cable designed for use in electrical installations where moisture and mechanical damage are concerns.

RV 0.6/1kV: A PVC-insulated cable suitable for low voltage applications, often used in residential and commercial setups.

d. Power Rating and Capacities: Power rating indicates the maximum amount of power (in watts or kilowatts) that a backup system can deliver continuously. Capacities often refer to the total energy storage available in a battery (measured in amp-hours or watt-hours). Selecting an appropriate power rating ensures that the system can handle the load requirements of connected devices during an outage.

Load Calculation: Calculate the total power requirements of all connected loads to determine the necessary capacity of the backup system. This involves assessing peak loads and continuous operating requirements.

System Selection: Choose a power backup system with a sufficient power rating and capacity to meet these demands during outages, ensuring it can handle both normal and peak loads effectively.

e. Environmental Compatibility: Environmental compatibility involves assessing how well a power backup system can operate under specific environmental conditions, such as temperature extremes, humidity, and exposure to dust or corrosive elements.

Systems designed for harsh environments may include protective enclosures or specialized cooling mechanisms.

Operating Conditions: Evaluate the environmental conditions where the power backup system will be installed, including temperature, humidity, and altitude. Systems must be rated for these specific conditions to operate reliably.

Durability Requirements: Consider whether additional protective measures are needed based on environmental factors, such as dust or moisture ingress.

f. Space Utilization: Space utilization refers to how efficiently the physical space occupied by the power backup system is used. This includes considerations for vertical stacking, modular designs, or compact configurations that maximize available space while maintaining accessibility for maintenance.

Available Space Assessment: Determine the total available space for installing the power backup system components, including generators, inverters, batteries, and control units.

Physical Dimensions: Measure the dimensions of each component. For instance, a typical UPS may have dimensions of approximately 19 inches wide x 3.5 inches high x 24 inches deep for rack-mounted units. Plan layouts that allow efficient use of space while ensuring accessibility for maintenance.

g. Cable Management: Cable management encompasses strategies for organizing and securing cables associated with the power backup system. Proper cable management reduces clutter, minimizes interference, and enhances safety by preventing tripping hazards or accidental disconnections.

Routing Requirements: Assess the routing needs for power and communication cables. Ensure that cables from the backup system to connected loads are organized to prevent clutter and interference.

Implement cable trays or conduits to manage cable runs effectively, reducing tripping hazards and maintaining a clean installation environment.

Organize cables in a way that promotes airflow around heat-generating components such as inverters and batteries. Avoiding Cable Bundling Near Heat Sources

h. Heat Dissipation: Heat dissipation is critical in ensuring that electronic components within a power backup system do not overheat during operation. Effective heat management strategies may include heat sinks, fans, or liquid cooling systems that maintain optimal operating temperatures and prolong equipment lifespan.

Ventilation and Cooling: Ensure adequate ventilation is provided around components. This may involve installing fans or vents to maintain optimal operating temperatures, typically between **20°C to 25°C (68°F to 77°F)**.

i. Weight Distribution: Weight distribution refers to how weight is balanced across the structure of the power backup system. Proper weight distribution is essential for stability during operation, especially in mobile or rack-mounted systems, where uneven weight can lead to tipping or mechanical failure.

Component Weight Consideration: Assess the weight of each component; for instance, a typical lead-acid battery can weigh between 40 kg to 100 kg (88 lbs to 220 lbs) depending on its capacity.

Structural Stability: Distribute weight evenly across the supporting structure to prevent sagging or structural damage. Use racks or shelves designed to support heavy equipment safely.

j. Future Expansion and Scalability: Future expansion and scalability involve designing a power backup system that can accommodate increased loads or additional features without requiring complete replacement. This could include modular designs that allow for adding extra batteries or upgrading components as technology advances or as energy demands grow.

Growth Potential Assessment: Evaluate potential future needs for additional loads or upgrades to the power backup system. For example, if current requirements are for a 5 kW system, consider whether there is space and capacity to expand to a 10 kW system later.

Flexible Component Placement: Design the layout with flexibility in mind, allowing for easy addition of components like extra batteries or larger inverters without requiring significant reconfiguration.



Practical Activity 4.3.2: Installing electrical power backup



Task:

1: Referring to the instruction from your trainer, you are requested to perform the given task below. The task should be done individually.

As a technician, you are asked to go in the electrical workshop install electrical power back up for a computer system.

2: Present your work to the trainer and whole class

3: Ask questions if any for clarification

4: Read key reading 4.3.2 and perform the task provided in application of learning 4.3 for more understanding.



Key readings 4.3.2:

Installing electrical power backup

Installing an electrical power backup system in a computer power system, such as a UPS (Uninterruptible Power Supply), involves several key steps to ensure reliability and efficiency.

To install the electrical power backup, follow the following steps:

Step1. Determine Power Requirements: Identify the devices that will be connected to the backup system and calculate their total power consumption (in watts or kilowatts). Consider both running and starting power requirements for devices like servers and workstations.

Prioritize which devices need backup power during an outage, ensuring that essential equipment is covered.

Step2. Choose the Right Backup System

- **Select Appropriate System:** Based on your power needs, choose between different types of backup systems.
- **Consider Features:** Look for features such as automatic transfer switches (ATS), monitoring capabilities, and expandability.

Step3. Prepare for Installation: Ensure you have all necessary tools (screwdrivers, pliers, multimeter) and safety gear (gloves, goggles) and choose a suitable location for the UPS or generator that allows for proper ventilation, easy access, and minimal exposure to moisture or heat.

Step4. Install the Backup System

- **Physical Installation:**

For a UPS, mount it securely in a designated area close to the devices it will support.

For generators, follow manufacturer guidelines for placement and ensure it is on a stable

surface.

- **Wiring Connections:**

Screw-Connectors and Crimping Ferrules: Ensure that the connectors used for the power backup system are compatible with the specific cable types and sizes being utilized.

Reliability and Durability: Choose connectors designed to withstand environmental stresses such as moisture, temperature fluctuations, and mechanical strain.

Connect the UPS to the main power supply and to the devices needing backup power.

If using a generator, install an Automatic Transfer Switch, ATS to manage the switch between utility and generator power seamlessly.

Step5. Cable Management

Wire Routing: Cables should be routed along walls or ceilings where possible, avoiding high-traffic areas, tripping or interference.

Cable Trays or Conduits: Mesh cable trays can be particularly effective in industrial settings, allowing for good ventilation while keeping cables accessible for maintenance.

Step6. Wire Termination

Proper Termination: Ensure that all wires are terminated correctly at both the power backup system and the connected loads.

Stripping Tools and Crimpers: Use high-quality wire stripping tools and crimpers designed for specific wire gauges (e.g., suitable for 6 mm² wires) to ensure a secure connection without damaging the wire insulation.

Step 7. Protection Devices

IP Rated Junction Boxes: For outdoor installations, use junction boxes with an IP rating of at least IP55 to protect against dust and water ingress.

Surge Protectors: Implement surge protection devices to safeguard the power backup system and connected equipment from voltage surges caused by lightning or other electrical disturbances.

Step 8. Grounding

Green/Yellow Insulated Cables: complying with electrical standards to ensure safety.

Grounding Electrode: a ground rod or grounding plate, ensuring that all components of the power backup system are properly grounded.

Step9. Test the Backup System

- **Power-On Self-Test:** Conduct initial tests to ensure that the system powers on correctly.
- **Load Testing:** Simulate a power outage by disconnecting from the main supply and verify that all critical devices receive power as intended.
- **Check Transfer Mechanisms:** For systems with ATS, confirm that it switches between power sources smoothly without interruption.

Step10. Regular Maintenance Plan

- **Routine Checks:** Schedule regular maintenance checks to inspect battery health (for UPS), fuel levels (for generators), and overall system performance.
- **Update Software/Firmware:** Keep any software associated with monitoring or managing the backup system up to date.

Step11. Documentation and Training

- **Label Circuits:** Clearly label circuits in your electrical panel that are connected to the backup system for easy identification during emergencies.
- **User Training:** Train relevant personnel on how to operate the backup system, including how to perform manual overrides if necessary.



Practical Activity 4.3.3: Applying communication interface



Task:

1. Referring to the instruction from your trainer, you are requested to perform the given task below. The task should be done individually.
As technician, you are tasked to go onto the workshop to apply communication interfaces for power back up system.
2. Read key reading 4.3.3 then perform the activity.
3. Present your work to the trainer and whole class
4. Ask questions if any for clarification
5. Perform the task provided in application of learning 4.3



Key readings 4.3.3

Applying communication interface

Introduction

When implementing a communication interface for power backup systems, it is essential to identify communication needs, select appropriate protocols, configure interfaces, and integrate with Battery Management Systems (BMS).

Applying a communication interface for a power backup system, such as an Uninterruptible Power Supply (UPS), is essential for effective monitoring, management, and integration within modern electrical systems.

Here's the steps and considerations involved in implementing communication interfaces for power backup systems:

Step1. Identify Communication Needs

- **Determine Requirements:** Assess the specific communication needs based on the application. Consider factors like remote monitoring, control capabilities, data logging, and integration with existing systems (e.g., SCADA).
- **Critical Parameters:** Identify which parameters need to be monitored (e.g., battery status, load levels, input/output voltage).

Step2. Choose Appropriate Communication Protocols

- **Common Protocols:** Select suitable communication protocols based on compatibility and performance:
 - ✓ **Modbus TCP/IP:** Widely used for industrial applications; allows for easy integration with various devices.
 - ✓ **Ethernet/IP:** Suitable for real-time control and monitoring in automation systems.
 - ✓ **PROFINET:** Ideal for high-speed data exchange in industrial environments.
 - ✓ **IEC 61850:** A standard for communication in substations and power systems, ensuring interoperability and vendor independence.

Step3. Integrate Communication Interfaces

- **Hardware Setup:** Install communication modules or interfaces on the UPS or backup power system. This may include Ethernet ports, RS-232/RS-485 interfaces, or wireless modules.
- **Wiring and Connections:** Ensure proper wiring between the UPS and the communication network. Use appropriate cabling to prevent interference and ensure reliable data transmission.

Step4. Configure Monitoring and Control Systems

- **Setup Monitoring Software:** Implement software that can interface with the UPS to monitor key parameters in real-time. Many advanced UPS systems come with proprietary software for this purpose.
- **Data Visualization:** Create dashboards or user interfaces that display critical information such as battery health, load status, and event logs.

Step5. Implement Automated Control Features

- **Remote Management:** Enable features that allow remote shutdown or management of connected devices during power failures. This helps ensure data integrity and prevents damage to equipment.
- **Alerts and Notifications:** Configure alerts for critical events (e.g., low battery, power failure) to notify users via email or SMS.

Step6. Test the Communication System

- **Conduct Functionality Tests:** Verify that the communication interface works correctly by simulating power outages and ensuring that all monitoring functions operate as expected.
- **Check Data Accuracy:** Ensure that the data reported by the system is accurate and reflects real-time conditions.

Step7. Regular Maintenance and Updates

- **Software Updates:** Keep communication software updated to enhance security and functionality.
- **Routine Checks:** Regularly test the communication interfaces to ensure they are functioning correctly and troubleshoot any issues promptly.

Step8. Documentation and Training

- **User Manuals:** Provide documentation detailing how to use the communication interface effectively.
- **Training Sessions:** Conduct training for relevant personnel on how to monitor and manage the power backup system using the communication interface.



Practical Activity 4.3.4: Integrating power backup to power supply unit



Task:

1: Referring to the instruction from your trainer, you are requested to perform the given task below. The task should be done individually.

You are tasked to go in the Electrical Workshop and Computer Lab and assess the compatibility between the computer system and power backup system by performing electrical connection, configuring the communication interface and protocols, labelling the components according to the performance, functionality and connectivity and test the system

2: Present your work to the trainer

3: Ask questions if any for clarification

4: Read the key reading 4.3.4 to get more clarifications and perform the task provided in application of learning 4.3



Key readings 4.3.4

Integrating power backup to power supply unit

Integrating a power backup system, such as an Uninterruptible Power Supply (UPS), into a power supply unit involves several critical steps.

Below are steps for Integrating Power Backup into a Power Supply Unit:

Step1. Planning and Assessment

- **Determine Power Requirements:** Calculate the total wattage of the devices that need backup power. This includes considering peak loads and the duration for which backup power is needed.
- **Select Backup Type:** Choose between different types of UPS systems (e.g., Standby, Online) based on your needs. Standby UPS is suitable for non-critical loads, while Online UPS provides continuous power without interruption.

Step2. Equipment Selection

- **Choose the Right UPS:** Select a UPS model that meets your calculated power requirements. Consider factors like battery capacity and additional features such as monitoring capabilities.
- **Battery Selection:** Decide on the type of battery (e.g., lead-acid, lithium-ion) based on your budget and maintenance preferences. Ensure compatibility with the chosen UPS.

Step3. Installation Preparation

- **Site Assessment:** Identify an appropriate location for the UPS installation, ensuring it has adequate ventilation and space for maintenance access. The environment should be cool and dry to prolong battery life.
- **Cabling Requirements:** Plan for necessary cabling, including AC input/output and

DC battery connections. Ensure that cables are appropriately rated for the load they will carry.

Step4. Physical Installation

- **Elevate the UPS:** If necessary, place the UPS on a sturdy platform to facilitate cable management and prevent overheating.
- **Connect Cables:**
 - ✧ Connect the AC input and output cables.
 - ✧ Connect the DC battery cables according to manufacturer specifications.
 - ✧ Ensure proper grounding of the unit.

Step5. Configuration

- **Set Up Control Circuits:** Configure any control circuitry required for monitoring and managing power flow between the grid and the UPS.
- **Testing:** Conduct a dry run of all UPS functions to ensure proper operation. Use manufacturer-provided testing software if available.

Step6. Maintenance Planning

- **Regular Testing and Monitoring:** Schedule periodic testing of battery health, including conductance and impedance tests, to ensure reliability during outages.
- **Filter and Fan Maintenance:** Regularly clean or replace filters and fans to maintain optimal operating conditions.

Step7. Documentation and Training

- **User Training:** Provide training for personnel on how to operate the UPS, including emergency procedures during power failures.

Maintain Documentation: Keep detailed records of installation procedures, equipment specifications, and maintenance schedules for future reference.



Points to Remember

- **Description of Power Backup Systems concepts**
 - a. Voltage Compatibility:** Voltage compatibility refers to the ability of a power backup system to operate within a specified voltage range.
 - b. Communication Protocols:** Communication protocols are standardized methods that allow different devices within a power backup system to exchange information.

- d. Power Rating and Capacities:** Power rating indicates the maximum amount of power (in watts or kilowatts) that a backup system can deliver continuously.
- e. Environmental Compatibility:** Environmental compatibility involves assessing how well a power backup system can operate under specific environmental conditions, such as temperature extremes, humidity, and exposure to dust or corrosive elements.
- f. Space Utilization:** Space utilization refers to how efficiently the physical space occupied by the power backup system is used.
- g. Cable Management:** Cable management encompasses strategies for organizing and securing cables associated with the power backup system.
- h. Heat Dissipation:** Heat dissipation is critical in ensuring that electronic components within a power backup system do not overheat during operation.
- i. Weight Distribution:** Weight distribution refers to how weight is balanced across the structure of the power backup system.
- j. Future Expansion and Scalability:** Future expansion and scalability involve designing a power backup system that can accommodate increased loads or additional features without requiring complete replacement.
- **To install the electrical power backup, follow the following steps:**
 - Start by** identifying the devices that will be connected to the backup system and calculate their total power consumption, **Choose the Right Backup System, Prepare for Installation, Install the Backup System,** cables should be routed along walls or ceilings where possible, avoiding high-traffic areas, tripping or interference, ensure that all wires are terminated correctly at both the power backup system and the connected loads, **then** implement surge protection devices to safeguard the power backup system and connected equipment from voltage surges caused by lightning or other electrical disturbances. Finally ensure that all components of the power backup system are properly grounded.
- **Applying communication interface**

Implementing a communication interface in a power backup system enhances monitoring capabilities, facilitates proactive maintenance, and ensures seamless integration with existing infrastructure. By following these steps—identifying needs, choosing protocols, integrating hardware, configuring systems, testing functionality, maintaining updates, and providing training—organizations can optimize their power backup solutions for reliability and efficiency.
- **Integrating power backup to power supply unit**

Integrating a power backup system, such as an Uninterruptible Power Supply (UPS), into a power supply unit involves several critical steps to ensure reliability and efficiency. The process begins with **planning and assessment**, where you determine the total wattage of devices requiring backup power and select the appropriate UPS type, whether Standby or Online, based on your needs. Next, in the **equipment selection** phase, you choose a UPS model that meets these power requirements and decide on the battery type, considering factors like capacity and maintenance preferences.

The **installation preparation** stage involves assessing the site for adequate ventilation and planning necessary cabling for both AC and DC connections. During **physical installation**, the UPS is securely mounted, cables are connected according to specifications, and proper grounding is ensured. Following this, **configuration** includes setting up control circuits and conducting tests to verify functionality. **Maintenance planning** is essential for ongoing reliability, involving regular testing of battery health and cleaning of equipment. Finally, **documentation and training** are crucial; personnel should be trained on UPS operation and emergency procedures while maintaining detailed records of installation and maintenance activities. This structured approach ensures that the power backup system operates effectively, safeguarding critical equipment during outages.



Application of learning 4.3.

You are an electronics engineer working for a company that designs high-performance custom gaming PCs. The company's latest project is to design a Power Supply backup system and integrate it onto an existing system which is currently having intermittent blackouts, various electrical components and batteries.

As a technician of the company, you are tasked with the following tasks:

- a) Assess the component placement and compatibility
- b) Install electrical power backup
- c) Apply communication interface
- d) Integrate power backup to power supply unit



Indicative content 4.4: Maintaining Power Backup



Duration: 4 hrs



Practical Activity 4.4.1: Implementing monitoring capabilities



Task:

1: Referring to the instruction from your trainer, you are requested to read the given task below. The task should be done individually.

As technician, go in the work environment to implement monitoring capabilities in a Battery Management System (BMS), focusing on real-time data collection and analysis for battery health and performance.

2: Perform the activity provided above

3: Present your work to whole class.

4: Read the key reading 4.4.1 from your manual to get more clarifications and perform the task provided in application of learning 4.4



Key readings 4.4.1

Implementing Monitoring Capabilities

Introduction

Implementing monitoring capabilities for maintaining power backup systems, such as Uninterruptible Power Supplies (UPS), is essential for ensuring reliability and preventing potential failures. One effective approach is to utilize **remote monitoring systems** that leverage Simple Network Management Protocol (SNMP) to provide real-time insights into UPS performance metrics, including battery charge, load levels, and voltage readings.

These systems can send automated alerts via SMS or email when issues arise, such as a failing battery or abnormal temperature readings, allowing for prompt corrective actions before problems escalate.

Steps for Implementing Monitoring Capabilities for Power Backup

Implementing monitoring capabilities for power backup systems, particularly Uninterruptible Power Supplies (UPS), is essential for ensuring reliability and minimizing downtime.

Here's a structured approach to effectively establish these monitoring capabilities.

Step1. Assessment of Current Infrastructure

- **Evaluate Existing Systems:** Begin by assessing the current UPS infrastructure to identify areas where monitoring can provide value. Consider factors such as the criticality of loads, frequency of outages, and maintenance needs.

Step2. Selecting the Monitoring Solution

- **Choose Appropriate Tools:** Select a monitoring solution that aligns with your specific requirements such as Power Quality Meters to Measure voltage, current, frequency, and harmonics to ensure power quality and also track individual cell voltages, temperature, and overall battery health using battery monitors.

Step3. Deployment and Integration

- **Install Monitoring Equipment:** Deploy compatible hardware or software, which may involve installing sensors, network adapters, or dedicated monitoring software on the UPS system
- **Network Configuration:** Integrate the monitoring system into your existing network infrastructure. Configure necessary network settings such as IP addresses and gateways to facilitate communication between the UPS and monitoring devices

Step4. Data Collection and Transmission

- **Real-Time Data Monitoring:** Once the system is operational, it should begin collecting data on key parameters like input/output voltage, battery health, load levels, and temperature in real-time
- **Secure Data Transmission:** Transmit the collected data securely to a central monitoring platform using Ethernet, Wi-Fi, or cellular networks

Step5. Analysis and Visualization

- **Data Interpretation:** Analyze the transmitted data to visualize it in an accessible format. This allows users to quickly assess the health and performance of their UPS systems.
- **Establish Alert Mechanisms:** Set up alerts and notifications for any anomalies detected by the monitoring system. This will enable proactive intervention before issues escalate

Step6. Testing and Optimization

- **Conduct Thorough Testing:** After installation, conduct tests to verify functionality

and reliability. Fine-tune settings as needed to optimize performance and minimize false alarms

Step7. Training and Documentation

- **Train Personnel:** Ensure that relevant staff are trained on how to use the monitoring system effectively. Provide documentation that outlines procedures for troubleshooting and maintenance

Step8. Continuous Monitoring and Improvement

Ongoing Evaluation: Regularly review the performance of the monitoring system to ensure it captures accurate data. Update software/firmware as necessary and incorporate feedback from maintenance activities to enhance reliability.



Practical Activity 4.4.2: Establishing Routine Maintenance Procedures



Task:

1: Referring to the instruction from your trainer, you are requested to perform the given task below. The task should be done individually.

After the work installing and integrating power backup about computer power system, you are tasked to establish a frequent monitoring schedule that would provide real time maintenance.

2: Read the key reading 4.4.2 from your manual to get more clarifications.

3: Perform the activity provided

4. Present your work to whole class.

5. Perform the task provided in application of learning 4.4



Key readings 4.4.2.

Establishing Routine Maintenance Procedures

Introduction

Establishing routine maintenance procedures involves creating a systematic approach to regularly scheduled tasks aimed at keeping equipment and facilities in optimal working condition. This proactive strategy is essential for preventing unexpected breakdowns and ensuring operational efficiency.

Maintaining power backup systems, such as Uninterruptible Power Supplies (UPS) and backup generators, is essential to ensure reliability during power outages.

Here are the key steps to establish effective routine maintenance procedures.

Step 1 : Scheduled Inspection

Frequency: Determine inspection frequency based on system size and criticality. For example, larger systems may require monthly inspections, while smaller systems might be inspected quarterly.

Create a detailed inspection checklist that includes:

- Visual inspection of physical components
- Checking connections for corrosion or wear
- Verifying battery health and charge levels
- Testing UPS functionality under load conditions.

Documentation: Document results of each inspection in a maintenance log. Record any issues found and corrective actions taken to maintain a history of system performance.

Step 2 : Battery Maintenance

Regular Testing: Conduct regular tests to assess battery capacity using methods such as load testing or impedance testing. This helps identify weak cells before they fail.

Equalization: Perform equalization cycles periodically to ensure all battery cells are charged uniformly, which helps prolong battery life and efficiency.

Cleaning: Clean battery terminals and connections regularly to prevent corrosion. Use appropriate cleaning agents and ensure connections are tight after cleaning.

Step 3 : Safety Precautions Sticker

Place safety precaution stickers on the power backup system to warn personnel of

potential hazards (e.g., high voltage) and provide instructions for safe operation.

Step 4 : Cleaning and Housekeeping

Regular Cleaning: Maintain cleanliness around the power backup system to prevent dust accumulation, which can affect performance. Schedule routine cleaning sessions as part of maintenance procedures.

Step 5 : Documentation and Record-Keeping

- Keep detailed maintenance logs that record all inspections, repairs, replacements, and any changes made to the system configuration.
- Document all maintenance procedures and guidelines for future reference. This ensures consistency in maintenance practices over time.

Step 6 : Training and Personnel Development

- **Regular Training:** Provide ongoing training for personnel responsible for maintaining the power backup system. Training should cover operational procedures, troubleshooting techniques, and safety protocols.
- **Certification:** Consider obtaining certifications for personnel involved in power backup system maintenance. Certifications can enhance skills and ensure compliance with industry standards



Points to Remember

- **Implementing Monitoring Capabilities**

Here's a structured approach to effectively establish these monitoring capabilities.

Step1. Begin by assessing the current UPS infrastructure to identify areas where monitoring can provide value.

Step2. Select a monitoring solution that aligns with your specific requirements such as Power Quality Meters to Measure voltage, current, frequency, and harmonics

Step3. Deploy compatible hardware or software, which may involve installing sensors, network adapters, or dedicated monitoring software on the UPS system

Step4. Once the system is operational, it should begin collecting data on key parameters like input/output voltage, battery health, load levels, and temperature in real-time

Step5. Analyze the transmitted data to visualize it in an accessible format. This allows users to quickly assess the health and performance of their UPS systems.

Step6. After installation, conduct tests to verify functionality and reliability.

Step7. Ensure that relevant staff are trained on how to use the monitoring system effectively.

Step8. Regularly review the performance of the monitoring system to ensure it captures accurate data.

- **Establishing Routine Maintenance Procedures**

Here are the key steps to establish effective routine maintenance procedures.

Step1. Schedule inspections based on manufacturer recommendations and specific system needs.

Step2. Check for signs of wear, corrosion, or leaks in all components.

Step3. Implement regular battery load tests to assess health and capacity.

Step4. Ensure proper airflow around the UPS and generator to prevent overheating.

Step5. Regularly simulate a power failure by applying a full load to the backup systems.

Step6. Regularly check for updates from manufacturers to enhance performance and security.

Step7. Keep detailed records of all maintenance activities, including inspections, repairs, and replacements.

Step8. Develop a plan outlining actions to take during power outages or equipment failures.

Step9. Regularly review maintenance procedures to adapt to changes in technology or operational requirements.



Application of learning 4.4.

You are an electronics engineer working for a company that designs high-performance custom gaming PCs. The company's latest project is to maintain the power backup system of a small data center. The power backup system includes a UPS (Uninterruptible Power Supply) and a battery bank.

As a technician of the company, you are tasked with implementing a monitoring system for the power backup and establishing a routine maintenance checklist to ensure its optimal performance.



Indicative content 4.5: Estimation of Power System Cost



Duration: 6 hrs



Theoretical Activity 4.5.1: Description of cost estimation



Tasks:

- 1: Answer the following questions related to description of cost estimation.
 - i. Define cost estimation
 - ii. Describe the types of cost estimation
 - iii. Describe the models of cost estimation.
- 2: Write the findings/answers on papers or flipchart.
- 3: Present your findings to the class.
- 4: Ask questions for more clarifications.
- 5: Read the key reading 4.5.1 from your manual to get more clarifications.



Key readings 4.5.1.:

Description of cost estimation

- Description of cost estimation

1. Definition of Cost Estimation

Cost estimation is the process of predicting the financial resources required to complete a project or task. It involves analysing historical data, performance metrics, and technical specifications to derive an estimate of future costs associated with materials, labour, overhead, and other expenses. The accuracy of a cost estimate is crucial for budgeting, financial planning, and project feasibility assessments.

2. Types of Cost Estimation

2.1 Analogous Estimating: This technique uses historical data from similar projects to estimate costs. It is quick but less accurate, making it suitable for initial project phases.

2.2 Parametric Estimating: This method relies on statistical relationships between historical data and project parameters (e.g., cost per square foot). It provides more accurate estimates than analogous estimating.

2.3 Bottom-Up Estimating: This detailed approach involves breaking down the project into smaller components and estimating costs for each part before aggregating them for a total estimate. It is time-consuming but highly accurate.

2.4 Three-Point Estimating: This technique considers three scenarios—optimistic, pessimistic, and most likely—to account for uncertainty in cost estimates.

2.5 Factor Estimating: This method uses predefined factors to estimate costs based on certain parameters or characteristics of the project.

3. Models of Cost Estimation

3.1 Least Squares Regression: A statistical method used to find the best-fit line for variable and fixed costs, facilitating future cost predictions based on production levels.

3.2 High-Low Method: This simple approach identifies the highest and lowest costs incurred to provide a preliminary cost range.

3.3 Statistical Modelling: This sophisticated method incorporates multiple economic factors affecting cost estimation, resulting in highly accurate projections but requiring extensive data.

4. Process of Cost Estimation

4.1 Define Project Scope: Clearly outline the objectives, deliverables, and requirements of the project.

4.2 Gather Historical Data: Collect relevant historical cost data from similar projects to inform estimates.

4.3 Select Estimation Method: Choose an appropriate estimation technique based on the project's complexity and available data.

4.4 Perform Cost Analysis: Analyse all components involved in the project, including labour, materials, overheads, and any other relevant costs.

4.5 Compile Estimates: Aggregate the individual component estimates into a comprehensive total cost estimate.

4.6 Review and Revise: Validate the estimates through peer review or expert judgment and make necessary adjustments based on feedback or new information.

4.7 Document Results: Maintain detailed records of the estimation process for future reference and auditing purposes.



Theoretical Activity 4.5.2: Description of bill of quantity



Tasks:

1: Answer the following questions related to description of cost estimation.

- i. Define bill of quantity
- ii. Describe elements of bill of quantity.

2: Write the findings/answers on papers or flipchart.

3: Present your findings to the class.

4: Ask questions for more clarifications.

5: Read the key reading 4.5.2 from your manual to get more clarifications.



Key readings 4.5.2.:

Description of bill of quantity

1. Definition of Bill of Quantity

A **Bill of Quantities (BoQ)** is a detailed document used in the construction industry that itemizes all materials, labour, and associated costs necessary for completing a project. It serves as a vital tool during the tendering process, providing contractors with a clear understanding of the work required and enabling them to submit competitive bids. The BoQ typically includes precise measurements and descriptions of each item, ensuring that all bidders are pricing the same quantities, which promotes fairness and accuracy in the tendering process.

2. Elements of Bill of Quantity

2.1 Item Number: A unique identifier for each entry in the BoQ.

2.2 Item Description: Detailed information about the work to be performed or materials to be supplied, ensuring clarity on what is required.

2.3 Quantity: The measured amount of each item, which can be in various units such as square meters, cubic meters, or number of units.

2.4 Unit of Measurement: The standard unit used to quantify each item (e.g., kg for weight, m² for area).

2.5 Unit Rate: The cost assigned per unit of measurement for each item, based on market rates or historical data.

2.6 Total Cost: The overall cost for each item, calculated by multiplying the quantity by the unit rate.

2.7 Remarks: Additional notes or specifications related to particular items that may require further clarification or context



Practical Activity 4.5.3: Design bill of quantity



Task:

1: Referring to the instruction from your trainer, you are requested to perform the given task below. The task should be done individually.

As a technician, you are tasked to establish a bill of quantity for a computer power system development project.

2: Perform the activity provided above

3: Present your work to whole class.

4: Read the key reading 4.5.3 from your manual to get more clarifications and perform the task provided in application of learning 4.5



Key readings 4.5.3

- Design bill of quantity

Introduction

Designing a Bill of Quantities (BoQ) is a structured process that ensures accurate estimation of costs and resources for construction projects.

Steps of Designing a Bill of Quantities:

Step1. Initial Project Documentation: Collect all relevant project documents, including architectural plans, engineering drawings, and specifications. This foundational step is crucial for understanding the scope and requirements of the project.

Step2. Define Scope of Work: Clearly outline the work to be performed, including detailed descriptions of tasks and materials needed. This helps ensure that all parties have a common understanding of project expectations.

Step3. Itemization and Categorization: Review the project documents to itemize all required materials and labour. Categorize these items as either materials or labour to facilitate clear differentiation in the BoQ.

Step4. Quantity Take-off: Measure and quantify each item based on project specifications. This involves calculating the total quantities needed for materials and estimating labour hours required for each task.

Step5. Cost Estimation: Determine the unit rates for each item based on market surveys, historical data, or current pricing schedules. Multiply these rates by the quantities to

calculate total costs for each item.

Step6. Quality Control: Review and cross-check the BoQ thoroughly for accuracy and consistency. It's advisable to have multiple stakeholders review the document to minimize errors.

Step7. Compliance with Standards: Ensure that the BoQ adheres to relevant industry standards and guidelines, such as the New Rules of Measurement (NRM) in the UK or other applicable regulations.

Step8. Final Compilation: Organize the BoQ in a clear format, typically in tabular form, including all necessary details like item numbers, descriptions, quantities, unit rates, and total costs. Include a cover page with project details if required.

Step9. Documentation and Distribution: Document the final BoQ comprehensively and distribute it to relevant stakeholders for review and use in tendering processes.

Sample Bill of Quantities

Here's a simplified example of a Bill of Quantities for a small construction project:

Item No.	Description	Unit	Quantity	Unit Rate (\$)	Total Cost (\$)	Remarks
1	Excavation for foundation	Cubic Meters	10	50	500	Includes disposal
2	Concrete for foundation	Cubic Meters	8	100	800	Mix design specified
3	Brickwork for walls	Square Meters	50	30	1500	Standard bricks
4	Roofing installation	Square Meters	40	25	1000	Includes insulation
5	Electrical wiring	Linear Meters	200	2	400	Basic wiring
Total					\$3200	



Practical Activity 4.5.4: Applying cost estimation



Task:

1: Referring to the instruction from your trainer, you are requested to perform the given task below.

As a technician, go in the working environment then you are tasked to design a powered battery charger for portable devices and also you are responsible for providing a cost estimation for building 10 prototypes of the charger. The goal is to present this estimation to potential investors, demonstrating the affordability and feasibility of the project.

2: Present your findings to the trainer

3: Read key readings 4.5.4 from trainee's manual and perform the application of learning 4.5



Key readings 4.5.4

Applying cost estimation

Introduction

Cost estimation in computer power systems involves a systematic approach to predict the financial resources required for projects such as system upgrades, new installations, or expansions.

Here are the steps to apply cost estimation in a project:

Step1. Define the Scope of the Project

- Clearly outline what the project entails, including the deliverables and objectives.
- Identify the work to be completed, the timeline, and the quality expectations.
- Define the quantity of products or services needed if applicable (e.g., the number of units to be produced).

Step2. List the Resources and Components Required

- Identify all the materials, tools, equipment, and resources needed for the project.
- Include direct materials (e.g., components, raw materials) and indirect materials

(e.g., office supplies, utilities).

- Consider any software, licenses, or technical support needed.

Step3. Determine the Cost of Each Resource

- **Material Costs:** Research the price of materials from suppliers, considering factors like shipping, bulk order discounts, and taxes.
- **Labor Costs:** Estimate labor costs by calculating the hourly rate or salary of workers multiplied by the number of hours needed for each task.
- **Fixed Costs:** These are costs that remain constant throughout the project (e.g., rent, equipment).
- **Variable Costs:** Costs that change based on the scale of production (e.g., material usage, energy consumption).

Step4. Estimate Time and Effort

- Estimate the time required to complete the project tasks and activities.
- Break down the tasks into smaller, manageable activities, and assign durations to each task.
- This is critical for labor cost calculation and scheduling.

Step5. Calculate Overhead Costs

- Include indirect costs such as administrative fees, utilities, and general office expenses.
- Consider factors like facility maintenance, support staff, and equipment depreciation.
- Overhead is often calculated as a percentage of labor or material costs.

Step6. Add Contingencies for Risk

- Include a contingency cost (usually 5–15%) to account for unexpected expenses, such as delays, material price fluctuations, or supply chain disruptions.
- This helps buffer the project from financial risks.

Step7. Summarize Total Costs

- Add up the material costs, labor costs, overhead, and contingency to get the total estimated project cost.
- If the project is large, break down costs by phase, activity, or deliverable.

Step8. Analyze the Cost Estimate

- Review the cost estimation to ensure its accuracy and completeness.
- Look for areas where costs can be optimized without sacrificing quality.
- Compare with similar past projects to validate estimates.

Step9. Present and Document the Cost Estimate

- Create a detailed cost estimation report that summarizes all costs and assumptions made.
- Include breakdowns of individual components, materials, labor, overhead, and contingencies.
- Provide explanations or justifications for any significant costs or assumptions.

Step10. Monitor and Adjust as Needed

- Once the project is underway, track actual expenses against your estimates.
- Adjust the cost estimation as the project progresses, especially if there are changes in scope, material costs, or unexpected delays.

Sample of cost estimation report:

Project Cost Estimation					
<u>Company Details</u>		Date: Quote Ref: Validity:			
<u>Client Details</u>					
<u>Job Detail</u>					
Phase - 1	Vendor	Labor	Material	Total Cost	Remark
Task- 1					
Task- 2					
Task- 3					
Phase - 2					

Here's a simplified example of a cost estimation for upgrading a computer power system:

Item No	Description	Unit	Quantity	Unit Rate (\$)	Total Cost (\$)	Remarks
1	UPS (Uninterruptible Power Supply)	Each	2	1,200	2,400	10 kVA capacity
2	Power Distribution Unit	Each	1	800	800	Includes installation
3	Backup Batteries	Each	4	150	600	For UPS
4	Installation Labor	Hours	20	50	1,000	Estimated hours
5	Testing and Commissioning	Each	1	500	500	Final system check
Total					\$5,300	



Points to Remember

Description of cost estimation

- **Cost estimation** is the process of forecasting the financial resources required to complete a project within a defined scope. It involves calculating the total costs associated with all elements necessary for the project, including materials, labor, and overheads. A cost estimate serves as a critical tool for decision-making, allowing organizations to evaluate project feasibility, secure funding, and establish budgets.

- **Types of Cost Estimation**

The types of cost estimation are: **Analogous Estimating, Parametric Estimating, Bottom-Up Estimating, Three-Point Estimating** and **Factor Estimating**.

- **Models of Cost Estimation**

- ✓ **Least Squares Regression:** A statistical method used to find the best-fit line for variable and fixed costs, facilitating future cost predictions based on production levels.

- ✓ **High-Low Method:** This simple approach identifies the highest and lowest costs incurred to provide a preliminary cost range.
- ✓ **Statistical Modeling:** This sophisticated method incorporates multiple economic factors affecting cost estimation, resulting in highly accurate projections but requiring extensive data.
- **Process of Cost Estimation**
 - ✓ Clearly outline the objectives, deliverables, and requirements of the project.
 - ✓ Collect relevant historical cost data from similar projects to inform estimates.
 - ✓ Choose an appropriate estimation technique based on the project's complexity and available data.
 - ✓ Analyze all components involved in the project, including labor, materials, overheads, and any other relevant costs.
 - ✓ Aggregate the individual component estimates into a comprehensive total cost estimate.

Description of bill of quantity

- **Bill of Quantities (BOQ)** is a detailed document that itemizes all the materials, labor, and other costs involved in an engineering project. It is commonly used in the construction industry to provide a clear breakdown of work that needs to be done, allowing contractors to price projects more accurately and fairly.

- **Elements of Bill of Quantity**

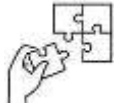
The elements of BoQ are: **Item Number, Item Description, Quantity, Unit of Measurement, Unit Rate, Total Cost** and **Remarks**.

- **Designing a Bill of Quantities**

The process of designing a Bill of Quantities involves several critical steps: gathering initial documentation, defining the scope of work, itemizing materials and labor, quantifying needs, estimating costs, ensuring quality control, complying with standards, compiling the final document, and distributing it to stakeholders. Following these steps ensures that the BoQ serves as an effective tool for accurate cost estimation and project management in construction.

- **Applying cost estimation**

Applying cost estimation involves a structured approach that begins with establishing a clear basis for estimation and collecting comprehensive project documentation. It progresses through quantifying scope, estimating direct and indirect costs, peer reviewing estimates, finalizing documentation, and obtaining approval. Continuous monitoring ensures that estimates remain relevant throughout the project lifecycle, facilitating effective financial management and decision-making.



Application of learning 4.5.

You are working as a project manager for a company tasked with developing a computer power system for a new computer lab in a technical secondary school. The lab requires a backup power system that can provide uninterrupted power to all computers and network equipment during outages. The solution will include UPS units, battery backups, and an energy monitoring system.

Your task is to establish the project's cost, perform a sensitivity analysis to assess the impact of fluctuating costs, and submit a report with your findings and recommendations.



Learning outcome 4 end assessment

Theoretical assessment

Q1. Multiple Choice Questions: Encircle the letter corresponding the correct answer.

1. What is the primary goal of a computer power management system?
 - a) To increase processing speed
 - b) To efficiently manage power consumption
 - c) To enhance graphics performance
 - d) To improve internet connectivity
2. Which of the following is NOT a power-saving technique?
 - a) Sleep mode
 - b) Dynamic Voltage and Frequency Scaling (DVFS)
 - c) Overclocking
 - d) Processor power states
3. What does the term "Hibernate" refer to in power management?
 - a) A low-power state where the system retains memory
 - b) A state where the system saves its state to disk and powers down
 - c) A full shutdown of the system
 - d) A mode where the system is fully operational
4. Which component is essential for monitoring battery health in a battery management system?
 - a) UPS
 - b) Power supply unit
 - c) Battery Management Unit (BMU)
 - d) Generator
5. In power management, what does DVFS stand for?
 - a) Dynamic Voltage and Frequency Scaling
 - b) Direct Voltage Frequency Standardization
 - c) Dynamic Variable Frequency Switching
 - d) Digital Voltage Frequency Scaling
6. What is a common method for calculating total power consumption in a computer system?
 - a) Total voltage multiplied by current
 - b) Sum of all components' wattage ratings
 - c) Average power consumption during idle state

- d) Total number of devices multiplied by usage hours
7. Which of the following statements about UPS (Uninterruptible Power Supply) is true?
- It provides backup power only during long outages.
 - It can deliver immediate power during outages.
 - It is not necessary for critical systems.
 - It only protects against voltage spikes.
8. What is the purpose of an Automatic Transfer Switch (ATS)?
- To monitor battery health
 - To switch between main and backup power sources automatically
 - To provide additional processing power
 - To optimize software performance
9. When estimating costs for a power management system, which of the following should be included?
- Only hardware costs
 - Installation and maintenance costs only
 - Both hardware and software costs, including installation and maintenance
 - Only software costs
10. Which state indicates that the system is fully operational?
- S1
 - S3
 - S0
 - S4

Q2. Matching Questions

- a. Match the following terms with their correct descriptions:

Term	Description
1. Sleep Mode	A. System saves its state to disk
2. Hibernate	B. Low-power state with RAM retention
3. ACPI	C. Standard for power management in computers
4. Power Consumption	D. Amount of energy used by a system
5. Battery Health Monitoring	E. Process of assessing battery performance

- b. Match the following power management techniques with their descriptions:

Term	Description
------	-------------

1. Dynamic Voltage Scaling	A. Adjusts voltage based on workload
2. Processor Power States	B. Defines different levels of CPU activity
3. Smart Power Management	C. Policies that optimize energy use
4. Energy Star Certification	D. Label indicating energy-efficient products
5. Battery Management System	E. Monitors and controls battery performance

c. Match the following components of cost estimation with their definitions:

Term	Description
1. Fixed Costs	A. Costs that remain constant regardless of output
2. Variable Costs	B. Costs that fluctuate with production levels
3. Direct Costs	C. Expenses directly tied to project execution
4. Indirect Costs	D. Overhead costs not directly linked to a project
5. Contingency Fund	E. Reserve funds set aside for unforeseen expenses

d. Match the following battery charging methods with their characteristics:

Term	Description
1. Constant Current Charging	A. Maintains a steady current until a set voltage is reached
2. Constant Voltage Charging	B. Maintains a steady voltage until the current drops below a threshold
3. Trickle Charging	C. Low-rate charging to maintain battery charge over time
4. Fast Charging	D. Rapid charging method to quickly replenish battery energy
5. Equalization Charging	E. Balances charge across all cells in a battery bank

e. Match the following components of a power backup system with their roles:

Term	Description
1. UPS	A. Provides immediate backup power
2. Generator	B. Supplies power for extended outages
3. Lithium Battery Bank	C. Stores energy for later use
4. Automatic Transfer Switch	D. Switches between main and backup power sources
5. Redundant Power Supply	E. Offers an alternative source in case of failure

f. Match the following monitoring tools with their functions:

Term	Description
1. Power Meter	A. Measures electrical consumption
2. Battery Monitor	B. Tracks battery health and status
3. Energy Management Software	C. Analyses overall energy usage
4. Environmental Sensors	D. Monitors temperature and humidity levels
5. Load Tester	E. Evaluates performance under specific loads

Q3. If a computer system consumes 400W during operation and requires a backup time of 5 hours during an outage, what is the total energy required from the backup system in watt-hours (Wh)?

Q4. A server has a total power consumption of 600W and is connected to a UPS that can deliver 2400Wh of energy. How long can the UPS support the server during an outage?

Q5. If a lithium battery bank has a capacity of 8000mAh at 12V, what is its total energy capacity in watt-hours (Wh)?

Q6. True/False Questions

- i. The sleep mode stops all actions on the computer.
- ii. In hibernate mode, the system consumes more power than in sleep mode.
- iii. DVFS can help extend battery life by reducing energy consumption.
- iv. The UPS provides backup power only during short outages.
- v. The ACPI standard defines how computers manage power consumption.
- vi. Battery Management Systems are not necessary for lithium batteries.
- vii. The total power delivered by a backup system is always greater than its total capacity.
- viii. Smart Power Management Policies can help reduce energy costs.
- ix. Redundant power supplies are used to increase system reliability.
- x. Regular maintenance procedures are unnecessary for effective power management systems.

Q7. Scenario Question1:

A company has implemented a new computer power management system that includes DVFS and sleep modes to save energy during non-peak hours. After six months of operation, they notice that their energy costs have decreased significantly, but some employees complain about slower performance during peak hours due to aggressive power-saving settings.

Question: What steps should the company take to balance energy savings with performance needs?

Q8. Scenario Question2:

A data centre relies on multiple UPS units and lithium battery banks for backup during outages but has recently experienced several failures due to poor monitoring practices and lack of routine maintenance. What monitoring capabilities and maintenance procedures should be established to ensure reliable operation of the backup systems?

Practical assessment

You are part of a team tasked with designing and implementing a comprehensive computer power management system for a mid-sized software development company. The company has been facing issues with high energy costs and inefficient power usage, especially during peak hours when employees are engaged in resource-intensive tasks such as software development and testing. Your team will also need to provide an accurate cost estimation for the implementation of this system.

As one of the team, at the end of this assessment, you are tasked with the following:

- a) Design computer power management system architecture
- b) Design charging controller
- c) Integrating a power backup system into power supply unit
- d) Implement user interface to monitoring and control of batteries
- e) Elaborate system cost estimation



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END



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